Algebraic Number Theory

— *EYNTKA* Series —

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Contents

1

Relating Algebra to Number Theory

A central object of study in mathematics is, naturally, the natural numbers $\mathbb{N}$ and, if the negatives are included, the integers $\mathbb{Z}$. So far, it is arguable to say that these objects have played a relatively minor algebraic role in this book: we know $\mathbb{N}$ (resp. $\mathbb{Z}$) is the initial abelian monoid (resp. ring) in their respective categories, that they are free objects, and that from $\mathbb{N}$ we may construct $\mathbb{Z}$, then $\mathbb{Q}$, and from there is any other field of characteristic 0. One aspect of $\mathbb{N}$ and $\mathbb{Z}$ we have barely touched on are their *primes* (recall the Fundamental Theorem of Arithmetic's, [5, $\mathbb{Z}$ is a UFD (Fundamental Theorem of Arithmetics)]). Exercise ref:HERE showed that there are infinitely many primes, and since $\mathbb{Z}$ is a PID we have that being irreducible is equivalent to being prime. However, we have explored little the nature of primes, and more generally have explored little of functions with integer solutions (elements of $f \in \mathbb{Z}[x_1, ..., x_n]$ are called *Diophantine equations*).

Unlike fields, it is in general much harder to ascertain whether an algebraic equation has solutions over the integers (or even integral extensions of the integers). Perhaps the most famous number theory problem demonstrating this difficulty is the following:

Show that for $n \geq 3$, the equation $x^n + y^n = z^n$ has no integer solutions

This problem is known as *Fermat's Last Theorem.* This is trivial if $n = 1$ (ex. $1 + 2 = 3$). It is more interesting to see the case of $n = 2$, that is we want to find all $(a, b, c) \in \mathbb{Z}^3$ ($\gcd(a, b, c) = 1$) such that $x^2 + y^2 = z^2$. Such a triple is called a *Pythagorean triple* and there are infinitely many. However, for $n \geq 3$, it is incredibly more difficult to find whether solutions exist or not, partly due to the fact that many rings that naturally arise in solving this case are *not* unique factorization domains, and partly because there turn out to be some "bad number" that make many of the number theory strategies fail. This particular problem has spurred a lot of the development of modern algebraic and analytic number theory, and shall be touched on later due to some interesting general theory that came from it. In this section we shall cover the mathematics that build built-up in solving Diophantine equations, including some special cases of Fermat's last theorem, and prepare the reader for Class Field theory which shall definitively answer the question of factoring integer-polynomials mod primes[1].

One final intuition that shall become more evident is that the theory of prime factorization in field extensions tie into solving irreducible polynomials mod some prime ideals. This theory extends results from the classical results of solutions or lack thereof for an irreducible polynomial over $\mathbb{Q}$ when considering it mod p for all primes $p \in \mathbb{Z}$. Recall that factorization and irreducibility of monic polynomials over $\mathbb{Q}$, more generally a finite extension $K/\mathbb{Q}$ is exactly reflected to the factorization and irreducibility of monic polynomials over $\mathbb{Z}$ (which will have a ring $\mathcal{O}_K$ that shall be the equivalent[2]) by Gauss's lemma ([5, Gauss's Lemma]). This was the historical origin of number theory, as it is much easier to work with polynomials finite fields (recall [5, Reduction Test] and [5, Limitation of reduction method]).

Material Overview

In this chapter, we shall start by exploring when do $\mathbb{Z}$-polynomials solutions over a field k. The space of solutions shall naturally be a subset of the field k. It shall be labeled as $\mathcal{O}_K$ and called *rings of integers*, and will have many important properties to determine. Similarly to how we extended fields to find solutions to polynomials over fields, rings of integers shall be extended (in parallel to a field extension) to find more solutions. Some key functionals such as the trace, norm, and discriminant shall become important tools to find properties of elements of $\mathcal{O}_K$ and the structure of $\mathcal{O}_K$ itself. As we classify $\mathcal{O}_K$, we shall see that we may consider $\mathcal{O}_K$ as a *lattice* when embedded in $\mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2}$ for appropriate $r_1, r_1 \in \mathbb{N}$. This geometric interpretation of $\mathcal{O}_K$ shall add some important new perspectives onto the elements of $\mathcal{O}_K$[3], and will allow us to define some important inequalities.

After having done a sweep over the structure of $\mathcal{O}_K$, we shall tackle some of it's algebraic structure. Namely, we shall see that they form a new type of ring called a *Dedekind domain.* Dedekind domains are very important types of rings; for those who are keen on algebraic geometry, Dedekind domains are the algebraic counter-part to 1-dimensional smooth curves (or schemes). In fact, we have seen Dedekind domains before in the previous chapter (for example, $\mathbb{Q}[x, y]/(x - y^2)$). Hence, Dedekind domains will be studied in their own right. They shall have a very rich structure: for example, many $\mathcal{O}_K$ will *not* be unique factorization domain (a fact that led to many wrong proofs of Fermat's last theorem), however we shall be able to define an abelian group structure on the ideals of $\mathcal{O}_K$ which *will* have a unique factorization (in the case where $\mathcal{O}_K$ is a PID, we shall see the ideal group I_K is isomorphic to $\mathbb{Q}^\times$, exactly mirroring the structure of Frac($\mathbb{N}$), however in general it may be much more complicated[4]). The discussion on Dedekind domains shall finish with a discussion of the unit's in Dedekind rings, which shall be important for finding the factorization of ideals.

After we realise we ought to be focusing on the *prime ideals* of $\mathcal{O}_K$ rather than the irreducible elements of $\mathcal{O}_K$ (for they are not necessarily prime), we shall start exploring how prime may further decompose as we extend to larger rings of integers. This helps in finding solutions to $\mathbb{Z}$-polynomials for the prime ideals in larger rings will have very rigid structure that can be taken advantage of. We have already seen this before: It shall turn out that $\mathcal{O}_{\mathbb{Q}(i)} = \mathbb{Z}[i]$ (the Gaussian integers). In the

[1]In particular, the *Artin Reciprocity Law* shall give the over-arching result showing how to factorize Diophantine equations

[2]See definition 1.1.3

[3]and of the norm

[4]which is measured by the *class group*, as we shall see

Gaussian integers, we saw that 2 is not prime, for

$$2 = (1 + i)(1 - i) = 1^2 - i^2 = 1 + 1$$

or in terms of ideals, we shall have $(2) = (1 + i)(1 - i)$. Hence, 2 has further factored into new primes. In [5, Application: Sums of Primes and Gaussian Integers], this was used to prove Fermat's Theorem for sums of squares. More generally, we shall see that any prime ideal $\mathfrak{p} \subseteq \mathcal{O}_K$ may factor:

$$\mathfrak{p}\mathcal{O}_L = \mathfrak{B}_1^{e_1} \cdots \mathfrak{B}_k^{e_k} \tag{1.1}$$

Though it may currently seem unrelated, the study of the factorization of primes *almost* mirrors that of the factorization of irreducible polynomials mod $\mathfrak{p}$ after extending fields. In fact, it shall turn out to be the case that *all but finitely many primes* shall factor exactly like how a polynomial factors, the "bad primes" essentially being primes for which an exponent in equation (1.1) is greater than 1, ≥ 2. This sort of exploration leads to a lot of very interesting theory that have applications beyond classical number theory, for example we shall be able to fully classify all abelian extensions of $\mathbb{Q}$.

For those who get motivation through algebraic geometry, algebraic number theory naturally ties into it; it may help to think of the main object of study of this chapter (rings of integers) as further expanding the dictionary between commutative algebra and geometry, namely they shall be important for *normal schemes* which links to the notion of a smooth curve[5]

At the end of this chapter, we shall briefly touch on some analytic number theory, for it's study is central to modern number theory and the author of this books considers that anyone that has reached this stage of the book would find it enlightening to see how the Riemann-zeta function play an important role in number theory.

1.0.1 Warning Some elementary analysis will be assumed in this section, including some properties of the real numbers.

1.1 Ring of Integers

In this chapter, whenever the notation $L/K/\mathbb{Q}$ is used for some letter's L, K, it is implied that L/K are both finite extensions of $\mathbb{Q}$, and L is an extension of K. Unless otherwise specified, the characteristic of any field is implicitly assumed to be 0. We start by defining our terms.

Definition 1.1.1: Algebraic Number Fields

Let K be a field. If $K/\mathbb{Q}$ is a finite extension of $\mathbb{Q}$, then K is called an *algebraic number field* or *number field* if unambiguous.

Example 1.1.2: Number Fields

1. Let $\zeta_n = e^{2\pi/n}$ be a root of unity. Then $\mathbb{Q}(\zeta_n)$ is a number field, called the *Cyclotomic field.*
2. Let $m \in \mathbb{Z}$ be not-a-square. Then Fields of the form $\mathbb{Q}(\sqrt{m})$ is called a *quadratic field.* Such fields always have degree 2. Notice that we can consider any $m \in \mathbb{Z}$, however we get many

[5] See a textbook in algebraic geometry for further details

equality of fields, for example $\mathbb{Q}(\sqrt{12}) = \mathbb{Q}(\sqrt{3})$. The field $\mathbb{Q}(i)$ is both a quadratic field and a Cyclotomic field since $i = \zeta_4$. As an exercise, show that $\mathbb{Q}(\sqrt{-3})$ is a Cyclotomic field.

3. The field $\mathbb{Q}(\sqrt{n}, \sqrt{m})$ is called a *biquadratic* field where n, m are both not-a-square.

4. The field $\mathbb{R}$ and $\mathbb{C}$ are *not* algebraic number fields, are they are not algebraic extensions of $\mathbb{Q}$ (they both contain transcendental elements).

5. Similarly, $k(x)$ for any field k is not an algebraic number field (it is an [algebraic] function field). However, in many ways this field is very similar to algebraic number fields, and shall be given some more attention in section 1.2.

Of key interest to this chapter the study of integer polynomials with solutions fields:

Definition 1.1.3: Ring of Integers (of Number Fields)

Let K be a field of characteristic 0 and identify $\mathbb{Z} \subseteq K$ via the (unique) map $\mathbb{Z} \hookrightarrow K$. Then the *ring of integers* of K is the integral closure of $\mathbb{Z}$ over K ([2, Integral Elements]):

$$\mathcal{O}_K := \overline{\mathbb{Z}}^K = \{a \in K \mid f(a) = 0,\ f(x) \in \mathbb{Z}[x]\}$$

As all fields in consideration shall be number fields $K/\mathbb{Q}$, there is already $\mathbb{Z} \subseteq K$ identified; the generality in the definition is to show that the ring of integers is defined over any field of characteristic 0. Note that there is an equivalent definition for fields of positive characteristic, however as $\mathrm{im}(\mathbb{Z}) \cong \mathbb{Z}/p\mathbb{Z}$ any integral closure of $\mathbb{Z}/p\mathbb{Z}$, $\overline{\mathbb{Z}/p\mathbb{Z}}^K$, is a field extension by [2, Integral Extension over Fields] and so there is no notion of prime ideals which essentially erases the key extra information that is studied. Instead, $\mathbb{F}_p[t]$ becomes the new $\mathbb{Z}$ and a parallel theory of finite extension $K/\mathbb{F}_p(t)$ is studied; these are called *algebraic function fields* and the corresponding $\overline{\mathbb{F}_p[t]}^K$ is the *ring of integers of function fields.*. The two theories have many parallels, but exploring these is beyond the scope of this book, see [9] for more details.

Although, once this chapter becomes it's own book, it may be a good idea to dedicate a section or chapter to this.

By [2, Integral Elements Form Rings], $\mathcal{O}_K \subseteq K$ is a ring. Since we will often work over $\mathbb{Z}$, it is useful to know that instead of taking all roots of polynomials, it is equivalent to take all the roots of *irreducible polynomials*; by using Gauss's lemma, if $f = gh$, $f \in \mathbb{Z}[x]$ and $g, h \in \mathbb{Q}[x]$, then in fact $g, h \in \mathbb{Z}[x]$. Hence, the smallest degree polynomial for which α is a root will be irreducible. An important ring of integer that is worth isolating is the following:

Definition 1.1.4: Algebraic Integers [of Number Fields]

The collection of elements

$$\mathcal{O}_{\overline{\mathbb{Q}}} = \mathcal{O}_{\mathbb{C}} := \overline{\mathbb{Z}}^{\mathbb{C}}$$

are called the *algebraic integers*, and is sometimes denoted $\mathbb{A}$[a].

[a] this notation comes from the theory of *idele rings* that is an important tool in class field theory, see [9]

Just like for fields, there may be multiple "integral closures", and they are all isomorphic which motivates saying they are *the* algebraic integers.

Lemma 1.1.5: Equivalence Definition

Let $K/\mathbb{Q}$ be a finite extension and let $\mathcal{O}_\mathbb{C}$ be the algebraic integers. Then:

$$\mathcal{O}_\mathbb{C} \cap K = \mathcal{O}_K$$

***Proof*:**
exercise.

Example 1.1.6: Ring of Integers

1. $\mathcal{O}_\mathbb{Q} = \mathbb{Z}$. For this reason, the integers in algebraic number theory are sometimes called the rational integers
2. $\mathcal{O}_{\mathbb{Q}(i)} = \mathbb{Z}[i]$. To see this, let $\alpha = a + bi$, $a, b \in \mathbb{Q}$. Then for $\alpha \in \mathcal{O}_{\mathbb{Q}(i)}$, we need to show that the minimal polynomial of α:
$$\alpha^2 - 2a\alpha + (a^2 + b^2) = 0$$
has integer coefficients, that is $-2a$ and $a^2 + b^2$ are integers. First, if $c, d \in \mathbb{Z}$, $c + di \in \mathcal{O}_{\mathbb{Q}(i)}$. Since $\mathcal{O}_{\mathbb{Q}(i)}$ is a ring, if $a + bi \in \mathcal{O}_{\mathbb{Q}(i)}$, then so is $a + bi - c - di = a' + b'i$ for $c, d \in \mathbb{Z}$ where $a', b' \leq 1$. But then from $(a')^2 + (b')^2 = k \in \mathbb{Z}$, we get $a', b' \in \{-1, 0, 1\}$. Then that implies $a, b \in \mathbb{Z}$, showing that $\mathcal{O}_{\mathbb{Q}(i)} = \mathbb{Z}[i]$.
3. In general, it is difficult to find the ring of integers. The difficulty of finding the ring of integers can even show up for quadratic extensions. Consider $F = \mathbb{Q}(\sqrt{-3})$. Then we might be tempted to say that $\mathcal{O}_F = \mathbb{Z}[\sqrt{-3}]$. However, this is not the case, take:
$$\omega = \frac{-1 + \sqrt{-3}}{2} \in \mathbb{Q}(\sqrt{-3})$$
Then $\omega^2 + \omega + 1 = 0$, however $\omega \notin \mathbb{Z}[\sqrt{-3}]$.
4. In general, we will need more tools to find $\mathcal{O}_k$, but in the quadratic case, since the polynomial are relatively simple to work with, we may find all rings of integers. Let $k = \mathbb{Q}(\sqrt{m})$. Then:
$$\mathcal{O}_k = \mathbb{Z}[\sqrt{m}] \qquad m \equiv 2, 3 \bmod 4$$
$$\mathcal{O}_k = \mathbb{Z}\left[\frac{1 + \sqrt{m}}{2}\right] \qquad m \equiv 1 \bmod 4$$
To see this, let $\alpha = r + s\sqrt{m}$, $r, s \in \mathbb{Q}$. If $s = 0$, we've done, hence if $s \neq 0$ we get the monic irreducible polynomial:
$$x^2 - 2rx + r^2 - ms^2$$
Hence, α is algebraic if and only if $2r$ and $r^2 - ms^2$ are integers. Since $2r \in \mathbb{Z}$, $r \in \frac{1}{2}\mathbb{Z}$. Considering now $r^2 - ms^2 \in \mathbb{Z}$, if $r = \alpha/2$ for $\alpha \in \mathbb{Z}$, then subtracting both sides by $(\alpha^2 - 1)/4$ we get that $1/4 - ms^2 \in \mathbb{Z}$ or $m^2 = \mathbb{Z} + 1/4$. Now, this equality holds true if $s = \beta/2$ and $m \equiv 1 \bmod 4$ or $s = \beta$ and $m \equiv 2, 3 \bmod 4$. This now gives our restrictions on our coefficients.

The ring of integers $\mathcal{O}_K/\mathbb{Z}$ hold the core information of $K/\mathbb{Q}$. In particular, if $\mathbf{Fld}_0^{\text{fin}}$ is the category whose objects are finite extensions of $\mathbb{Q}$ and morphisms are field morphisms and $\mathbf{AlgInt}_0^{\text{fin}}$ is the category whose objects are the rings of integers $\mathcal{O}_K$ of finite extensions $K/\mathbb{Q}$ and whose morphisms are ring homomorphisms, these two categories are *isomorphic*. From here onto section 1.1.1 this correspondence is proven. Some of the key facts will require results that are proved later in this section (ex. that all non-trivial ring homomorphism $\mathcal{O}_K \to \mathcal{O}_L$ are injective), however knowing this correspondence is valuable enough that it is worth introducing this result early.

First, if $\varphi : F \to L$ be a field homomorphism (not necessarily a k-homomorphism), then $\varphi|_{\mathcal{O}_F} : \mathcal{O}_F \to \mathcal{O}_L$ is certainly well defined since if $\alpha \in \mathcal{O}_F$, then:

$$a_0 + a_1\alpha + \cdots + a_{n-1}\alpha^{n-1} + \alpha^n = 0 \qquad a_i \in \mathbb{Z}$$

Then $\varphi(a_i\alpha^i) = a_i\varphi(\alpha)^i$, since $a_i \in \mathbb{Z}$. Hence $\varphi(\alpha) \in \mathcal{O}_L$. From this, define $\mathcal{O}_- : \mathbf{Fld}_0^{\text{fin}} \to \mathbf{AlgInt}_0^{\text{fin}}$ which gives the commutative diagram:

$$\begin{tikzcd} F \arrow[r, hook, "\varphi"] \arrow[d, "\mathcal{O}_-"] & L \arrow[d, "\mathcal{O}_-"] \\ \mathcal{O}_F \arrow[r, "\varphi|_{\mathcal{O}_F}"] & \mathcal{O}_L \end{tikzcd}$$

Now given a map $\psi : \mathcal{O}_F \to \mathcal{O}_K$, can we find a map $\Psi : F \to K$ which is furthermore an inverse to the above functor? Yes; to find this functor, we first find how to retrieve the field from the ring of integers:

Proposition 1.1.7: Field of Fraction of Integral Closures of Extensions

Let R be an integral domain, $K = K(R)$ its field of fraction, and $S = \overline{R}^L$ where L/K is an algebraic extension. Then $L = K(S)$. Equivalently, $L = \langle S \rangle_{K(R)}$.

If $R = \mathbb{Z}$ and $K(R) = \mathbb{Q}$ so that $S = \overline{\mathbb{Z}}^L = \mathcal{O}_L$, then $L = K(\mathcal{O}_L)$.

***Proof*:**
Let R, K, S and L be defined as in the question. We shall first show that if $\alpha \in L$ is algebraic over K, then there exists a $d \in R$ such that $d\alpha$ is integral over R. Since α is algebraic over K,

$$\alpha^m + a_1\alpha^{m-1} + \cdots + a_m = 0 \qquad a_i \in K$$

Let d be the least common multiple of the denominators of a_i so that $da_i \in R$ for all i. Then multiplying the above equation by d^m, we get:

$$d^m\alpha^m + d^m a_1\alpha^{m-1} + \cdots + d^m a_m = 0 \qquad a_i \in K$$

re-writing:

$$(d\alpha)^m + da_1(d\alpha)^{m-1} + \cdots + d^m a_m = 0 \qquad a_i \in K,$$

where each $a_i d, ..., a_m d^m \in R$, hence $d\alpha$ is integral over R. Note that this immediately gives the equivalence $L = \langle S \rangle_{K(S)}$.

With that, to see that $L = K(S)$, notice that every $\alpha \in L$ can be written as $\alpha = b/d$ for $d \in R$, hence $L \subseteq K(S)$. Conversely, if $a/b \in K(S)$, where $a, b \in S$ and each satisfy an R-polynomial, then they are certainly satisfy a K-polynomial, and furthermore S is all elements of L that satisfy an R-polynomial, hence $a/b \in L$, giving equality.

Corollary 1.1.8: Basis For Number Field

Let $F/\mathbb{Q}$ be a finite extension of degree n and $(\alpha_1, \alpha_2, ..., \alpha_n)$ an $\mathbb{Q}$-basis for F. Then there exists an $m_i \in \mathbb{Z}$ such that $m_1\alpha_1, m_2\alpha_2, ..., m_n\alpha_n \in \mathcal{O}_F$

***Proof*:**
let $R = \mathbb{Z}$ in proposition 1.1.7.

Theorem 1.1.9: Isomorphic Categories: Fields and Rings of Integers

Let $K/\mathbb{Q}$ be a number field and $\mathcal{O}_K$ the ring of integers of K. Then:

1. If L/K, then $\mathcal{O}_L \cap K = \mathcal{O}_K$. In particular $\mathcal{O}_K \cap \mathbb{Q} = \mathbb{Z}$.
2. $\mathcal{O}_K \otimes_{\mathbb{Z}} \mathbb{Q} \cong K$. Namely, $\mathcal{O}_{-} \otimes_{\mathbb{Z}} \mathbb{Q} : \mathbf{AlgInt}_0^{\text{fin}} \to \mathbf{Fld}_0^{\text{fin}}$ is a functor.

Giving us a correspondence that is covariant under inclusion:

$$\left\{ \begin{matrix} \varphi : F \to K \\ \text{char}(F) = \text{char}(K) = 0 \end{matrix} \right\} \underset{(-)\otimes_{\mathbb{Z}}\mathbb{Q}}{\overset{\overline{\mathbb{Z}}^{(-)}}{\rightleftarrows}} \left\{ \begin{matrix} \varphi : \mathcal{O}_F \to \mathcal{O}_K \\ K(\mathcal{O}_F) = F,\ K(\mathcal{O}_K) = K \end{matrix} \right\}$$

which gives a bijection on Hom's:

$$\text{Hom}_{\mathbf{Fld}_0^{\text{fin}}}(L_1, L_2) \xrightarrow{\sim} \text{Hom}_{\mathbf{AlgInt}_0^{\text{fin}}}(B_1, B_2)$$

And hence there is an isomorphism of categories that preserves the lattice structure (meets and joins):

$$\mathbf{Fld}_0^{\text{fin}} \cong \mathbf{AlgInt}_0^{\text{fin}}$$

***Proof*:**
For the first two points:

1. This is essentially by definition: $\mathcal{O}_k = \overline{\mathbb{Z}}^k$ are all integral elements over $\mathbb{Z}$ that belonging k. Hence,$\mathcal{O}_k \cap \mathbb{Q}$ must be integral over k and also rational. But the only integral elements that are rational are the integers, hence we have equality. Similarly, $\mathcal{O}_L \cap k$ are all integral elements over $\mathbb{Z}$ that also belong to k, which by definition is $\mathcal{O}_k$
2. This comes from [2, Characterization of Localization] and its corollary.

To show the hom's are bijective, we would have to show that a non-trivial ring homomorphism $\varphi : \mathcal{O}_K \to \mathcal{O}_L$ for some two fields K, L is injective and that $L/K/\mathbb{Q}$ is a finite extension: both of these are a direct consequence of the proof of theorem 1.2.6. The lattice structure is preserved as $-\cap K$ and $-\otimes \mathbb{Q}$ preserves the lattice structures. Finally, composing these two functors certainly give the identity on $\mathbf{Fld}_0^{\text{fin}}$ and $\mathbf{AlgInt}_0^{\text{fin}}$, showing they are isomorphic (not just equivalent!) as we sought to show.

Note that finiteness is important: it shall be shown that $\mathcal{O}_K$ for $K/\mathbb{Q}$ a finite extension is a domain with factorization (or atomic domain, recall [5, Atomic Domains]), however $\mathcal{O}_{\mathbb{C}}$ is *not* an atomic domain as shown in [5, Domain with no Factorization] which is certainly desirable as we are generalizing the factorization of primes.

This shows how to relate $\mathcal{O}_K$ with K and a bit about how to move down the lattice, but we have not said much about the structure of $\mathcal{O}_K$. We know it is a ring, but is it a PID? A UFD? Every ring is a $\mathbb{Z}$-module; it is a finitely generated $\mathbb{Z}$-module? Noetherian $\mathbb{Z}$-module? Projective? The strategy in example 1.1.6 to find the structure of $\mathcal{O}_K$ gets much harder for more complicated roots of irreducible, even once we have degree 3, and especially beyond degree 4. Our first goal is to establish the structure of $\mathcal{O}_K$. To find the structure, we will require some tools for detecting elements of $\mathcal{O}_K$: we shall find numerical values for the elements of K to determine whether they are elements of $\mathcal{O}_K$, i.e. we shall find functionals. We shall see that $\mathcal{O}_K$ will be a free $\mathbb{Z}$-module of finite rank, in fact it shall be a particular $\mathbb{Z}$-module known as a *complete* lattice, as shall be demonstrated in section 1.1.4.

1.1.1 Functionals: Field Trace and Field Norms

(this link is useful: here)

Let M/k be a finite extension, and take $\alpha \in M$. We shall define two invariances on α that will detect whether α is algebraic over k or not, detect when it is an algebraic integer, and eventually help us find the structure of $\mathcal{O}_M$ as well as the factorization of elements of $\mathcal{O}_M$.

Define $L_\alpha : M \to M$ by $m \mapsto \alpha m$. This is a k-linear transformation, since:

$$L_\alpha(a + kb) = \alpha(a + kb) = \alpha a + k\alpha b = L_\alpha(a) + kL_\alpha(b)$$

Thus, we can represent this linear transformation in a matrix form (usually with basis given by α). Since the trace and determinant are basis independent calculations, it is meaningful to ponder what the trace/determinant of this transformation and what information it may provide us:

Definition 1.1.10: Trace And Norm

Let M/k be a finite extension, $L_\alpha : M \to M$ be a linear transformation defined by multiplication Then we call the trace of this linear transformation the *trace* of α and the determinant of this linear transformation the *norm* of α.

Example 1.1.11: Trace And Norm

1. Let $\mathbb{C}/\mathbb{R}$ be the extension and $\alpha = x + iy$. Pick basis $\{1, i\}$. Then multiplication is:

$$\begin{pmatrix} x & y \\ -y & x \end{pmatrix}$$

and so the trace is $2x = 2\mathrm{Re}\alpha$ and the norm is $x^2 + y^2 = \|\alpha\|^2$. If you know some complex analysis, then we see that the norm does indeed match with our usual understanding of the norm, and the trace can be thought of as a generalization of taking the real part of the complex number (up to a factor).

2. Let $\alpha = a + b \cdot \left(\frac{1+\sqrt{-7}}{2}\right) \in \mathbb{Z}\left[\frac{1+\sqrt{-7}}{2}\right]$. Then:

$$\mathrm{Nm}_{\mathbb{Q}(\sqrt{-7})/\mathbb{Q}}(\alpha) = \alpha^2 + \alpha\beta + 2\beta^2$$

3. Let $M = k(\alpha)$ be a finite extension, and let

$$m_{\alpha,k}(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$$

be the minimal polynomial. Then this is the same as the characteristic polynomial (since plugging in L_α into the minimal polynomial produces the zero transformation by Cayley Hamilton [a]). Then the trace of α is $-a_{n-1}$ and the norm is $\pm a_0$ depending on whether $m_{\alpha,k}(x)$ is even or odd.

[a] Alternatively, choose basis $1, \alpha, ..., \alpha^{n-1}$ and compute

Naturally, if F/k, $a_1, a_2 \in F$ and $b \in k$:

$$\mathrm{tr}_{F/k}(a_1 + a_2) = \mathrm{tr}_{F/k}(a_1) + \mathrm{tr}_{F/k}(a_2) \quad \mathrm{tr}_{F/k}(ba_1) = b\mathrm{tr}_{F/k}(a_1) \quad \mathrm{tr}_{F/k}(a) = na$$

and:

$$\mathrm{Nm}_{F/k}(a_1a_2) = \mathrm{Nm}_{F/k}(a_1)\mathrm{Nm}_{F/k}(a_2) \quad \mathrm{Nm}_{F/k}(ba_1) = b^n\mathrm{Nm}_{F/k}(a_1) \quad \mathrm{Nm}_{F/k}(b) = b^n$$

Most of the properties of the trace and norm come from linear algebra, and furthermore since every finite separable extension has a primitive element generating the entire field, the norm and trace always show up in the minimal polynomial. In [5, Symmetric Functions], we saw another way of characterizing the trace and norm, which will be emphasized here due to its importance:

Proposition 1.1.12: Trace and Norm via Galois Conjugates

Let M/k be a finite extension where $[M : k] = n$ and let $\alpha \in M$ be algebraic over k. Then:

$$\mathrm{tr}_{M/k}(\alpha) = [M : k(\alpha)] \cdot \sum_{\substack{r \\ \text{root of } m_{\alpha,k}}} r \qquad \mathrm{Nm}_{M/k}(\alpha) = \left(\prod_{\substack{r \\ \text{root of } m_{\alpha,k}}} r \right)^{[M:k(\alpha)]}$$

***Proof*:**
Let $\sigma_i(\alpha) \in \overline{k}$ be the roots of $m_{\alpha,k}(x)$ where $\sigma_i : M \to \overline{k}$ are the embeddings of M into the algebraic closure of k. Suppose first that $M = k(\alpha)$ and is a separable extension. Pick a basis $1, \alpha, ..., \alpha^{n-1}$. Then concretely computing the trace, we get the $n-1$th coefficient of

$$\alpha^n - a_{n-1}\alpha^{n-1} - \cdots - a_1\alpha - a_0 = 0$$

which, if we factor out the above polynomial in a normal closure and collect the terms of the roots as we've done in [5, Symmetric Functions], we see that

$$\alpha_{n-1} = \sum_i \sigma_i(\alpha)$$

that is, it is the sum of the roots. Similarly, we get that $\mathrm{Nm}_{F/k}(\alpha) = a_0 = \prod_i \sigma_i(\alpha)$, giving us both results.

For the general finite extension case, we may use induction and transitivity. The inseparable case is left as an exercise.

Corollary 1.1.13: Trace and Norm Via Embeddings

Let L/k be a separable extension of degree n, an $\sigma_1, \sigma_2, ..., \sigma_n$ the embedding of L into it's Galois closure or $\overline{k}$. Then for $\alpha \in L$:

$$\operatorname{tr}_{L/k}(\alpha) = \sum_{i}^{n} \sigma_i(\alpha) \qquad \operatorname{Nm}_{L/k}(\alpha) = \prod_{i}^{n} \sigma_i(\alpha)$$

***Proof*:**
this is just a re-phrasing of the previous result. Note that there may be $\sigma_i(\alpha)$ that repeat since σ_i may be fixed in $k(\alpha)$.

This immediately shows us that for algebraic elements, $\operatorname{tr}_{M/k}(\alpha), \operatorname{Nm}_{M/k}(\alpha) \in k$, since $\pm a_{n-1}, a_0 \in k$. An easy way to remember that that the trace gives a_{n-1} and Norm gives a_0 is that it is not possible that a minimal polynomial have 0 constant term since

$$x^n + a_{n-1}x^{n-1} + \cdots + a_1 x$$

is reducible. Hence, $\operatorname{Nm}_{M/k}(\alpha) = a_1$ while $\operatorname{tr}_{M/k}(\alpha) = a_{n-1}$. To relate this back to our goal of understanding $\mathcal{O}_M$, we would like to identify when $\alpha \in M$ belongs in $\mathcal{O}_M$ using tr and Nm. This is exactly when the coefficients are integers, that is $\operatorname{tr}_{M/k}(\alpha), \operatorname{Nm}_{M/k}(\alpha) \in \mathbb{Z}$. We've in fact proved something slightly more general that is important to point out for when we are working with intermediary extensions:

Corollary 1.1.14: Trace and Norm on Rings of Integers

Let R be an integrally closed integral domain ($\overline{R}^K = R$), and let $F/K(R)$ be a finite field extension of the field of fraction of R. Then if $\alpha \in F$ is integral over R,

$$\operatorname{tr}_{F/K(R)}(\alpha) \in R \qquad \operatorname{Nm}_{F/K(R)}(\alpha) \in R$$

A particular consequence of this is $\operatorname{tr}_{L/k}(\mathcal{O}_L) \subseteq \mathcal{O}_k$

***Proof*:**
If α is integral, then certainly so are all its Galois conjugates. But then adding and multiplying all the Galois conjugates remains integral, meaning the coefficients of the corresponding polynomials remains in R, hence the trace and norm of α is in R.

In particular, if $\alpha \in \mathcal{O}_L = \overline{\mathbb{Z}}^L \subseteq \overline{k}^L$, $\operatorname{tr}_{L/k}(\alpha) \in \mathcal{O}_L \cap k = \mathcal{O}_k$ by theorem 1.1.9.

If $R = \mathbb{Z}$ and $F/\mathbb{Q}$ is a finite extension, we get can detect potential integral elements using the trace and norm! Since we may be working with intermediate fields $M/L/k$, it is useful to know how to compute the trace and norm given information about the trace and norm of the intermediate fields:

Proposition 1.1.15: Trace and Norm w.r.t. Towers

Let $M/L/K$ be finite extensions. Then:

$$\mathrm{tr}_{M/K}(\alpha) = \mathrm{tr}_{L/K}(\mathrm{tr}_{M/L}(\alpha)) \qquad \mathrm{Nm}_{M/K}(\alpha) = \mathrm{Nm}_{L/K}(\mathrm{Nm}_{M/L}(\alpha))$$

***Proof*:**
Let $\sigma_1, \sigma_2, ..., \sigma_n$ be the embeddings of L in $\mathbb{C}$ which fix K pointwise and let $\tau_1, \tau_2, ..., \tau_m$ be the embeddings in $\mathbb{C}$ fixing L pointwise. We will want to compose σ_i with the τ_j, so we will extend our embeddings as follows: Let N/M be the normal closure of M so that all σ_i, τ_j can be extended to automorphisms of N; relabel our extensions σ_i, τ_j (i.e. keep the same labeling). Then:

$$\mathrm{tr}_{L/k}(\mathrm{tr}_{M/L}(\alpha) = \sum_i^n \sigma_i \left(\sum_j^m \tau_j(\alpha) \right) = \sum_{i,j} \sigma_i \tau_j(\alpha)$$

$$\mathrm{Nm}_{L/k}(\mathrm{Nm}_{M/L}(\alpha) = \prod_i^n \sigma_i \left(\prod_j^m \tau_j(\alpha) \right) = \prod_{i,j} \sigma_i \tau_j(\alpha)$$

It only remains to show that the mappings $\sigma_i \tau_j$ when restricted to M give the embeddings of M in $\mathbb{C}$ fixing k pointwise. Certainly they fix k pointwise and there is the right number of them, so this follows by showing they are all distinct (TBD).

Note that it is not necessarily sufficient that the trace and norm is integral:

Example 1.1.16: Trace And Norm Insufficient To Detect Integral Elements
Let $L = \mathbb{Q}(\alpha)$ where $\alpha^4 + 2\alpha^3 + 1/2\alpha^2 + 1/2\alpha + 2\alpha = 0$. Note that $x^4 + 2x^3 + 1/2x^2 + 1/2x + 3$ is irreducible, hence this is a degree 4 extension. Furthermore,

$$\mathrm{tr}_{L/k}(\alpha) = 2 \qquad \mathrm{tr}_{L/K}(\alpha) = 3$$

But certainly $\alpha \notin \mathcal{O}_L$.

A notable exception to this is for quadratic extensions, since any extension of degree 2 is quadratic, giving only 2 possible coefficients:

Example 1.1.17: Rings of Integers of Quadratic Extensions
Let $K = \mathbb{Q}(\sqrt{n})$ where n is square free, and consider $\alpha = a + b\sqrt{n} \in \mathcal{O}_K$ where a, b are square free. Then:

$$\mathrm{tr}_{K/\mathbb{Q}}(\alpha) = 2a \in \mathbb{Z} \qquad \mathrm{Nm}_{K/\mathbb{Q}}(\alpha) = (a + b\sqrt{n})(a - b\sqrt{n}) = a^2 - nb^2 \in \mathbb{Z}$$

We may ask when $2a \in \mathbb{Z}$, that is when is it possible that $a = 1/2$? Let's say it was, so that

$$\left(\frac{1}{2}\right)^2 - nb^2 \in \mathbb{Z} \qquad \Leftrightarrow \qquad \frac{b^2 n}{4} \in \mathbb{Z} + \frac{1}{4}$$

hence, $4b^2 n \in \mathbb{Z}$. Since n is square-free, $n \neq m^2$ implying that we cannot have $b^2 = p^2/m^2 \cdot m^2$, so $2b \in \mathbb{Z}$. Thus, if $a = 1/2$, it must be that $b = 1/2$. Similarly, if b is an integer, a has to be an integer.

Now, is $\alpha = \frac{1}{2} + \frac{1}{2}\sqrt{n}$ an algebraic integer? For this to be the case, it would have to solve the equation:

$$\alpha^2 - \alpha + \frac{1-n}{4} = 0$$

or equivalently, $\mathrm{Nm}(\alpha) = \frac{1-n}{4}$. Hence, we need $n \equiv 1 \bmod 4$. Thus, we have

$$\mathcal{O}_{\mathbb{Q}(\sqrt{d})} = \begin{cases} \mathbb{Z}[\sqrt{d}] & d \not\equiv 1 \bmod 4 \\ \mathbb{Z}\left[\frac{1+\sqrt{n}}{2}\right] & d \equiv 1 \bmod 4 \end{cases}$$

A final result in this section is to find the group of units of $\mathcal{O}_K$. Note that in general, if $\alpha \in \mathcal{O}_K$, then $1/\alpha \notin \mathcal{O}_K$; for example $2 \in \mathcal{O}_{\mathbb{Q}}$, but certainly $1/2 \notin \mathcal{O}_{\mathbb{Q}}$. In fact, the only units in $\mathcal{O}_{\mathbb{Q}}$ are ± 1. If $\mathcal{O}_K$ contains roots of unity, then certainly their inverse is in $\mathcal{O}_K$ (for if $\alpha = \zeta_n$ $1/\zeta = \zeta_n^{n-1}$). There are in general more units in $\mathcal{O}_K$, and the group of units of a ring of integers shall be fully characterized in section 1.2.4. With the norm, we have a way of detecting units:

Proposition 1.1.18: Norm And Units

Let L/K be a finite field extension. Then

$$\alpha \in \mathcal{O}_L^* \qquad \Leftrightarrow \qquad \mathrm{Nm}_{L/K}(\alpha) \in \mathcal{O}_K^*$$

In the space case where $K = \mathbb{Q}$, then $\alpha \in \mathcal{O}_L^*$ if and only if $\mathrm{Nm}_{L/\mathbb{Q}}(\alpha) = \pm 1$.

***Proof*:**
Let $\alpha \in \mathcal{O}_L^*$ so that $\mathrm{Nm}_{L/K}(\alpha) \in \mathcal{O}_K$. If α is a unit in $\mathcal{O}_L$, then $1/\alpha \in \mathcal{O}_L$ exists. Then:

$$1 = \mathrm{Nm}_{L/K}(1) = \mathrm{Nm}_{L/K}(\alpha \cdot 1/\alpha) = \mathrm{Nm}_{L/K}(\alpha)\mathrm{Nm}_{L/K}(1/\alpha)$$

hence, $\mathrm{Nm}_{L/K}(\alpha), \mathrm{Nm}_{L/K}(1/\alpha) \in \mathcal{O}_K^\times$. Conversely, if $\mathrm{Nm}_{L/K}(\alpha) \in \mathcal{O}_K$, then there exists a β such that

$$\beta \mathrm{Nm}_{L/K}(\alpha) = 1$$

Then by definition of the norm:

$$1 = \beta \mathrm{Nm}_{L/K}(\alpha) = \beta \prod_{\sigma} \sigma(\alpha) = \alpha \left(\beta \prod_{\sigma \neq \mathrm{id}} \sigma(\alpha) \right) = \alpha y$$

where $y \in \mathcal{O}_L$ since each $\sigma(\alpha) \in \mathcal{O}_L$, which gives the result.

Proposition 1.1.19: Norm And Irreducible

Let $L/\mathbb{Q}$ be a finite extension and $\alpha \in \mathcal{O}_L$. Then if p is prime:

$$\mathrm{Nm}_{L/\mathbb{Q}}(\alpha) = p \implies \alpha \text{ is irreducible}$$

***Proof*:**
Suppose α where reducible so that $\alpha = \beta\gamma$ where β, γ or not units, hence have norm greater than 1. Then:

$$\begin{aligned}\mathrm{Nm}_{L/\mathbb{Q}}(\alpha) &= \mathrm{Nm}_{L/\mathbb{Q}}(\beta\gamma)\\ &= \mathrm{Nm}_{L/\mathbb{Q}}(\beta)\mathrm{Nm}_{L/\mathbb{Q}}(\gamma)\\ &= p\end{aligned}$$

a contradiction.

Note that the converse is not true

Example 1.1.20: Prime, But Norm Not Prime
Let $K = \mathbb{Q}(\sqrt{2})$ so that $\mathcal{O}_K = \mathbb{Z}[\sqrt{2}]$. Then $\mathrm{Nm}_{K/\mathbb{Q}}(5) = 5^2 = 25$, which is certainly not prime, however 5 is prime in $\mathbb{Z}[\sqrt{2}]$. This is at this stage a tedious calculation: assume that it factors $5 = \alpha\beta$ where $\alpha, \beta \in \mathbb{Z}[\sqrt{2}]$, $\alpha = a + bi$, $\beta = c + di$, and:

$$\mathrm{Nm}_{K/\mathbb{Q}}(\alpha) = a - 2b = 5 = c - 2d = \mathrm{Nm}_{K/\mathbb{Q}}(\beta)$$

but no such a, b, c, d exist. Later, we shall be able to deduce this almost immediately using Dedekind-Kummer's Theorem (see theorem 1.4.7).

In lemma 1.2.27, we shall give a partial converse which shall as a corollary imply:

$$\alpha \text{ prime} \implies \mathrm{Nm}_{K/\mathbb{Q}}(\alpha) = p^n$$

for appropriate n. Note how α must be prime, which is *stronger* than irreducible. However, we will often be working over rings where we have few prime elements but many irreducible elements. In section 1.2 we shall show how to "fix" this problem to make the norm more useful.

1.1.2 Functionals: Discriminant

As we saw, if $\alpha \in F/\mathbb{Q}$ is integral, then $\mathrm{tr}(\alpha), \mathrm{Nm}(\alpha) \in \mathbb{Z}$. However, the converse is not necessarily true as we saw in example 1.1.16. We will require a more powerful result, one that will verify that an element is an element of a ring of integers. For this, we shall bring in some multilinear algebra and define a new tool that shall become invaluable finding a *basis* for $\mathcal{O}_K$ (recall that it was mentioned that $\mathcal{O}_K$ shall be proven to be a finitely generated free $\mathbb{Z}$-module). Given a basis can be found, then an element can be determined to be integral using the basis. The tool that shall be used to detect a basis for $\mathcal{O}_K$ is called the *field discriminant*. The norm and the trace shall both be used in the definition of the field discriminant.

Some Multi-Linear Algebra

Recall from [5, Multilinear Forms] that multi-linear forms have matrix representations and so the determinant of multi-linear maps is well-defined:

Definition 1.1.21: [General] Discriminant

Let $\varphi : V \times V \to k$ be a symmetric k-bilinear map ($\varphi(x,y) = \varphi(y,x)$). Then given a basis $v_1, v_2, ..., v_n$, the *discriminant* of φ is:

$$\operatorname{disc}(\varphi) = \det \begin{pmatrix} \varphi(v_1,v_1) & \varphi(v_1,v_2) & \cdots & \varphi(v_1,v_n) \\ \varphi(v_2,v_1) & \varphi(v_2,v_2) & \cdots & \varphi(v_2,v_n) \\ \vdots & \ddots & \vdots & \\ \varphi(v_n,v_1) & \varphi(v_n,v_2) & \cdots & \varphi(v_n,v_n) \end{pmatrix}$$

and is denoted $\operatorname{disc}(\varphi)$.

Let $\{\beta_1, \beta_2, ..., \beta_n\}$ be a basis for V and $\{v_1, v_2, ..., v_n\}$ be some collection of elements so that

$$v_i = \sum_i a_i \beta_i$$

Then:

$$\varphi(v_k, v_l) = \sum_{i,j} \varphi(a_{ki}\beta_i, a_{lj}\beta_j) = a_{ki}\varphi(\beta_i, \beta_j)a_{kl}$$

hence, as matrices, we have the (component-wise) equality:

$$(\varphi(v_k, v_l)) = A \cdot (\varphi(e_i, e_j)) \cdot A^{\mathrm{tr}}$$

where $A = (a_{ij})$. Then if we have a change of basis, we have the determinant of the matrix is:

$$\det((\varphi(f_i, f_j)) = \det(A)^2 \cdot \det((\varphi(\beta_i, \beta_j))) \tag{1.2}$$

Since $A \in \mathrm{GL}(k)$, we see that the discriminant is defined up to the square of a unit. Hence, if $\operatorname{disc}(\varphi) \neq 0$, $\operatorname{disc}(\varphi)$ is well-defined in $k^\times/(k^\times)^2$ If $K = \mathbb{Q}$, then $(\mathbb{Q}^\times)^2 = 1$, hence the discriminant is well-defined in $\mathbb{Q}^\times$.

Invertible matrices allowed us to define an isomorphism between V and $V^\vee$ when V is finite dimensional. We shall require this same property for bilinear forms. For this, the notion of non-generate form needs to be upgraded for bilinear forms:

Definition 1.1.22: Non-degenerate Form

let φ be a bilinear form. Then φ is *non-generate* if φ has a non-zero discriminant

Proposition 1.1.23: Non-degenerate Form Equivalences

let φ be a bilinear form. Then φ is *non-generate* if any of the following equivalent conditions hold:

1. φ has a non-zero discriminant
2. the left kernel is zero: $\{v \in V \mid \varphi(v,x) = 0, \ \forall x \in V\}$
3. the right kernel is zero: $\{v \in V \mid \varphi(x,v) = 0, \ \forall x \in V\}$

> ***Proof*:**
> exercise

If φ is non-degenerate, it always induces an isomorphism

$$V \mapsto V^\vee \qquad v \mapsto (x \mapsto \varphi(v, x))$$

Showing that non-degenerate maps φ give a "representation" of the hom-dual. The particular symmetric k-bilinear form that will be useful in finding the rank of the ring of integers is the one built from the trace, [5, Trace of a Finite Field Extension]; in this chapter we call it the *trace pairing* and write it with the lower-case tr used throughout,

$$\operatorname{tr}_{F/k} : F \times F \to k \qquad (\alpha, \beta) \mapsto \operatorname{tr}_{F/k}(\alpha\beta)$$

Its nondegeneracy is a statement about *separability*, not about F alone: if F/k is separable then the trace pairing is nondegenerate ([5, The Trace Form of a Separable Extension Is Nondegenerate]). The hypothesis is not decorative. Over $k = \mathbb{F}_p(t)$ the extension $F = k(t^{1/p})$ has $\operatorname{tr}_{F/k} \equiv 0$, so the pairing vanishes identically. Nor can separability be dodged by arguing that $\operatorname{tr}_{F/k}(\alpha\alpha^{-1}) = \operatorname{tr}_{F/k}(1) = [F : k] \neq 0$ for $\alpha \neq 0$: that degree is computed *in* k, and it dies whenever $\operatorname{char} k \big| [F : k]$ — already for $\mathbb{F}_4/\mathbb{F}_2$, which is separable and whose trace pairing is nondegenerate all the same.

Every extension we pair below is separable, so by proposition 1.1.23 the left kernel is trivial and we have an isomorphism

$$F \xrightarrow{\cong} \operatorname{Hom}_k(F, k) = F^* \qquad x \mapsto (y \mapsto \operatorname{tr}_{F/k}(xy))$$

The submodule of F that this isomorphism carries onto a prescribed integrality condition is the *integral trace dual*, defined in definition 1.1.35 once we have the ring of integers in hand to state it for.

Field Discriminant

With a general overview of discriminants, we return to detecting algebraic integers more precisely using the discriminant. First the determinant, det, is an alternating n-form, meaning $(\det)^2$ is a symmetric n-form. Note that this is the same discriminant introduced in [5, Important Functional: Resultant and Discriminant]; the equivalence shall soon be shown.

Definition 1.1.24: [Field] Discriminant

Let L/k be a finite extension of degree n, and let $\sigma_1, \sigma_2, ..., \sigma_n \in \operatorname{Emb}_k(L)$ be the k-embeddings of L in $\mathbb{C}$. Then for any n-tuple of elements $\alpha_1, \alpha_2, ..., \alpha_n \in L$, define the *discriminant* of $\alpha_1, \alpha_2, ..., \alpha_n$ to be:

$$\operatorname{disc}_{L/k}(\alpha_1, \alpha_2, ..., \alpha_n) = \det \begin{pmatrix} \sigma_1(\alpha_1) & \cdots & \sigma_1(\alpha_n) \\ \vdots & \ddots & \vdots \\ \sigma_n(\alpha_1) & \cdots & \sigma_n(\alpha_n) \end{pmatrix}^2 = \det((\sigma_i(\alpha_j))_{ij})^2$$

Notice that the square makes the discriminant independent of the ordering of the σ_i and α_j. If we chose a different set of elements $(\beta_1, \beta_2, ..., \beta_n)$ that differ from $(\alpha_1, \alpha_2, ..., \alpha_n)$ by a change of basis, that is $\vec{\alpha} = A\vec{\beta}$, then as we observed in equation 1.2, we get

$$\mathrm{disc}_{L/k}(\beta_1, \beta_2, ..., \beta_n) = \det(A)^2 \cdot \mathrm{disc}_{L/k}(\alpha_1, \alpha_2, ..., \alpha_n)$$

meaning the discriminant is not quite basis independent, meaning it is properly defined for operators in $L^\times / L^{\times 2}$ (where $L^\times$ are the units of L, since det maps invertible matrices to units, see [5, Invertability And Determinants]).

Example 1.1.25: Field Discriminant

1. Let $K = \mathbb{Q}(\sqrt{m})$ where m is square free. Then K has a basis $A = \{1, \sqrt{m}\}$ and $B = \left\{1, \frac{1+\sqrt{m}}{2}\right\}$. If $m \equiv 2, 3 \bmod 4$, then the former is a basis for $\mathcal{O}_K$ and if $m \equiv 1 \bmod 4$, then then the latter is a basis for $\mathcal{O}_K$. Computing both, we get:

$$\mathrm{disc}_{K/\mathbb{Q}}(A) = \det \begin{pmatrix} 1 & \sqrt{m} \\ 1 & -\sqrt{m} \end{pmatrix}^2 = (-\sqrt{m} - \sqrt{m})^2 = 4m$$
$$\mathrm{disc}_{K/\mathbb{Q}}(B) = \det \begin{pmatrix} 1 & \frac{1+\sqrt{m}}{2} \\ 1 & \frac{1-\sqrt{m}}{2} \end{pmatrix}^2 = \left(\frac{-2\sqrt{m}}{2}\right)^2 = m$$

Yet another basis for $\mathbb{Q}(\sqrt{m})$ is $C = \{1, \sqrt{m}/10\}$. In that case:

$$\mathrm{disc}_{K/\mathbb{Q}}(C) = \det \begin{pmatrix} 1 & \frac{\sqrt{m}}{10} \\ 1 & -\frac{\sqrt{m}}{10} \end{pmatrix} = \left(\frac{-\sqrt{m}}{5}\right)^2 = \frac{m}{25}$$

However, C is not a basis for $\mathcal{O}_K$, namely since the denominator is too big. Notice that the first two discriminant integers, while the third one is not.

2. Let $K = \mathbb{Q}(\sqrt[3]{2})$. Then $A = \{1, \sqrt[3]{2}, \sqrt[3]{3}\}$ is a basis for K, and:

$$\mathrm{disc}_{K/\mathbb{Q}}(A) = \begin{pmatrix} 1 & \sqrt[3]{2} & \sqrt[3]{4} \\ 1 & \sqrt[3]{2}\zeta_3 & \sqrt[3]{3}\zeta_3^2 \\ 1 & \sqrt[3]{2}\zeta_3^2 & \sqrt[3]{3}\zeta_3 \end{pmatrix}^2 = -108 = -2^2 \cdot 3^3$$

showing that the discriminant can be negative. We shall come back to seeing how this is useful for finding whether $\mathcal{O}_K = \mathbb{Z}[\sqrt[3]{2}]$.

The following alternative form is useful for an alternate way of computing the discriminant as well as important for deducing some theoretical properties of the elements inputted into the discriminant:

Proposition 1.1.26: Discriminant Via Trace-Pairing

$$\mathrm{disc}_{L/k}(\alpha_1, \alpha_2, ..., \alpha_n) = \det(\mathrm{tr}_{L/k}(\alpha_i \alpha_j))$$

***Proof*:**
This follows immediately from the matrix equation. If $[\sigma_i(\alpha_j)]$ represents the matrix:

$$[\sigma_j(\alpha_i)][\sigma_i(\alpha_j)] = [\sigma_1(\alpha_i\alpha_j) + \cdots + \sigma_n(\alpha_i\alpha_j)] = [\mathrm{tr}_{L/k}(\alpha_i\alpha_j)]$$

Using the fact that the determinant is multiplicative and invariant under transposes, we get our result.

In particular, the above form allows for the detection of sets of algebraic integers:

Corollary 1.1.27: Detecting Algebraic Integers using Discriminant

Let K be a number field and $\alpha_1, \alpha_2, ..., \alpha_n$ a $\mathbb{Q}$-basis. Then $\text{disc}_{K/\mathbb{Q}}(\alpha_1, ..., \alpha_n) \in \mathbb{Q}$, and if all $\alpha_1, \alpha_2, ..., \alpha_n$ are algebraic integers, then $\text{disc}(\alpha_1, \alpha_2, ..., \alpha_n) \in \mathbb{Z}$.

***Proof*:**
we know that $\text{tr}_{K/\mathbb{Q}}(\alpha_i\alpha_j) \in \mathbb{Q}$ by proposition 1.1.12 and are integers via corollary 1.1.14, hence $\text{disc}_{K/\mathbb{Q}}(\alpha_1, \alpha_2, ..., \alpha_n) \in \mathbb{Q}$, and if they are all algebraic integers then $\text{disc}_{K/\mathbb{Q}}(\alpha_1, \alpha_2, ..., \alpha_n) \in \mathbb{Z}$.

Corollary 1.1.28: Discriminant And Linear Independence

$\text{disc}_{K/\mathbb{Q}}(\alpha_1, \alpha_2, ..., \alpha_n) = 0$ if and only if $\alpha_1, \alpha_2, ..., \alpha_n$ are $\mathbb{Q}$-linearly dependent.

***Proof*:**
If the $\alpha_1, \alpha_2, ..., \alpha_n$ are $\mathbb{Q}$-linearly dependent, then two of the columns would be linearly dependent, hence $\text{disc}_{K/\mathbb{Q}}(\alpha_1, \alpha_2, ..., \alpha_n) = 0$. Conversely, assuming $\text{disc}_{K/\mathbb{Q}}(\alpha_1, \alpha_2, ..., \alpha_n) = 0$, then the rows of the matrix $[T(\alpha_i\alpha_j)]$ are linearly dependent; let's label them R_i so that for some $a_1, a_2, ..., a_n$ which are not all zero:

$$a_1R_1 + \cdots + a_nR_n = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

For the sake of contradiction, let's say that the $\alpha_1, \alpha_2, ..., \alpha_n$ are $\mathbb{Q}$-linearly independent. Then they form a $\mathbb{Q}$-basis for K^n. Consider:

$$\alpha = a_1\alpha_n 1 + \cdots + a_n\alpha_n$$

where $\alpha \neq 0$ since some $a_i \neq 0$ and $(\alpha_1, \alpha_2, ..., \alpha_n)$ is linearly independent. Then $(\alpha\alpha_1, ..., \alpha\alpha_n)$ also forms a $\mathbb{Q}$-basis for K. By the linear dependence of the row space, we have that for each j

$$\text{tr}_{K/\mathbb{Q}}(\alpha\alpha_j) = \sum_i a_i \text{tr}_{K/\mathbb{Q}}(\alpha_i\alpha_j) = 0$$

But not for any $t \in K$, $t = \sum b_i\alpha\alpha_j$, and by our above observation that implies that $\text{tr}_{K/\mathbb{Q}}(t) = 0$, however that is a contradiction since we know that $\text{tr}_{K/\mathbb{Q}}(1) = n$. Hence, it must be that $(\alpha_1, \alpha_2, ..., \alpha_n)$ are linearly independent, completing the proof.

The corollary can be thought of as motivation for the name discriminant: If $K/\mathbb{Q}$ is a number field of dimension n, the discriminant will "discriminate" sets that are bases for $K/\mathbb{Q}$.

Since number fields are finite separable extensions, we always find a primitive element for a separable algebraic extension, the following theorem gives another important interpretation of the discriminant:

Proposition 1.1.29: Discriminant via Primitive Element

Let L/K be a finite separable field extension, with α a primitive element and $m_{\alpha,k}(x) \in K[x]$ the minimal polynomial with roots $\alpha_1, ..., \alpha_n$ in the splitting field of L. Then the *[field] discriminant* of L[a] is:

$$\Delta_K^2 = \left(\prod_{1 \le i < j \le n} (\alpha_i - \alpha_j) \right)^2 = \mathrm{Nm}_{L/K}(m'_{\alpha,k}(\alpha))$$

where $m'_{\alpha,k}(x)$ is the derivative of the minimal polynomial. If unambiguous the discriminant is often denoted

$$\mathrm{disc}_{L/K}(\alpha)$$

[a] In [5, Galois Groups of Polynomials], we called this the discriminant of $m_{\alpha,k}(x)$

***Proof*:**
Take as basis $1, \alpha, \alpha^2, ..., \alpha^{n-1}$. This comes down to noticing that $(\mathrm{tr}_{B/A}(\alpha^i \alpha^j))$ form the matrix:

$$\begin{pmatrix} 1 & \mathrm{tr}(\alpha) & \mathrm{tr}(\alpha^2) & \cdots & \mathrm{tr}(\alpha^{n-1}) \\ \mathrm{tr}(\alpha) & \mathrm{tr}(\alpha^2) & \mathrm{tr}(\alpha^3) & \cdots & \mathrm{tr}(\alpha^n) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \mathrm{tr}(\alpha^{n-1}) & \mathrm{tr}(\alpha^n) & \mathrm{tr}(\alpha^{n+1}) & \cdots & \mathrm{tr}(\alpha^{2n-2}) \end{pmatrix}$$

gives us:

$$\begin{pmatrix} \sum_i \alpha_i^0 & \sum_i \alpha_i^1 & \sum_i \alpha_i^2 & \cdots & \sum_i \alpha_i^{n-1} \\ \sum_i \alpha_i^1 \alpha & \sum_i \alpha_i^2 & \sum_i \alpha_i^3 & \cdots & \sum_i \alpha_i^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \sum_i \alpha_i^{n-1} & \sum_i \alpha^n & \sum_i \alpha_i^2 & \cdots & \sum_i \alpha_i^{2n-2} \end{pmatrix}$$

This matrix can be split into:

$$\begin{pmatrix} 1 & 1 & \cdots \\ \alpha_1 & \alpha_2 & \cdots \\ \alpha_1^2 & \alpha_2^2 & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} 1 & \alpha_1 & \alpha_1^2 & \cdots \\ 1 & \alpha_2 & \alpha_2^2 & \cdots \\ \vdots & \vdots & \ddots & \end{pmatrix}$$

which are the Vandermonde matrices we saw when dealing with the symmetric group. Then by simple computation we get:

$$\det(\mathrm{tr}_{B/A}(\alpha_i \alpha_j)) = \prod_{1 \le i < j \le n} (x_i - x_j)$$

For the final equality, notice that:

$$\mathrm{Nm}_{L/K}(m_{\alpha,k}(\alpha)) = \prod_{r=1}^{n} \sigma_r(m'_{\alpha,k}(\alpha)) = \prod_{r=1}^{n} m'_{\alpha,k}(\alpha_r) \stackrel{!}{=} \prod_{s \ne r} (\alpha_r - \alpha_s)$$

where $\stackrel{!}{=}$ comes from differentiating $m_{\alpha,k}(x)$ and plugging in α_r, completing the proof.

If $L = K[\alpha]$, then we shall often write $\mathrm{disc}_{L/K}(\alpha)$ instead of $\mathrm{disc}_{L/K}(\alpha, \alpha_2, .., \alpha_n)$. Note that this theorem shows that the discriminant cannot detect a basis on a non-separable extension:

Corollary 1.1.30: Discriminant And Separability

Let F/k be a finite extension. Then $\mathrm{disc}_{F/k}$ is a nonzero function if and only if F is separable

If you recall that the resultant is zero if a two polynomials share a root, then this shows how the discriminant is also related to the resultant.

***Proof*:**
By proposition 1.1.29, we have the discriminant is the product of the differences of roots. If all the roots are distinct, the discriminant is nonzero, and similarly if the extension is inseparable so that there are repeated roots, the discriminant is zero.

Example 1.1.31: Discriminant On Cyclotomic extension
Let ζ_n be an pth primitive root of unity and consider $K = \mathbb{Q}[\zeta_p]$. Then recall from [5, Cyclotomic Polynomials and Fields] that it's minimal polynomial is:

$$f(x) = 1 + x + x^2 + \cdots + x^{p-1}$$

Then to compute $f'(\zeta_n)$, recall that

$$x^p - 1 = (x-1)f(x) \implies px^{p-1} = f(x) + (x-1)f'(x) \implies f'(\zeta_n) = \frac{p}{\zeta_p(\zeta_p - 1)}$$

Taking the norm, we get:

$$\mathrm{Nm}_{K/\mathbb{Q}}(f'(\zeta_p)) = \frac{\mathrm{Nm}_{K/\mathbb{Q}}(p)}{\mathrm{Nm}_{K/\mathbb{Q}}(\zeta_p)\mathrm{Nm}_{K/\mathbb{Q}}(\zeta_p) - 1} = \frac{p^{p-1}}{1 \cdot p}$$

where $\mathrm{Nm}_{K/\mathbb{Q}}(\zeta_p - 1) = p$ is left as an exercise. Hence:

$$\mathrm{disc}_{K/\mathbb{Q}}(\zeta_p) = \mathrm{Nm}_{K/\mathbb{Q}}(f'(\zeta_p)) = p^{p-2}$$

The precise value of $\mathrm{disc}(\zeta_n)$ shall be covered in lemma 1.4.65, however it is relatively easy to see that for arbitrary n, $\mathrm{disc}(\zeta_n)\big|n^{\varphi(n)}$ where φ is the Euler totient. If f is the [monic] irreducible polynomial for ζ_n over $\mathbb{Q}$ for arbitrary $n \in \mathbb{N}$, we get that

$$x^n - 1 = f(x)g(x) \qquad g(x) \in \mathbb{Z}[x]$$

Differentiating and evaluating at ζ_n, we get $n = \zeta_n f'(\zeta_n)g(\zeta_n)$. Taking the norm, we get:

$$n^{\varphi(n)} = \pm\mathrm{disc}(\zeta_n)\mathrm{Nm}(\zeta_n g(\zeta_n))$$

which since $\mathrm{Nm}(\zeta_n g(\zeta_n)) \in \mathbb{Z}$, we get that $\mathrm{disc}(\zeta_n)\big|n^{\varphi(n)}$.

Exercise 1.1.1

1. Give a second proof of [5, The Trace Form of a Separable Extension Is Nondegenerate], independent of the Vandermonde argument used there: show that if F/K is finite and separable, then $\mathrm{tr}_{F/K}(a)$ cannot be 0 for all $a \in F$ (Hint: use Dedekind's lemma, [5, linear independence of Automorphisms (Dedekind's Lemma)]). Deduce that the bilinear form $(x, y) \mapsto \mathrm{tr}_{F/K}(xy)$ is non-degenerate.

2. (series of question for Stickelberger's theorem, that $d_K \equiv 0, 1 \bmod 4$)

1.1.3 Abelian Group Structure

We now prove that $\mathcal{O}_K$ is a free $\mathbb{Z}$-module of finite rank. From the above discussion, we can start seeing how this is useful in finding elements of $\mathcal{O}_K$: given it is finitely generated and we know the rank, we can look for generating linearly independent sets, i.e. a basis, for $\mathcal{O}_K$ (since $\mathcal{O}_K$ is a module over a PID).

We in fact can prove something slightly stronger that shall be useful later in this chapter:

Proposition 1.1.32: Generation of Ring of Integers

Let R be a normal integral domain, ($\overline{R}^{K(R)} = R$), $K = K(R)$ the field of fraction of R, F/K a separable extension of degree n, and let $S = \overline{R}^F$ where F/K. Then there exists free R-submodules M' of F such that if $M = R^n$,

$$M \subseteq S \subseteq M'$$

Hence, S is a finitely generated R-module if R is Noetherian, and it is free of rank n if R is a Principal Ideal Domain.

Hence, the integral closure over a larger field that is a finite extension shall only extend the integral closure by a relatively controlled manner; the more condition on R the better the control.

We are usually considering $\mathcal{O}_F$ for some number field $F/\mathbb{Q}$, the generality is to work over the cases that are less nice (as shall be done in algebraic geometry[6], see [3]). To see this result more concretely, it is worth stating the following immediate corollary:

Corollary 1.1.33: $\mathcal{O}_K$ is a Free $\mathbb{Z}$-Module

Let $F/\mathbb{Q}$ be a number field of degree n. Then $\mathcal{O}_F$ is a free $\mathbb{Z}$-module of degree n.

***Proof*:**
Of proposition 1.1.32
Let $K = K(R)$ and let $B = \{\beta_1, \beta_2, ..., \beta_n\}$ be a basis for F over K. We shall use this basis to squish S between two finitely generated R-modules (note that S is an R-module since S/R is an integral extension)

To define the R-module M, as we saw in the proof of proposition 1.1.7, there exists a nonzero

[6] The integral points of curves shall often have more nuanced behaviour

$d \in R$ such that $d\beta_i \in S$. Clearly $dB = \{d\beta_1, d\beta_2, ..., d\beta_n\}$ is still a K-basis for F. Now, consider

$$\langle d\beta_1, d\beta_2, ..., d\beta_n \rangle_R = M$$

Since S is naturally an R-module and $d\beta_i \in S$, we have that $M \subseteq S$. Without loss of generality, we may now assume that $\beta_i \in S$ for all $1 \leq i \leq n$, that is $B \subseteq S$ and is a K-basis for F.

For the second R-module, define: the *integral trace dual*

$$^*M = M^{\mathrm{tr}\vee} = \left\{ b \in F \;\middle|\; \forall \ell \in M, \mathrm{tr}_{F/K}(b\ell) \in R \right\}$$

The extension F/K is separable by hypothesis, so the trace pairing is nondegenerate ([5, The Trace Form of a Separable Extension Is Nondegenerate]) and there is a dual K-basis $B' = \{\beta_1', ..., \beta_n'\}$ of F given by $\mathrm{tr}(\beta_i \beta_j') = \delta_{ij}$. Both modules are free of rank n over R, so in particular $M \cong_R M^{\mathrm{tr}\vee}$. This $M^{\mathrm{tr}\vee}$ is the module called M' in the statement.

The key result we shall prove is:

$$M = R\beta_1 + R\beta_2 + \cdots + R\beta_n \subseteq S \subseteq R\beta_1' + R\beta_2' + \cdots + R\beta_n' = M^{\mathrm{tr}\vee} \tag{1.3}$$

The first inclusion has already been proved, hence we show the second inclusion. Let $b \in S$ be arbitrary. Since $S \subseteq F$ and B' is a basis for F, we may write b as

$$b = \sum_i b_i \beta_i' \qquad b_i \in K$$

If we show that each $b_i \in R$, we are done. First, Since $\beta_i, b \in S$, their product is in S since S is a ring, $b\beta_i \in S$, hence $\mathrm{tr}_{F/K}(b\beta_i) \in R$ for each i since S is integral over R. Computing, we get:

$$\mathrm{tr}(b\beta_i) = \mathrm{tr}(\sum_j b_j \beta_j' \beta_i) = \sum_j b_j \mathrm{tr}(\beta_j' \beta_i) = \sum_j b_j \delta_{ij} = b_i$$

Hence, $b_i \in R$, showing equation (1.3). If R is Noetherian, then $M^{\mathrm{tr}\vee}$ is a Noetherian R-module, and hence S would be finitely generated. If furthermore R is a Principal Ideal Domain, then any submodule of a free module (over a PID) is free, and S is free of rank less than or equal to n. Since it contains a free R-module of rank n, it has to be free rank greater than or equal to n, and so it has rank n as an R-module, completing the proof.

Note that this implies that if $\mathcal{O}_K$ is a PID and L/K a separable extension where $K = K(\mathcal{O}_K)$, then *every* nonzero finitely generated $\mathcal{O}_L$-module is a free $\mathcal{O}_K$-module of rank $[L : K]$.

Corollary 1.1.34: Maximality of Ring Of Integers for Number Fields

The ring of integers of a number field L is the largest subring that is finitely generated as a $\mathbb{Z}$-module.

***Proof*:**
By the above proposition, $\mathcal{O}_L$ is finitely generated as a $\mathbb{Z}$-module If $B \subseteq L$ is a subring that is also a finitely generated $\mathbb{Z}$-submodule, then every element of B is integral over $\mathbb{Z}$ ([2, Integral Element Equivalences](3)). Hence, $B \subseteq \mathcal{O}_L$.

With 1.1.32, we now have an "upper bound" on how far away a ring of integers is form simply being $\mathbb{Z}[\alpha]$ where $K = \mathbb{Q}(\alpha)$.

Definition 1.1.35: Integral Trace Dual

Let R be a normal integral domain, $K = K(R)$, L/K a finite separable extension, $S = \overline{R}^L$. Then the *integral trace dual* of S is

$$^*S = \left\{ b \in L \;\middle|\; \forall \ell \in S, \operatorname{tr}_{L/K}(b\ell) \in R \right\}$$

Example 1.1.36: Finding Integral Trace Dual

Let $K = \mathbb{Q}(\sqrt{5})$ considered as an extension $K/\mathbb{Q}$ so that $R = \mathbb{Z}$. Then we know that $L = \langle 1, \sqrt{5} \rangle \subseteq \mathcal{O}_K$. Let's try to bound by above the elements of $\mathcal{O}_K$. Let $b = r + s\sqrt{5} \in \mathbb{Q}(\sqrt{5})$ be an arbitrary element. Then:

$$\operatorname{tr}(b \cdot 1) = 2r \qquad r(b\sqrt{5}) = 10s$$

Hence, if $2r \in \mathbb{Z}$ and $10s \in \mathbb{Z}$, we get:

$$^*\mathcal{O}_K = \left\langle \frac{1}{2}, \frac{\sqrt{5}}{10} \right\rangle_{\mathbb{Z}}$$

Note that $^*\mathcal{O}_K$ is bigger than $\mathcal{O}_K$, for example $1/2 \notin \mathcal{O}_K$. By similar reasoning, we can check that $\frac{1}{\sqrt{5}} \in {}^*\mathcal{O}_K$, but it can be verified that it is not in $\mathcal{O}_K$, and so $\mathcal{O}_K^\vee \neq \mathcal{O}_K$. This does not contradict corollary 1.1.34 as $^*\mathcal{O}_K$ is *not* a subring (note that $1/2 \cdot 1/2 \notin {}^*\mathcal{O}_K$). Note however that they are both still rank 2.

Note that separability is a necessary hypothesis to insure the non-degeneracy of the trace-pairing, meaning this result works at least over over number fields, function fields, and perfect fields[7]. Naturally, if R is not a Principal Ideal Domain, then S need not be a free R-module. For now, we shall always let $R = \mathbb{Z}$ so that $K = \mathbb{Q}$ and $F/\mathbb{Q}$ is a number field. Note that we are only working with the addition structure of $\mathcal{O}_K$, while $\mathcal{O}_K$ is a ring. Thus, when talking about generators as a $\mathbb{Z}$-module, it must not be confounded with a set of generators for the ring. For example $(\alpha) \subseteq \mathcal{O}_K$ is a principal ideal and hence generated by a single element, but it may be generated by many elements as a $\mathbb{Z}$-module.

Definition 1.1.37: Integral Basis

Let $K/\mathbb{Q}$ be a number field. Then a $\mathbb{Z}$-basis for $\mathcal{O}_K$ is called an *integral basis* for K.

The previous proposition also gives more motivation for squaring the determinant in the definition of the discriminant. By corollary 1.1.27, if $(\alpha_1, \alpha_2, ..., \alpha_n)$ is a $\mathbb{Z}$-basis for $\mathcal{O}_F$, then $\operatorname{disc}_{F/\mathbb{Q}}(\alpha_1, \alpha_2, ..., \alpha_n) \in \mathbb{Z}$ and any change of basis A must map to a unit, hence $\det(A) \in \mathbb{Z}^\times = \{-1, 1\}$. Squaring thus guarantees that the discriminant is basis independent for rings of integers. Now that we have explicitly found that $\mathcal{O}_K$ is a finitely generated $\mathbb{Z}$-module in the case of K being a number field, the term *field discriminant*, or just *discriminant* for short, shall imply that the input is an integral basis for $\mathcal{O}_K$, unless explicitly stated otherwise:

[7] fields of positive characteristic may require some extra work

Definition 1.1.38: Discriminant For Ring of Integers

Let $K/\mathbb{Q}$ be a number field. Then if $A = \{\alpha_1, \alpha_2, ..., \alpha_n\}$ is an integral basis for $\mathcal{O}_K$, then:

$$d_K := \text{disc}(\mathcal{O}_K) := \text{disc}_{K/\mathbb{Q}}(A) = \text{disc}(\mathcal{O}_K)$$

is the *field discriminant* of K. It is sometimes also denoted $\text{disc}(K)$ if clear from context.

The discriminant can be thought of as capturing some information on how far $\mathcal{O}_K$ is from being Thus, in example 1.1.25 where $K = \mathbb{Q}(\sqrt{m})$, we have:

$$d_K = \begin{cases} 4m & m \equiv 2, 3 \text{ mod } 4 \\ m & m \equiv 1 \text{ mod } 4 \end{cases}$$

As we saw, we may have many bases for K, not all of them necessarily basis for $\mathcal{O}_K$. Given $K/\mathbb{Q}$ of dimension n, we may have many $M \subseteq B$ of rank n. Since M is of rank n, then if $\mathcal{O}_K \subseteq M$, $\mathcal{O}_K/M$ is of rank 0, and if $M \subseteq \mathcal{O}_K$, then $M/\mathcal{O}_K$ is also of rank 0 (some elementary group theory can be used to prove this, see exercise ref:HERE (fund thm of fin gen ab grp)). This naturally leads to asking how the discriminants of such $\mathbb{Z}$-modules compare:

Proposition 1.1.39: Discriminant And Comparing Basis

Let $B/\mathbb{Z}$ be a free $\mathbb{Z}$-module of rank n, and let $A \subseteq B$ be a $\mathbb{Z}$-submodule of finite index (i.e. $[B : A] < \infty$). Then if $\text{disc}_{B/\mathbb{Z}}(\gamma_1, \gamma_2, ..., \gamma_n) \neq 0$, then:

$$\text{disc}(A) = [B : A]^2 \text{disc}(B)$$

***Proof*:**
Let $\beta_1, \beta_2, ..., \beta_n$ is a basis for B. By the previous corollary, if $N \subseteq B$ is of finite index, then it must be a free module of rank n and so the discriminant is nonzero, and similarly the other way around.

Then there exists a change of basis matrix A, and we get $\text{disc}(N) = \det(A)^2 \text{disc}(B)$. It is not hard to see that $\det(A) = [B : N]$ (it will be easier to visualize after finishing section 1.1.4), which complete the proof.

Corollary 1.1.40: Subset And Same Discriminant, Then Equal

Let $F/\mathbb{Q}$ be number field, Let $A, B \subseteq F$ be free $\mathbb{Z}$-modules, $B \subseteq A$ and $\text{disc}(A) = \text{disc}(B)$. Then $B = A$. Hence, we may use discriminants to identify free $\mathbb{Z}$-modules of finite rank.

***Proof*:**
Since the discriminants match and

$$\text{disc}(B) = [B : A]^2 \text{disc}(N)$$

we have $[B : A] = 1$, meaning $B = N$.

Corollary 1.1.41: Isomorphism Check Of Number Fields

Let $\mathbb{Q}(\alpha), \mathbb{Q}(\beta)$ be two number fields. Then if the discriminant is different, the number fields are not isomorphic

***Proof*:**
If the rings of integers were the same, their index would be 1, but since their discriminant is not equal this is not the case.

Note that it is possible that there are multiple fields with the same discriminant:

Example 1.1.42: Same Discriminant, Different Fields
Take $\mathbb{Q}(\alpha), \mathbb{Q}(\beta), \mathbb{Q}(\gamma)$ where

$$a^3 - 18\alpha - 6 = 0 \qquad \beta^3 - 36\beta - 78 = 0 \qquad \gamma^3 - 58\gamma - 150 = 0$$

Then each has discriminant 22356, but these fields are not isomorphic (as shall be shown in example 1.4.20)

It should be noted that there are only finitely many fields for each discriminant value, however we shall not prove this fact[8]

The next couple of theorems give some computational tools on finding the ring of integers:

Proposition 1.1.43: Computation Of Dual For Ring Of Integers

Let K be a number field with basis $(\alpha_1, \alpha_2, ..., \alpha_n)$ of algebraic integers and let $d = \text{disc}_{K/\mathbb{Q}}(\alpha_1, \alpha_2, ..., \alpha_n)$. Then every $\alpha \in \mathcal{O}_k$ can be expressed as:

$$\frac{m_1\alpha_1}{d} + \cdots + \frac{m_k\alpha_k}{d}$$

where $m_i \in \mathbb{Z}$ and each m_i^2 is divisible by d $(d \big| m_i^2)$

***Proof*:**
First, $d \neq 0$ since $\alpha_1, \alpha_2, ..., \alpha_n$ forms a basis for K and $d \in \mathbb{Z}$ since the α_i are algebraic integers (which exists by corollary 1.1.8).

Let $\alpha = x_1\alpha_1 + \cdots + x_n\alpha_n$ for some $x_i \in \mathbb{Q}$. Letting $\sigma_1, \sigma_2, ..., \sigma_n$ denote the embeddings of K into $\mathbb{C}$, we get

$$\sigma_i(\alpha) = x_1\sigma(\alpha_1) + \cdots + x_n\sigma(\alpha_n)$$

for each $1 \leq i \leq n$. We can solve for the x_j's using Cramers rule, namely

$$x_j = y_j/\delta$$

where $\delta = \det(\sigma_i(\alpha_j))$ and y_j is obtained by replacing the jth column of the previous determinant by $\sigma_i(\alpha)$. Clearly y_j and δ are algebraic integers, and furthermore $\delta^2 = \text{disc}_{K/\mathbb{Q}}(\alpha_1, \alpha_2, ..., \alpha_n)$.

[8] For those interested, it is the *Hermite-Minkowski's Theorem*

Hence, letting $d = \text{disc}_{K/\mathbb{Q}}(\alpha_1, \alpha_2, ..., \alpha_n)$ we have

$$dx_j = \delta y_j$$

showing that the rational number dx_j is an algebraic integer, and so $dx_j \in \mathbb{Z}$. Let $m_i = dx_j$. What's left to show is that $m_j^2/d \in \mathbb{Z}$, so it suffices to show that it is an algebraic integer. But $m_j^2/d = y_j^2$, which completes the proof.

Hence, for any $\mathcal{O}_K$, if $\alpha_1, \alpha_2, ..., \alpha_n$ is a basis for $K/\mathbb{Q}$

$$\bigoplus \alpha_i \mathbb{Z} \subseteq \mathcal{O}_K \subseteq \bigoplus \frac{\alpha_i}{d}\mathbb{Z}$$

And if $K = \mathbb{Q}(\alpha)$ so that $1, \alpha, ..., \alpha^{n-1}$ is a basis for K, then by proposition 1.1.29 $d = \left(\prod_{i\neq j}(\alpha_i - \alpha_j)\right)^2$ and:

$$\mathbb{Z}[\alpha] \subseteq \mathcal{O}_K \subseteq \mathbb{Z}\left[\frac{\alpha}{d}\right]$$

Note that in general, $\mathbb{Z}[\alpha] \neq \mathcal{O}_K$; a counter-example shall be given in example 1.4.29. In the special case where $\mathbb{Z}[\alpha] = \mathcal{O}_K$, we give such a number ring a name:

Definition 1.1.44: Monogenic Number Ring

Let $K/\mathbb{Q}$ be a number field, $\mathcal{O}_K$ be the associated number ring, and α the primitive element so that $K = \mathbb{Q}(\alpha)$. Then if

$$\mathbb{Z}[\alpha] = \mathcal{O}_K$$

Then $\mathcal{O}_K$ is called *monogenic* and is said to be a *monogenic number ring*.

Since all number fields are monogenic, that is $K = \mathbb{Q}(\alpha)$ for some appropriate α, it is reasonable to ask if $\mathcal{O}_K$ can be expressed using α. The following theorem gives this structure:

Theorem 1.1.45: Basis For Ring Of Integers

Let $\alpha \in \mathcal{O}_k$ and suppose α has degree n over $\mathbb{Q}$. Then there exists $d_1, d_2, ..., d_{n-1} \in \mathbb{Z}$, and monic polynomial $f_i \in \mathbb{Z}[x]$ of degree i for $1 \leq i \leq n-1$ such that

$$\mathcal{O}_K = \mathbb{Z} \oplus \frac{f_1(\alpha)}{d_1}\mathbb{Z} \oplus \cdots \oplus \frac{f_{n-1}(\alpha)}{d_{n-1}}\mathbb{Z}$$

is an integral basis where

$$d_1 \big| d_2 \big| \cdots \big| d_{n-1}$$

and the d_i's are unique.

***Proof*:**
Marcus p.26

Example 1.1.46: Ring Of Integers Of Cubic Extension

Let $K = \mathbb{Q}(\sqrt[3]{2})$ and $\alpha = \sqrt[3]{2}$. We want to find $\mathcal{O}_K$. First $\operatorname{rank}\mathcal{O}_K = 3$ since $[K : \mathbb{Q}] = 3$, and $L = \mathbb{Z}[\sqrt[3]{2}] \subseteq \mathcal{O}_K$. If A is the change of basis from L to $\mathcal{O}_K$, then $\det(A) = [\mathcal{O}_K : \mathbb{Z}[\alpha]]$[a]. Thus, by doing a change of basis from $\mathbb{Z}[\alpha]$ to $\mathcal{O}_K$ by A, we get that:

$$\operatorname{disc}(\mathbb{Z}[\alpha]) = [\mathcal{O}_K : \mathbb{Z}[\alpha]]^2 \operatorname{disc}(\mathcal{O}_K)$$

Thus, computing the discriminant of the left hand side we get:

$$\operatorname{disc}(\mathbb{Z}[\alpha]) = \det \begin{pmatrix} 3 & 0 & 0 \\ 0 & 0 & 6 \\ 0 & 6 & 0 \end{pmatrix} = -3 \cdot 36 = -(2^2 \cdot 3^2) \cdot 3 = -108$$

giving us that $[\mathcal{O}_K : \mathbb{Z}[\alpha]]^2 \big| 108 \Rightarrow [\mathcal{O}_K : \mathbb{Z}[\alpha]] \big| 2 \cdot 3 = 6$. If we show the index is 1, the proof is done. We shall prove the contradiction for the case of $[\mathcal{O}_K : \mathbb{Z}[\alpha]] = 3$, the proof for $[\mathcal{O}_K : \mathbb{Z}[\alpha]] = 2$ is similar, and from that we can also conclude that $[\mathcal{O}_K : \mathbb{Z}[\alpha]] \nmid 6$.

Let's say $[\mathcal{O}_K : \mathbb{Z}[\alpha]] = 3$. Then $\mathcal{O}_K/\mathbb{Z}[\alpha] \cong \mathbb{Z}/3\mathbb{Z}$. Any element $\beta \in \mathcal{O}_K$ lands in one of the cosets of $\mathcal{O}_K/\mathbb{Z}[\alpha]$ which implies that $a, b, c \in \frac{1}{3}\mathbb{Z}$. We want to show that a, b, c must be integers (lands in the zero coset). Since the ring of integers is closed under subtraction, we may without loss of generality subtract the non-fractional part of any element β we choose so that $a, b, c \in \{-1, 0, 1\}$[b]. Hence we take the norm of $\beta = (a/3) + (b/3)\alpha + (c/3)\alpha^2$

$$N(\beta) = \det \begin{pmatrix} a/3 & 2c/3 & 2b/3 \\ b/3 & a/3 & 2c/3 \\ c/3 & b/3 & a/3 \end{pmatrix} = \frac{1}{27}(a^3 + 2b^2 + 4c^3 - 6abc)$$

Since β is algebraic, $N(\beta) \in \mathbb{Z}$ meaning $27 \big| (a^3 + 2b^2 + 4c^3 - 6abc)$. But, unless $a = b = c = 0$:

$$|a^3 + 2b^2 + 4c^3 - 6abc| \leq 15$$

showing that no such combination of a, b, c exist, meaning β is an algebraic integer if $a, b, c \in \mathbb{Z}$. A similar argument can be made in the case of $[\mathcal{O}_K : \mathbb{Z}[\alpha]] = 2$, hence we have that $\mathcal{O}_K = \mathbb{Z}[\alpha]$, completing the proof.

[a] since the index is finite, a basis for $\mathbb{Z}[\alpha]$ may be scaled to form a basis for $\mathcal{O}_K$, which translates to the determinant result

[b] Equivalently, any coset can be chosen to have such representatives

In some special cases, the ring of integers of more composite fields may be found given ring of integers of each respective field:

Theorem 1.1.47: Ring of Integers of Composites

Let $K_1/K, K_2/K$ be Galois with degree n and m and with coprime discriminants $d_{K_1} = d$ and $d_{K_2} = d'$ respectively. Let $L = K_1K_2$ and $K_1 \cap K_2 = K$. Let $\omega_1, \omega_2, ..., \omega_n$ and $\omega_1', \omega_2', ..., \omega_m'$ be an integral basis for K_1/K and K_2/K. Then $\omega_i\omega_j'$ is an integral basis for K_1K_2 with discriminant $(d_{K_1})^m(d_{K_2})^n = d^m(d')^n$, namely

$$\mathcal{O}_L = \mathcal{O}_{K_1}\mathcal{O}_{K_2}$$

***Proof*:**
Since $K_1 \cap K_2 = K$, $[K_1K_2 : K] = nm$, so the nm products $\omega_i\omega'_j$ form a basis of K_1K_2/K. Now, let α be an integral element of K_1K_2 over K, and write:

$$\alpha = \sum_{i,j} a_{ij}\omega_i\omega'_j \qquad a_{ij} \in K$$

We must show that $a_{ij} \in \mathcal{O}_K$. Let

$$\beta_j = \sum_i a_{ij}\omega_i$$
$$\mathrm{Gal}_{K_2}(L) = \{\sigma_1, \sigma_2, ..., \sigma_n\}$$
$$\mathrm{Gal}_{K_1}(L) = \{\tau_1, \tau_2, ..., \tau_m\}$$

Then since $[L : K] = nm$, by [5, Galois Composite Fields II]$\mathrm{Gal}_K(L) = \{\sigma_k\tau_\ell \mid 1 \leq k \leq n \; 1 \leq \ell \leq m\}$. Furthermore, let:

$$T = \tau_\ell(\omega'_j) \qquad a = (\tau_1(\alpha), ..., \tau_m(\alpha))^t \qquad b = (\beta_1, ..., \beta_m)^t$$

Then by definition we have $\det(T)^2 = d'$ and $a = Tb$. If T^* is the adjoint of T, then some linear algebra manipulation gives:

$$\det(T)b = T^*a$$

Now, since T^* and a have integral entries (in L), $d'b$ has integral entries in K_1< i.e.

$$d'\beta_j = \sum_k d/a_{ij}\omega_i$$

Hence, $d'a_{ij} \in \mathcal{O}_K$. Now, changing (ω_i) and $\omega'_j)$, we get that $da_{ij} \in A$, hence since the discriminants are relatively prime, we get for appropriate x, x':

$$a_{ij}xda_{ij} + x'd'a_{ij} \in \mathcal{O}_K$$

Hence, $\omega_i\omega'_j$ form an integral basis for L/K, completing the first part of the proof.

We next find the discriminant the ring of integers of L/K. Since we have the Galois group of L over K, take the $(nm \times m)$ matrix

$$M = (\sigma_k\tau_\ell(\omega_i\omega'_j)) = (\sigma_k(\omega_i)\tau_\ell(\omega'_j))$$

Note that M can be interpreted as an $(m \times m)$-matrix with entries being $(n \times n)$-matrices where each (ℓ, j)-entry is the matrix $Q\tau_\ell\omega'_j$ where $Q = (\sigma_k\omega_i)$:

$$M = \begin{pmatrix} Q & & 0 \\ & \ddots & \\ 0 & & Q \end{pmatrix} \begin{pmatrix} E\tau_1\omega'_1 & \cdots & E\tau_m(\omega'_1) \\ \vdots & \ddots & \vdots \\ E\tau_1(\omega'_m) & \cdots & E\tau_m(\omega'_m) \end{pmatrix}$$

where E is an $(n \times n)$-unit matrix. Now, changing indices to compare the two representations of M, we get:

$$\mathrm{disc}(\mathcal{O}_L) = \det(M)^2 = \det(Q)^{2m} = \det\left((\tau_\ell(\omega'_j))\right)^{2n} = d^m(d')^n$$

as we sought to show.

Note that for the general separable case, we need to replace our assumption with

$$K_1 \cap K_1 = K \qquad K_1,\ K_2 \text{ are linearly disjoint}$$

If the discriminants are not relatively prime, then there is a partial generalization in the case where $K = \mathbb{Q}$:

Corollary 1.1.48: Ring of Integers of Composite, Partial Generalization

Let $K_1/\mathbb{Q}$ and $K_2/\mathbb{Q}$ be finite separable extensions, $d_1 = \operatorname{disc}(\mathcal{O}_{K_1})$ and $d_2 = \operatorname{disc}(\mathcal{O}_{K_2})$. Then if $L = K_1K_2$:

$$\mathcal{O}_L \subseteq \frac{1}{\gcd(d_1, d_2)}\mathcal{O}_{K_1}\mathcal{O}_{K_2}$$

***Proof*:**
Marcus p.25

Example: Ring of Integers of Cyclotomic Field

Let's use the above results to show that if $K = \mathbb{Q}(\zeta_n)$ then $\mathcal{O}_K = \mathbb{Z}[\zeta_n]$, that is the ring of integers of K is monogenic. Besides quadratic extensions, this is so far the only other ring of integers we have classified (in particular since they are rather difficult to classify). We first require some lemmas:

Lemma 1.1.49: Cyclotomic Field Lemmas

Let $n = p^r$ where $n \geq 3$. Then:

1. $\mathbb{Z}[\zeta_n] = \mathbb{Z}[1-\zeta_n]$ and $\operatorname{disc}(\zeta_n) = \operatorname{disc}(1-\zeta_n)$
2. $\prod_k (1-\zeta_n^k) = p$

***Proof*:**

1. Certainly $\mathbb{Z}[\zeta_n] = \mathbb{Z}[1-\zeta_n]$ since $\zeta_n = 1-(1-\zeta_n)$, implying the discriminants are equal. Similarly, by proposition 1.1.29

$$\operatorname{disc}(\zeta_n) = \prod_{r<s}(\alpha_r - \alpha_s)^2 = \prod_{r<s}((1-\alpha_r)-(1-\alpha_s))^2 = \operatorname{disc}(1-\zeta_n)$$

2. Let

$$f(x) = \frac{x^{p^r-1}}{x^{p^{r-1}}-1} = 1 + x^{p^{r-1}} + x^{2p^{r-1}} + \cdots x^{(p-1)p^{r-1}}$$

Then for $p \nmid k$, ζ_n^k are the roofs of f, since they are roots of $x^{p^r}-1$, but not of $x^{p^{r-1}}-1$. Thus:

$$f(x) = \prod_{p \nmid k}(x - \zeta_n^k)$$

Since there is exactly $\varphi(p^r) = (p-1)p^{r-1}$ where φ is the Euler totient. Thus, setting $x = 1$ gives us the desired result.

Proposition 1.1.50: Ring Of Integers Of Prime-Power Cyclotomic Field

Let $n = p^r$ and $K = \mathbb{Q}(\zeta_n)$. Then:

$$\mathcal{O}_K = \mathbb{Z}[\zeta_n]$$

***Proof*:**
We shall show that $\mathcal{O}_K = \mathbb{Z}[1-\zeta_n]$. By theorem 1.1.45, any $\alpha \in \mathcal{O}_K$ can be written as:

$$\alpha = \frac{n_1 + n_2(1-\zeta_n) + \cdots + n_m(1-\zeta_n)^{m-1}}{\operatorname{disc}(1-\zeta_n)}$$

where $m = \varphi(p^r)$, $n_i \in \mathbb{Z}$, and $\operatorname{disc}(1-\zeta_n) = \operatorname{disc}(\zeta_n)$. As we saw in example 1.1.31, if $n = p$ is a prime, $\operatorname{disc}(\zeta_n) = p^{p-2}$. Since $n = p^r$, by example 1.1.31, d is a power of p.

Let's now assume for the sake of contradiction that $\mathcal{O}_K \neq \mathbb{Z}[1-\zeta_n]$, so that there must exist some $\alpha \in \mathcal{O}_K$ such that not all n_i are divisible by d, namely $\mathcal{O}_K$ contains an element of the form:

$$\beta = \frac{n_i(1-\zeta_n)^{i-1} + n_{i+1}(1-\zeta_n)^i + \cdots + n_m(1-\zeta_n)^{m-1}}{p}$$

where $i \leq n$, $n_i \in \mathbb{Z}$, not all n_i are divisible by p. Now, by lemma 1.1.49 since $(1-\zeta_n)\big|(1-\zeta_n)^k$ in $\mathbb{Z}[\zeta_n]$,

$$\frac{p}{(1-\zeta_n)^m} \in \mathbb{Z}[\zeta_n] \implies \frac{p}{(1-\zeta_n)^i} \in \mathbb{Z}[\zeta_n] \implies \frac{\beta p}{(1-\zeta_n)^i} \in \mathcal{O}_K$$

Subtracting the terms that are in $\mathcal{O}_K$ (by closure of multiplication) we get

$$\frac{n_i}{1-\zeta_n} \in \mathcal{O}_K \implies \operatorname{Nm}_{K/\mathbb{Q}}\left(\frac{n_i}{1-\zeta_n}\right) \in \mathbb{Z} \implies \operatorname{Nm}_{K/\mathbb{Q}}(1-\zeta_n)\big|\operatorname{Nm}_{K/\mathbb{Q}}(n_i)$$

However, this is a contradiction since $\operatorname{Nm}_{K/\mathbb{Q}}(n_i) = n_i^m$, but by lemma 1.1.49 $\operatorname{Nm}_{K/\mathbb{Q}}(1-\zeta_n) = p$, Hence

$$\mathcal{O}_K = \mathbb{Z}[\zeta_n]$$

as we sought to show.

We now look to generalize to any cyclotomic field. The key idea is that the ring of integers of cyclotomic fields will be the composite of simpler ring of integers

Proposition 1.1.51: Ring of Integers of Cyclotomic Field

Let $K = \mathbb{Q}(\zeta_n)$ for any $n \in \mathbb{N}$. Then:

$$\mathcal{O}_K \cong \mathbb{Z}[\zeta_n]$$

***Proof*:**
We shall do a form of induction. This was proven if n is a prime power, so let's assume it's not so that we may write $n = n_1 n_2$ for relatively prime $n_1, n_2 > 1$. We shall work by induction on

n_1, n_2. The base case was proposition 1.1.50, hence set:

$$\begin{aligned} \omega_1 &= \zeta_{n_1} & \omega_2 &= \zeta_{n_2} \\ K_1 &= \mathbb{Q}(\omega_1) & K_2 &= \mathbb{Q}(\omega_2) \\ \mathcal{O}_{K_1} &= \mathbb{Z}[\omega_1] & \mathcal{O}_{K_2} &= \mathbb{Z}[\omega_2] \end{aligned}$$

First, it is clear that

$$\zeta_n^{n_1} = \omega_2 \qquad \zeta_n^{n_2} = \omega_1 \implies \zeta_n = \omega_1^r \omega_2^s,\ r, s \in \mathbb{Z}$$

Thus, $L = K_1 K_2$. Next, as we saw in example 1.1.31,

$$\operatorname{disc}(\omega_1)\big| n_1 \qquad \operatorname{disc}(\omega_2)\big| n_2$$

since $\gcd(n_1, n_2) = 1$, we see the discriminants are relatively prime. Since $\varphi(n) = \varphi(n_1)\varphi(n_2)$, we by theorem 1.1.47 that

$$\mathcal{O}_K = \mathcal{O}_{K_1}\mathcal{O}_{K_2} = \mathbb{Z}[\omega_1]\mathbb{Z}[\omega_2] = \mathbb{Z}[\zeta_n]$$

as we sought to show.

Exercise 1.1.2

1. Generalize proposition 1.1.39 for modules over $\mathcal{O}_K$.
2. Give a counter-example where $\mathcal{O}_{KL} = \mathcal{O}_K \mathcal{O}_L$?

1.1.4 Lattice Structure: Geometry of Number Rings

An important word of warning must be given about free $\mathbb{Z}$-modules due to the existence of non-invertible elements. It is possible that A, B are two free $\mathbb{Z}$-module of rank 2, one contains the basis of the other, however $A \subsetneq B$! This is in contract to the case of modules over fields where if A, B are two k-vector-spaces of the same dimension and $A \subseteq B$, then $A = B$[9]. Even more behavior can happen: consider $\mathbb{Q} \otimes_{\mathbb{Z}} A$. Then this is a vector-space with a finite dimension. However, the dimension need not match the rank of A:

Example 1.1.52: Behavior of $\mathbb{Z}$-modules

1. Let $A = \langle (1, 0), (0, 1)\rangle_{\mathbb{Z}}$ and $B = \langle (1/2, 0), (0, 1/2)\rangle_{\mathbb{Z}}$. Then by construction they are both isomorphic as abelian groups to $\mathbb{Z}^2$, however $A \subsetneq B$.
2. Let A be a free $\mathbb{Z}$-module of rank 2. Then we may extend scalar to make A into a free $\mathbb{R}$-module; either take $\mathbb{R} \otimes_{\mathbb{Z}} A$ or consider $\langle a_1, a_1\rangle_{\mathbb{R}}$ for some free generating set $\{a_1, a_2\}$. Then $\mathbb{R} \otimes_{\mathbb{Z}} A$ *need not be dimension 2*. For example, take

$$A = \langle (1, 1), (\sqrt{2}, \sqrt{2})\rangle_{\mathbb{Z}}$$

Then $\mathbb{R} \otimes_{\mathbb{Z}} A$ is one dimensional

[9] $b \in B$ can be appropriately shrunk to a nonzero size so that its in A (since the dimensions are the same), and then we use closure of scalar multiplication

3. By construction of $\mathcal{O}_k$, if $\langle \alpha_1, \alpha_2, ..., \alpha_n \rangle_{\mathbb{Z}} = \mathcal{O}_k$, then $\langle \alpha_1, \alpha_2, ..., \alpha_n \rangle_{\mathbb{Q}} = k$, Thus $\mathcal{O}_k$ is a little more than just a $\mathbb{Z}$-module. We shall explore this in this section.

The third example shows that the rank of the ring of integers more coincides with the dimension of the natural vector-space when extending the actions to a field. As was explored in [5, Vector Space], [finite dimensional] vector spaces have easy visualizations using orthonormal bases. In [5, Exploring Integral Domains: Factorization and Valuation], this was used implicitly when it was intuited that $\mathbb{Z}[i]$ forms a nice square lattice in $\mathbb{Q}^2$. More generally, ring of integer $\mathcal{O}_K$ will form a lattice when embedded into an appropriate vector-space. To better visualize this lattice, it is easer to see it embedded into a direct sum of $\mathbb{R}$ and $\mathbb{C}$[10]. This has the extra advantage that these spaces have a well-defined notion of *measure* that gives us an area. We shall take advantage of tools from analysis to further enlighten the structure of rings of integers, however as a consequence an elementary level of topology and analysis will be assumed.

Let $n = [K : \mathbb{Q}]$ so that $\operatorname{Hom}_{\mathbb{Q}}(K, \mathbb{C}) = n$. Then every embedding of K into $\mathbb{C}$ is either embeds into $\mathbb{R}$ or does not (that is, either $K \cap (\mathbb{C} \setminus \mathbb{R}) = \emptyset$ or $\neq 0$); let's call the former *real embeddings* and the latter *complex embeddings*. For a complex embedding $\tau \in \operatorname{Hom}_{\mathbb{Q}}(K, \mathbb{C})$, we may always find another embedding $\overline{\tau}$ since for any α where $p(\alpha) = 0$ for appropriate $p(x) \in \mathbb{Q}[x]$

$$0 = \overline{\tau}(0) = \overline{\tau}(p(\alpha)) = \tau(\overline{p(\alpha)}) = \tau(p(\overline{\alpha}))$$

showing that $\overline{\tau}$ maps roots to roots. Hence the degree n may be split into $n = r_1 + 2r_2$ where r_1 represents the amount of embeddings where the image σ_i is real:

$$\alpha_1, \alpha_2, ..., \alpha_{r_1}$$

while r_2 represents the embeddings that embedded into $\mathbb{C}$ is not strictly in $\mathbb{R}$, giving us a total list of embeddings:

$$\sigma_1, \sigma_2, ..., \sigma_{r_1}, \tau_1, \tau_2, ..., \tau_{r_2}, \overline{\tau_1}, ..., \overline{\tau_{r_2}}$$

Definition 1.1.53: Archimedean Embedding

Let $K/\mathbb{Q}$ be a number field. Then its *Archimedean embedding* is defined as

$$\begin{aligned} \sigma : & K \hookrightarrow \mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2} \\ & \alpha \mapsto (\sigma_1(\alpha), ..., \tau_{r_2}(\alpha)) \end{aligned}$$

The space $\mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2}$ is called the *Minkowski space*, and will be labeled $K_{\mathbb{R}}$.

Example 1.1.54: Archimedean Embedding

1. Trivially, $\mathbb{Q}_{\mathbb{R}} = \mathbb{R}$, and $\mathbb{C}_{\mathbb{R}}$ is not defined since $\mathbb{C}/\mathbb{Q}$ is not a finite extension.
2. Take $K = \mathbb{Q}(i)$ so that $\mathcal{O}_K = \mathbb{Z}[i]$. Then $\operatorname{Gal}_{\mathbb{Q}}(\mathbb{Q}(i)) = \{\mathrm{id}, c\}$ where c is the conjugate. Then K is simply the embedding of K into $\mathbb{C}$

$$\sigma : K \hookrightarrow \mathbb{C}^1 \qquad k \mapsto k$$

[10] For those who may prefer $\overline{\mathbb{Q}}$ over $\mathbb{R}$ or $\mathbb{C}$, I would stipulate $\mathbb{R}$ and $\mathbb{C}$ are is a very natural object in this context: $\mathbb{R}$ is perhaps the only "natural" object for which a finite extension is an algebraic closure, this will have algebraic and geometric consequence when counting the number of embeddings of a field into the algebraically closed field $\mathbb{C}$

Notice the gaussian integers form the usual grid in $\mathbb{C}$.

3. Take $K = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ Then $\mathrm{Gal}_{\mathbb{Q}}(K) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ corresponds to the conjugating each root separately. Then

$$\begin{gathered}\sigma : K \mapsto \mathbb{R}^4 \\ (a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6}) \\ \mapsto \\ (a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6}, a - b\sqrt{2} + c\sqrt{3} + d\sqrt{6}, \\ a + b\sqrt{2} - c\sqrt{3} + d\sqrt{6}, a - b\sqrt{2} - c\sqrt{3} + d\sqrt{6})\end{gathered}$$

Our goal is to show that $\sigma(\mathcal{O}_K)$ behaves like the just like $\mathbb{Z}[i]$ when embedded in $\mathbb{C}$. To characterize a lattice in higher dimensions, we define it in the following way:

Definition 1.1.55: [Integral] Lattice

Let V be a finite dimensional vector space over$\mathbb{R}$. Then a *lattice* in V is a $\mathbb{Z}$-submodule satisfying any two of the following 3 properties;

1. Λ spans V as a k-vector space ($\langle \Lambda \rangle_{\mathbb{R}} = V$)
2. Λ is discrete in the topological space V (there is an open ball near the origin that intersects only the origin) and the rank of the lattice is the dimension of V[a]
3. Λ is a free $\mathbb{R}$-module on n generators

[a]some authors call lattices where $n = m$ *complete lattices*, and if $n < m$ a *lattices*

Lemma 1.1.56: Equivalence Of Conditions

Any two of these conditions implies the third

***Proof*:**
Let $L \cong_{\mathbf{Ab}} \mathbb{Z}^n$. We'll show that (2) $\Leftrightarrow$ (3). For (2) $\Rightarrow$ (1), namely

(2) L is discrete

(3) $\langle L \rangle_{\mathbb{R}} = \mathbb{R}^n$

let $\ell_1, \ell_2, ..., \ell_n$ be a $\mathbb{Z}$-basis for L. Then there exists an invertible matrix $A \in \mathrm{GL}_n(\mathbb{R})$ such that $A = (a_{ij})$ where for each e_i of the standard basis for $\mathbb{R}^n$,

$$e_i = \sum_j a_{ij} \ell_j$$

Let $v = (v_1, v_2, ..., v_n) \in \mathbb{Z}^n$. Then

$$\sum_i^n v_i \ell_i = \sum_i^n w_i e_i$$

where $w = A^{-1}v$. We have now converted our number into one in the standard basis, hence we have a notion of size we may borrow. In particular, we have that if $v = Aw$, then

$$\|w\| \geq \frac{\|v\|}{n\|A\|}$$

where we can choose our favourite matrix norm, say the max norm on entries. hence, as v grows, w get's further from 0.

Conversely, we'll do the contrapositive, so suppose $\langle \ell_1, \ell_2, ..., \ell_n \rangle_{\mathbb{R}} \neq \mathbb{R}^n$. Then for some $a_i \in \mathbb{R}$ not all zero

$$\sum_i^n a_i \ell_i = 0$$

By the Pigeon hole Principle[a] (also known as Dirichlet's box principle), there exists $N \in \mathbb{N}, N > 0$ such that $|Na_i - [Na_i]| < \epsilon$ where $[Na_i]$ is the closest integer. Thus

$$0 = \sum Na_i\ell_i = \underbrace{\sum [Na_i]\ell_i}_{\in L} + \underbrace{\sum \{Na_i\}\ell_i}_{\in B_\epsilon(0)}$$

where $\{Na_i\}$ denote the fractional part, completing the proof.

[a] see [7, Pigeon-hole Principle]

The discrete condition need only be checked at the origin due to linearity[11]. An example of a free $\mathbb{Z}$-module in $\mathbb{R}^2$ that is not discrete is the one elaborated in example 1.1.52

$$\langle (1,1), (\sqrt{2}, \sqrt{2}) \rangle \subseteq \mathbb{R}^2$$

In particular, we may get as close as we want to $(0,0)$ by adding integer-multiples these two elements.

Proposition 1.1.57: Archimedean Embedding of $\mathcal{O}_K$ is Lattice

Let $K/\mathbb{Q}$ be a number field and σ the Archimedean embedding. Then $\sigma(\mathcal{O}_k)$ is a lattice in $\mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2}$, namely

$$\langle \sigma(\mathcal{O}_K) \rangle_{\mathbb{R}} = \mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2}$$

***Proof*:**
We may prove this result in using any of the equivalent definitions.

1. Let's show $\sigma(\mathcal{O}_K)$ is discrete. Let $\alpha \in K$ so that

$$\mathrm{Nm}_{K/\mathbb{Q}}(\alpha) = \prod_i^{r_1} \sigma_i(\alpha) \cdot \prod_j^{r_2} \|\tau_j(\alpha)\|^2 \in \mathbb{Q}$$

Then if $\alpha \in \mathcal{O}_K$, $\mathrm{Nm}_{K/\mathbb{Q}}(\alpha) \in \mathbb{Z}$. Therefore, if $\alpha \in \mathcal{O}_K \setminus \{0\}$, we have $|\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)| \geq 1$. But then $|\sigma(\alpha)| \geq 1$. Thus, taking $B \subseteq \mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2}$,

$$B = \{\alpha \mid |\alpha_i| < 1,\ 1 \leq i \leq r_1 + r_2\}$$

[11] This is reminiscent to how many conditions for topological groups need only be checked at the origin, including many proofs in real and complex analysis

Then $\sigma(\mathcal{O}_K) \cap B = \{0\}$

2. Let's show that the $\mathbb{R}$-span of $\sigma(\mathcal{O}_K)$ is our entire space. Since $\mathcal{O}_K$ is $\mathbb{Z}$-finitely generated we have
$$\langle \alpha_1, \alpha_2, ..., \alpha_n \rangle_{\mathbb{Z}} = \mathcal{O}_K$$
We want to show that
$$\text{span}_{\mathbb{R}}(\sigma_1(\alpha_i), ..., \sigma_{r_1}(\alpha_i), .., \tau_{r_2}(\alpha_i))_{i\in\{1,...,n\}} = \mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2}$$
Then the determinant of these values would be nonzero:
$$0 \neq \det \begin{pmatrix} \sigma_1(\alpha_1) & \cdots & \sigma_{r_1}(\alpha_1) & \text{Re}\tau_1(\alpha_1) & \text{Im}(\tau_1(\alpha_1)) & \cdots & \text{Im}\tau_{r_2}(\alpha_1) \\ \sigma_1(\alpha_2) & \cdots & \sigma_{r_1}(\alpha_2) & \text{Re}\tau_2(\alpha_2) & \text{Im}(\tau_2(\alpha_2)) & \cdots & \text{Im}\tau_{r_2}(\alpha_2) \\ & & & \vdots & & & \\ & & & \vdots & & & \\ & & & \vdots & & & \end{pmatrix}$$
By doing some elementary operations, we get the matrix to be of the form:
$$0 \neq \det \begin{pmatrix} \sigma_1(\alpha_1) & \cdots & \sigma_{r_1}(\alpha_1) & \tau_1(\alpha_1) & \overline{\tau_1(\alpha_1)} & \cdots & \overline{\tau_{r_2}(\alpha_1)} \\ \sigma_1(\alpha_2) & \cdots & \sigma_{r_1}(\alpha_2) & \tau_2(\alpha_2) & \overline{\tau_2(\alpha_2)} & \cdots & \overline{\tau_{r_2}(\alpha_2)} \\ & & & \vdots & & & \\ & & & \vdots & & & \\ & & & \vdots & & & \end{pmatrix}$$
Now, choose a primitive element $\alpha \in K$ so that $K = Q(\alpha)$. Then we know that $\mathbb{Z}[\alpha] = \langle 1, \alpha, ..., \alpha^{n-1} \rangle_{\mathbb{Z}} \subseteq \mathcal{O}_K$. We know this is of finite index. Letting $\alpha_i = \alpha^{i-1}$, we get a Vandermonde matrix. If M is the above matrix:
$$\det(M) = \prod_{\substack{\sigma \neq \sigma' \\ \sigma,\sigma' \in \text{Emb}_K(\mathbb{C})}} (\sigma(\alpha) - \sigma'(\alpha)) \neq 0$$
But then this implies that the volume of $\sigma(\mathcal{O}_K)$ is non-zero, as we sought to show.

Corollary 1.1.58: Sign Of Discriminant

Let $K/\mathbb{Q}$ be a number field, $\mathcal{O}_K$ its ring of integers, its discriminant. Then:
$$\text{sign}(d_K) = (-1)^{r_2}$$

***Proof*:**
Simply keep track of the elementary operations when transitioning between both matrices in

proposition 1.1.57

$$\det\begin{pmatrix} \sigma_1(\alpha_1) & \cdots & \sigma_{r_1}(\alpha_1) & \tau_1(\alpha_1) & \overline{\tau_1(\alpha_1)} & \cdots & \overline{\tau_{r_2}(\alpha_1)} \\ \sigma_1(\alpha_2) & \cdots & \sigma_{r_1}(\alpha_2) & \tau_2(\alpha_2) & \overline{\tau_2(\alpha_2)} & \cdots & \overline{\tau_{r_2}(\alpha_2)} \\ & & & \vdots & & & \\ & & & \vdots & & & \\ & & & \vdots & & & \end{pmatrix}$$

$$= \frac{1}{(2i)^{r_2}}\det\begin{pmatrix} \sigma_1(\alpha_1) & \cdots & \sigma_{r_1}(\alpha_1) & \mathrm{Re}\tau_1(\alpha_1) & \mathrm{Im}(\tau_1(\alpha_1)) & \cdots & \mathrm{Im}\tau_{r_2}(\alpha_1) \\ \sigma_1(\alpha_2) & \cdots & \sigma_{r_1}(\alpha_2) & \mathrm{Re}\tau_2(\alpha_2) & \mathrm{Im}(\tau_2(\alpha_2)) & \cdots & \mathrm{Im}\tau_{r_2}(\alpha_2) \\ & & & \vdots & & & \\ & & & \vdots & & & \\ & & & \vdots & & & \end{pmatrix}$$

completing the proof.

Hence, any ring of integers $\mathcal{O}_K$ forms a lattice in $K_{\mathbb{R}}$, and any ideal $\mathfrak{a} \subseteq \mathcal{O}_K$ will also form a lattice in $K_{\mathbb{R}}$. Since $K_{\mathbb{R}}$ is an inner product and a lattice is discrete, it is meaningful to define the volume of a lattice:

Definition 1.1.59: Volume of Lattice

let $\mathcal{O}_K$ be a ring of integers. Then if $\Gamma = (v_1, v_2, ..., v_n)$ is an integral basis for $\mathcal{O}_K$, then the *volume* of $\mathcal{O}_K$ is defined to be

$$\mathrm{vol}(\mathcal{O}_K) := \mathrm{vol}(K_{\mathbb{R}}/\Gamma) = \mathrm{coVol}(\Gamma) = |\det(\langle v_i, v_j\rangle)|^{1/2} =$$

where $\frac{1}{2^{r_2}}$ comes from the extra complex embeddings, as seen in the proof of corollary 1.1.58.

If you know some differential geometry, we may equivalently take the canonical differential form on $\mathbb{R}^n \cong \mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2}$:

$$\omega = dx_1 \wedge \cdots \wedge dx_n$$

and write:

$$\mathrm{vol}(\mathcal{O}_K) = \mathrm{coVol}(K_{\mathbb{R}}/\mathcal{O}_K) = \frac{1}{2^{r_2}}\left|\int_{K_{\mathbb{R}}/\mathcal{O}_K}\omega\right| = \frac{1}{2^{r_2}}\left|\int_P \omega\right|$$

where the $1/2^{r_2}$ comes from eliminating the extra complex dimension, and:

$$P = \left\{\sum a_i v_i \;\middle|\; 0 \leq a_i \leq 1\right\}$$

This also motivates the vocabulary of covolume. Note that any different choice of a basis will produce the same result since the change of basis matrix B will have determinant ± 1 which in the absolute value becomes 1, and hence it is well-defined. Note that every ideal $\alpha \subseteq \mathcal{O}_K$ will also produce a lattice, since it's a sublattice of $\mathcal{O}_K$, and hence we may find the volume of any sublattice. The way we shall find the volume of the lattice is by taking advantage of the discriminant and the index of the ideal:

Proposition 1.1.60: Volume of Ideal Lattice

Let $0 \neq \mathfrak{a} \subseteq \mathcal{O}_K$ be an ideal. Then $\Gamma_{\mathfrak{a}} = \sigma(\mathfrak{a})$ is a [complete] lattice in K_R and its fundamental mesh has volume

$$\mathrm{vol}(\Gamma_{\mathfrak{a}}) = \frac{1}{2^{r_2}}\sqrt{|\mathrm{disc}(K)}[\mathcal{O}_K : \mathfrak{a}]$$

As a consequence:

$$\mathrm{vol}(\mathcal{O}_K) = \frac{1}{2^{r_2}}\sqrt{|\mathrm{disc}(\mathcal{O}_K)}$$

Thus, the discriminant provides volume information for the lattice given by the ring of integers.

The theorem can equivalently be stated as:

$$\mathrm{vol}(\Gamma_{\mathfrak{a}}) = \left|\int_{\mathbb{R}^{r_1}\oplus\mathbb{R}^{r_2}/\Gamma_{\mathfrak{a}}} \omega\right|$$

***Proof*:**
Let $\alpha_1, \alpha_2, ..., \alpha_n$ be a $\mathbb{Z}$-basis of $\mathfrak{a}$ so that $\Gamma_{\mathfrak{a}} = \mathbb{Z}\sigma(\alpha_1) + \cdots + \mathbb{Z}\sigma(\alpha_n)$ is a lattice. Numbering the embeddings $\tau K \to \mathbb{C}$ as $\tau_1, \tau_2, ..., \tau_n$, define the matrix $A = (\tau_\ell \alpha_i)$ so that we get by proposition 1.1.39,

$$\mathrm{disc}(\mathfrak{a}) = \mathrm{disc}(\alpha_1, \alpha_2, ..., \alpha_n) = \det(A)^2 = [\mathcal{O}_K : \mathfrak{a}]^2\mathrm{disc}(\mathcal{O}_K) = [\mathcal{O}_K : \mathfrak{a}]^2 d_K$$

On the other hand, we have:

$$(\langle\sigma(\alpha_i), \sigma(\alpha_j)\rangle)_{ij} = \left(\sum_{\ell=1}^n \tau_\ell(\alpha_i)\overline{\tau}_\ell\alpha_j\right) = A\overline{A}^t$$

Thus:

$$\mathrm{vol}(\Gamma_{\mathfrak{a}}) = \left|\det\left(\langle\sigma(\alpha_i), \sigma(\alpha_j)\rangle\right)_{ij}\right|^{1/2} = |\det(A)| = \sqrt{|d_K|}[\mathcal{O}_k : \mathfrak{a}]$$

as we sought to show.

Thus, if $K = \mathbb{Q}(\sqrt{m})$ where m is square-free:

$$\mathrm{vol}(\mathcal{O}_K) = \begin{cases} \sqrt{m} & m \equiv 1 \bmod 4 \\ 2\sqrt{m} & m \equiv 2, 3 \bmod 4 \end{cases}$$

Absolute Norm and Geometric Inequalities

Besides introducing a geometric intuition for number rings and their ideals, the key reason to introduce volumes is that they will allow us make many bounding arguments on ideals using the lattice structure of $\mathcal{O}_K$. To do this, we introduce a new "norm" on the ideals of $\mathcal{O}_K$ that shall represent the size of the ideal with respect to the lattice. This will be helpful in many algebraic arguments, but also in upgrading many clasical results about primes (such as their distribution[12], see [9]).

[12] the number of primes p where $|p| \leq X$ tends asymptotically to $\frac{X}{\log(X)}$

Definition 1.1.61: Absolute Norm

Let $0 \neq \mathfrak{a} \subseteq \mathcal{O}_K$ be a nonzero ideal of $\mathcal{O}_K$. Then the *absolute norm* of $\mathfrak{a}$ is given by

$$\mathfrak{N}_K(\mathfrak{a}) := [\mathcal{O}_K : \mathfrak{a}]$$

If unambiguous, the notation $\mathfrak{N}(\mathfrak{a})$ is also used.

The absolute norm is inspired by the fact that ideals of $\mathcal{O}_K$, when thought of as representing a sublattice of a certain sparsity as compared to the lattice of $\mathcal{O}_K$. The norm of an ideal is always finite by proposition 1.1.39. Note that the notation $\mathfrak{N}_{K/\mathbb{Q}}$ should not be used, for it shall have a different meaning than $\mathfrak{N}$[13]

Lemma 1.1.62: Norm Of Principal Ideal

Let $(\alpha) \subseteq \mathcal{O}_K$ be a principal ideal. Then:

$$\mathfrak{N}((\alpha)) = \left|N_{K/\mathbb{Q}}(\alpha)\right|$$

***Proof*:**
Let $\beta_1, \beta_2, ..., \beta_n$ be a $\mathbb{Z}$-basis for $\mathcal{O}_K$. Then $\alpha\beta_1, \alpha\beta_2, ..., \alpha\beta_n$ is a $\mathbb{Z}$-basis for $(\alpha) = \alpha\mathcal{O}_K$. Then if $A = (a_{ij})$ is the change of basis formula so

$$\alpha w_i = \sum_j a_{ij}\omega_j$$

by proposition 1.1.39 $|\det(A)| = [\mathcal{O}_K : \alpha]$. But by definition of the basis of A, namely it is left multiplication, we get the definition of the norm:

$$|\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)| = \det(A)$$

giving us equality, as we sought to show.

Lemma 1.1.63: Ideal Divides Absolute Norm

Let $I \subseteq \mathcal{O}_K$ be an ideal. Then:

$$I \big| (\mathfrak{N}(I))$$

***Proof*:**
If we show $\mathfrak{N}(I) = [\mathcal{O}_K : I] \in I$, then $(\mathfrak{N}(I)) \subseteq I$, which by corollary 1.2.19 implies we're done. But this is simply some ring theory: Take $1 \in \mathcal{O}_K$. Then

$$\overline{n} = \underbrace{\overline{1 + \cdots + 1}}_{n \text{ times}} \in \mathcal{O}_K/I$$

Since $|\mathcal{O}_K/I| = n$, it must be that $\overline{n} = \overline{0}$, in particular $n \in I$. Since $n = [\mathcal{O}_K : I]$, we have the desired result.

[13] see definition 1.3.13

Lemma 1.1.64: Volume Form Reformulation

Let $I \subseteq R$ be an ideal of a number field K. Then:

$$\mathrm{vol}(I) = \mathfrak{N}(I)\mathrm{vol}(\mathcal{O}_K)$$

***Proof*:**
This is just reformulating proposition 1.1.60.

Theorem 1.1.65: Bounding Ideals

Let $K/\mathbb{Q}$ be a number field and $\mathcal{O}_K$ its number ring. Then there exists $\lambda \in \mathbb{R}_{>0}$ such that for every $0 \neq I \subseteq \mathcal{O}_K$, there exists a $\alpha \in I$ such that

$$|\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)| \leq \lambda \mathfrak{N}(I)$$

***Proof*:**
Let $\alpha_1, \alpha_2, ..., \alpha_n$ be an integral basis for R, and let $\sigma_1, \sigma_2, ..., \sigma_n$ denote the embeddings of K into $\mathbb{C}$. I claim that:

$$\lambda = \prod_{i=1}^{n} \left(\sum_{j=1}^{n} |\sigma_i(\alpha_j) \right)$$

Let I be any ideal, and m the unique positive integer satisfying:

$$m^n \leq \mathfrak{N}(I) < (m + 10^n$$

Consider the $(m+1)^n$ elements of $\mathcal{O}_K$ where:

$$\sum_{j=1}^{n} m_j \alpha_j \qquad m_j \in \mathbb{Z} \qquad 0 \leq m_j \leq m$$

Then two of these *must* be congruent $\mod I$ since there are more than $\mathfrak{N}(I)$ of them. Taking the difference of these two elements, we get a nonzero member of I of the form:

$$\alpha = \sum_{j=1}^{n} m_j \alpha_j \qquad m_j \in \mathbb{Z} \qquad |m_j| \leq m$$

Hence:

$$\left|\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)\right| \leq \prod_{i=1}^{n} |\sigma_i(\alpha)| \leq \prod -i = 1^n \left(\sum_{j=1}^{n} m_j |\sigma_i(\alpha_j)| \right) \leq m^n \lambda \leq \mathfrak{N}(I)\lambda$$

as we sought to show.

Proposition 1.1.66: Image Of Norm Is Finite Up To Units

Let K be a field, $\mathcal{O}_K$ its ring of integers, and $a \in \mathbb{Z}_{>0}$. Then up to multiplication by units, there are only finitely many element $\alpha \in \mathcal{O}_K$ such that

$$\mathrm{Nm}_{K/\mathbb{Q}}(\alpha) = a$$

***Proof*:**
Let $a \in \mathbb{Z}_{>0}$. Recall that $\mathcal{O}_K/a\mathcal{O}_K$ is finite. I claim that, up to multiplication by units, there is at most one element α in each coset such that $|\mathfrak{N}(\alpha)| = |\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)| = a$. For, if $\beta = \alpha + a\gamma$ where $\gamma \in \mathcal{O}_K$, then since $N(\beta) = a$, we can write $\beta = \alpha + N(\beta)\gamma$. Dividing both sides by β then re-arranging we get:

$$\frac{\alpha}{\beta} = 1 \pm \frac{\mathrm{Nm}_{K/\mathbb{Q}}(\beta)}{\beta}\gamma$$

since $\mathrm{Nm}_{K/\mathbb{Q}}(\beta) = \beta \prod_{\tau \neq \mathrm{id}} \tau(\beta)$, we have that $\alpha/\beta \in \mathcal{O}_K$. By diving by α, rearranging, and substituting $a = N(\alpha)$, we also get:

$$\frac{\beta}{\alpha} = 1 \pm \frac{\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)}{\alpha}\gamma \in \mathcal{O}_K$$

hence, α and β are associates. Thus, up to multiplication by units, there is at most $[\mathcal{O}_K : a\mathcal{O}_K]$ elements of norm $\pm a$.

For computational purposes, we add a tighter bound on the norms of ideals. We start with the following lemma:

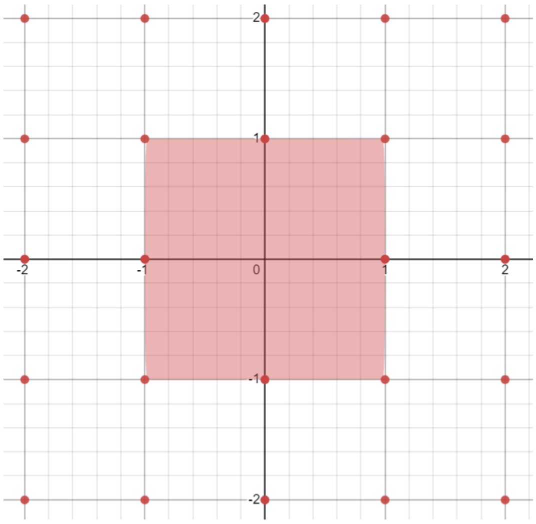
Lemma 1.1.67: Minkowski's Lattice Point Theorem

Let Γ be a discrete complete lattice in the Euclidean vector space V of dimension n and let X be a centrally symmetric convex subset of V. Then if

$$\mathrm{vol}(X) > 2^n \mathrm{vol}(\Gamma)$$

then X contains at least one nonzero lattice point $\gamma \in \Gamma$

Before proving this result, consider $\sigma(\mathbb{Z}[i])$. The smallest centrally symmetric set would



it is clear this square have volume arbitrarily close to 2^2, any large and it would intersect $\sigma(\mathbb{Z}[i])$. Similarly, if there was a long stretched area that tried to avoid the points of $\sigma(\mathbb{Z}[i])$, do to the size constraint, it will necessarily be to large to avoid intersecting the lattice:

***Proof*:**
It suffice to show that there exists two distinct lattice points $\gamma_1, \gamma_2 \in \Gamma$ such that

$$\left(\frac{1}{2}X + \gamma_1\right) \cap \left(\frac{1}{2}X + \gamma_2\right) \neq \emptyset$$

To show this, let's first say there exists such a point so that

$$\frac{1}{2}x_1 + \gamma_1 + \frac{1}{2}x_2 + \gamma_2$$

Then we may re-write to obtain:

$$\gamma = \gamma_1 - \gamma_2 = \frac{1}{2}x_1 - \frac{1}{2}x_2$$

Notice that this is in the center o the line segment joining $-x_1$ and x_2, and hence belongs in $X \cap \Gamma$.

Now, if the set $\frac{1}{2}X + \gamma$, $\gamma \in \Gamma$ were all pairwise disjoint, then the same is the case for the intersections $\Phi \cap (\frac{1}{2}X\gamma)$ for fundamental mesh Φ of Γ, hence we would have

$$\mathrm{vol}(\Phi) \geq \sum_{\gamma \in \Gamma} \mathrm{vol}\left(\Phi \cap \left(\frac{1}{2}X + \gamma\right)\right)$$

However, translation so $\Phi \cap (\frac{1}{2}X + \gamma)$ by $-\gamma$ creates the set $(\Phi - \gamma) \cap \frac{1}{2}X$ of equal volume, implying $\Phi - \gamma$ covers all of V, and hence $\frac{1}{2}X$ It follows from our assumption that

$$\mathrm{vol}(\Phi) \geq \sum)\gamma \in \Gamma \mathrm{vol}\left((\varphi - \gamma) \cap \frac{1}{2}X\right) = \mathrm{vol}(\frac{1}{2}X) = \frac{1}{2^n}X$$

contradicting the hypothesis.

Theorem 1.1.68: Minkowski's Bound

Let $[\mathfrak{a}] \in Cl(K)$ be an ideal class of degree n. Then there exists an integral ideal $\mathfrak{a}$ such that

$$\mathfrak{N}(\mathfrak{a}) \leq \frac{n!}{n^n}\left(\frac{4}{\pi}\right)^{r_2}\sqrt{|d_K|}$$

***Proof*:**
Let $\mathbb{R}^N \cong_{\mathbb{R}} \mathbb{R}^{r_1} \oplus \mathbb{R}^{r_2}$. Define the following norm on $\mathbb{R}^n$:

$$N(x) = x_1 \cdots x_r(x_{r_1+1}^2 + x_{r_1+2}^2)\cdots(x_{n-1}^2 + x_n^2)$$

(actually, type this up later, Marcus p.95)

Notice that is we re-arrange, we get:

$$\frac{n^n}{n!}\left(\frac{\pi}{4}\right)^{r_2} \leq \sqrt{|d_K}$$

Thus, we get a result that may seem obvious, but is in general rather difficult to prove:

Corollary 1.1.69: Discriminant Nonzero

Let $\mathcal{O}_K \neq \mathbb{Z}$. Then:

$$|d_K| > 1$$

***Proof*:**
immediate consequence of above since $n > 1$

1.2 Ring Structure and Dedekind Domains

So far, we have shown that rings of integers are lattices whose rank corresponds to the dimension of their corresponding field extension. Since they are rings, we may naturally ask what type of ring are they. Using example 1.1.17, we can look at the particular example of $\mathcal{O}_{\mathbb{Q}(\sqrt{-5})} = \mathbb{Z}[\sqrt{-5}]$. Is this ring a Euclidean Domain, Principal Ideal Domain, or Unique Factorization Domain? It is certainly Noetherian, being a finitely generated $\mathbb{Z}$-module (where $\mathbb{Z}$ is Noetherian). However, not all irreducible elements are prime, for example:

$$6 = 2 \cdot 3 = (1+\sqrt{-5})(1-\sqrt{-5})$$

showing that 2 is not prime. This may seem troubling since we may have to keep track of multiple factorizations showing there is more structure or tools that would need to be added/used to keep track of this information. Fortunately, there is a way to recover factorization. The goal of this section is to define a new type of ring for which the notion of Unique factorization is recovered, and show that all rings of integers are this type of ring. First, we may use the norm to see that a factorization into irreducible is always certainly possible in $\mathcal{O}_k$. For, if $\alpha = \beta\gamma$, then if β, γ are non-units:

$$1 < |N_{K/\mathbb{Q}}(\beta)| < |N_{K/\mathbb{Q}}(\alpha)| \qquad 1 < |N_{K/\mathbb{Q}}(\gamma)| < |N_{K/\mathbb{Q}}(\alpha)|$$

showing that it is a factorization domain. However this factorization need not be unique as we have demonstrated with $\mathcal{O}_{\mathbb{Q}(\sqrt{-5})} = \mathbb{Z}[\sqrt{-5}]$, another example being $21 = 3 \cdot 7 = (1 + 2\sqrt{-5})(1 - 2\sqrt{-5})$. We may see these are irreducible elements by using the norm. Doing the case of 3, if $3 = \alpha\beta$ for non-units α, β, then $9 = N_{K/\mathbb{Q}}(\alpha)N_{K/\mathbb{Q}}(\beta)$ would give us that $N_{K/\mathbb{Q}}(\alpha) = \pm 3$. Given $\alpha = a + b\sqrt{-5}$ we get

$$N_{K/\mathbb{Q}}(a + b\sqrt{-5}) = a^2 + 5b^2 = \pm 3$$

But this certainly has no solutions in $\mathbb{Z}$, and hence 3 is irreducible in $\mathcal{O}_K$. A similar argument can be made for $7, 1 \pm 2\sqrt{-5}$. Furthermore, the numbers 3 and 5 are not associated to $1 \pm 2\sqrt{-5}$, for we would require that

$$\frac{1 \pm 2\sqrt{-5}}{3}, \frac{1 \pm 2\sqrt{-5}}{7} \in \mathcal{O}_K$$

both of which we can check are not using the norm. Hence, we certainly have different factorizations.

The failure of unique factorization has lead Eduard Kummer to try to embed the ring of integers of K into a bigger domain of "ideal numbers" where unique factorization into "ideal prime numbers" would hold (which was inspired by the discovery of complex numbers somehow being "ideal" for roots polynomial roots). As an example, our original decomposition in $\mathbb{Z}[\sqrt{-5}]$

$$21 = 3 \cdot 7 = (1 + 2\sqrt{-5})(1 - 2\sqrt{-5})$$

would factor into ideal prime numbers $\mathfrak{p}_1, \mathfrak{p}_2, \mathfrak{p}_3, \mathfrak{p}_4$ with the relations

$$3 = \mathfrak{p}_1\mathfrak{p}_2 \qquad 7 = \mathfrak{p}_3\mathfrak{p}_4 \qquad 1 + 2\sqrt{-5} = \mathfrak{p}_1\mathfrak{p}_3 \qquad 1 - 2\sqrt{-5} = \mathfrak{p}_2\mathfrak{p}_4$$

Then we would have fixed non-uniqueness since we would get the factorization:

$$21 = (\mathfrak{p}_1\mathfrak{p}_2)(\mathfrak{p}_3\mathfrak{p}_4) = (\mathfrak{p}_1\mathfrak{p}_3)(\mathfrak{p}_2\mathfrak{p}_4)$$

Some key properties that these ideals numbers should have is that for $a, b, \lambda \in \mathcal{O}_K$, an ideal number $\mathfrak{a}$ should satisfy the following natural divisibility restrictions:

$$\mathfrak{a}\big|a \text{ and } \mathfrak{a}\big|b \Rightarrow \mathfrak{a}\big|a \pm b \qquad a\mathfrak{a}\big|a \Rightarrow \mathfrak{a}\big|\lambda a$$

An furthermore an ideal number $\mathfrak{a}$ ought to be determinable by the set of its divisors in $\mathcal{O}_K$, that is

$$\underline{\mathfrak{a}} = \left\{a \in \mathcal{O}_K \;\middle|\; \mathfrak{a}\big|a\right\}$$

But given these rules, this set is simply an ideal of $\mathcal{O}_K$ as we know the notion of ideal. It was Richard Dedekind that took the notion of ideal numbers that Kummer invented and invented the notion of *ideals* of $\mathcal{O}_K$ giving the modern definition of ideals. Then division $\mathfrak{a}\big|a$ is replaced by belonging $a \in \mathfrak{a}$, and divisibility $\mathfrak{a}\big|\mathfrak{b}$ is replaced by containment $\mathfrak{b} \subseteq \mathfrak{a}$.

Hence, we shall seek a unique factorization of *ideals* into *prime ideals* in $\mathcal{O}_K$. The 3 key properties of $\mathcal{O}_K$ that require this factorization was singled out by Dedekind and form a class of rings larger than rings of integers and are worth taking a look at on their own:

Definition 1.2.1: Dedekind Domain

Let R be an integral domain. Then if:

1. R is a Noetherian ring
2. every nonzero prime ideal of R is maximal
3. R is normal ($\overline{R}^{K(R)} = R$)

then R is called a *Dedekind Domain*

The Noetherian condition can be thought of as a natural eliminator of any infinites lurking around, while the prime implies maximal condition can be thought of as making sure that there can be a prime ideal that further contains prime ideals. In particular, we shall see that $(a) \subseteq (b)$ if and only if $b|a$, which doesn't make sense if we allow for prime ideals being contained in larger prime ideals.

1.2.2 Foreshadowing in Algebraic Geometry Notice that we have already seen a ring with these conditions before, namely if V is a smooth 1-dimensional variety, then the associated k-algebra is Noetherian, 1 [Krull] dimensional and is normal (normal implies smoothness[a]). In Algebraic Geometry, Dedekind domains can be thought of as representing 1-dimensional smooth schemes (in particular, 1-dimensional smooth $\mathbb{Z}$-curves)

[a]But only necessarily in 1 dimension; in higher dimension normalness need not imply smoothness, see [1]

Lemma 1.2.3: PID's, UFD's, and Dedekind Domain

Let R be a ring. Then:

1. If R is a PID, R is a Dedekind Domain
2. If R is a UFD, R need not be a Dedekind Domain
3. If R is a Dedekind domain and a UFD, it is a PID

The following image demonstrates the realation:

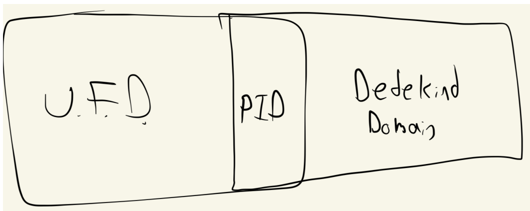


Figure 1.1: Venn diagram

***Proof*:**
R is trivially Noetherian and all prime ideals are maximal. For normalness, since R is a Unique Factorization Domain, $\overline{R}^{K(R)} = R$, hence R is a Dedekind Domain.

For the counter-example, take $\mathbb{Z}[\sqrt{-13}]$ and find two ways of decomposing 14. The last is an exercise.

For algebro-geometric purposes, the following characterizes Dedekind domains locally:

Lemma 1.2.4: Dedekind Domain As DVR

Let R be commutative ring. Then R is a Dedekind domain if and only if for each prime P, R_P is a DVR. Thus, Dedekind domains are local DVR's.

***Proof*:**
exercise.

Lemma 1.2.5: Dedekind Domains And Localization

Let R be a Dedekind domain and $S \subseteq R \setminus \{0\}$ is a multiplicative subset. Then $S^{-1}R$ is also a Dedekind Domain.

***Proof*:**
The localization of Noetherian rings is Noetherian, and by the correspondence of ideals in localizations, we have that prime implies maximal in $S^{-1}R$. Finally, $S^{-1}R$ is integrally closed since if $x \in K = K(R)$ satisfies:

$$x^n + \frac{a_1}{s_1}x^{n-1} + \cdots + \frac{a_n}{s_n} = 0 \qquad \frac{a_i}{s_i} \in S^{-1}R$$

Then multiplying with the nth power of $s_1 \cdots s_n$ gives that sx is integral over R, hence $sx \in R$, and so $x \in S^{-1}R$, as we sought to show.

Importantly for number theory, all rings of integers are Dedekind domains:

Theorem 1.2.6: Ring Of Integers Are Dedekind Domains

Let $\mathcal{O}_K$ be a ring of integers. Then it is a Dedekind domain.

***Proof*:**
By proposition 1.1.32, $\mathcal{O}_K$ is Noetherian, and by definition $\mathcal{O}_K$ is normal, namely $\mathcal{O}_K = \overline{\mathbb{Z}}^K$ meaning by [2, Integral Elements Form Rings]:

$$\overline{\mathcal{O}_K}^K = \overline{\overline{\mathbb{Z}}^K}^K = \overline{\mathbb{Z}}^K$$

What's left to show is that every prime ideal is maximal, which is equivalent to showing $\mathcal{O}_K/\mathfrak{p}$ is a field. In particular, since $\mathcal{O}_K/\mathfrak{p}$ is always an integral domain, if we show that it is finite then by [5, Finite Integral Domain is a Field] we shall have that it's a field. The key intuition comes from the fact that $\mathcal{O}_K$ and its ideals form a lattice, meaning when we quotient we necessarily get a finite ring. Furthermore, the volume of the ideal is determined by the discriminant and the index, which will determine the cardinality of the quotient ring[a]

Let $0 \neq \alpha \in \mathfrak{p}$. Then $(p) = \mathfrak{p} \cap \mathbb{Z}$ is also a prime ideal. Since $\mathfrak{p} \subseteq \mathcal{O}_K$, we know α is associated to a minimal polynomial $m_{\alpha,\mathbb{Z}}(x)$ where

$$m_{\alpha,\mathbb{Z}}(\alpha) = \alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha + a_0 = 0 \qquad a_i \in \mathbb{Z}$$

Then $a_0 = -(\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha) \in \mathfrak{p} \subseteq \mathcal{O}_K$, and since $m_{\alpha,\mathbb{Q}}(x)$ is irreducible with integer coefficients, $a_0 \neq 0$ and $a_0 \in (p) = p\mathbb{Z}$. From here, we shall give two proofs: first, we may directly appeal to the fact that

$$(\mathcal{O}_K/\mathfrak{p})/(\mathbb{Z}/p\mathbb{Z})$$

is an integral extension (see [2, Integral Extensions]) where the base ring is a field, hence $\mathcal{O}_K/\mathfrak{p}$ is a field by [2, Integral Extension over Fields]. Another is the following: we have a non-trivial ideal with chains of inclusions $(a_0) \subseteq \mathfrak{p} \subseteq \mathcal{O}_K$ implying

$$|\mathcal{O}_K/\mathfrak{p}| \leq |\mathcal{O}_K/(a_0)| = |\mathcal{O}_K/a_0\mathcal{O}_K|$$

Now, since we are quotienting ideals, this is a $\mathbb{Z}$-module quotient, and since we are quotienting n-rank free $\mathbb{Z}$-modules, we get that

$$|\mathcal{O}_K/(a_0)| = p^n$$

But then $\mathcal{O}_K/\mathfrak{p}$ is finite (even better, its order divides $|a_0|^n$). Thus, $\mathfrak{p}$ must be maximal, and since $\mathfrak{p}$ was arbitrary every prime ideal is maximal, completing the proof.

[a]We shall soon see that the index can be thought of as the "norm" of a general ideal; this will mean the constant term of a minimal polynomial represent the "size", sheading some light on the proof

Since in general $\mathcal{O}_K$ is not a Unique Factorization Domain, rings of integers represent a class of rings where prime implies maximal, but factorization is not unique. In this way, we see that the integral closure of $\mathbb{Z}$ with respect to K may be far from having the nice element-wise properties we are used to in PIDs, but this shall be salvaged through having an understandable ideal structure and a way to measure how far away our ring is from being a PID.

Another important example of a Dedekind domain that is not a number ring is the ring is $k[x]$ (since it's a PID). We shall later see that integral extensions of Dedekind domains are Dedekind domains, an so these types of rings will form a basis for another large class of Dedekind domains that are not number rings.

1.2.1 Fractional Ideals in Rings

Following from the idea that ideals are the generalisation of numbers, we may define the following divisibility notions usually used for the [algebraic] lattice of integers induced by divisibility. These concepts shall be defined for general commutative rings:

Definition 1.2.7: GCD, LCM, Divisibility Of Ideals

let R be a commutative ring, $\mathfrak{a}, \mathfrak{b} \subseteq R$ ideals of R. Then:

$$\gcd(\mathfrak{a}, \mathfrak{b}) = (\mathfrak{a}, \mathfrak{b}) = \mathfrak{a} + \mathfrak{b} \qquad \operatorname{lcm}(\mathfrak{a}, \mathfrak{b}) = \mathfrak{a}\mathfrak{b}$$

And we can say that $\mathfrak{a}, \mathfrak{b}$ are relatively prime when $\gcd(\mathfrak{a}, \mathfrak{b}) = (1)$. Divisibility $\mathfrak{a}\big|\mathfrak{b}$ is defined as $\mathfrak{b} \subseteq \mathfrak{a}$, and hence we have an order and the ideals similar to that given by divisibility.

As we've seen in [5, Exploring Integral Domains: Factorization and Valuation], gcd and lcm are usually well-behaved in special classes of rings, usually UFDs. We shall show they are well-behaved in Dedekind domains since the ideals shall have unique factorizations. Let's now prove some lemmas for this result.

One finiteness fact is needed throughout, and it is the reason the Noetherian hypothesis appears in the definition of a Dedekind domain at all: in a Noetherian domain every nonzero proper ideal *contains* a product of nonzero prime ideals, each of which contains it ([2, Nonzero Ideals Contain Products of Nonzero Primes]). An ideal need not factor into primes — that is the theorem we are working toward, and it needs the full strength of a Dedekind domain — but it always contains such a product, and that much is available from the Noetherian condition alone.

For the following lemmas, we bring back the notion of fractional ideal that was defined in [2, Noetherian Rings, Hilbert's Basis Theorem, and Localization]. These ideals were at the time a way of extending ideals into the ring of fractions, and were shown to work well with localization. Due to all the properties of Dedekind domains, the product of ideals can be made into a *group*, even an *abelian group*. Since the fractional ideal structure will have such a strong restriction, this will produce incredible results on the relations between ideals. To begin, the following lemma shows that no prime ideal acts like the identity:

Lemma 1.2.8: Fractional Ideals

Let $P \subseteq R$ be a prime ideal of the Dedekind domain R, $K = K(R)$. Define

$$(R :_K P) = P^{-1} = \{x \in K \mid xP \subseteq R\}$$

Then

$$AP^{-1} = \left\{ \sum_i a_i x_i \;\middle|\; a_i \in A,\; x_i \in P^{-1} \right\} \neq A$$

for every ideal $0 \neq A \subseteq R$

Note that by definition $R \subseteq P^{-1}$ and so P^{-1} is not in general an ideal. The simplest "fractional ideal" would be those of $\mathbb{Z}$, namely $(n)^{-1} = \frac{1}{n}\mathbb{Z}$.

***Proof*:**
Note that when we work with the inverse of elements, we are considering them as being part of $K(R)$.

We shall first show the special case of $P^{-1} \neq R$. Let $a \in P$ with $a \neq 0$. The ideal (a) is nonzero and proper, so it contains a product of nonzero primes by the fact recalled above ([2, Nonzero Ideals Contain Products of Nonzero Primes]); consider such a product $P_1 P_2 \cdots P_r \subseteq (a) \subseteq P$

with r minimal. Then one of the P_i, is containing in P, without loss of generality say $P_1 = P$ (since P_l is maximal). Indeed, if non of the P_i were contained in P, then for every i there would exist $a_i \in P_i \setminus P$ such that $a_1 \cdots a_r \in P$, but P is maximal. Next, by minimality $P_2 \cdots P_r \not\subseteq (a)$, so there exists $b \in P_2 \cdots P_r$ such that $b \notin (a) = aR$, that is $a^{-1}b \notin R$. On the other hand, since $P = P_1$, $P_1 b \in P_1 P_2 \cdots P_n r$, hence we have $bP \subseteq (a) = aR$ that is $a^{-1}bP \subseteq R$, and so $a^{-1}b \in P^{-1}$. It follows that $P^{-1} \neq R$, namely $P^{-1} \supsetneq R$.

Next, let $0 \neq A \subseteq R$ be an ideal. Since R is Noetherian, let

$$A = (\alpha_1, \alpha_2, ..., \alpha_n)$$

For the sake of contradiction, say $AP^{-1} = A$. Then for every $x \in P^{-1}$

$$xa_i = \sum_j a_{ij}\alpha_j \qquad a_{ij} \in R$$

Letting M be the matrix $(x\delta_{ij} - a_{ij})$, we get $M(\alpha_1, \alpha_2, ..., \alpha_n)^t = 0$. By Cramers rule, if $d = \det(M)$, then $d\alpha_1 = \cdots = d\alpha_n = 0$, and so $d = 0$. Then x is integral over R, since its' the zero of the monic polynomial

$$f(X) = \det(X\delta_{ij} - a_{ij}) \in R[x]$$

Thus, since R is integrally closed $x \in R$. But then $P^{-1} = R$; a contradiction to your first part, as we sought to show.

Definition 1.2.9: Fractional Ideal and Quotient ideal

Let R be an integral domain and $K = K(R)$. Then a *fractional ideal* J of K is a finitely generated non-trivial R-submodule of K where there exists a $r \in R$ such that $rJ \subseteq R$ is an ideal.
If $I, J \subseteq R$ are ideals, then the *quotient ideal* of I with respect to J is:

$$(I : J) = \{x \in R \mid xJ \subseteq I\}$$

If $I = R$,then $(R : J)$ is denoted J^{-1}

Fractional ideals are multiplied in the similar way ideals are multiplied (treat them as sub-algebra multiplication). Since all ideals of R are rank 1 submodules of R, they are all fractional ideals. For those who are motivated by algebraic geometry, fractional ideals shall be shown to form line bundles over the smooth $\mathbb{Z}$-curves $\mathcal{O}_K$ [14].

Quotient ideals have a lot of interesting theory, for example to motivate their name if J is principle and $I \subseteq J$, then $I = J \cdot (I : J)$. There is no room in this chapter to explore these in detail, hence they are left as exercises 1.2.1(1.2.1 (6)).

Fractional ideals that are ideals are given a name:

[14]See section 1.7 to see how to interpret rings of integers as smooth $\mathbb{Z}$-curves, and ref:HERE for interpreting fractional ideals as line bundles

Definition 1.2.10: Integral Ideals

Let R be an integral domain and $K = K(R)$. Then a fractional ideal $I \subseteq K$ that is of rank 1 is called an *integral ideal*, and $I \subseteq R$.

Example 1.2.11: Fractional Ideal

1. Since $\mathbb{Z}$ is a Principal Ideal Domain, every ideal is of the form $a\mathbb{Z}$ for $a \in \mathbb{Z}$. Then $(a)^{-1} = a^{-1}\mathbb{Z}$. Hence, the fractional ideals are in bijective correspondence with $\mathbb{Q}^{\times}_{>0}$. Naturally, all ideals of $\mathbb{Z}$ are fractional ideals, in particular they are integral ideals.

2. More generally, if $a \in K^{\times}$, then we get fractional principal ideals $(a) = aR$, and we have a bijective correspondence between the fractional ideals and $K^{\times}/R^{\times}$.

The condition that there must exist an $r \in R$ such that $rJ \subseteq R$ is superfluous in Noetherian domains:

Corollary 1.2.12: Fractional Ideal Representative

Let R be a Noetherian Domain, $K = K(R)$, and $\mathfrak{a} \subseteq K$ a fractional ideal. Then there exists a $r \in R \setminus \{0\}$ and ideal $I \subseteq R$ such that

$$\mathfrak{a} = r^{-1}I$$

Stated differently, $\mathfrak{a} \cong_R I$

If $R = \mathcal{O}_K$, we may take $r \in \mathbb{Z}$. Furthermore, the above is an if and only if statement.

***Proof*:**
Use the fact that R is Noetherian. Let $\mathfrak{a} = (\alpha_1, \alpha_2, ..., \alpha_n)$. In particular, since $\mathfrak{a}$ is a finitely generated $\mathbb{Z}$-module:

$$\mathfrak{a} = \sum R\alpha_i \qquad\qquad \alpha_i \in K(R)$$

by definition, $\alpha_i = a_i/b_i$ for $a_i, b_i \in R$. Taking $b = b_1 b_2 \cdots b_n$, we get

$$b\mathfrak{a} \subseteq R$$

Hence, $b\mathfrak{a}$ is an ideal, say $b\mathfrak{a} = I$ Then $\mathfrak{a} \subseteq b^{-1}I$.

This immediately implies that $\mathfrak{a} \cong_R I$, for if $\mathfrak{a} = (\alpha_1, \alpha_2, ..., \alpha_n)$ so that $I = (r^{-1}\alpha_1, ..., r^{-1}\alpha_n)$, then define $\varphi(\alpha_i) = r^{-1}\alpha_i$ is an R-module isomorphism.

Notice that this generalizes the definition of ideals, if R is a Dedekind ring over itself, then the ideals of R are the finitely generated sub-modules. Then if $K = \text{Frac}(R)$, the fractional ideals are the finitely generated K-modules.

Corollary 1.2.13: Inverse of Prime Ideals

Let R be a Dedekind domain and $\mathfrak{p} \subseteq R$ be a prime ideal. Then:

$$\mathfrak{p}\mathfrak{p}^{-1} = (1)$$

***Proof*:**
If $x \in \mathfrak{p}\mathfrak{p}^{-1}$, then $x = \sum p_i k_i$ for $p_i \in \mathfrak{p}$ and $k_i \in \mathfrak{p}^{-1}$. But by definition of $\mathfrak{p}^{-1}$, we have that $p_i k_i \in R$, hence x element is in R, and so $\mathfrak{p}\mathfrak{p}^{-1} \subseteq R$. Next, certainly $\mathfrak{p} \subseteq \mathfrak{p}\mathfrak{p}^{-1} \subseteq R$. Since $\mathfrak{p}$ is maximal, either $\mathfrak{p}\mathfrak{p}^{-1} = \mathfrak{p}$ or $\mathfrak{p}\mathfrak{p}^{-1} = R$. The former is a contradiction by lemma 1.2.8, and hence $\mathfrak{p}\mathfrak{p}^{-1} = (1)$, completing the proof.

Lemma 1.2.14: Invariance Number Ring Multiplication

Let $t \in K$, $\mathfrak{a} \subseteq K$ a fractional ideal. If $tI = I$, then $t \in R$

***Proof*:**
Since I is a finitely generated R module, there exists a surjective R-module homomorphism $\varphi : R^n \to I$. Next, since $tI = I$, there exists an R-module homomorphism $t : I \to I$ sending $i \mapsto ti$. Then $t \in \text{Hom}_R(I, I)$. Since R^n is free, if we consider the following lift:

$$\begin{tikzcd}
R^n \arrow[r, two heads] & I \\
R^n \arrow[u, dashed, "\text{lift}"] \arrow[ur, two heads] \arrow[r, two heads, "\varphi"] & I \arrow[u, "t"']
\end{tikzcd}$$

Then $xt \in \text{Hom}_R(R^n, I)$. Taking $\alpha \in \text{Hom}_R(R^n, R^n)$:

$$\det(xI - \alpha)(\alpha) = 0 \Rightarrow \det(xI - \alpha)(t) = 0$$

now using that R is a Dedekind domain, namely that it's integrally closed, we get that $t \in R$, completing the proof.

1.2.2 Prime Decomposition in Dedekind Domain

With these properties, we may show the factorization of prime ideals in Dedekind domains.

Theorem 1.2.15: Unique Factorization Of Ideals In Dedekind Domains

Let R be a Dedekind Domain and $\mathfrak{a} \subseteq R$ a non-trivial proper ideal. Then there exists unique $\mathfrak{p}_1, \mathfrak{p}_2, ..., \mathfrak{p}_n$ such that

$$\mathfrak{a} = \mathfrak{p}_1\mathfrak{p}_2 \cdots \mathfrak{p}_n$$

***Proof*:**
We start with the existence of such a factorization. Let $\mathfrak{M}$ be the set of all non-trivial proper ideals which do not admit a prime ideal decomposition. If $\mathfrak{M}$ is non-empty, then since R is Noetherian there exists a maximal element, say $\mathfrak{m}$. Then this ideal is contained in some maximal ideal $\mathfrak{p}$. Since $R \subseteq \mathfrak{p}^{-1}$, we get the following chain of inclusions:

$$\mathfrak{m} \subseteq \mathfrak{m}\mathfrak{p}^{-1} \subseteq \mathfrak{p}\mathfrak{p}^{-1} \subseteq R$$

by lemma 1.2.8, we have $\mathfrak{m} \subsetneq \mathfrak{m}\mathfrak{p}^{-1}$ and $\mathfrak{p} \subsetneq \mathfrak{p}\mathfrak{p}^{-1} \subseteq R$. By maximality of $\mathfrak{p}$, $\mathfrak{p}\mathfrak{p}^{-1} = R$, and by maximality of $\mathfrak{m}$ in $\mathfrak{M}$, $\mathfrak{m}\mathfrak{p}^{-1}$ admits a prime ideal decomposition, say

$$\mathfrak{m}\mathfrak{p}^{-1} = \mathfrak{p}_1 \cdots \mathfrak{p}_r$$

But then, since $\mathfrak{p}\mathfrak{p}^{-1} = R$, $\mathfrak{m}R = \mathfrak{m}$ and so

$$\mathfrak{m} = \mathfrak{m}\mathfrak{p}^{-1}\mathfrak{p} = \mathfrak{p}_1 \cdots \mathfrak{p}_r\mathfrak{p}$$

contradicting the definition of $\mathfrak{m}$. Hence $\mathfrak{M}$ is an empty-set

For uniqueness, let

$$\mathfrak{a} = \mathfrak{p}_1\mathfrak{p}_2 \cdots \mathfrak{p}_r = \mathfrak{q}_1\mathfrak{q}_2 \cdots \mathfrak{q}_s$$

be two prime ideal factorizations of $\mathfrak{a}$. Then by the repeated application of property of prime ideals ($\mathfrak{a}\mathfrak{b} \subseteq P \Rightarrow \mathfrak{a} \subseteq P$ or $\mathfrak{b} \subseteq P$), we have that $\mathfrak{p}_1$ divides a factor of the right hand side, say $\mathfrak{q}_1$, and by maximality is thus equal to $\mathfrak{q}_1$. Then we can multiply both sides by $\mathfrak{p}_1^{-1}$ and since $\mathfrak{p}_1 \neq \mathfrak{p}_1\mathfrak{p}_1^{-1} = R$ we get

$$\mathfrak{p}_2 \cdots \mathfrak{p}_r = \mathfrak{q}_2 \cdots \mathfrak{q}_s$$

Then proceeding inductively, we will get that $r = s$ and $\mathfrak{p}_i = \mathfrak{q}_i$, up to renumbering, completing the proof.

1.2.16 Note It is in fact equivalent to define a Dedekind domain as ring in which each ideal factors uniquely, showing that this theorem characterizes Dedekind domains, see exercise ref:HERE.

From now on, we shall group the prime ideals to write

$$\mathfrak{a} = \mathfrak{p}_1^{n_1} \cdots \mathfrak{p}_r^{n_r} \qquad n_i > 0$$

If $\mathfrak{a} = (a)$ is a principal ideal, we shall use the slight abuse of notation and write

$$a = \mathfrak{p}_1^{n_1} \cdots \mathfrak{p}_r^{n_r} \qquad n_i > 0$$

Similarly, we shall often write $\mathfrak{a}\big|a$ instead of $\mathfrak{a}\big|(a)$. An important result worth emphasizing is the structure of R/I for an ideal $I \subseteq R$. First, note that if $\mathfrak{p}_i + \mathfrak{p}_j = (1)$, then $\mathfrak{p}_i^{n_i} + \mathfrak{p}_j^{n_j} = (1)$ (this was exercise ref:HERE (in CRT section)); the trick is to consider $1 = 1^{n_i+n_j+1} = (ax+by)^{n_i+n_j+1}$, use the binomial theorem, and group appropriate elements. Then by the Chinese Remainder Theorem:

$$R/I = \bigoplus_i R/P_i^{n_i}$$

By the 4th isomorphism theorem, $\overline{I} \subseteq R/P_i^{n_i}$ if and only if $P_i^{n_i} \subseteq I$, that is $I\big|P_i^{n_i}$. But by unique factorization, that means that $I = P_i^k$ for $1 \leq k \leq n_i$. Hence, every quotient has only finitely many ideals, each ideal I corresponding to a set of prime power of the decomposition of I via the isomorphism given by the CRT. This matches with the result that $R/(a)$ is a finite ring, showing there can only be finitely many ideals, and the prime factorization theorem for ideals gives us the structure of the ideals of $R/(a)$ through the correspondence given by the 4th isomorphism theorem. This also immediately shows that R/I is *Artinian*[15].

Using the fact that ideals factor and "behave" like the positive rational numbers ($-\mathfrak{p} = \mathfrak{p}$ since -1 is a unit), we can conclude some powerful results. We have shown that every ideal has a unique inverse. This immediately shows that:

$$\mathfrak{ab} = \mathfrak{ac} \quad \implies \quad \mathfrak{b} = \mathfrak{c}$$

by the Cancellation law. Next, it is easy to see is that we may generalize the notion of finding GCD and LCM of UFD's to Dedekind domains:

Corollary 1.2.17: GCD, LCM, and Divisibility In Dedekind Domains

Let $\mathfrak{a}, \mathfrak{b} \subseteq R$ be ideals in a Dedekind domain R. Then

1. $\operatorname{lcm}(\mathfrak{a}, \mathfrak{b}) = P_1^{\max(a_1,b_1)} \cdots P_k^{\max(a_k,b_k)}$
2. $\gcd(\mathfrak{a}, \mathfrak{b}) = P_1^{\min(a_1,b_1)} \cdots P_k^{\min(a_k,b_k)}$
3. $\mathfrak{a}\big|\mathfrak{b}$ if and only if there exists a $\mathfrak{c}$ such that $\mathfrak{ac} = \mathfrak{b}$

***Proof*:**
For the first two, work with the factorization and property of ideals.

For 3, if $\mathfrak{a}\big|\mathfrak{b}$, we have that $\mathfrak{b} \subseteq \mathfrak{a}$. In a Dedekind domain, multiplying by $\mathfrak{a}^{-1}$, we get $\mathfrak{a}\mathfrak{a}^{-1} = (1)$ by extending lemma 1.2.8 with the above theorem. Hence:

$$\mathfrak{b}\mathfrak{a}^{-1} \subseteq \mathfrak{a}\mathfrak{a}^{-1} \subseteq R$$

showing that $\mathfrak{b}\mathfrak{a}^{-1} = \mathfrak{c} \subseteq R$ is an ideal, and $\mathfrak{b} = \mathfrak{ca}$, which properly generalizes the notion factorizing elements to the level of ideals.

We know that every ideal has an inverse. However, that inverse is a fractional ideal. Fortunately, there always exists an integral ideal that is "almost" the inverse, in that the product is a principal ideal[16]:

Corollary 1.2.18: Exists J such that IJ is Principal

Let R be a Dedekind Domain and $I \subseteq R$ an ideal. Then there exists an ideal $J \subseteq R$ such that $IJ = (a)$ for some element $a \in I$. In other words, there exists an ideal J such that the product IJ is principal.

[15] Equivalently, $\operatorname{kdim}(R/I) = 0$, and hence it is Artinian

[16] this almost inverse can be made into a concrete inverse when we show the principal ideals form a normal subgroup formed by the group of fractional ideals

***Proof*:**
Let $I = P_1^{n_1} \cdots P_k^{n_k}$. For each i, choose a nonzero element $r_i \in P_i^{n_i} \setminus P_i^{n_i+1}$. By the Chinese Remainder Theorem, there exists an $a \in R$ such that

$$a \equiv r_i \bmod P_i^{n_i+1} \qquad 1 \leq i \leq k$$

Thus, $a \in P_i^{n_i} \setminus P_i^{n_i+1}$ for each i, and so since $a \in P_i^{n_i}$, the power of P_i in the factorization is n_i. Now, taking the prime factorization of (a), we get:

$$(a) = P_1^{n_1} \cdots P_k^{n_k} P_{n+1}^{n_{k+1}} \cdots P_m^{n_m}$$

Letting $J = P_{n+1}^{n_{k+1}} \cdots P_m^{n_m}$ gives us

$$(a) = IJ$$

This same process can be repeated for fractional ideals; for any fractional ideal I, take $r \in R$ such that $rI = J \subseteq R$. Then proceed with the proof above, giving us an ideal L such that $(a) = JL = rIL$. Then $(r^{-1}a) = IL$, completing the proof.

In section 1.3.1, we shall find a more systematic way of finding the prime ideals that multiply to get a principal ideal using Galois groups. As seen with rings of integers, not all Dedekind domains are PIDs since they are not all UFDs. With the factorization theorem, we get that they are the next best thing:

Corollary 1.2.19: PIR, and Generation Of Ideals In Dedekind Domains

Let R be a Dedekind domain and $I \subseteq R$ an ideal. Then R/I is a Principal Ideal Ring, and I is generated by *at most* 2 elements.

***Proof*:**
For any ideal $I \subseteq R$, by the Chinese remainder theorem and prime factorization of I:

$$R/I = \bigoplus_i R/P_i^{n_i}$$

We shall show that each $R/P_i^{n_i}$ is a principal ideal ring, from which we may conclude by the isomorphism that R/I is a principal ideal ring. This shall give the desired result, for if $I \subseteq R$, then for any $J \subseteq I$ we may take R/J where $\overline{I} \subseteq R/(\alpha)$. Then since all ideals of $R/(\alpha)$ are principal, $\overline{I} = (\overline{\beta})$, that is $I = J + (\beta) = (\alpha, \beta)$. If we take $\alpha \in I$, then we would get $(\alpha) \subseteq I$ and $R/(\alpha)$, and so $I = (\alpha) + (\beta) = (\alpha, \beta)$, showing each ideal is generated by at most 2 elements.

Hence, we prove $R/P_i^{n_i}$ is a Principal ideal ring (PIR). If $\overline{J} \subseteq R/(P_i^{n_i})$, then $P_i^{n_i} \subseteq J \subseteq R$, meaning $J|P_i^{n_i}$, and so by unique factorization it must be that $J = P_i^k$ for some $1 \leq k \leq n_i$, which has image $\overline{P_i^k}$ (i.e., there are finitely many ideals).

Now, if $\overline{P_i} = \overline{P_i^2}$, then $\overline{P_i} = \overline{P_i^n} = \overline{0}$, and so $P_i = P_i^n$ implying that $R/P_i^{n_i}$ is a field, and it then automatically a PIR. If $\overline{P_i} \neq \overline{P_i^2}$, take $\alpha \in \overline{P_i} \setminus \overline{P_i^2}$. Then $\overline{\alpha} \notin \overline{P_i^2} \supseteq \overline{P_i^3} \supseteq \cdots$. Thus it must be that

$$\overline{P_i} \big| (\overline{\alpha})$$

But since the only ideals of $R/P_i^{n_i}$ are $\overline{P_i}, ..., \overline{P_i^{n_i-1}}$, it must follow that $(\overline{\alpha}) = \overline{P_i}$. From that it follows that $(\overline{\alpha}^k) = \overline{P_i^k}$, showing that $R/P_i^{n_i}$ is a PIR. But then R/I is a PIR since the isomorphism maps principal ideals to principal ideals, which then as we saw completes the proof.

As a consequence of this, if $\mathfrak{p}\big|(p)$ for some $p \in \mathbb{Z}$, then by using the construction of the proof above, we may always find an element $\alpha \in \mathfrak{p}$ such that $\mathfrak{p} = (p, \alpha)$. Hence, factorization of a prime $p \in \mathbb{Z}$ can always by recovered by adding an appropriate element to (p). More generally, if $I \subseteq J$, then $J = I + (a)$ (can you think of any conditions a needs to satisfy?). Our final result shows all pairs of ideals are, up to a multiple of $K(R)$, relatively prime. This mirrors the property that if $\gcd(a, b) = c$, we can divide a by c to get $\gcd(a/c, b) = 1$:

Corollary 1.2.20: Always Almost Relatively Prime

Let A, B be fractional ideals of the Dedekind Domain R. Then there exists elements $x, y \in K$ such that xA and yB are relative prime.

***Proof*:**
By multiplying A, B by appropriate scalars, we may assume they are integral ideal. We shall find an element c such that $cA + B = R$. First, let $a \in A$, and I be the ideal such that $AI = (a)$. Then since $BI \subseteq I$, consider:

$$\overline{I} \subseteq R/BI$$

Then since R/BI is principal, $I = BI + (b)$ for some $b \in R$. Multiplying by A, dividing by a, and letting $c = b/a$, we get

$$cA + B = R$$

as we sought to show.

With the division of ideals defined and mirroring the structure of $K^\times/R^\times$ ($\mathbb{Q}^\times_{>0}$ when working with $\mathbb{Z}$), it is tempting to say that they form a group structure, and this is indeed the case and is the focus in the remainder of this section.

One of the main metrics to measure is how many more "ideal numbers" need to be added to recover unique factorization into primes. If $\mathfrak{p}$ and $\mathfrak{q}$ are added, then if for some $t \in K$, $t\mathfrak{p} = \mathfrak{q}$, then $\mathfrak{p}$ and $\mathfrak{q}$ don't actually differ much in structure: it can be shown that $I = aJ$ ($a \in K(R)$) if and only if $I \cong_{\mathcal{O}_K} J$). Hence, we may classify "different" classes of ideals up to multiplication by a principal ideal. The more classes there are, the more a ring R fails to be a unique factorization domain. When R is a number ring, we shall show that there are only finitely many such classes.

The first result to prove is that the collection of fractional ideals of a Dedekind domain does indeed form a group:

Theorem 1.2.21: Ideal Group

Let R be a Dedekind domain, $K = K(R)$, and let J_K be the collection of fractional ideals. Then $(J_K, \cdot)$ has an abelian group structure via fractional ideal multiplication, and is called the *ideal group* of K

***Proof*:**
The operation is certainly well-defined (the product of fractional ideals is a fractional ideal), associativity and commutativity come for free, and the identity is $R = (1)$ since for any $\mathfrak{a} \in J_K$

$$\mathfrak{a}(1) = (1)\mathfrak{a} = \mathfrak{a}$$

What's left to check for is the existence of an inverse. For prime ideals, we have that $\mathfrak{p} \subsetneq \mathfrak{p}\mathfrak{p}^{-1}$, an hence by maximality we have $\mathfrak{p}\mathfrak{p}^{-1} = (1)$. For an integral ideal, if $\mathfrak{a} = \mathfrak{p}_1\mathfrak{p}_2 \cdots \mathfrak{p}_r$, then its inverse is

$$\mathfrak{b} = \mathfrak{p}_1^{-1} \cdots \mathfrak{p}_r^{-1}$$

We want to show that $\mathfrak{b} = \mathfrak{a}^{-1}$. Since $\mathfrak{b}\mathfrak{a} = (1)$, then by definition of fractional ideals $\mathfrak{b} \subseteq \mathfrak{a}^{-1}$. Conversely, to show $\mathfrak{a}^{-1} \subseteq \mathfrak{b}$, if $x\mathfrak{a} \subseteq (1)$, then $x\mathfrak{a}\mathfrak{b} \subseteq \mathfrak{b}$. Since $\mathfrak{a}\mathfrak{b} = R$, we have $aR \subseteq \mathfrak{b}$, that is $(a) \subseteq \mathfrak{b}$, and so $x \in \mathfrak{b}$. Hence, we have $\mathfrak{b} = \mathfrak{a}^{-1}$. Finally, if $\mathfrak{a}$ is an arbitrary fractional ideal and $0 \neq c \in R$ such that $c\mathfrak{a} \subseteq R$, then $(c\mathfrak{a})^{-1} = c^{-1}\mathfrak{a}^{-1}$ is the inverse of $c\mathfrak{a}$, so then $\mathfrak{a}\mathfrak{a}^{-1} = R$, completing the proof.

Corollary 1.2.22: Fractional Ideal Prime Decomposition

Every fractional ideal $\mathfrak{a}$ admits a unique representation as the product

$$\mathfrak{a} = \prod_{\mathfrak{p}} \mathfrak{p}^{n_{\mathfrak{p}}}$$

where $n_p \in \mathbb{Z}$ and $n_p = 0$ for almost all $\mathfrak{p}$. Thus, J_K is the free abelian group generated by the set of nonzero prime ideals $\mathfrak{p}$ of R

***Proof*:**
Notice that every fractional ideal $\mathfrak{a}$ is the quotient of two integral ideals,

$$\mathfrak{a} = \mathfrak{b}/\mathfrak{c}$$

which both have a prime decomposition. Thus, $\mathfrak{a}$ has a prime decomposition as stated in the corollary. Uniqueness follows the same argument as for integral ideals.

Definition 1.2.23: Ideal Group

Let R be a Dedekind Domain. Then J_R denotes the *ideal group* given by ideal multiplication.

We shall often write J_R interchangeably with J_K

Example 1.2.24: Ideal Group

1. The canonical example would be $J_{\mathbb{Z}} \cong \mathbb{Q}_+^{\times}$ with the bijection

   $$q\mathbb{Z} = (q) \mapsto q$$

   Note that it is important to think of the multiplicative structure of $\mathbb{Q}$, $(\mathbb{Q}, +)$ is not a free abelian group, having no two distinct elements be linearly dependent and not being cyclic.

2. More generally, let R be a Principal Ideal Domain. Then:

$$J_R \cong K^\times / R^\times$$

This isomorphism does not work in general, namely since not all ideals of R are principal.

A natural subgroup of J_K is formed by the principal ideal (a) where $a \in K^\times$. This subgroup can be thought of as representing our usual notion of factorization into elements instead of ideals As some foreshadowing, notice that the principal ideal The quotient of J_K by the subgroup of principle ideals can be thought of as measuring how far away the ideal group is from being a PID:

Definition 1.2.25: Ideal Class Group

Let J_K be the class group of K. Then

$$Cl(K) = J_K / P_K$$

is called the *ideal class group*, *class group*, or *Picard group* of K.

1.2.26 Note on Nomenclature The term Picard group shall not be used in this book, as the nomenclature is more so associated within algebraic geometry. It is pointed out here that some call the class group the Picard group as the class group is a special case of the Picard group.

These naturally fit in the following exact sequence:

$$1 \to R^\times \to K^\times \to J_K \to Cl(K) \to 1$$

where $K^\times \to J_K$ is the map $a \mapsto (a)$. Hence, we may think of the class group $Cl(K)$ as a measure of the expansion that happens when we pass from numbers to ideals, where the unit group $R^\times$ measure the contraction of the same process. Thus, we may want to understand the groups $R^\times$ and $Cl(K)$ more thoroughly. In general, these can be arbitrary abelian groups[17], but for rings of integers $\mathcal{O}_K$ of number fields K, we get that the size of the class group is always finite and that $\mathcal{O}_K^\times$ is a free $\mathbb{Z}$-module along with roots of unity contained in K, as will soon be shown.

1.2.3 Class Group of Ring of Integer

Recall that ideals $\mathfrak{a} \subseteq \mathcal{O}_K$ form a sub-lattice of $\mathcal{O}_K$ in the Archimedean embedding. The following theorem is key bound on the covolume of lattice. Using theorem 1.1.65, we may rather easily show that the class group is finite. The way we shall find the size of the class group is by showing that every element in a class group will have absolute norm less than a bound (say M), and that only finitely many ideals can have norm equal to $m < M$. Then, by considering the principal ideals that have norm less than M, we shall be able to find all prime ideals that compose these principal ideals, which shall give us an estimate on the size of the group.

[17] There are "enough" 1-dimensional smooth schemes to make arbitrary class groups on Dedekind domains

Lemma 1.2.27: Absolute Norm of Prime

Let $0 \neq \mathfrak{p} \subseteq \mathcal{O}_K$ be a prime ideal of $\mathcal{O}_K$ and let $\mathcal{O}_K$ be rank n. Then

$$\mathfrak{N}(\mathfrak{p}) = p^n$$

***Proof*:**
By [2, Integral Extension Prime Correspondence (Cohen-Seidenberg Theorem)], $\mathfrak{p} \cap \mathbb{Z} = p\mathbb{Z}$. Since $\mathcal{O}_K/\mathbb{Z}$ is an integral extension, $\mathcal{O}_K/\mathfrak{p}$ is a finite field extension of $\mathbb{Z}/p\mathbb{Z}$ of degree n by [2, Integral Extensions](3). Hence:

$$\mathfrak{N}(\mathfrak{p}) = p^n$$

Proposition 1.2.28: Absolute Norm for Ring of Integers Is Multiplicative

Let $\mathfrak{a} = \mathfrak{p}_1^{n_1} \cdots \mathfrak{p}_k^{n_s}$ be the prime Factorization of the nonzero ideal $\mathfrak{a}$. Then:

$$\mathfrak{N}(\mathfrak{a}) = \mathfrak{N}(\mathfrak{p}_1)^{n_1} \cdots \mathfrak{N}(\mathfrak{p}_k)^{n_s}$$

From this, we may conclude that

$$\mathfrak{N}(\mathfrak{a}\mathfrak{b}) = \mathfrak{N}(\mathfrak{a})\mathfrak{N}(\mathfrak{b})$$

***Proof*:**
By the Chinese remainder theorem:

$$\mathcal{O}_K/\mathfrak{a} \cong \mathcal{O}_K/\mathfrak{p}_1^{n_1} \oplus \cdots \oplus \mathcal{O}_K/\mathfrak{p}_s^{n_s}$$

Hence, we see that the index $[\mathcal{O}_K : \mathfrak{a}]$ is the product of the indexes $[\mathcal{O}_K : \mathfrak{p}_k^{n_i}]$. To find this, notice that the chain

$$\mathfrak{p}^n \subseteq \mathfrak{p}^{n-1} \subseteq \cdots \mathfrak{p} \subseteq \mathcal{O}_K$$

is a sequence of integral extensions[a], so we may use the fact that the index is multiplicative, namely:

$$[\mathcal{O}_K : \mathfrak{p}^n] = [\mathcal{O}_K : \mathfrak{p}][\mathfrak{p}^2 : \mathfrak{p}] \cdots [\mathfrak{p}^n : \mathfrak{p}^{n-1}]$$

We must now show each index has the same size. We shall take advantage of the fact that $\mathcal{O}/\mathfrak{p}$ is finite, and show that each $\mathfrak{p}^i/\mathfrak{p}^{i+1}$ is isomorphic to $\mathcal{O}/\mathfrak{p}$ (meaning they are all of the same size, hence the same index). I claim that $\mathfrak{p}^i/\mathfrak{p}^{i+1}$ is an $(\mathcal{O}_K/\mathfrak{p})$-vector space of dimension 1 with the natural left-multiplication action; this will come down to the multiplicative structure of J_K. To see this, let $a \in \mathfrak{p}^i \setminus \mathfrak{p}^{i+1}$ and take $\mathfrak{b} = (a) + \mathfrak{p}^{i+1} = \gcd((a), \mathfrak{p}^{i+1})$ so that $\overline{\mathfrak{b}}_{\mathfrak{p}^{i+1}} = \overline{(a)}_{\mathfrak{p}^{i+1}}$. Then by construction

$$\mathfrak{p}^{i+1} \subsetneq \mathfrak{b} \subseteq \mathfrak{p}^i$$

From this, we must conclude that $\mathfrak{p}^i = \mathfrak{b}$; this can be seen immediately by realising that $\mathfrak{b} = \gcd((a), \mathfrak{p}^{i+1}) = \mathfrak{p}^i$ by definition of (a). Therefore $\overline{\mathfrak{b}} = a \bmod \mathfrak{p}^{i+1}$ is a basis for the $\mathcal{O}_K/\mathfrak{p}$-vector-space $\mathfrak{p}^i/\mathfrak{p}^{i+1}$, hence $\mathfrak{p}^i/\mathfrak{p}^{i+1} \cong \mathcal{O}_K/\mathfrak{p}$. This can also be demonstrated by taking

$$\mathcal{O} \to \mathfrak{p}^i/\mathfrak{p}^{i+1} \qquad \alpha \mapsto \overline{\alpha a}$$

and seeing the map is surjective since $\mathfrak{p}^i = \gcd(\mathfrak{b}, \mathfrak{p}^{i+1})$ and has kernel $\mathfrak{p}$. Since this is true for all i, $1 \leq i \leq n$:

$$\mathfrak{N}(\mathfrak{p}^n) = [\mathcal{O}_K : \mathfrak{p}^n] = [\mathcal{O}_K : \mathfrak{p}][\mathfrak{p} : \mathfrak{p}^2] \cdots [\mathfrak{p}^{n-1} : \mathfrak{p}^n] = \mathfrak{N}(\mathfrak{p})^n$$

Thus more generally:

$$\mathfrak{N}(\mathfrak{a}\mathfrak{b}) = \mathfrak{N}(\mathfrak{a})\mathfrak{N}(\mathfrak{b})$$

as we sought to show.

[a]This is a *rng* integral extension. Show every $x \in \mathfrak{p}^i$ is a solution to a $\mathfrak{p}^{i+1}$ polynomial

You may ask what is the absolute norm of a prime $\mathfrak{p}$? If $\mathfrak{p}$ were principal, we know it is p^f for appropriate f. For the non-principal case, in corollary 1.3.16, we shall show that if $\mathfrak{p}_k$ lies over p, then $\mathfrak{N}(\mathfrak{p}_k) = p^f$ for an appropriate number f that we shall determine later.

Since the absolute norm is multiplicative and the index is always positive, and fractional ideals also form lattices, we may extend it to be defined on fractional ideals and get a group homomorphism:

$$\mathfrak{N} : J_K \to \mathbb{R}^{\times}_{>0} \qquad \mathfrak{a} \mapsto \mathfrak{N}(\mathfrak{a})$$

where $\mathfrak{a} = \prod_{\mathfrak{p}} \mathfrak{p}^{n_\mathfrak{p}}$ with $n_\mathfrak{p} \in \mathbb{Z}$.

Corollary 1.2.29: Volume Form Reformulation

Let $I \subseteq K$ be a fractional ideal of a number field K. Then:

$$\text{vol}(I) = \mathfrak{N}(I)\text{vol}(\mathcal{O}_K)$$

***Proof*:**
This is just reformulating proposition 1.1.60 or lemma 1.1.64

Perhaps include the proof in MIT p.18?

Theorem 1.2.30: Class Number

Let K be a number field and let $Cl(K) = J_K/P_k$ be the class group of K. Then $Cl(K)$ is finite.

***Proof*:**

Let $0 \neq \mathfrak{p} \subseteq \mathcal{O}_K$ be a prime ideal of $\mathcal{O}_K$. By lemma 1.2.27,$\mathfrak{N}(\mathfrak{p}) = p^n$, where $n = [\mathcal{O}_K/\mathfrak{p} : \mathbb{Z}/p\mathbb{Z}]$. Given p, there are only finitely many prime ideals $\mathfrak{p}$ such that $\mathfrak{p} \cap \mathbb{Z} = p\mathbb{Z}$ by proposition 1.1.66 (equivalently, because $p\mathcal{O}_K$ must have a number of terms in it's factorization and $\mathfrak{p} \big| p\mathcal{O}_K$). Thus, there are only finitely many prime ideal of bounded absolute norm. Since every integral ideal factors into prime ideals, we have

$$\mathfrak{N}(\mathfrak{a}) = \mathfrak{N}(\mathfrak{p}_1)^{n_1} \cdots \mathfrak{N}(\mathfrak{p}_k)^{n_k}$$

It follows that there are only a finite number ideals $\mathfrak{a}$ of $\mathcal{O}_K$ with bounded absolute norm $\mathfrak{N}(\mathfrak{a}) \leq M$.

Thus, it suffices to show that each class $[\mathfrak{a}] \in Cl(K)$ contains an integral ideal $\mathfrak{a}_1$ satisfying

$$\mathfrak{N}(\mathfrak{a}_1) \leq \lambda$$

for if there were infinitely many classes, then there would be infinitely many distinct ideals with absolute norm less than λ, contradicting our earlier discussion. By theorem 1.1.65, for any ideal $I \in [\mathfrak{a}]^{-1}$, pick $\alpha \in \mathfrak{a}$ such that

$$|\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)| \leq \lambda \mathfrak{N}(I)$$

By corollary 1.2.18 there exists a J such that $(\alpha) = IJ$. By definition, $J \in [\mathfrak{a}]$. Then:

$$\mathfrak{N}(I)\mathfrak{N}(J) = \mathfrak{N}(\alpha) = |\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)| \leq \lambda \mathfrak{N}(I) \implies \mathfrak{N}(J) \leq \lambda$$

but this now gives that there are only finitely many classes, and hence the class group is finite, as we sought to show.

Definition 1.2.31: Class Number

Let $Cl(K)/ = J_K/P_k$ be the class group of K. Then $h_K = [J_K : P_K]$ is called the *class number* of $Cl(K)$ or K.

A consequence of there being a finite class number is that for every ideal $I \subseteq \mathcal{O}_K$, there exists a minimal number n such that $I^n = (\alpha)$ for some $\alpha \in \mathcal{O}_K$. There is also a minimal number n' such that for *any* ideal of $\mathcal{O}_K$, the ideal to the power of n' is principal. This may encourage thoughts of converting the problem on ideals into one on principal ideals, given there is a way of going back to the original ideals. We shall see that this shall sometimes be the case.

Example 1.2.32: Class Groups

1. Naturally, $|Cl(\mathbb{Q})| = 1$ since $\mathbb{Z}$ is a PID
2. Similarly, $|Cl(\mathbb{Q}(i)| = |Cl(\mathbb{Q}(\sqrt{-3})| = 1$ since $\mathbb{Z}[i]$ and $\mathbb{Z}\left[\frac{1+\sqrt{-3}}{2}\right]$ are PID.
3. Every DVR has ideals of the form (π^n) for appropriate uniformizer π. Thus, the ideal group is isomorphic to $\mathbb{Z}$, and the class group is trivial.
4. Consider $Cl(\mathbb{Q}(\sqrt{-5})$. Since $\mathbb{Z}[\sqrt{-5}]$ is not a Unique Factorization Domain (hence Principal Ideal Domain), we will get a non-trivial class group. Let $K = \mathbb{Q}(\sqrt{-5})$. Choose an integral basis

$$\mathcal{O}_K = \langle 1, \sqrt{-5}\rangle_{\mathbb{Z}}$$

so that $D_k = -20$. Hence, the Minkowski bound gives us:

$$M_k = \sqrt{20}\frac{2}{2^2}\frac{4}{\pi} < 4$$

In fact, we have < 3, but we shall purposefully miss-guess the bound to show how to determine when two ideals belong in the same class group. Consider the ideal $(2), (3)$. Assume that $\mathfrak{p}_2 \big| (2)$. Then we know $\mathfrak{N}(\mathfrak{p}_2) \big| \mathfrak{N}((2)) = 4$. To find $\mathfrak{p}_2 \subseteq \mathcal{O}_K$ consider the map $\varphi : \mathcal{O}_K \to \mathcal{O}_K/(2)$. Since $\mathcal{O}_K/(2)$ is finite the pre-image of a prime ideal of a surjective map is prime, we may find prime ideals of $\mathcal{O}_K/(2)$. By the 4th isomorphism theorem, if

$\overline{I} \subseteq \mathcal{O}_K/(2)$ if and only if $(2) \subseteq I$, that is $I\big|(2)$, and hence any prime ideal corresponding to prime ideal of $\mathcal{O}_K/(2)$ divides (2); thus we look for $\varphi^{-1}(P) = \mathfrak{p}_2$. To find such a $\mathfrak{p}_2$, think of $\mathcal{O}_K$ as $\mathcal{O}_K \cong \mathbb{Z}[x]/(x^2+5)$[a] so that we can take:

$$\mathcal{O}_K/(2) \cong \mathbb{Z}[x]/(x^2+5,2) \cong \mathbb{F}_2[x]/(x^2-1) = \mathbb{F}_2[x]/(x+1)^2)$$

Recall that the prime ideals of a ring are in corresponding with the prime ideals of the ring quotiented by the nilradical (see [6, Radical Ideals]). But then we immediately see that the only prime ideal in $\mathcal{O}_K/(2)$ is $(1+x)$. Hence

$$\mathfrak{p}_2 = (1+\sqrt{-5}, 2)$$

Since $\mathfrak{N}(\mathfrak{p}_2)\big|4$, the only possibility is that $\mathfrak{p}^2 = 2$, and indeed:

$$\mathfrak{p}_2^2 = (4, 2+2\sqrt{-5}, -4+2\sqrt{-5}) = (2)$$

Is $\mathfrak{p}_2$ principal? If it was, then $\mathfrak{p}_2 = (a+b\sqrt{-5})$. Then:

$$2 = \mathfrak{N}(\mathfrak{p}_2) = |\mathrm{Nm}_{K/\mathbb{Q}}(a+b\sqrt{-5})| = a^2+5b^2$$

but it's evident this is not possible, hence it is not principal! Similarly with (3), we get:

$$\mathcal{O}_K/(3) \cong \mathbb{Z}[x]/(x^2+5,3) = \mathbb{F}_3[x]/(x^2-1) = \mathbb{F}_2 \oplus \mathbb{F}_3$$

From this, we can see that $3 = \mathfrak{p}_3\mathfrak{p}_3'$ where

$$\mathfrak{p}_3 = (3, 1+\sqrt{-5}) \qquad \mathfrak{p}_3' = (3, 1-\sqrt{-5})$$

Finally, notice that $[\mathfrak{p}_2] = [\mathfrak{p}_3]$. To see this, notice their product is in fact principal:

$$\mathfrak{p}_2\mathfrak{p}_3 = (2, 1+\sqrt{-5})(3, 1+\sqrt{-5}) = (6, 1+\sqrt{-5}, (1+\sqrt{-5})^2) = (1+\sqrt{-5})$$

Hence, $\mathfrak{p}_3 = (1+\sqrt{-5})^{-1}\mathfrak{p}_2$. Evidently, $\mathfrak{p}_3, \mathfrak{p}_3'$ are in the same class. Since there are only two elements, it must be that $Cl(K) \cong \mathbb{Z}/2\mathbb{Z}$.

5. Let $K = \mathbb{Q}(\sqrt{-23})$. Choose $\mathcal{O}_K = \langle 1, \frac{1+\sqrt{-23}}{2}\rangle_{\mathbb{Z}}$ so that it's discriminant is $d_k = -23$ and the Minkowski bound is $M_K = \sqrt{23}\frac{4}{\pi}\frac{1}{2} < 4$. Hence, we again check the prime ideals (2) and (3), noting that the minimal polynomial of the generator is $x^2 - x + 6$:

$$\mathcal{O}_K/(2) \cong \mathbb{F}_2[x]/(t^2 - t)$$

Thus, if $t = \frac{1+\sqrt{-23}}{2}$:

$$(2) = (2, t-1)(2, t)$$

and similarly

$$(3) = (3, t-1)(3, t)$$

Certainly, each prime multiple of (2) and (3) are in the same equivalence class. Is $[(2,t)] = [(3, t-1)]$? Choosing good representatives and multiplying, we get

$$(3,t)(2,t) = (6,t) = (t)$$

hence, we again get that $Cl(K) = \mathbb{Z}/2\mathbb{Z}$.

6. Let $K = \mathbb{Q}(\sqrt{15})$ and $\mathcal{O}_K = \langle 1, \sqrt{15}\rangle_{\mathbb{Z}}$. Then doing the same procedure, we get:

$$(3, \sqrt{15})^2 = (3)$$

How do we know that $(3, \sqrt{15})$ is principal or not? We can see that $a^2 - 15b^2 = 3$ has no solutions using modular arithmetic's, but it will not always be feasible to use this method, so we introduce another. Let's say $(3, \sqrt{15}) = (\alpha)$. Then

$$(\alpha^2) = (3)$$

meaning they are associate, hence $\alpha^2 = 3u$ for $u \in \mathcal{O}_K^\times$. Does there exist such a unit? In section 1.2.4, we shall find this structure. For now, it can be taken for granted that $\mathcal{O}_K^\times = \pm\langle 4 + \sqrt{15}\rangle$ (notice the norm of all such numbers is 1). Then from this we see that no such unit exist, hence $(3, \sqrt{15})$ is not principal.

Notice that all of these examples are for quadratic extensions. It is much more difficult to find prime-factorization for non-quadratic extension since the ring of integers is not easy to find. In the next section, we shall see how we may get some information on the factorization of primes when $\mathcal{O}_K$ is not available

[a] Note how this relied on knowing the structure of $\mathcal{O}_K$! See comment at the end of this example

In general, we have $h_K > 1$. If $h_K = 1$ then we can think of the prime factorization of ideals being the same as the prime factorization of elements as is already well-understood. If $K = \mathbb{Q}(\sqrt{d})$ where d is square-free and negative, then the quadradic extensions in which the class number is 1 is

$$-1, -2, -3, -7, -11, -19, -43, -67, -163$$

For real quadratic extensions, there are a lot more, for example:

$$2, 3, 5, 6, 7, 11, 13, 14, 17, 19, 22, 23, 29, 31, 33, \cdots$$

it is unknown where there are infinitely many real quadratic fields of class number 1, or even if there are infinitely many algebraic number fields (of any degree) with class number 1. In general, it is in fact very difficult to determine some structure in the class group given some class of number fields, with an important notable exception cyclotomic extensions which due to the work of Kenkichi Iwasawa we know behave in a regular way.

The importance of studying cyclotomic extensions links back to Fermat's last theorem which the reader surely recalls states that the equation $x^p + y^p = z^p$ has no nonzero integer solutions for $p \geq 3$. For $p = 2$, we take the cyclotomic extension $\mathbb{Q}(\zeta_4) = \mathbb{Q}(i)$ to get

$$x^2 + y^2 = (x + iy)(x - iy)$$

and then we get the Fermat's twin prime theorem that fully solved the problem in this case. If p is prime, then considering the equation $y^p = z^p - x^p$ over $\mathbb{Q}(\zeta_p)$, we may factor this in $\mathbb{Z}[\zeta_p]$ as

$$\begin{aligned} y^p &= x^p - z^p \\ &= z^p \left(\left(\frac{x}{z} \right)^p - 1 \right) \\ &= z^p \prod_{k=0}^{p-1} \left(\frac{x}{z} - \omega_k \right) \\ &= \prod_{k=0}^{p-1} z \left(\frac{x}{z} - \omega_k \right) \\ &= \prod_{k=0}^{p-1} (x - \omega_k z) \end{aligned}$$

where $\omega_k = \zeta_p^k$. Thus, if we assume an existence of a solution, we obtain two multiplicative decompositions of the same number in $\mathbb{Z}[\zeta_p]$. Using this, we can show it contradicts unique factorization *provided* that $\mathbb{Z}[\zeta_p]$ is a unique factorization domain. It was due to some mathematician erroneously assuming unique factorization (equivalently for us, that the class number is always 1) that lead to many fake proofs of Fermat's last theorem[18]. Kummer fixed this by introducing the notion of ideal primes, and translated the statement into one where we must show that $p \nmid h_p$ instead of assuming $h_p = 1$. If this is the case, such a prime is called *regular*, and *irregular* otherwise. Unfortunately, not all primes are regular. Of the first 25 primes less than 100, the primes 37, 59, and 67 are irregular. As of writing this book, it is unknown whether there are infinitely many regular primes.

The regular and irregular prime are going to play important roles in our coming theory, but they do not play much of a role in the eventual proof of Fermat's Last theorem. Due to the work of Gerhard Frey, there was a reformulation of Fermat's last theorem into a statement about elliptic curves. This lead to a new development of mathematics which is beyond the scope of this book, but the interested reader may feel free to search up the *Taniyama-Shimura-Weil Conjecture* to see how Fermat's Last Theorem got translated into a problem of elliptic curves.

We end this section with showing a certain Diophantine equation has no integer solution as it combines nicely all the machinery we have done, and some generalizations that is motivation for class field theory. Starting with the Diophantine equation:

Theorem 1.2.33: No Solution To Diophantine Equation

The equation

$$y^2 = x^3 - 5$$

has no integer solutions

***Proof*:**
Suppose the pair (x, y) is an integer solution. If x is even, then $x^3 - 5$ is odd, and we get

$$y^2 \equiv -1 \bmod 4 \equiv 3 \bmod 4$$

which is easy to check is impossible. Next, x and y must be coprime, for if they weren't then

[18] It is speculated that this is the proof Fermat was referring to as being too long for the margins of his book

their common factor must be 5 (since then $p \big| y^2 - x^3 = -5$), so

$$5^2 y_0^2 = 5^3 x_0 - 5 = 5y_0^2 = 5^2 x_0 - 1$$

But then mod 5 the equation has no solution, hence it has no solution in general.

Now, factoring in $\mathbb{Z}[\sqrt{-5}]$, we get

$$(y - \sqrt{-5})(y + \sqrt{-5}) = (x)^3$$

The ideals $(y - \sqrt{5})$ and $(y + \sqrt{5})$ are coprime; for is $\mathfrak{p}$ divides both, then $\mathfrak{p}\big|(x)^3$, so $\mathfrak{p}\big|(x)$, and since $(y - \sqrt{5}), (y + \sqrt{5}) \subseteq \mathfrak{p}$ then $(y - \sqrt{-5}) + (y + \sqrt{-5}) = 2y \in \mathfrak{p}$ so $\mathfrak{p}\big|(2y)$. Since x is odd and $\mathfrak{p}\big|(x)$, $\mathfrak{p}\big|(y)$. Then since $\mathfrak{p}\big|(y^2 - x^3) = (5)$, we must have $\mathfrak{p} = (\sqrt{-5})$. So $(\sqrt{-5})\big|(y + \sqrt{-5})$, implying $(5)\big|(x)$. But now we can again work in mod 5 and reach a contradiction.

Hence, they are co-prime. Thus, as the left hand side is the cube of (x), we must have:

$$(y - \sqrt{-5}) = \mathfrak{a}^3 \qquad (y + \sqrt{-5}) = \mathfrak{b}^3$$

Finally, since the class group is of size 2 and a third power produces a principal ideal, it must be that $\mathfrak{a}, \mathfrak{b}$ are both principal. Given that the only units are ± 1 (something we shall prove in the next section), we have that:

$$y + \sqrt{-5} = (a + b\sqrt{-5})^3 = a^3 + 3a^2 b\sqrt{-5} - 3ab^2 \cdot 5 - b^4 \cdot 5\sqrt{-5} = (a^3 - 15ab^2) + (3a^2 b - 5b^3)\sqrt{-5}$$

Manipulating and comparing the coefficients of $\sqrt{-5}$, we get that:

$$1 = 3ba^2 - 5b^3 = b(3a^2 - 5b^2)$$

implying $b = \pm 1$ since $a, b \in \mathbb{Z}$. But then:

$$3a^2 - 5 = \pm 1$$

but now this can be inspected to see that it's impossible, hence no such integer solution exists.

Note that a few things lined up well here:

- There were only two units, if there are more then the considerations are more complicated.
- The class group happened to let us deduce the ideal is principal; if the ideal was not principal there would be more to consider.
- The equation was easily factorizable in the extension (ex. $y^2 = x^5 + x + 1$ is a good deal more difficult)
- The factors are coprime.

Exercise 1.2.1

1. Let $A, B \subseteq R$ with $B \subseteq A$ and $A \neq R$. Show there exists a $\gamma \in K$ such that $\gamma B \subseteq R$, but $\gamma B \not\subseteq A$

2. Show that $IJ = (I+J)(I \cap J) = \gcd(I,J)\text{lcm}(I,J)$
3. Let $\alpha \in \mathcal{O}_K \setminus \{0\}$. Show that there exists a $\beta \in \mathcal{O}_K$ st $\alpha\beta \in \mathbb{Z} \setminus \{0\}$
4. Let $K = \mathbb{Q}(\sqrt{-3})$. Show that $\mathcal{O}_K \neq \mathbb{Z}[\sqrt{-3}]$. Show that
5. Show that $3 = x^2 + 5y^2$ has no integer solutions. Note that 3 splits in $\mathbb{Z}[\sqrt{-3}]$ showing that if the ringer of integers is not UFD then it is more complicated to use factorizations.
6. exercise for quotient ideals
   a) here

1.2.4 Dirichlet's Unit Theorem

(Current intuition when it comes to solving equations: we are working with ideals, but equations use elements. We must be concious of the units at play to reduce back to elements)

As we saw in example 1.2.32, it is important to know the unit group of $\mathcal{O}_K$ to find elements in the class group. We start this section by showing that we may embed $\mathcal{O}_K^\times$ to form a lattice, and then use this to find the structure of $\mathcal{O}_K$.

The group $\mathcal{O}_K^\times$ may have some torsion elements, namely all roots of unity that lie in K; this collection is labeled as $\mu(K)$. This group will in fact be determined by the number of embeddings into Minkowski space. On a high-level, the structure of this group is within this diagram:

$$\begin{tikzcd}
K^\times \arrow[r, "\sigma"] \arrow[rr, bend left, "\sigma^\times"] \arrow[d, "N_{K/\mathbb{Q}}"] & K_\mathbb{R}^\times \arrow[r, "\log"] \arrow[d, "\text{nm}"] & {[\prod_\tau \mathbb{R}]^+} \arrow[d, "\text{Tr}"] \\
\mathbb{Q}^* \arrow[r, hook] & \mathbb{R}^* \arrow[r, "\log|\ |"] & \mathbb{R}
\end{tikzcd}$$

where we shall write $K_\mathbb{R}^\times = (\mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2})^\times$. In particular, the top row of the above commutative diagram can be replaced by:

$$\begin{aligned}
\mathcal{O}_K^\times &= \left\{ r \in \mathcal{O}_K \;\middle|\; N_{K/\mathbb{Q}(n)=\pm 1} \right\} \\
S &= \left\{ y \in K_\mathbb{R}^\times \;\middle|\; N(y) = \pm 1 \right\} && \text{norm-one surface} \\
H &= \left\{ x \in \left[\prod_\tau \mathbb{R}\right]^+ \;\middle|\; \text{tr}(x) = 0 \right\} && \text{trace-zero hyperplane}
\end{aligned}$$

Then we get the homomorphism:

$$\mathcal{O}_K^\times \xrightarrow{\sigma} S \xrightarrow{\log} H$$

where $\sigma^\times = \log \circ \sigma$.

Proposition 1.2.34: Roots Of Unity In Group Of Units

Let $\Gamma_K^\times = \sigma^\times(\mathcal{O}_K^\times) \subseteq H$. Then the sequence

$$1 \to \mu(K) \to \mathcal{O}_K^\times \xrightarrow{\sigma^\times} \Gamma_K^\times \to 0$$

is exact.

***Proof*:**
Certainly $\mu(K) \subseteq \ker(\sigma^\times)$. Conversely, say $r \in \mathcal{O}_K^\times$ where $\sigma^\times(r) = 0$. Then $|\tau(r)| = 1$ for each embedding $\gamma : K \to \mathbb{C}$, hence $\sigma(r) = (\tau(r))$ is in a bounded domain of the $\mathbb{R}$-vector space $K_\mathbb{R}$. On the other hand, $\sigma(r)$ is a point of the lattice $\sigma(\mathcal{O}_K)$ of $K_\mathbb{R}$. Thus, the kernel of $\sigma^\times$ can contain only a finite number of elements, and so since it's a finite group, it contains only the roots of unity of $K^\times$.

We now build towards the structure of $\Gamma_K^\times$. Under the right adjustments $\mathcal{O}_K^\times$ can be embedded in a similar way to $\mathcal{O}_K$. Take $\sigma(\mathcal{O}_K^\times) \subseteq \mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2}$. the way a mesh is formed is by taking the logarithm, for $\log(ab) = \log(a) + \log(b)$

$$\sigma^\times : K^\times \to \mathbb{R}^{r_1} \oplus \mathbb{R}^{r_2} \qquad \sigma^\times = \log \circ \sigma$$

Proposition 1.2.35: Multiplicative Group Forms Lattice

Let H be the trace-zero hyperplane. Then $\sigma^\times(\mathcal{O}_K^\times) \subseteq W$ and it is discrete.

***Proof*:**
First, the map $(\mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2})^\times \to \mathbb{R}^{r_1+r_2}$ mapping

$$(a_1, a_2, ..., a_{r_1}, b_1, b_2, ..., b_{r_2}) \mapsto (\log(a_1), ..., \log(a_{r_1}), ..., \log(b_{r_2}))$$

is a proper topological map, in particular the image of a discrete set is discrete. Since $\sigma(\mathcal{O}_K) \subseteq \mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2}$ is discrete. $\sigma(\mathcal{O}_K \setminus \{0\}) \subseteq (\mathbb{R}^{r_1 \oplus r_2})^\times$ is discrete, Next, for $\epsilon > 0$, if $\sigma^\times(\alpha) \subseteq \overline{B_\epsilon}$, then $\forall \varphi \in \mathrm{Hom}_\mathbb{Q}(K, \mathbb{C})$, $e^{-\epsilon} < |\varphi(\alpha)| < e^\epsilon$. Now, since the ball is compact, by properness there are only finitely many α, but that completes the proof.

Theorem 1.2.36: Dirichlet Unit Theorem

The group of units $\mathcal{O}_K^\times$ is the direct product of the finite cyclic group $\mu(K)$ and a free abelian group of rank $r_1 + r_2 - 1$. The free-generators of $\mathcal{O}_K^\times$ are called the *fundamental units* of $\mathcal{O}_K^\times$:

$$\mathcal{O}_K^\times \cong_\mathbb{Z} \mathbb{Z}^{r_1+r_2-1} \oplus \mu(K)$$

We may also characterize $\mathcal{O}_K^\times$ by the following short exact sequence:

$$1 \to \mu(K) \to \mathcal{O}_K^\times \to \Gamma_K^\times \to 0$$

where since the right term is a free $\mathbb{Z}$-module, the sequence splits giving us $\mathcal{O}_K^\times \cong \mu(K) \oplus \Gamma_K^\times$.

***Proof*:**
(Marcus p.100 looks good)

Thus, there exist units $u_1, u_2, ..., u_t$, $t = r_1 + r_2 - 1$ such that for any other unit $u \in \mathcal{O}_K^{\times}$, it can be written as

$$u = \zeta u_1^{n_1} \cdots u_t^{n_t}$$

where $\zeta \in \mu(K)$ is a root of unity.

Example 1.2.37: Unit Group And Class Group

1. Let $K = \mathbb{Q}(\sqrt{2})$. Then $1 + \sqrt{2}$ generates the free part of the group of units. Similarly, for $K = \mathbb{Q}(\sqrt{3})$, $2 + \sqrt{3}$ generates the free part of the group of units.
2. Let us compute the class group and unit group of $\mathbb{Q}(\sqrt[3]{7})$ Let $K = \mathbb{Q}(\sqrt[3]{7})$ and $s = \sqrt[3]{7}$. We again look for the Minkowski bound. Doing a similar computation as before, we may workout

$$\mathcal{O}_K = \langle 1, \sqrt[3]{7}, \sqrt[3]{7}^2 \rangle$$

Hence, the discriminant is:

$$\det \begin{pmatrix} 1 & \sqrt[3]{7} & \sqrt[3]{7}^2 \\ 1 & \sqrt[3]{7}\zeta_3 & \sqrt[3]{7}^2\zeta_3^2 \\ 1 & \sqrt[3]{7}\zeta_3^2 & \sqrt[3]{7}^2\zeta_3 \end{pmatrix} = -1323$$

Thus:

$$M_K = \sqrt{1323}\frac{6}{27}\frac{4}{\pi} < 11$$

Hence, we check prime ideal $(2), (3), (5), (7)$ of $\mathbb{Z}$. Since the minimal polynomial of $\sqrt[3]{7}$ is $x^3 - 7$, we see that in $\mathbb{F}_p[x]/(x^3 - 7_p)$ we get:

(2) $(x+1)(x^2+x+1)$, hence

$$\mathfrak{p}_2 = (2, s+1) \qquad \mathfrak{p}_2' = (2, s^2+s+1)$$

(3) $(x-1)^3$, hence

$$\mathfrak{p}_3 = (3, s-1)$$

(5) $(x+2)(x^2+3x+4)$, hence

$$\mathfrak{p}_5 = (5, s+2) \qquad \mathfrak{p}_5' = (5, s^2+3s+4)$$

(7) x^3, hence

$$\mathfrak{p}_7 = (7, s) = (s)$$

Thus, just like before we have that $[\mathfrak{p}_2]$, $[\mathfrak{p}_3]$, and $[\mathfrak{p}_5]$ generate $Cl(K)$ (since $[\mathfrak{p}_7] = [1]$). Next, we see that

$$\mathfrak{p}_2\mathfrak{p}_3 = (6) \qquad \mathfrak{p}_3^2\mathfrak{p}_5 = (15)$$

Hence, $[\mathfrak{p}_3] = [\mathfrak{p}_2^{-1}]$ and $[\mathfrak{p}_3^2] = [\mathfrak{p}_5^{-1}]$. Then, with some computation we see that:

$$[\mathfrak{p}_3^3] = [\mathfrak{p}_3]^3 = [\text{id}]$$

Thus, we see that $Cl(K)$ is generated by $[\mathfrak{p}_3]$ and has order 3, hence $CL(K) \cong \mathbb{Z}/3\mathbb{Z}$. However, what if $\mathfrak{p}_3$ is in fact principal? For the sake of contradiction, let's say

$$\mathfrak{p}_3 = (\alpha) \implies \mathfrak{p}_3^3 = (3) = (\alpha^3)$$

Then $\alpha^3 = 3u$ for some unit u. To find u, we find the units of the ring of integers. To find its units, first notice that if:

$$1 = x^3 - 7 = (x - s)(x^2 + sx + s^2)$$

Then $x - s$ is a unit. But then, $x = 2$ works since $8 - 7 = 1$. Then by Dirichlet's Unit Theorem, we have that the rank of the unit group is:

$$r_1 + r_2 - 1 = 1 + 1 - 1 = 1$$

Hence, the group of units is

$$\langle 2 - \sqrt[7]{2} \rangle$$

which is the second object asked of us to find. Using this, we can finally conclude that there is no unit u that satisfies the equation $\alpha^3 = 3u$ (if $\alpha = a + bs + cs^2$, in the process of finding the coefficients a, b, c we shall be forced to have $b = c = 0$). Hence, we may properly conclude that

$$Cl(K) \cong \mathbb{Z}/3\mathbb{Z}$$

completing the proof.

In general, there is an algorithm for computing the ring of units that involved continued fractions. It is not covered in this book, but for those interested see *Borevich and Shafarevich, Number Theory* section 7.3 of chapter 2.

Proposition 1.2.38: Volume Of Unit Lattice

Let $\mathcal{O}_K$ be a ring of integers with unit group $\mathcal{O}_K^*$. Then the volume of the unit lattice $\sigma^\times(\mathcal{O}_K^\times)$ in the hyper-plane H is:

$$\operatorname{vol}\left(\sigma^\times(\mathcal{O}_K^\times)\right) = \sqrt{r_1 + r_2} R$$

where R is the absolute value of the determinant of an arbitrary minor rank $t = r + s - 1$ of the following matrix:

$$\begin{pmatrix} \sigma_1^\times(u_1) & \cdots & \sigma_1^\times(u_t) \\ \vdots & \ddots & \ddots \\ \sigma_{t+1}^\times(u_1) & \cdots & \sigma_{t+1}^\times(u_t) \end{pmatrix}$$

***Proof*:**
computations

Definition 1.2.39: Regulator

The value of R in the above proof is called the *regulator* of $\mathcal{O}_K$.

Exercise 1.2.2

1. What number rings $\mathcal{O}_K$ have finitely many units?

1.3 Extension of Dedekind Domains and Number Fields

In [2, Integral Extensions and Prime Ideals], we covered how prime ideals of a base ring correspond to prime ideals in an integral extension. Since all extensions that we are working with are integral extensions, the same theory applies. Working over Dedekind domains, we may further ask if the integral extension of a Dedekind domain is a Dedekind domain. This will be the case, and so we may further factor prime ideals in the extended ring, namely if a prime ideal $\mathfrak{p} \subseteq R$ factors in S where S/R is an integral extension. When focusing on rings of integers $\mathcal{O}_L$, the exploration of how primes factor into larger primes has a lot of consequences on the structure of $\mathcal{O}_L$ itself, and perhaps a bit surprisingly at first sight on the field extension L/K itself. We shall see in the next section that we can use the theory of factorization of prime ideals to classify all abelian extension over $\mathbb{Q}$, this is the *Kronecker-Weber Theorem* promised in [5, Composite Fields Determining Abelian Extensions].

First, if R is a Dedekind domain with field of fraction K, $I \subseteq R$ an ideal, through them map $K \hookrightarrow L$ we get $I^e \subseteq \overline{R}^L$. We further generalize this for fractional ideals:

Definition 1.3.1: Extension Of Fractional Ideals

Let R be a Dedekind domain, K it's field of fractions, and L/K be a finite field extension. Then if $I \subseteq K$ is a fractional ideal and $\mathcal{O} = \overline{R}^L$,

$$I_L := I\mathcal{O} = \left\{ \sum a_i b_i \;\middle|\; a_i \in I,\ b_i \in \mathcal{O} \right\}$$

When necessary, the notation $I_{L/K}$ will be used to specify the base field over which I is taken.

It can be verified that $I_L \subseteq L$ is the minimal fractional ideal containing I, that if $M/L/K$, then $(I_L)_M = I_M$, and that $I_L = I \otimes_{\mathcal{O}_K} \mathcal{O}$. The results of [2, Extension And Contraction With Common Operations] follow quickly giving all the same results. Furthermore since $(I+J)_L = I_L + J_L$:

$$(\gcd_K(I,J))_L = \gcd_L(I_L, J_L)$$

However, by [2, Extension And Contraction With Common Operations], we so far only have

$$(\mathrm{lcm}_K(I,J))_L \subseteq \mathrm{lcm}_M(I_L, J_L)$$

Note that for $\alpha \in K$, $(\alpha)_L = (\alpha\mathcal{O}_K)_L = \alpha\mathcal{O}_L$, hence principal ideals extend to principal ideals. For reasons that shall become clearer later[19], the map

$$J_K \to J_L \qquad I \mapsto I_L$$

[19] we shall possibly increase the number of primes, hence change the "basis"

is called the *base change map*. To study the fractional ideals of L, the first immediate result we may want is that $\overline{R}^L$ remains a Dedekind domain:

Lemma 1.3.2: Extending Dedekind Domains

Let R be a Dedekind domain with field of fraction K and let L/K be a finite field extension of K. Then $\mathcal{O} = \overline{R}^L$ is a Dedekind domain.

***Proof*:**
By definition, $\mathcal{O}$ is integrally closed, and if $\mathfrak{p} \subseteq \mathcal{O}$ is prime, then $P = \mathfrak{p} \cap R$ is also prime (the contraction of a prime ideal is prime), but P is then maximal in R; hence $(\mathcal{O}/\mathfrak{p})/(R/P)$ is an integral extension over a field meaning $\mathcal{O}/\mathfrak{p}$ is a field, thus $\mathfrak{p}$ is maximal. It remains to show that $\mathcal{O}$ is Noetherian.

For now, we shall prove this for the separable case (the non-separable case has an easier proof with more machinery, and the separable case is all that is needed for number fields). Let $\alpha_1, \alpha_2, ..., \alpha_n$ be a K -basis for L. By multiplying by an appropriate constant, we may let $\alpha_1, \alpha_2, ..., \alpha_n \in \mathcal{O}$ (see proposition 1.1.7). By proposition 1.1.32,

$$\mathcal{O} \subseteq \frac{R\alpha_1}{d} + \cdots + \frac{R\alpha_n}{d}$$

Now, for any ideal $I \subseteq \mathcal{O}$, by the above it is contained in a finitely generated R-module, hence since R is Noetherian I (as a R-module) is finitely generated. But then *a fortiori*, it is a finitely generated $\mathcal{O}$-module, hence a finitely generated ideal, making $\mathcal{O}$ Noetherian.

Example 1.3.3: Dedekind Domains Via Extension
The result above shows that all rings of integers of a finite extension $L/K/\mathbb{Q}$ is a ring of integers. For an example that is not a number ring, take $R = k[x]$ for some field k. As R is a PID, it is a Dedekind domain. Then if $L/k(x)$ is an integral extension, the integral closure of R over L, $S = \overline{R}^L$, is a Dedekind domain. As $L \cong k(x)[y]/(p(y))$ for some irreducible polynomial over $k[x]$ L represents 1-dimensional curves. In section ref:HERE, we shall briefly show these curves are all 1-dimensional smooth curves.

Extending Number Fields

An immediate corollary of lemma 1.3.2 is the following:

Corollary 1.3.4: Extending Number Fields

Let L/K be a finite extension of numbers fields so that $\mathcal{O}_K = \overline{\mathbb{Z}}^K$ and $\mathcal{O} = \overline{\mathcal{O}_K}^L$. Then

$$\mathcal{O} = \mathcal{O}_L = \overline{\mathbb{Z}}^L$$

***Proof*:**
We have to show that $\mathcal{O}_L = \overline{\mathbb{Z}}^L \stackrel{!}{=} \overline{\mathcal{O}_k}^L = \mathcal{O}$, namely the $\stackrel{!}{=}$ equality. The $\subseteq$ direction is trivial, so consider the other direction: let $\alpha \in \mathcal{O}$. Then since $\mathcal{O}$ is Dedekind, it is Noetherian, it is finitely generated as a $\mathcal{O}_K$-module. Since $\mathbb{Z}[\alpha] \subseteq \mathcal{O}$:

$$\mathbb{Z}[\alpha] \subseteq \mathcal{O}_K a_1 + \cdots + \mathcal{O}_K a_n$$

Since each $\mathcal{O}_K$ is a finitely generated free $\mathbb{Z}$-module, we have that $\mathbb{Z}[\alpha]$ is a Noetherian $\mathbb{Z}$-module, hence finitely generated. But then by [2, Integral Element Equivalences] α is integral over $\mathbb{Z}$, and so $\alpha \in \overline{\mathbb{Z}}^L$, completing the equality.

Thus, when working with finite extensions of number ring L/K, there is a corresponding integral extensions are those of the ring of integers $\mathcal{O}_L/\mathcal{O}_K$ creating the diagram:

$$\begin{tikzcd} K \arrow[r, "\iota"] \arrow[d] & L \arrow[d] \\ \mathcal{O}_K \arrow[r, "\iota"] & \mathcal{O}_L \end{tikzcd}$$

As we saw, principal ideals extend to principal ideals. As it turns out, all non-principal ideals extend to be principal given a large enough extension. This result shall underpin a lot of future simplification and is good to see early on:

Lemma 1.3.5: Number Fields, Extension To Principal

Let $I \subseteq K$ be a fractional ideal where $K/\mathbb{Q}$ is a number field. Then in some finite extension L/K, I_L becomes a principal ideal.

***Proof*:**
As $K/\mathbb{Q}$ is a number fields, by theorem 1.2.30, $Cl(K)$ is finite. Thus, for every $I \in J_K$, there exists an $n \in \mathbb{N}$ such that $I^n = \alpha\mathcal{O}_K$. Let $L = K(\sqrt[n]{\alpha})$. letting $\beta = \sqrt[n]{\alpha}$, we get

$$(I_L)^n = (I_L^n) = (I^n\mathcal{O}_L) = (\alpha\mathcal{O}_L) = (\beta\mathcal{O}_L)^n$$

Thus, by unique factorization:

$$I_L = \beta\mathcal{O}_L$$

as we sought to show.

This result is quite useful; in the right cases we can pass to the field in which all our ideals become principal ideals, proof what we want for principal ideals, then try to come back down (coming back down is often called the descent argument). The following lemma gives a very common descent argument, and is important for computing the absolute norm of extended ideals: since the norm of an ideal gives a measure how space a given ideal is within its Dedekind, extending the domain shall make the given ideal sparser[20]:

[20]This means that the ideal doesn't "extend enough" to cancel the added density given by extending the ring

Lemma 1.3.6: Number Fields, Extension And Absolute Norm

Let $L/K/\mathbb{Q}$ be a finite extension of number fields and $I \subseteq K$ a fractional ideal. Then:

$$\mathfrak{N}_{L/\mathbb{Q}}(I_L) = \mathfrak{N}_{K/\mathbb{Q}}(I)^{[L:K]}$$

Proof:
If $I = (\alpha) = \alpha\mathcal{O}_K$ is principal, then $\mathrm{Nm}(I) = \left|\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)\right|$ and $\mathrm{Nm}(I_L) = \left|\mathrm{Nm}_{L/\mathbb{Q}}(\alpha)\right|$. Since $\alpha \in K$, by proposition 1.1.12

$$\mathrm{Nm}_{L/\mathbb{Q}}(\alpha) = \mathrm{Nm}_{K/\mathbb{Q}}(\alpha)^{[L:K]}$$

In general, $I^n = \alpha\mathcal{O}_K$, thus using the multiplicity of the absolute norm:

$$\mathfrak{N}_{L/\mathbb{Q}}(I_L)^n = \mathfrak{N}_{L/\mathbb{Q}}(I_L^n) = \mathrm{Nm}_{L/\mathbb{Q}}(\alpha\mathcal{O}_L) = \mathrm{Nm}_{L/\mathbb{Q}}(\alpha\mathcal{O}_K)^{[L:K]} = \mathfrak{N}(I^n)^{[L:K]} = \mathfrak{N}_{K/\mathbb{Q}}(I)^{n[L:K]}$$

taking the nth root, and we're done

1.3.1 Ideal Correspondence in Extension

Let L/K be a finite extension, J_K and J_L be the corresponding ideal group. Then we have a map $J_K \to J_L$ given by $I \mapsto I_L$. Do we have a converse map, and if so is it an inverse of the natural extension map? If not, to what degree is it an inverse map? Recall in [2, Properties of Extensions and Contractions] that $I \subseteq SI \cap R$, in other words $I \subseteq I_S \cap R$. This can be made an equality for when extending number fields, and further generalized to fractional ideals. Recall that this is not true in general (a counter-example for the general case is $(2) \subsetneq \mathbb{Z} = (2)_{\mathbb{Q}} \cap \mathbb{Z}$), hence this is special for extension of Dedekind domains:

Proposition 1.3.7: Ideal Correspondence In Extensions Of Dedekind Domains

let L/K be a finite field extension with Dedekind domain $\mathcal{O}$ and o respectively. Let $I \subseteq K$ a fractional ideal. Then:

$$I_L \cap K = (I^e)^c = I$$

Hence, unlike for general integral extensions, the contraction is an inverse map to the extension.

Proof:
Let $\beta \in J = I\mathcal{O} \cap K$ so that $\beta = \sum_i a_i k_i$ where $a_i \in I_L$ and $k_i \in \mathit{o}$ (since $\mathcal{O} \cap K = \mathit{o}$). Then the following equality holds:

$$\beta^n = \mathrm{Nm}_{L/K}(\beta) = \prod_{\tau} \tau(\beta) = \prod_{\tau} \left(\sum_i k_i \tau(a_i) \right)$$

where the first equality comes from $\beta \in K$, the second comes from the equivalence definition of norm (given an appropriate normal closure N/L), and the third is expanding the definition of β. On the right hand side, we have that the sum is invariant under the Galois group $\mathrm{Gal}_K(N)$ for appropriate normal closure N/L from which we took each τ. Thus, the sum is an element in o, so the product is in $\mathcal{o}$. Then we have $\beta^n \in I^n$, so $\beta \in I$, completing the proof.

Due to the importance of this new paradigm of thinking, two other proofs shall be presented. Notice that $IJ^{-1} \subseteq \mathcal{o}$ and:

$$\mathfrak{N}(IJ^{-1}) = 1$$

giving that $IJ^{-1} = \mathcal{o}$, i.e. $I = J$.

Another proof goes as follows: Since $I \subseteq J$, $I_L \subseteq J_L$. Since $I_L \subseteq J_L \subseteq I_L$ by properties of extensions, $I_L = J_L$. Thus:

$$\mathfrak{N}(J) = \mathfrak{N}(I_L)^{1/[L:K]} = \mathfrak{N}(I)$$

and since $I \subseteq J$, the norms being equal means the index between the two are the same, hence $I = J$.

Note that in the case where $\mathfrak{p}$ is prime, the result can be essentially immediately proven since by [2, Properties of Extensions and Contractions] $\mathfrak{p} \subseteq \mathfrak{p}\mathcal{O}_L \cap K$, but since $\mathfrak{p}$ is maximal we get equality.

Corollary 1.3.8: Homomorphism between Extensions of Ideal Group

Let J_K be the ideal group of a Dedekind domain R and let L/K be a finite field extension. Then the map $J_K \to J_L$ mapping $I \mapsto I_L$ is an injective group homomorphism that preserves gcd and lcm.

On the other hand, the map $Cl(K) \to Cl(L)$ is a homomorphism that need not be injective.

For class groups of number fields:

$$\varinjlim_{\substack{K \\ [K:\mathbb{Q}]<\infty}} (Cl(K)) = 0$$

***Proof*:**
By the previous proposition, $I_L = J_K$ if $I = J$, hence the map $\varphi : \mathscr{F}(K) \to \mathscr{F}(L)$ mapping $I \mapsto I_L$ is injective and is certainly a homomorphism by a simple generalization of [2, Extension And Contraction With Common Operations] (recall that $(\mathfrak{a}\mathfrak{b})^e = \mathfrak{a}^e\mathfrak{b}^e$ for ideals).

For the map between class groups of number fields, since the extension for every fractional ideal $I \subseteq \mathbb{Q}$, there exists an extension in which it is principle, we see that the colimit can only be satisfied by the trivial group, hence $\varinjlim_{\substack{K \\ [K:\mathbb{Q}]<\infty}} (Cl(K)) = 0$, as we sought to show.

A side consequence of the colimit definition is that we may define

$$Cl(\overline{\mathbb{Q}}) := \varinjlim(Cl(K)) = 0$$

This shows that to some degree, the lack of principal ideals is tied to what algebraic extension we choose. Though the extension map is injective, it is not in general surjective:

Example 1.3.9: Lack of Surjectivity

Let $L = \mathbb{Q}(\sqrt{-5})$ and $K = \mathbb{Q}$ so $\mathcal{O}_L = \mathbb{Z}[\sqrt{-5}]$. Then the (non-principal) ideal $I = (1+\sqrt{-5}, 1-\sqrt{-5})$ are all numbers of the form $m+n\sqrt{-5}$ where m,n are even, so $I \cap \mathbb{Q} = (2)$, hence I lies over (2). However, $I \neq (2)_L$ since I is not principal (recall $(-)_L$ maps principals to principals), and no other ideal can map to I or else I would lie over some different prime, hence I is not an extension. The ideal (2) is still related to I, namely $I^2 = (2)$ (this can be used to show I is not principal).

With these results, we have better control over the extension of primes and primes lying over (resp. lying under) relate:

Proposition 1.3.10: Dedekind Domain, Lying Over And Lying Under

Let $\mathcal{O}/\mathfrak{o}$ be a finite extension of Dedekind domains given by L/K, $0 \neq \mathfrak{p} \subseteq \mathfrak{o}$ a prime, and $\mathcal{O} = \overline{\mathfrak{o}}^L$, $\mathfrak{p}$, where $\mathfrak{p}$ may factor further. Then $\mathfrak{p}$ will never be trivialized in an extension, namely

$$\mathfrak{p}\mathcal{O} \neq \mathcal{O} = (1)$$

***Proof*:**
Let $p \in \mathfrak{p} \setminus \mathfrak{p}^2$. Then

$$(p) = \mathfrak{p}\mathfrak{a} \qquad \mathfrak{p} \nmid \mathfrak{a}$$

Hence, $\mathfrak{p} + \mathfrak{a} = (1)$ ($\gcd(\mathfrak{p}, \mathfrak{a}) = (1)$), and so $a + b = 1$ for $a \in \mathfrak{p}$ and $b \in \mathfrak{a}$. Certainly $b \notin \mathfrak{p}$ so $b\mathfrak{p} \subseteq \mathfrak{p}\mathfrak{a} = (p)$. Now, if $\mathfrak{p}\mathcal{O} = \mathcal{O}$, then $b\mathcal{O} = b\mathfrak{p}\mathcal{O} \subseteq p\mathcal{O}$. hence $b\mathcal{O}_K \subseteq \mathfrak{p}$ so $b = px$ for some $x \in \mathcal{O} \cap K = \mathcal{O}_K$, implying $b \in \mathfrak{p}$ – a contradiction.

Proposition 1.3.11: Equivalence Of Prime Division In Dedekind Domain

Let $\mathfrak{p} \subseteq R$, $\mathfrak{B} \subseteq \mathcal{O}$, $\mathcal{O}/R$ an integral extension where R is a Dedekind Domain. Then the following are equivalent:

1. $\mathfrak{B} \big| \mathfrak{p}\mathcal{O}$
2. $\mathfrak{B} \supseteq \mathfrak{p}\mathcal{O}$
3. $\mathfrak{B} \supseteq \mathfrak{p}$
4. $\mathfrak{B} \cap \mathcal{O} = \mathfrak{p}$
5. $\mathfrak{B} \cap K = \mathfrak{p}$

***Proof*:**
Since we are in a Dedekind domain, $A \big| B$ if and only if $A \supseteq b$, giving us $1 \Leftrightarrow 2$. Since $\mathfrak{B}$ is an ideal of $\mathcal{O}_L$, certainly $2 \Leftrightarrow 3$. We got $3 \Leftarrow 4 \Leftrightarrow 5$ by the above injectivity argument. For $3 \implies 4$, since $\mathfrak{p}$ is maximal and $\mathfrak{B} \cap \mathcal{O}_K \supseteq \mathfrak{p}$, we have either equality or $\mathcal{O}_K$, if $\mathfrak{p} \cap \mathcal{O}_K = \mathcal{O}_K$, then $1 \in Q$, implying $Q = \mathcal{O}_L$, contradicting it's maximality.

Corollary 1.3.12: Lying Over Unique Prime

Let $\mathfrak{p}, \mathfrak{q} \subseteq \mathcal{O}$, $\mathfrak{p} \neq \mathfrak{q}$. Let:

$$\mathfrak{p}\mathcal{O}_L = \mathfrak{B}_1^{e_1} \cdots \mathfrak{B}_n^{e_n} \qquad \mathfrak{q}\mathcal{O}_L = \mathfrak{C}_1^{f_1} \cdots \mathfrak{C}_m^{f_m}$$

Then:

$$\{\mathfrak{B}_1, ..., \mathfrak{B}_n\} \cap \{\mathfrak{C}_1, ..., \mathfrak{C}_n\} = \emptyset$$

that is, extended prime ideals lie over a single prime

***Proof*:**
Let's say the intersection is non-empty, so that $\mathfrak{a}$ is a common factor. Then write:

$$\mathfrak{p}\mathcal{O}_L = \mathfrak{a}^{e_1} \cdots \mathfrak{B}_n^{e_n} \qquad \mathfrak{q}\mathcal{O}_L = \mathfrak{a}^{f_1} \cdots \mathfrak{C}_m^{f_m}$$

Then:

$$(\mathfrak{p})_L^{-1}\mathfrak{p}^{e_1-1}\mathfrak{B}_2^{-e_2} \cdots \mathfrak{B}_n^{-e_n} = \mathfrak{a} = (\mathfrak{q})_L^{-1}\mathfrak{p}^{f_1-1}\mathfrak{C}_2^{-f_2} \cdots \cdots \mathfrak{C}_m^{-f_m}$$

Thus we get:

$$\mathfrak{q}\mathcal{O}_L = (\mathfrak{p})_L\mathfrak{p}^{e_1-1}\mathfrak{B}_2^{-e_2} \cdots \mathfrak{B}_n^{-e_n}\mathfrak{p}^{f_1-1} \cdots \mathfrak{C}_m^{f_m}$$

And in particular:

$$\mathfrak{C}_1^{f_1} \cdots \mathfrak{C}_m^{f_m} = q\mathcal{O}_L \subseteq p\mathcal{O}_L = \mathfrak{B}_1^{e_1} \cdots \mathfrak{B}_n^{e_n}$$

But then, since $\mathfrak{p}\mathcal{O}_L \cap K = \mathfrak{p}$ and $\mathfrak{q}\mathcal{O}_L \cap K = \mathfrak{q}$, we have by corollary 1.3.8 that:

$$\mathfrak{q} \subseteq \mathfrak{p}$$

which since $\mathfrak{q}$ is maximal and $\mathfrak{p} \neq \mathcal{O}_K$ we have $\mathfrak{q} = \mathfrak{p}$, contradicting the fact that $\mathfrak{p} \neq \mathfrak{q}$, hence, it must be that they share no common factors.

Another proof goes as follows: if $\mathfrak{p} \subseteq \mathcal{O}_K$ is a prime and $\mathfrak{p}_L = \mathfrak{B}_1^{e_1} \cdots \mathfrak{B}_k^{e_k}$ is the factorization of $\mathfrak{p}_L$ in $\mathcal{O}_L$, then if there was a prime ideal $\mathfrak{q}$ such that $\mathfrak{q} \mapsto \mathfrak{B}_i$ so that by the above for any $1 \leq i \leq k$, then since $\mathfrak{p} \subseteq \mathfrak{B}_i \cap \mathcal{O}_k$, by maximality $\mathfrak{p} = \mathfrak{B}_i \cap \mathcal{O}_K$. Hence if $\mathfrak{q} = \mathfrak{B}_I \cap \mathcal{O}_k$, by injectivity $\mathfrak{p} = \mathfrak{q}$.

Another consequence of the above theorem is that if two prime factorizations of a prime share a factor in an extended ring of integers, they are equal.

Relative Norm

Since the ideals of Dedekind domains share factorization structure to numbers, we may hope there is an equivalent of the field norm $\mathrm{Nm}_{L/K}$ for ideals. by lemma 1.3.5, we may hope that our new relative norm map gives us some information on the power to which we may rise our ideal to get a principal ideal. We now endeavour to find a map $J_L \to J_K$ that behaves more like the field norm and generalizes the absolute norm, which can be thought of as a map:

$$\mathfrak{N}_{L/\mathbb{Q}} : J_L \to \mathbb{Q}_L \cong \mathbb{Q}_{>0}^{\times}$$

In particular, if we work of *Galois* extensions L/K, then we may hope to translate some of the fact that

$$\mathrm{Nm}_{L/K}(\alpha) = \prod_{\sigma} \sigma(\alpha)$$

to prime ideals! In particular, this allows us to bring in the Galois theory in the study of prime ideals!

We start by showing the case for Galois extensions, the general case can be inferred from the Galois case. Given a Galois extension L/K with Galois group $G = \mathrm{Gal}_K(N)$, then we can recover L from K using G by taking $L^G = K$, hence $\mathcal{O}_L^G = \mathcal{O}_K$. Certainly, if $\tau \in G$ and $\mathfrak{a} \subseteq L$, then $\tau(\mathfrak{a}) \subseteq L$ Since L/K is normal, and hence $\tau(\mathfrak{a})$ is well-defined. It may be tempting to then say that $J_L^G = J_K$ and hence defining $J_L \to J_K$ by taking

$$I \mapsto \prod_{\tau} \tau(I)$$

However, it is not the case that $J_L^G = J_K$; the ideal in example 1.3.9 is invariant under the Galois group, but certainly an ideal of $\mathbb{Z}$. We fix this by the following:

Definition 1.3.13: Relative Norm

Let L/K be a finite field extension and $I \subseteq K$ a fractional ideal in L. Then the *relative norm* or *ideal norm* of L with respect to K is:

$$\mathfrak{N}_{L/K}(I) = \left(\left\{\mathrm{Nm}_{L/K}(\alpha) \;\middle|\; \alpha \in I\right\}\right) \in J_K$$

namely, $\mathrm{Nm}_{L/K}(I)$ is the ideal generated by all norms of elements in I.

Note that this is well-defined since $\mathrm{Nm}_{L/K}(\alpha) \in K$ as every minimal polynomial $m_{\alpha,K}(x) \in K[x]$. The norm as defined above gives us a map $\mathfrak{N}_{L/K} : J_L \to J_K$. The field norm can be thought of as the ideal norm, for example take

$$\mathfrak{N}_{\mathbb{Q}(\sqrt{2})/\mathbb{Q}}((\sqrt{2})) = \left(\left\{\mathrm{Nm}_{L/K}\left(\sqrt{2}b\right) \;\middle|\; \sqrt{2}b \in \sqrt{2}\mathbb{Z}\right\}\right) = (2)$$

suggesting that $(\mathrm{Nm}_{L/K}(\alpha)) = \mathfrak{N}_{L/K}((\alpha))$, which will indeed be the case and shall be proved after the next theorem. An important reason to choose this definition is that it works well with extension: Let $J_L \xrightarrow{\mathrm{Nm}_{L/K}} J_K \xrightarrow{I_L} J_L$. What will be the image of an ideal in this chain? In the simplest case, we may take principal ideal $\alpha\mathcal{O}_L \subseteq \mathcal{O}_L$ where $\alpha \in \mathcal{O}_K$. Then it's image is $\alpha\mathcal{O}_K$, and it's extension is $\alpha\mathcal{O}_L$. Thus for the simplest type of integral principal ideal, the map is a right-inverse. For more general cases, it is not necessarily a right inverse (it cannot be, since I_L is not surjective). However, the behavior of the image is well-behaved, and in fact tied with the "almost" definition of the ideal norm presented earlier:

Theorem 1.3.14: Ideal Correspondence in Extension

Let $N/L/K$ be finite field extension where N/K is a normal extension so that $G = \mathrm{Gal}_K(N)$. Then:

$$(\mathfrak{N}_{L/K}(I))_L = \prod_{\tau \in G} (\tau I)_{L/\tau K} =: \mathrm{Nm}'_{L/K}(I)$$

In particular

$$\begin{Bmatrix} \text{Fractional ideals of} \\ \mathcal{O}_K \end{Bmatrix} \underset{\mathfrak{N}_{L/K}}{\overset{I_L}{\rightleftarrows}} \begin{Bmatrix} \text{Fractional Ideals of} \\ \mathcal{O}_L \end{Bmatrix}$$

***Proof*:**
Let $I \subseteq L$ be a fractional ideal. Without loss of generality, we may make I integral. Then, certainly an ideal is a subset of it's extension (given the natural embedding) $\sigma_i I \subseteq (\sigma_i I)_{L/\sigma_i K}$, and so by definition of $\mathrm{Nm}'_{L/K}(I)$, forall $\alpha \in I$:

$$\mathrm{Nm}_{L/K}(\alpha) = \prod_{\tau} (\tau(\alpha)) \in \mathrm{Nm}'_{L/K}(I)$$

Hence $\mathrm{Nm}_{L/K}(I) \subseteq \mathrm{Nm}'_{L/K}(I)$. Extending both sides, we get $(\mathrm{Nm}_{L/K}(I))_L = (\mathrm{Nm}'_{L/K}(I))_L = \mathrm{Nm}'_{L/K}(I)$. Hence, since we are in a Dedekind domain, there exists an [integral] ideal J such that

$$\mathrm{Nm}_{L/K}(I)_L = J\mathrm{Nm}'_{L/K}(I)$$

We want to show that $J = (1)$. We shall use lemma 1.6.1, namely that there exists an ideal $B \subseteq K$ relatively prime to $J \cap K$ such that JB is principal, say $JB = (b)$ $(b \in K)$. Then since J is fixed under $\mathrm{Gal}_K(L)$, each $(\tau B)_{L/\tau K})$ is relatively prime to J for each τ. Then since $b \in J$:

$$\begin{aligned} \mathrm{Nm}_{L/K}(I) &\supseteq \left(\prod_{\tau} \tau(b) \right) \\ &= \left(\prod_{\tau} (\tau(B))_{L/\tau K)} \right) \left(\prod_{\tau} (\tau(J))_{L/\tau K} \right) \end{aligned}$$

Thus, $\prod_{\tau} (\tau B)_{L/\tau K}$ is a multiple of J. However, $(\tau B)_{L/\tau K}$ is relatively prime to J, hence it must be that $J = (1)$, as we sought to show.

Lemma 1.3.15: Ideal Norm Properties

Let $M/L/K$ be algebraic number fields, $\mathfrak{a}, \mathfrak{b} \subseteq M$ be ideals, $I \subseteq K$ an ideal, and $a \in M$ an element. Then:

1. $\mathfrak{N}_{M/K}(a) = (\mathrm{Nm}_{M/K}(a)) = \left(\prod_{\sigma \in \mathrm{Gal}_K(M)} \sigma(a)\right)$
2. $\mathfrak{N}_{L/K}(\mathfrak{N}_{M/L}(\mathfrak{a})) = \mathfrak{N}_{M/K}(\mathfrak{a})$
3. $\mathfrak{N}_{M/k}(\mathfrak{a}\mathfrak{b}) = (\mathrm{Nm}_{M/k}(\mathfrak{a}))\mathrm{Nm}_{M/k}(\mathfrak{b})$
4. $I^{[M:K]} = \mathfrak{N}_{M/K}((I)_M)$
5. $(\mathfrak{N}_K(I)) = \mathfrak{N}_{K/\mathbb{Q}}(I)$

***Proof*:**
Note that for all of these equalities, we immediately have $\supseteq$, hence we want to show $\subseteq$ for each of these.

1. comes straight from the definition
2. exercise
3. exercise
4. Use lemma 1.3.6 and the finiteness of the class group to make an argument on principal ideals, we see that if we take $J_K \xrightarrow{I_L} J_L \xrightarrow{\mathrm{Nm}_{L/K}} J_K$, we pick up a $[L : K]$ exponent.
5. use part (1)

Corollary 1.3.16: Relative Norm Of Prime Ideal

Let L/K be a finite extension of number fields and let $\mathfrak{B}$ lie above $\mathfrak{p}$. Then if $f = [\mathcal{O}_L/\mathfrak{B} : \mathcal{O}_K/\mathfrak{p}]$, then:

$$\mathfrak{N}_{L/K}(\mathfrak{B}) = \mathfrak{p}^f$$

***Proof*:**
Using the above lemma:

$$\begin{aligned}(\mathfrak{N}_L(\mathfrak{p}_L)) &= \mathfrak{N}_{L/\mathbb{Q}}(\mathfrak{p}_L)\\ &= \mathfrak{N}_{K/\mathbb{Q}}(\mathfrak{p}^{[L:K]})\\ &= (\mathfrak{N}_K(\mathfrak{p}))^{[L:K]}\end{aligned}$$

Thus, $\mathfrak{N}_{L/K}(\mathfrak{B})$ is a power of $\mathfrak{p}$ by the 4th point in the above lemma. and by the 2nd we get:

$$\mathfrak{N}_{K/\mathbb{Q}}(\mathfrak{N}_{L/K}(\mathfrak{B})) = \mathfrak{N}_{K/\mathbb{Q}}(\mathfrak{p}^f) \implies \mathfrak{N}_{L/K}(\mathfrak{B}) = \mathfrak{p}^f$$

as we sought to show.

The relative norm can in fact be defined to be the unique group homomorphism where

$$\mathfrak{N}(\mathfrak{B}) = \mathfrak{p}^{[\mathcal{O}_L/\mathfrak{B}:\mathcal{O}_K/\mathfrak{p}]}$$

which is left as an exercise. By multiplicity, we get that in general:

$$\mathfrak{N}_{L/K}(\mathfrak{p}\mathcal{O}_L) = \prod_i \mathfrak{N}_{L/K}\mathbb{F}_{L/K}(\mathfrak{B}_i)^{e_i} = \prod_i \mathfrak{p}^{f_i e_i}$$

Corollary 1.3.17: Ideal Norm Correspondence Not Galois

Let $M/L/K$ were field extensions, M/K normal but not necessarily L/K. Then

$$(\mathrm{Nm}_{L/K}(I))_M = \prod_{\tau \in \mathrm{Gal}_K(M)/\mathrm{Gal}_L(M)} \tau(I_M)$$

***Proof*:**
exercise

With the relative norm defined, it is perhaps natural to think there is a relative discriminant:

Definition 1.3.18: Relative Discriminant

Let L/K be a finite extension of number fields. Then:

$$\mathrm{disc}'_{L/K}(\mathcal{O}_L) = \left(\left\{\mathrm{disc}_{L/K}(\alpha_1, \alpha_2, ..., \alpha_n) \;\middle|\; \alpha_1, \alpha_2, ..., \alpha_n \text{ basis for } \mathcal{O}_L \;\right\}\right)$$

is the *relative discriminant* of L with respect to K.

This won't come up much, but is useful in a couple of tower arguments later on.

Though we have characterized a correspondence of the ideals, it is more difficult to characterize I_L. In the following section, we characterize prime extensions, and take advantage of the fact that the factorization to give us a characterization of I_L.

1.4 Prime Factorization in Extension

Since $\mathcal{O}$ is a Dedekind domain by lemma 1.3.2, we have that $\mathfrak{p}\mathcal{O}$ decomposes uniquely into a product of prime ideals:

$$\mathfrak{p}\mathcal{O} = \mathfrak{B}_1^{e_1} \cdots \mathfrak{B}_r^{e_r}$$

If it is clear from context, we shall often write $\mathfrak{p}$ instead of $\mathfrak{p}\mathcal{O}$. Sometimes, we shall also write $\mathfrak{p}_L$ for $\mathfrak{p}\mathcal{O}_L$. We shall *almost always* find exactly how $\mathfrak{p}\mathcal{O}$ decomposes, sometimes without requiring knowledge of the ring of integers. When the extension L/K is Galois, we shall have even more control, and if L/K is not Galois, we can take $M/L/K$ to be a Galois extension, work within the Galois, and see if the results can be translated back to L/K.

Let's say $\mathfrak{B}$ lies over $\mathfrak{p}$ and that $\mathfrak{B}$ is in the prime decomposition of $\mathfrak{B}$ [21]. Then we have two important quantities associated to $\mathfrak{B}$: it's power in the prime decomposition, and the index $[\mathcal{O}/\mathfrak{B} : \mathcal{O}_K/\mathfrak{p}]$, that is the degree of the extension of the residue field. These quantities are given a name:

Definition 1.4.1: Ramification Index and Inertia Degree

Let R be a Dedekind domain, $K = \mathrm{Frac}(R)$, L/K be a finite field extension, $\mathcal{O} = \overline{R}^L$ and $\mathfrak{p} \subseteq R$ a prime ideal. If $\mathfrak{B}_i$ is a prime in $\mathcal{O}$ lying above $\mathfrak{p}$, then it's degree is called the *ramification index* and is often denoted

$$e(\mathfrak{B}_i/\mathfrak{p}) \qquad \text{where} \qquad \mathfrak{B}^{e(\mathfrak{B}_i/\mathfrak{p})} \,\|\, \mathfrak{p}\mathcal{O}$$

or e_i for short, where $\|$ means *maximally divides*. The degree of the field extension

$$f(\mathfrak{B}_i/\mathfrak{p}) = [\mathcal{O}/\mathfrak{B}_i : R/\mathfrak{p}]$$

or f_i for short, and is called is called the *inertia degree* or *residue field degree* of $\mathfrak{B}_i$ over $\mathfrak{p}$.

For future reference, it will be important to keep track to "how much" a prime ideal splits. This will share strong similarities to the splitting of polynomials[22] (recall how "rare" an irreducible polynomial is inseparable)? This prompts the following vocabulary:

Definition 1.4.2: Split, Unramified, and Totally Ramified Primes

Let $\mathfrak{p} \subseteq R$ be a prime ideal of a Dedekind domain R, $K = K(R)$, L/K a finite extension, and $\mathcal{O} = \overline{R}^L$. Then given

$$\mathfrak{p}\mathcal{O} = \mathfrak{B}_1^{e_1} \cdots \mathfrak{B}_g^{e_g}$$

If $g = [L : K]$, then $\mathfrak{p}$ is said to *split completely* ($e_i = f_i = 1$), giving:

$$\mathfrak{p}\mathcal{O} = \mathfrak{B}_1 \cdots \mathfrak{B}_n$$

If $e_i = 1$ for all i (but some f_i are not necessarily 1), then we sy that $\mathfrak{p}$ is *unramified* and (if the residue field is separable):

$$\mathfrak{p}\mathcal{O} = \mathfrak{B}_1 \cdots \mathfrak{B}_g$$

for some $g \leq n$. If $g = 1$, then $\mathfrak{p}$ is said to be *totally ramified*, *indecomposable*, or *nonsplit* if:

$$\mathfrak{p} = \mathfrak{B}_1^{e_1}$$

and we know there is only a single prime ideal $\mathfrak{B}_i$ lying over $\mathfrak{p}$.

Most of the time, we shall work in the case of character zero extensions, hence L/K is separable. In this case, the values e_i and f_i are connected; if $\mathfrak{p}$ factors in an extension, then there is a simple equation the ramification and inertia degree must satisfy which equals the degree of the extension:

[21] we shall soon show all primes lying above $\mathfrak{p}$ will be part of the prime decomposition

[22] Though there shall be important failures when this comparison will *not* hold! This shall be elaborated soon

Proposition 1.4.3: Fundamental Identity

Let L/K be a finite separable extension, $n = [L : K]$, $\mathcal{O} = \overline{\mathcal{O}_K}^L$, and $\mathfrak{p} \subseteq \mathcal{O}_K$ a prime ideal. If

$$\mathfrak{p}\mathcal{O} = \mathfrak{B}_1^{e_1} \cdots \mathfrak{B}_g^{e_g}$$

is the prime decomposition of $\mathfrak{p}\mathcal{O}$ and f_i is the inertia degree of $\mathfrak{B}_i$ over $\mathfrak{p}$, then

$$\sum_i^g e_i f_i = n$$

***Proof*:**
Using absolute norms, this theorem becomes almost immediate, since:

$$\begin{aligned}\mathfrak{N}(\mathfrak{p})^{[L:K]} &\overset{\text{lem 1.3.6}}{=} \mathfrak{N}(\mathfrak{p}\mathcal{O}) = \mathfrak{N}\left(\prod_i \mathfrak{B}_i^{e_i}\right) \overset{\text{prop 1.2.28}}{=} \prod_i \mathfrak{N}(\mathfrak{B}_i)^{e_i} \\ &\overset{\text{cor 1.3.16}}{=} \prod_i \mathfrak{N}(\mathfrak{p})^{e_i f_i} = \mathfrak{N}(\mathfrak{p})^{\sum_i e_i f_i}\end{aligned}$$

taking log of both sides gives us:

$$n = [L : K] = \sum_I e_i f_i$$

This can also be proven directly: first by the CRT we have

$$\mathcal{O}/\mathfrak{p}\mathcal{O} \cong \bigoplus_i^r \mathcal{O}/(\mathfrak{B}_i^{e_i})$$

Next, we have that $\mathcal{O}/\mathfrak{p}\mathcal{O}$ and $\mathcal{O}/\mathfrak{B}_i^{e_i}$ are vector-spaces over $k = \mathcal{O}_K/\mathfrak{p}$. Hence, it suffices to show that

$$\dim_k(\mathcal{O}/\mathfrak{p}\mathcal{O}) = n \qquad \dim_k(\mathcal{O}/(\mathfrak{B}_i^{e_i})) = e_i f_i$$

To show the first equality, since $\mathcal{O}$ is a finitely generated $\mathcal{o}$-module, pick $x_1, x_2, ..., x_m \in \mathcal{O}$ such that $\overline{x_1}, ..., \overline{x_n}$ is a k-basis for $\mathcal{O}/\mathfrak{p}\mathcal{O}$. If $x_1, x_2, ..., x_m$ is a basis for L/K, then we're done. Let's first show they are linearly independent. For the sake of contradiction, let's say they weren't so that $x_1, x_2, ..., x_m$ are linearly dependent over K. Then they are linearly dependent over $\mathcal{O}_K$ by Gauss's lemma. Hence, there are elements $a_1, a_2, ..., a_m \in \mathcal{O}_K$ that are not all zero such that

$$a_1 x_1 + \cdots + a_m x_m = 0$$

We want to use this to show that the congruent mod $\mathfrak{p}$ is also linearly dependent. To that end, consider $\mathfrak{a} = (a_1, a_2, ..., a_m)$. Take $a \in \mathfrak{a}^{-1}$ such that $a \notin \mathfrak{a}^{-1}\mathfrak{p}$ so that $a\mathfrak{a} \not\subseteq \mathfrak{p}$. Hence, the element $aa_1, ..., aa_m$ lie in $\mathcal{O}_K$, but are not all in $\mathfrak{p}$. Hence, we have:

$$aa_1 x_1 + \cdots + aa_m x_m = 0 \text{ mod } \mathfrak{p}$$

giving linear dependence of the elements $\overline{x_1}, ..., \overline{x_m}$ over k – a contradiction.

Next, for spanning, consider the $\mathcal{O}_K$-module

$$M = \mathcal{O}_K x_1 + \cdots \mathcal{O}_K x_m$$

We shall show we can "upgrade" this to $L = Kx_1 + \cdots + Kx_m$. Consider $N = \mathcal{O}/M$. Since $\mathcal{O} = M + \mathfrak{p}\mathcal{O}$, $\mathfrak{p}N = N$. Since $\mathcal{O}$ is finitely generated (being a Dedekind domain), N is finitely generated. If $\alpha_1, \alpha_2, ..., \alpha_s$ are generators of N, then since $\mathfrak{p}N = N$:

$$\alpha_i = \sum_j a_{ij}\alpha_j \qquad a_{ij} \in \mathfrak{p}$$

Now, define the matrix $A = (a_{ij}) = I$ where I is the identity matrix of rank s. Let $B = A^*$ be the adjoint of A whose entries are the minors of rank $(s-1)$ of A. Then

$$A(\alpha_1, ..., \alpha_s)^t = 0 \qquad BA = \det(A)I = dI$$

Thus:

$$0 = BA(\alpha_1, \alpha_2, ..., \alpha_s)^t = d\alpha_1, ..., d\alpha_s)^t$$

thus, $dN = 0$, implying $d\mathcal{O} \subseteq M = \mathcal{O}_K x_1 + \mathcal{O}_K x_2 + \cdots + \mathcal{O}_K x_m$. Importantly, $d \neq 0$ since $d = \det((a_{ij}) = I) \equiv (-1)^s \bmod p$ since $a_{ij} \in \mathfrak{p}$. Thus:

$$L = dL = Kx_1 + Kx_2 + \cdots + Kx_m$$

showing that it spans L/K, and hence it is a basis, establishing the first equality.

For the second, recall in proposition 1.2.28 that each $\mathfrak{B}_i^{n+1}/\mathfrak{B}_i^n \cong_{\mathcal{O}/\mathfrak{B}_i} \mathcal{O}/\mathfrak{B}_i$. Hence, since $f_i = [\mathcal{O}/\mathcal{B}_i : \mathcal{O}_k/\mathfrak{p}]$, we get

$$n = \dim_k(\mathcal{O}/\mathfrak{p}\mathcal{O}) \sum_i \dim_k(\mathcal{O}/(\mathfrak{B}_i^{e_i})) = \sum_i^r e_i f_i$$

as we sought to show.

Corollary 1.4.4: Multiplicity of Ramification And Inertia

Let $M/L/K$ be a finite extension of number fields. Then:

$$e_{M/K} = e_{M/L}e_{L/K} \qquad f_{M/K} = f_{M/L}f_{L/K}$$

***Proof*:**
immediate consequence of the arguments in the above theorem

Hence, we have a measure of how much a prime ideal splits in an extension.

Example 1.4.5: Decomposing Prime

1. It is possible that $\mathfrak{p}$ is totally ramified where $e_1 > 1$ and $f_1 = 1$: take $L = \mathbb{Q}(\sqrt{3})$ and $K = \mathbb{Q}$ with $I = (3)$. Then $(3)\mathcal{O} = (\sqrt{3})^2$

2. Similarly, it is possible that $\mathfrak{p}$ is totally ramified and $e_1 = 1$ and $f_1 > 1$: take $K = \mathbb{Q}$, $L = \mathbb{Q}(\sqrt{3})$, but $I = (5))$.

3. Let us consider the inertia degree to get some structure of the Dedekind domain. Let $L = \mathbb{Q}(\sqrt[3]{2})$ and $L/\mathbb{Q}$ be a finite extension. Since it is always the case that $\mathbb{Z}[\sqrt[3]{2}] \subseteq \mathcal{O}_L$ [a], Take $I = (\sqrt[3]{2}) \subseteq \mathcal{O}_L$, $I \cap \mathbb{Q} = (2)$, and $\mathfrak{N}((\sqrt[3]{2})) = |\mathrm{Nm}_{L/\mathbb{Q}}(\sqrt[3]{2})| = 2$, and $2 = (\sqrt[3]{2})^3$. Hence, $f_1 = 1$, meaning
$$[\mathcal{O}_L/\sqrt[3]{2}\mathcal{O}_L : \mathbb{Z}/2\mathbb{Z}] = 1$$
implying both fields only have two elements, hence
$$\mathcal{O}_L/\sqrt[3]{2}\mathcal{O}_L \cong \mathbb{Z}/2\mathbb{Z}$$
Notice how this also gave information on the ring of integers,

4. Let $\mathcal{O}_K = \mathbb{Z}[i]$ so that $2 = (1+i)(1-i)$. Then $\mathcal{O}_K/2\mathcal{O}_K$ has order 4 (look at it'snorm), and certainly $2\mathcal{O}_K \subseteq (1-i)$, hence $|\mathcal{O}_K/(1-i)| \big| 4$, which by direct computation we may see that $|\mathcal{O}_K/(1-i)| = 2$, hence $f = 1$. On the other hand, we can see that 3 is prime in $\mathcal{O}_K$, and $|\mathcal{O}_K/3\mathcal{O}_K| = 9$, giving $f_{3\mathcal{O}_K/3} = 2$.

[a] In fact, they are equal in this case, which is not needed at the moment

Since the factorization of $\mathfrak{p} \subseteq K$ in L/K is tied to the degree of L/K, we may venture into hoping that the structure of L/K may reflect something about the factorization of $\mathfrak{p}$. A natural question we may ask here is how do we find the prime factors in the extension? Finding them is split into two cases: the "good" primes and the "bad' primes. We shall see that the power of the exponentials has a lot to do with it. This is because we shall see that most primes shall decompose with ramification index 1, since it will correspond to the splitting of a polynomial in separable extension (note that all number field extensions are separable). However, as we saw it is possible that prime may have higher ramification index, however it is impossible that a polynomial splits irreducibly with multiplicity higher than 1 over a separable extension. Hence, we may think of such prime as "bad" primes. In the following we shall show how to find primes whose factorization mirrors that of a minimal polynomial, and how to detect when it is not the case.

let L/K be a finite separable extension. Then by the primitive extension theorem, there exists a $\theta \in K$
$$L = K(\theta)$$
Usually, we shall let L be Galois so that we may computationally find this element, namely it must not be fixed by any element of $\mathrm{Gal}_K(L)$ (hence, not in any intermediate extension). In fact, we may find θ so that $\theta \in \mathcal{O}_K$: if $L = K(\alpha)$, since $L = \overline{\mathcal{O}_K}^L$ we may multiply out the denominator by d to get $\alpha = \theta$; let us simply re-define θ to be in $\mathcal{O}_L$. Now we have a single element, θ with minimal polynomial $m_{K,\theta}(x)$ which characterizes L. It would be nice to say something about the nature of the decomposition of $\mathfrak{p}$ in $\mathcal{O}_L$ given this information. In the good case, we shall be able to do so, and in the bad case we shall require a bit more work. In particular, a finite number of primes ideals will be exclude; only those that are relatively prime to an ideal known as the *conductor* of the ring $\mathcal{O}_K[\theta]$ will be considered:

Definition 1.4.6: Conductor

Let $\mathcal{O}_K[\theta]$ be the ring of integers extended by θ. Then the conductor of $\mathcal{O}_K[\theta]$, is the largest ideal $\mathscr{F}_\theta \subseteq \mathcal{O}_L$ which is contained in $\mathcal{O}_K[\theta]$:

$$\mathscr{F}_\theta = \{\alpha \in \mathcal{O}_L \mid \alpha\mathcal{O}_L \subseteq \mathcal{O}_K[\theta]\}$$

In the case where $K = \mathbb{Q}$ and $\mathcal{O}_\mathbb{Q} = \mathbb{Z}$, then $\mathscr{F}_\theta = \{\alpha \in \mathcal{O}_L \mid (\alpha)_L \subseteq \mathbb{Z}[\theta]\}$. Since $\mathcal{O}_L$ is a finitely generated $\mathcal{O}_K$ module, $\mathscr{F}_\theta \neq 0$. The conductor is a measure of the "bad primes" (hence the ramification). In our case, the conductor is never 0, but it may be that $\mathscr{F}_\theta = (1)$, in which case the bad primes will be "tame" (or the field is "tamely ramified")

Theorem 1.4.7: Dedekind-Kummer Theorem

Let L/K be a separable extension, $L = K(\theta)$ where θ is a primitive element where $m_{\theta,K}(x) \in \mathcal{O}_K[x]$. Let $\mathfrak{p}$ be a prime ideal of $\mathcal{O}_K$ which is relatively prime to the conductor $\mathscr{F}$ of $\mathcal{O}_K[\theta]$ and let $m(x) = m_{\theta,K}(x)$ and consider:

$$\overline{m}(x) = \overline{m}_1(x)^{e_1} \cdots \overline{m}_r(x)^{e_r} \bmod \mathfrak{p}$$

be the factorization of the polynomial $\overline{m}(x) = m(x) \bmod \mathfrak{p}$ into irreducible over the residue field $\mathcal{O}_K/\mathfrak{p}$. Then

$$\mathfrak{B}_i = \mathfrak{p}\mathcal{O} + m_i(\theta)\mathcal{O} = (\mathfrak{p}\mathcal{O}, m_i(\theta))_L$$

are different prime ideals of $\mathcal{O}$ over $\mathfrak{p}$. The inertia degree f_i of $\mathfrak{B}_i$ is the degree of $\overline{m}_i(x)$, and

$$\mathfrak{p} = \mathfrak{B}_1^{e_1} \cdots \mathfrak{B}_r^{e_r}$$

Note that if $\mathfrak{p}$ is not relatively prime to $\mathscr{F}$, then $\mathfrak{p}$ is not relatively prime to the relative discriminant, hence $\mathfrak{p}$ will surely ramify, and hence this is a problem for primes whose ramification is somehow "more bad"; such primes are often said to have *wild ramification* and they shall be studied in section 1.4.3.

***Proof*:**
Let $\mathcal{O}' = \mathcal{O}_K[\theta]$ and $\overline{\mathcal{O}_K} = \mathcal{O}_K/\mathfrak{p}$. The key isomorphism that we can establish since $\mathfrak{p}$ is relatively prime to the conductor $\mathscr{F}_\theta$ is:

$$\mathcal{O}_L/\mathfrak{p}\mathcal{O}_L \overset{!}{\cong} \mathcal{O}_K[\theta]/\mathfrak{p}\mathcal{O}_K[\theta] \cong \overline{\mathcal{O}_K}[x]/(\overline{m}(x))$$

where the $\overset{!}{\cong}$ isomorphism is the hard one to prove; the 2nd one comes from:

$$\mathcal{O}_K[\theta]/\mathfrak{p} \cong \mathcal{O}_K[x]/(\mathfrak{p}, m(x)) \cong \overline{\mathcal{O}}_K[x]/(\overline{m}(x))$$

For the $\overset{!}{\cong}$ isomorphism, from the relative primality of $\mathfrak{p}$ and $\mathscr{F}$ we get $\mathfrak{p}\mathcal{O}_L + \mathscr{F} = \mathcal{O}_L$. Since $\mathscr{F} \subseteq \mathcal{O}' = \mathcal{O}_k[\theta]$, we get that $\mathcal{O}_L = \mathfrak{p}\mathcal{O}_L + \mathcal{O}'$. From this, we get that the homomorphism

$$\mathcal{O}' \to \mathcal{O}_L/\mathfrak{p}\mathcal{O}_L$$

is surjective (every element of $\mathcal{O}_L$ is a linear combination of an element in $\mathcal{O}'$ and $\mathfrak{p}\mathcal{O}_L$) with kernel $\mathfrak{p}\mathcal{O}_L \cap \mathcal{O}'$. What's left to show is that this kernel equals $\mathfrak{p}\mathcal{O}'$. $\mathfrak{p}\mathcal{O}' \subseteq \mathfrak{p}\mathcal{O}_L \cap \mathcal{O}'$ is immediate, so for the other direction since $\mathfrak{p} + \mathscr{F} \cap \mathcal{O}_K = \gcd(\mathfrak{p}, \mathscr{F} \cap \mathcal{O}_K) = (1)$, $1 = pq + fs$ for some $p \in \mathfrak{p}$, $f \in \mathscr{F} \cap \mathcal{O}_K$. Then certainly $1 = pq + fs \in \mathfrak{p} + \mathscr{F}$, hence $\mathcal{O}' = \mathfrak{p} + \mathscr{F}$. Then:

$$\begin{aligned}\mathfrak{p}\mathcal{O}_L \cap \mathcal{O}' &= (\mathfrak{p} + \mathscr{F})(\mathfrak{p}\mathcal{O}_L \cap \mathcal{O}') \\ &= \mathfrak{p}(\mathfrak{p}\mathcal{O}_L \cap \mathcal{O}') + \mathscr{F}(\mathfrak{p}\mathcal{O}_L \cap \mathcal{O}') \\ &\subseteq \mathfrak{p}\mathcal{O}'\end{aligned}$$

where the last inclusion comes from the fact that any element of in the 2nd equality has the form $p(p\ell + f\ell')$ where $p\ell, f\ell' \in \mathcal{O}'$ by definition (namely the intersection), and so $\mathcal{O}_L/\mathfrak{p}\mathcal{O}_L \cong \mathcal{O}'/\mathfrak{p}\mathcal{O}'$. Then by our above argument we have $\mathcal{O}'/\mathfrak{p}\mathcal{O}' \cong \overline{\mathcal{O}}_K/(\overline{m}(x))$, hence we have:

$$\mathcal{O}_L/\mathfrak{p}\mathcal{O}_L \cong \overline{\mathcal{O}_K}[x]/(\overline{m}(x))$$

Now looking at the ring on the right, since in $\bmod \mathfrak{p}$ we have $\overline{m}(x) = \prod_i^r \overline{m}_i(x)^{e_i}$, by the CRT we get:

$$\overline{\mathcal{O}}_K[x]/(\overline{m}(x)) \cong \bigoplus_i^r \overline{\mathcal{O}}_K[x]/(\overline{m}_i(x))^{e_i}$$

But now we know the structure of each direct summand, $\overline{\mathcal{O}}_k[x]/(\overline{m}_i(x))^{e_i}$ is a principal ideal ring, and by the isomorphism we may deduce the prime ideal of $R = \overline{\mathcal{O}_K}[x]/(\overline{m}(x))$ are the principal ideal

$$(\overline{m_i}) = (\overline{m}_i(x) \bmod \overline{m}(x)) \qquad 1 \leq i \leq r$$

By ring extension theory, the degree of $[R/(\overline{m}_i) : \overline{\mathcal{O}}_K]$ equals the degree of the polynomial $\overline{m}_i(x)$, and by definition of the $(\overline{m}_i)$:

$$(0) = \bigcap_i^r (\overline{m}_i)^{e_i}$$

Now, through the isomorphism $\overline{\mathcal{O}}_K[x]/(\overline{m}(x)) \cong \mathcal{O}_L/\mathfrak{p}\mathcal{O}_L$ given by $\overline{f}(x) \mapsto \overline{f}(\theta)$, we get the same result for the primes ideals of $\overline{\mathcal{O}}_L = \mathcal{O}_L/\mathfrak{p}\mathcal{O}_L$. Namely, the prime ideals $\overline{\mathfrak{B}}_i$ of $\overline{\mathcal{O}}_L$ correspond to the prime ideals

$$\overline{\mathfrak{B}}_i = (\overline{m}_i) = (m_i(\theta) \bmod \mathfrak{p}\mathcal{O}_L)$$

with degrees $[R/\overline{\mathfrak{B}}_i : \mathcal{O}_K/\mathfrak{p}]$ where $R = \overline{\mathcal{O}_K}[x]/(\overline{m(x)})$ equalling the degrees of the polynomials $\overline{m}_i(x)$. Furthermore:

$$(0) = \bigcap_i^r \overline{\mathfrak{B}}_i^{e_i}$$

Now, given the canonical projection $\mathcal{O}_L \to \mathcal{O}_L/\mathfrak{p}\mathcal{O}_L$, let $\mathfrak{B}_i = \mathfrak{p}\mathcal{O}_L + m_i(\theta)\mathcal{O}_L$, (the preimage of $\overline{\mathfrak{B}}_i$). Then since the pre-image of a prime ideal is prime, each $\mathfrak{B}_i$ is prime, and

$$\mathfrak{B}_i \cap \mathcal{O}_K = \mathfrak{p}$$

since, when written in element form, we have that

$$p\ell + m_i(\theta)\ell' \in \mathcal{O}_K$$

which is true if and only if $\ell' = 0$ and $p\ell \in \mathcal{O}_K$, showing that $\mathfrak{B}_i$, $i \leq i \leq r$, varies over the prime ideals of $\mathcal{O}_L$ above $\mathfrak{p}$, with $f_i = [\mathcal{O}_L/\mathfrak{B}_i : \mathcal{O}_K/\mathfrak{p}]$ being the degree of the polynomial $\overline{m}_i(x)$. Furthermore, $\mathfrak{B}_i^{e_i}$ is the pre-image of $(\overline{\mathfrak{B}}_i^{e_i})$ (since $e_i = \#\left\{\overline{\mathfrak{B}}^n \;\middle|\; n \in \mathbb{N}\right\}$) and

$$(0) = \cap_{i=1}^r \mathfrak{B}_i^{e_i} = \prod \mathfrak{B}_i^{e_i} \subseteq \mathfrak{p}\mathcal{O}_L$$

so that

$$\mathfrak{p}\mathcal{O}_L \mid \prod_{i=1}^{r} \mathfrak{B}_i^{e_i}$$

since $\deg(m(x)) = [L : K] = n = \sum e_i f_i$, we in fact has no other factors or powers, showing us that in fact:

$$\mathfrak{p}\mathcal{O}_L = \prod_{i=1}^{r} \mathfrak{B}_i^{e_i}$$

as we sought to show.

Corollary 1.4.8: Factorization For Number Rings (Dedekind-Kummer Theorem)

Let L/K be a finite extension of number fields, $L = K(\theta)$ for a primitive element θ where $m_{\theta,K}(x) \in \mathcal{O}_K[x]$. Let $\mathfrak{p} \subseteq \mathcal{O}_K$ a prime ideal lying over $p \in \mathbb{Z}$. If $p \nmid [\mathcal{O}_L : \mathcal{O}_K[\alpha]]$, then

$$\mathfrak{B}_i = \mathfrak{p}\mathcal{O}_L + m_i(\theta)\mathcal{O}_L = (\mathfrak{p}\mathcal{O}_L, m_i(\theta))_L$$

are different prime ideals of $\mathcal{O}$ over $\mathfrak{p}$. The inertia degree f_i of $\mathfrak{B}_i$ is the degree of $\overline{m}_i(x)$, and

$$\mathfrak{p} = \mathfrak{B}_1^{e_1} \cdots \mathfrak{B}_r^{e_r}$$

***Proof*:**
In the proof of theorem 1.4.7 we used the conductor to show that:

$$\mathcal{O}_L/\mathfrak{p}\mathcal{O}_L \cong \mathcal{O}'/\mathfrak{p}\mathcal{O}' \cong \overline{\mathcal{O}_K}[x]/(\overline{m}(x))$$

In particular, we could have used the CRT earlier and shown that:

$$\frac{\mathcal{O}_L}{\mathfrak{B}} \cong \frac{\overline{\mathcal{O}_K}[x]}{(\overline{m_i(x)})}$$

Hence, if we can show this condition, we are done. Take the map $\varphi : \mathcal{O}_K[x] \to \mathcal{O}_L/\mathfrak{B}_i$ mapping $p(x) \mapsto p(\theta)$. We claim that this map is surjective. To see this, it suffices to show that

$$\mathcal{O}_L = \mathcal{O}_K[\theta] + \mathfrak{B}_i$$

We know that $p \in \mathfrak{p} \subseteq \mathfrak{B}_i$, hence $p\mathcal{O}_L \subseteq \mathfrak{B}_i$. We now claim the slightly stronger:

$$\mathcal{O}_L = \mathcal{O}_K[\theta] + p\mathcal{O}_L$$

Now, by assumption $p \nmid |\mathcal{O}_L : \mathcal{O}_K[\theta]|$. Since the index of $\mathcal{O}_K[\theta] + p\mathcal{O}_L$ in $\mathcal{O}_L$ is the common divisor of $[\mathcal{O}_L : \mathcal{O}_K[\theta]]$ and $[\mathcal{O}_L : p\mathcal{O}_L]$, and these *must* be relatively prime since

$$[\mathcal{O}_L : p\mathcal{O}_L] = p^k$$

for some $k \in \mathbb{N}$. Thus $\mathcal{O}_K[x] \to \mathcal{O}_L/\mathfrak{B}_i$ is surjective. Now Certainly $(\mathfrak{p}, m_i) \subseteq \ker\varphi$. Furthermore, it is clear that $(\mathfrak{p}, m_i) \subseteq \mathcal{O}_K[x]$ is a maximal ideal hence $\ker\varphi$ is either $(\mathfrak{p}, m_i)$ or $\mathcal{O}_K[x]$. since the map is surjective, then either $\mathfrak{B}_i = \mathcal{O}_L = (1)_L$ (i.e. $\mathfrak{B}_i$ is *not* a prime, a contradiction), or it is indeed the desired kernel, meaning:

$$\frac{\mathcal{O}_L}{\mathfrak{B}} \cong \frac{\overline{\mathcal{O}_K}[x]}{(\overline{m_i(x)})}$$

but now the proof follows from the rest of the proof of theorem 1.4.7, as we sought to show.

Corollary 1.4.9: Monogenic Number Rings

Let L/K be a finite extension of number rings such that $\mathcal{O}_L = \mathcal{O}_K[\alpha]$ for some $\alpha \in \mathcal{O}_L$. Then all primes split as given in the above

***Proof*:**
Choose $\theta = \alpha$. Then $[\mathcal{O}_L : \mathcal{O}_K[\alpha]] = [\mathcal{O}_K[\alpha] : \mathcal{O}_K[\alpha]] = 1$. Since certainly no prime divides 1, we have that all primes factor in given their minimal polynomials and modular factorization, as we sought to show.

Note that not all number rings are of the form $\mathcal{O}_L = \mathcal{O}_K[\alpha]$, that is not all number rings are monogenic. We shall show a counter-example in example 1.4.29. This contrasts to finite extension of field where every separable finite extension of is a simple extension by [5, Primitive Element Theorem]. Note how this is dependent on finding a θ and $\mathfrak{p}$ being relatively prime to $\mathscr{F}_\theta$ (or $[\mathcal{O}_L : \mathcal{O}_K[\theta]]$ in the case of number fields)[23].

As it turns out, the power of an extended prime has important geometric and algebraic consequences; it can be thought of as the measure of how much the Galois group moves a field around. We hence give it an name:

Definition 1.4.10: Ramified and Unramified Extensions

Let $\mathfrak{p} \subseteq R$ be a prime ideal, $K = K(R)$, L/K a finite extension, and $\mathcal{O} = \overline{K}^L$. Then given

$$\mathfrak{p}\mathcal{O} = \mathfrak{B}_1^{e_1} \cdots \mathcal{B}_r^{e_r}$$

Then the prime ideal $\mathfrak{B}_i$ is called *unramified* over $\mathcal{O}_k$ (or over K) if $e_i = 1$ and if the residue fields $(\mathcal{O}/\mathfrak{B}_i)/(\mathcal{O}_K/\mathfrak{p})$ is separable[a]:

$$\mathfrak{p}\mathcal{O} = \mathfrak{B}_1 \cdots \mathfrak{B}_r$$

Otherwise, $\mathfrak{B}_i$ is called *ramified*, and *totally ramified* if furthermore $f_i = 1$. The prime ideal $\mathfrak{p}$ is called *unramified* if all $\mathfrak{B}_i$ are unramified, and $\mathfrak{p}$ is ramified otherwise. The extension L/K is called *unramified* or an *unramified extension* if all prime ideals $\mathfrak{p}$ of K are unramified.

[a]which for number fields this is always the case

It is in fact rare that a prime ideal $\mathfrak{p}$ of K is ramified in L[24], however every [nontrivial] extension of $\mathbb{Q}$ is ramified (i.e. some primes ramify). Later, we shall see that there do exist [nontrivial] unramified extension L/K where $K \neq \mathbb{Q}$. To understand ramification in extension, we first show how we can factor most $\mathfrak{B}_i$. The term "ramification" comes from complex analysis, where if a holomorphic function f has a root, $f(a) = 0$, the order of the root is the ramification index[25].

[23]However, if K is a local field, as defined in [9], every extension *is* monogenic. This property, among many other similar niceties, makes developing number theory over local fields much more streamlined. The interested teacher should see [9] as a follow up to this chapter

[24]This correspond to the algebro-geometric fact that all but finitely many points of curve are *nonsingular*

[25]if $f(a) = 0$ has order n, there exists a g such that $f = g^n$ in some neighbourhood, which shows how much f "winds" around the root

Theorem 1.4.11: Counting Ramified Primes

Let L/K be a separable extension where $K = \text{Frac}(R)$ or a Dedekind domain R. Then there are only finitely many prime ideals of K which are ramified in L, in particular the ideals $\mathfrak{p} \subseteq K$ that are *not* relatively prime to the discriminant of $m_{K,\theta}(x)$ and $\mathscr{F}_\theta$.

Note that it is still possible that primes factor as given in theorem 1.4.7 since a prime may be relatively prime to $\mathscr{F}_\theta$ but not the discriminant of $m_{K,\theta}(x)$.

***Proof*:**
Let $\theta \in \mathcal{O}$ be a primitive element for L, $L = K(\theta)$, and let $m(x) = m_{K,\theta}(x) \in R[x]$ be the minimal polynomial. Take the discriminant of $m(x)$

$$\text{disc}(m_{\theta,K}(x)) = \text{disc}_{L/K}(1, \theta, ..., \theta^{n-1}) = \prod_{i<j}(\theta_i - \theta_j)^2 \in R$$

We shall show every prime (fractional) ideal $\mathfrak{p} \subseteq K$ which is relatively prime to d and to the conductor $\mathscr{F}_\theta$ of $R[\theta]$ is unramified. Note that there are finitely many primes $\mathfrak{p} \subseteq K$ that are not relatively prime to the conductor and the discriminant since both have finitely many factors, hence this proofs the result.

By theorem 1.4.7, $\mathfrak{p}\mathcal{O}$ factors in the same way $\overline{m}_{\theta,K}(x)$ factors (mod $\mathfrak{p}$). Then the ramification indices e_i equal 1 when $\overline{m}(x)$ has no multiple roots. But this is the case since $\overline{0} \neq \overline{d} = d \bmod \mathfrak{p}$ is nonzero by assumption of being relatively prime, and the discriminant is nonzero if and only if all the roots are distinct. Thus, the residue class field extension $(\mathcal{O}/\mathfrak{B}_i)/(R/\mathfrak{p})$ are generated by $\overline{\theta} \equiv \theta \bmod \mathfrak{B}_i$, and thus are separable, showing $\mathfrak{p}$ is unramified, as we sought to show.

We can in fact be more precise than the above result: the ramified primes ideals will be given by the *relative discriminant* of $\mathcal{O}_L/\mathcal{O}_K$, which will be given by the ideal $\mathfrak{o} \subseteq \mathcal{O}_K$ which is generated by the discriminants $d(\omega_1, \omega_2, ..., \omega_n)$ for all bases $\omega_1, \omega_2, ..., \omega_n$ of L/K contained in $\mathcal{O}_L$. A special case is that of the base-field being $K = \mathbb{Q}$.

Proposition 1.4.12: Detecting Ramification In Number Fields

Let p be a prime in $\mathbb{Z}$. Suppose p is ramified. Then:

$$p \big| \text{disc}(\mathcal{O}_K)$$

In proposition 1.4.16, we will show the converse, showing this is an iff.

***Proof*:**
Let $\mathfrak{B}$ lie over $\mathfrak{p}$ and $e_{\mathfrak{B}/\mathfrak{p}} > 1$. Thus

$$p\mathcal{O}_K = \mathfrak{B}I$$

where I is divisible by *all* primes of $\mathcal{O}_K$ lying over p. Let $\sigma_1, \sigma_2, ..., \sigma_n$ denote the embeddings of K in $\mathbb{C}$, and extend each σ_i to automorphisms of some extension of L/K which is normal over $\mathbb{Q}$. Let $\alpha_1, \alpha_2, ..., \alpha_n$ be some integral basis for R. What we shall do is replace one of the α_i be a suitable element that will show that $p \big| \text{disc}(\mathcal{O}_K)$.

Take $\alpha \in I - p\mathcal{O}_K$. Then certainly α is in every prime ideal lying over p but *not* in $p\mathcal{O}_K$. writing

$$\alpha = m_1\alpha_1 + \cdots m_n\alpha_n$$

for appropriate $m_i \in \mathbb{Z}$, then since $\alpha \notin p\mathcal{O}_K$, we have that p does not divide each m_i. Re-arrange the terms so that $p \nmid m_1$. Then if d is the discriminant of $\mathcal{O}_K$, through direct computation, we can see that

$$\operatorname{disc}(\alpha, \alpha_2, ..., \alpha_n) = m_1^2 d$$

Since $p \nmid m_1$, we shall show that $p \big| \operatorname{disc}(\alpha, \alpha_2, ..., \alpha_n)$

Given $L/K/\mathbb{Q}$, since α is in every prime of $\mathcal{O}_K$ lying over p, it certainly is in every prime in $\mathcal{O}_L$ lying over p. Fixing any $\mathfrak{B}$ of S lying over p, we see that for any automorphism σ_i, $\sigma(\alpha) \in \mathfrak{B}$. To see this ,$\sigma^{-1}(\alpha)$ is a prime of $\sigma^{-1}(S) = S$ lying over p, hence must contain α. In particular, this gives $\sigma_i(\alpha) \in \mathfrak{B}$ for all i. But then $\mathfrak{B}$ contains $\operatorname{disc}(\alpha, \alpha_2, ..., \alpha_n)$. Since the discriminant is in $\mathbb{Z}$, we get that:

$$\mathfrak{B} \cap \mathbb{Z} = p\mathbb{Z}$$

but then $p \big| \operatorname{disc}(\mathcal{O}_K)$, as we sought to show.

Corollary 1.4.13: Ramification Over Number Fields

Let $\alpha \in R$ so that $K = \mathbb{Q}(\alpha)$ and let f be a monic polynomial over $\mathbb{Z}$ such that $f(\alpha) = 0$. Then if p is a prime such that

$$p \not\big| \operatorname{Nm} f'(\alpha)$$

Then p is unramified in K

***Proof*:**
This follows from the above an the fact that if α is an algebraic integer and $f \in \mathbb{Z}[x]$ monic such that $f(\alpha) = 0$, then by proposition 1.1.29:

$$\operatorname{disc}(\alpha) \big| \operatorname{Nm}_{\mathbb{Q}(\alpha)/\mathbb{Q}}(f'(\alpha))$$

Using this result, we have an easier verification for ramification of primes of $\mathbb{Z}$ to number fields, since they must divide the discriminant, but the discriminant may only have finitely many factors. In fact, it is easy to generalize this result for extension of number fields

Corollary 1.4.14: Ramification In Number Fields

Let L/K be an extension of number fields Then only finitely many primes of $\mathcal{O}_K$ are ramified in $\mathcal{O}_L$.

***Proof*:**
Let $\mathfrak{p}$ be a prime of $\mathcal{O}_K$ which is ramified in $\mathcal{O}_L$. Then $\mathfrak{p} \cap \mathbb{Z} = p\mathbb{Z}$ is ramified in S. Hence there are only finitely many possible primes p that ramify, meaning there are only finitely many possible $\mathfrak{p}$.

The converse is also true, namely

$$p\Big|\text{disc}(\mathcal{O}_K) \Leftrightarrow \ p \text{ ramifies in } K$$

The full proof shall be given in greater generality in 1.4.3 where it is generalized for primes $\mathfrak{p} \subseteq \mathcal{O}_K$ being ramified in $\mathcal{O}_L$ for algebraic extension L/K. However, it is instructive for the reader to prove it in the following special case that is particularly important for the study of local fields (see [9]):

Lemma 1.4.15: Partial Converse For Ramification

Let $K = \mathbb{Q}(\alpha)$ where $\alpha \in \mathcal{O}_K$ and $p \nmid [\mathcal{O}_K : \mathbb{Z}[\alpha]]$. Then if $p\Big|\text{disc}(\mathcal{O}_K)$, p ramifies in K

***Proof*:**
This is a good exercise!

For the result when *only* considering primes of $\mathbb{Z}$, the result is easier:

Proposition 1.4.16: Characterizing Ramification In Number Fields

Let p be a prime in $\mathbb{Z}$. Suppose $p\Big|\text{disc}(\mathcal{O}_K)$. Then p is ramified.

***Proof*:**
(difficult) exercise; or see[8, p. 79]

1.4.1 Example: Eisenstein Polynomials

We say $f \in \mathbb{Z}[x]$ is *Eisenstein* with respect to p is p divides each coefficient but p^2 does not divide the constant coefficient. Then by Eisenstein's criterion ([5, Eisenstein's Criterion]), such a polynomial is irreducible in $\mathbb{Q}[x]$. We shall use these to quickly determine a few useful ring of integer.

Lemma 1.4.17: Eisenstein Polynomial Modular Roots

Let $K/\mathbb{Q}$ be a number field of degree n and assume $K = \mathbb{Q}(\alpha)$ for $\alpha \in \mathcal{O}_K$ with minimal polynomial $m_{\alpha,\mathbb{Q}}$ Eisenstein at p. Then then for some $f(x) \in \mathbb{Z}[x]$ of degree at most $n-1$ with coefficients $a_0, a_1, ..., a_{n-1}$, if:

$$f(\alpha) \equiv 0 \bmod p\mathcal{O}_K$$

Then:

$$a_i \equiv 0 \bmod p\mathbb{Z} \qquad 1 \leq i \leq n-1$$

***Proof*:**
we shall do an argument by induction. Certainly if $a_0 \equiv 0 \bmod p\mathcal{O}_K$ then $a_0 \equiv 0 \bmod p\mathbb{Z}$ (since $a_0 \in \mathbb{Z}$). Thus, assume for $i < j$, $j \in \{0, 1, ..., n-1\}$ that $a_i \equiv 0 \bmod p\mathbb{Z}$. Then for $i < j$:

$$a_j\alpha^j + a_{j+1}\alpha^{j+1} + \cdots + a_{n-1}\alpha^{n-1} \equiv \bmod p\mathcal{O}_K$$

Multiply the above by α^{n-1-j} to make all but the first term $a_j\alpha^{n-1}$ a multiple of α^n. Since α is a root of an Eisenstein polynomial at p, we have

$$\alpha^n \equiv 0 \bmod p\mathcal{O}_K$$

Hence:

$$a_j\alpha^{n-1} \equiv p\mathcal{O}_K$$

Then $a_j\alpha^{n-1} = p\gamma$, for $\gamma \in \mathcal{O}_K$. Taking the norm, we get:

$$a_j^n \mathrm{Nm}_{K/\mathbb{Q}}(\alpha)^{n-1} = p^n \mathrm{Nm}_{K/\mathbb{Q}}(\gamma) \in \mathbb{Z}$$

the left hand side is divisible by the constant term of the minimal polynomial (since that is the value of $\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)$). By Eisenstein, p divides $\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)$ exactly once, so since p^n must divide the left hand side gives that $p\big|a_j^n$, thus $p\big|a_j$, thus $a_i \equiv 0 \bmod p\mathbb{Z}$ for $i < j+1$, giving us the result.

Lemma 1.4.18: Eisenstein Polynomial divisibility by p

Let $K/\mathbb{Q}$ be a number field of degree n, $K = \mathbb{Q}(\alpha)$ where $\alpha \in \mathcal{O}_K$ with minimal polynomial Eisenstein at p. Then if

$$r_0 + r_1\alpha + \cdots + r_{n-1}\alpha^{n-1} \in \mathcal{O}_K \qquad r_i \in \mathbb{Q}$$

then each r_i has no p in its denominator

***Proof*:**
For the sake of contradiction, assume some r_i has a p in its denominator. Let d be the lcm of the denominators so the r_i's so $p\big|d$ so that we may write $r_i = a_i/d$ for appropriate $a_i \in \mathbb{Z}$ where $a_i \nmid p$. Then:

$$r_0 + r_1\alpha + \cdots + r_{n-1}\alpha^{n-1} \in \mathcal{O}_K \implies \frac{a_0 + a_1\alpha + \cdots + a_{n-1}\alpha^{n-1}}{d} \in \mathcal{O}_K$$

multiplying by d,w e get:

$$a_0 + a_1\alpha + \cdots + a_{n-1}\alpha^{n-1} \in d\mathcal{O}_K \subseteq p\mathcal{O}_K$$

Then by lemma 1.4.17 we have that $a_i \in p\mathbb{Z}$ for each i, a contradiction!

Theorem 1.4.19: Eisenstein Polynomial Divisibility Test

Let $K = \mathbb{Q}(\alpha)$ where $\alpha \in \mathcal{O}_K$ a root of an Eisenstain polnomial at p of degree n. Then:

$$p \nmid [\mathcal{O}_K : \mathbb{Z}[\alpha]]$$

***Proof*:**
For the sake of contradiction, suppose $p\big|[\mathcal{O}_K : \mathbb{Z}[\alpha]]$. Then $\mathcal{O}_K/\mathbb{Z}[\alpha]$ has an element of order p, that is there is some $\gamma \in \mathcal{O}_K$ such that $\gamma \notin \mathbb{Z}[\alpha]$ but $p\gamma \in \mathbb{Z}[\alpha]$. Now, given basis $\{1, \alpha, ..., \alpha^{n-1}\}$ for $K/\mathbb{Q}$, then

$$\gamma = r_0 + r_1\alpha + \cdots + r_{n-1}\alpha^{n-1}$$

where $r_i \in \mathbb{Q}$. Since $\gamma \notin \mathbb{Z}[\alpha] <$ not all $r_i \in \mathbb{Z} >$ Since $p\gamma \in \mathbb{Z}[\alpha]$, $pr_i \in \mathbb{Z}$. Thus, r_i has p in its denominator, but that contradicts lemma 1.4.18, as we sought to show.

Example 1.4.20: Using Eisenstein Test

1. Let $K = \mathbb{Q}(\sqrt[3]{2})$. We shall quickly find $\mathcal{O}_K$. Recall that:

   $$-2^2 3^3 = -108 = \mathrm{disc}(\mathbb{Z}[\sqrt[3]{2}]) = [\mathcal{O}_K : \mathbb{Z}[\sqrt[3]{2}]]^2 \mathrm{disc}(\mathcal{O}_K)$$

   Since $\sqrt[3]{2}$ is a root of $x^3 - 2$, which is eisenstien at 2, we see that $2 \nmid [\mathcal{O}_K : \mathbb{Z}[\sqrt[3]{2}]]$. Furthermore, $1 + \sqrt[3]{2}$ is the root of

   $$(x-1)^3 - 2 = x^3 - 3x^2 + 3x - 3$$

   which is eisenstein at 3 s o

   $$3 \nmid [\mathcal{O}_K : \mathbb{Z}[1 + \sqrt[3]{2}]] = [\mathcal{O}_K : \mathbb{Z}[\sqrt[3]{2}]$$

   since $\mathbb{Z}[\sqrt[3]{2}] = \mathbb{Z}[1 + \sqrt[3]{2}]$. THus, the index must be 1, giving us:

   $$\mathcal{O}_K = \mathbb{Z}[\sqrt[3]{2}]$$

2. Let $K = \mathbb{Q}(\sqrt[4]{2})$. Then:

   $$-2^{11} = \mathrm{disc}(\mathbb{Z}[\sqrt[4]{2}]) = [\mathcal{O}_K : \mathbb{Z}[\sqrt[4]{2}]]^2 \mathrm{disc}(\mathcal{O}_K)$$

   and $\sqrt[4]{2}$ is a root of $x^4 - 2$ which is eisenstein at 2, hence by the eisenstein divisibility test the idnex is 1 and

   $$\mathcal{O}_K = \mathbb{Z}[\sqrt[4]{2}]$$

   Let $K = \mathbb{Q}(\sqrt[5]{2})$. Then

   $$2^4 5^4 = \mathrm{disc}(\mathbb{Z}[\sqrt[5]{2}]) = [\mathcal{O}_K : \mathbb{Z}[\sqrt[5]{2}]]^2 \mathrm{disc}(\mathcal{O}_K)$$

   Certainly 2 doesn't divide the index. Next, the minimal polynomial of $\sqrt[5]{2} - 2$ is

   $$(x+2)^5 - 2 = x^5 + 10x^4 + 40x^3 + 80x^2 + 80x + 30$$

   which is Eisenstein at 5 and $\mathbb{Z}[\sqrt[5]{2} - 2] = \mathbb{Z}[\sqrt[5]{2}]$ Hence:

   $$\mathcal{O}_K = \mathbb{Z}[\sqrt[5]{2}]$$

3. show that $\mathcal{O}_{\mathbb{Q}(\sqrt[3]{3})} = \mathbb{Z}[\sqrt[3]{3}]$ and $\mathcal{O}_{\mathbb{Q}(\sqrt[3]{5})} = \mathbb{Z}[\sqrt[3]{5}]$.

4. Recall the three cubic extensison K_1, K_2, K_3 in example 1.1.42 with the same discriminant, but it was claimed that they where not isomorphic as fields. The three polynomials in question where:

$$f_1(x) = x^3 - 18x - 6 \qquad f_2(x) = x^3 - 36x - 78 \qquad f_3(x) = x^3 - 54x - 150$$

each having discriminant $22345 = 5 \cdot 41 \cdot 109$. It can be verified that these three polynomials have real roots (by either graphing or doing some guess-work with derivative test and evaluating), and from the above methods we can check that

$$\mathcal{O}_{K_i} = \mathbb{Z}[\alpha_i]$$

for appropriate root α_i. Since the ring of integers is Monogenic, by Dedekind-Kummers theorem the factorization of primes p mirrors the factorization of $f_i(x)$ mod p. Then it can be verified that

(a) f_1, f_2 are irreducible mod 5, but

$$f_3(x) \equiv x(x-2)(x-3) \bmod 5$$

(b) f_2, f_3 are irreducible mod 11, but

$$f_1(x) \equiv (x-3)(x-9)(x-10) \bmod 11$$

Thus, we see that these fields cannot be isomorphic, or else the prime factorization would have been equivalent (recall that ring homomorphisms map primes to primes)

It may be asked if $K = \mathbb{Q}(\sqrt[n]{2})$

$$\mathcal{O}_K = \mathbb{Z}[\sqrt[n]{2}]$$

This is true for $n \leq 1000$, but is *not always true*. For those interested, they may search for *Wieferich primes* (see Keith Conrad *The ring of integers in radical extensions*).

Eisenstein polynomials also allow us to identify when a polynomial totally ramifies:

Theorem 1.4.21: Eisenstein Polynomial And Total Ramification

Let $K = \mathbb{Q}(\alpha)$. Then α is a root of a $\mathbb{Z}$-polynomial that is Eisenstein at p iff p is totally ramified in K.

Note how there is no reference to the ring of integers

***Proof*:**
(*must* add this here! this link

Example 1.4.22: Dececting Ramification Using Eisenstein

Since $\alpha = \sqrt[3]{10}$ is a root of $x^3 - 10$, which is eisenstein at 2 and 5, then in $K = \mathbb{Q}(\alpha)$, $(2) = \mathfrak{p}^3$ and $(5) = \mathfrak{q}^3$. Note however htat

$$\mathcal{O}_K \neq \mathbb{Z}[\sqrt[3]{10}]$$

It is in fact equal to

$$\mathcal{O}_K = \mathbb{Z}[\beta] \qquad \beta = \frac{1}{3}\left(1 + \sqrt[3]{1-} + \sqrt[3]{100}\right)$$

which is a root of $x^3 - x^2 - 3x - 3$.

1.4.2 Primes in Galois Extensions

The question of prime decomposition in extension becomes more interesting when L/K is also a Galois extension since $\mathrm{Gal}_K(L)$ can act on the set of primes.

Lemma 1.4.23: Galois Group Acts On Primes

Let L/K be a finite galois extension with galois group $G = \mathrm{Gal}_K(L)$, $\mathfrak{p} \subseteq \mathcal{O}_K$ a prime ideal of $\mathcal{O}_K$, and $\mathfrak{P} = \{\mathfrak{B}_1, \mathfrak{B}_2, ..., \mathfrak{B}_n\}$ the set of primes lying above $\mathfrak{p}$. Then:

$$G \curvearrowright \mathfrak{P} \qquad \text{via} \qquad \sigma \cdot \mathfrak{B}_i = \sigma(\mathfrak{B}_i)$$

***Proof*:**
let $G = \mathrm{Gal}_K(L)$. Then for each $a \in \mathcal{O}_L$, $\tau(a) \in \mathcal{O}_L$ for all $\tau \in G$ since K is invariant under τ and all conjugates have an integral minimal polynomial (i.e. the same minimal polynomial with coefficients in $\mathbb{Z} \subseteq K$), so there is a natural action of G on $\mathcal{O}_L$; $G \curvearrowright \mathcal{O}_L$. Next, given that τ is a field isomorphism, $\tau : \mathcal{O}_L \to \mathcal{O}_L$ is a ring isomorphism, importantly it sends primes to primes. If $\mathfrak{B}$ is a prime ideal of $\mathcal{O}_L$ above $\mathfrak{p}$, then $\tau(\mathfrak{B})$ is also a prime ideal above $\mathfrak{p}$ for each prime $\mathfrak{B}$ since

$$(\tau(\mathfrak{B})) \cap \mathcal{O}_K = \tau(\mathfrak{B} \cap \mathcal{O}_K) = \tau(\mathfrak{p}) = \mathfrak{p}$$

since τ is K-invariant, hence $\mathcal{O}_K \subseteq K$ is invariant.

Definition 1.4.24: Prime Ideal Conjugate

Let $\mathcal{O}_L/\mathcal{O}_K$ be an extension of rings of integers where L/K is Galois, and $\mathfrak{B}$ lie above $\mathfrak{p}$. An ideal $\tau(\mathfrak{B})$ is called a *prime ideal conjugate* to $\mathfrak{B}$.

As we saw, most primes split similarly to a minimal polynomial. The following is further parallel between prime numbers and polynomials: just like conjugate elements, the Galois group acts transitively:

Proposition 1.4.25: Galois Group Transitive Action On Ideals

Let $G = \mathrm{Gal}_K(L)$. Then G acts transitively on the set of all ideal $\mathfrak{B}$ of $\mathcal{O}$ lying above $\mathfrak{p}$, that is the prime ideals are all conjugates of each other.

***Proof*:**
Say $\mathfrak{B}, \mathfrak{B}'$ are two primes ideals lying above $\mathfrak{p}$ where, for the sake of contradiction, $\sigma(\mathfrak{B}) \neq \mathfrak{B}'$

for any $\sigma \in G$. By the CRT, there exists an $x \in \mathcal{O}$ such that

$$x \equiv 0 \bmod \mathfrak{B}' \qquad x \equiv 1 \bmod \sigma(\mathfrak{B}) \qquad \forall \sigma \in G$$

Then

$$\mathrm{Nm}_{L/K}(x) = \prod_{\sigma \in G} \sigma(x) \overset{!}{\in} \mathfrak{B}' \cap \mathcal{O}_K = \mathfrak{p}$$

On the other hand, $x \notin \sigma(\mathfrak{B})$ for all $\sigma \in G$, hence $\sigma(x) \notin \mathfrak{B}$ for all $\sigma \in G$ since if $\sigma(x) = b \in \mathfrak{B}$, then $x = \sigma^{-1}(b) = \tau(b)$ for some $\tau \in G$. Thus,

$$\prod_{\sigma \in G} \sigma(x) \overset{!}{\notin} \mathfrak{B} \cap \mathcal{O}_K = \mathfrak{p}$$

Hence, $\prod_{\sigma \in G} \sigma(x)$ is both in and not in $\mathfrak{p}$, a contradiction.

Using the Galois group, we may encode in group-theoretic terms the number of different primes ideals a prime ideal $\mathfrak{p} \subseteq \mathcal{O}_K$ decompose to in $\mathcal{O}_L$. One of the properties of an element being acted on by a [transitive] group is it *stabalizer*:

Definition 1.4.26: Decomposition Group And Field

Let $G = \mathrm{Gal}_K(L)$. Let $\mathfrak{B} \subseteq \mathcal{O}$ be a prime ideal. Then the subgroup

$$D_{\mathfrak{B}/\mathfrak{p}} = D_{\mathfrak{B}} = \{\tau \in G \mid \tau(\mathfrak{B}) = \mathfrak{B}\} = \mathrm{stab}_G(\mathfrak{B})$$

is called the *decomposition group* of $\mathfrak{B}$ over K. The fixed field

$$L^{D_{\mathfrak{B}}} = \{x \in L \mid \tau(x) = x \ \forall \tau \in G_{\mathfrak{B}}\}$$

is called the *decomposition field* of $\mathfrak{B}$ over K, and is often labeled as $L_{\mathfrak{B}}$ or $L_{\mathfrak{B}}$ if unambiguous.

Notice that the less primes that are fixed, the larger the decomposition field is, hence there is a contravariant relation between the size of the group and the extension of the field, just like in Galois theory. Due to transitivity and the introduction of Galois theory, we immediately get many counting arguments. If $\mathfrak{B}$ is a prime ideal lying over $\mathfrak{p}$, and we let σ vary over a set of representatives from the cosets in $G/D_{\mathfrak{B}}$, then by transitivity $\sigma(\mathfrak{B})$ varies over the different primes ideals above $\mathfrak{p}$, each occurring exactly once, hence the number of prime ideals is equal to the index $[G : D_{\mathfrak{B}}]$. For reference, we shall write:

Definition 1.4.27: Prime Counter

Let L/K Then the number of prime numbers is equal to:

$$g_{L/K}(\mathfrak{p}) = [G : D_{\mathfrak{B}}]$$

for any prime $\mathfrak{B}$ over $\mathfrak{p}$. If clear from context, we shall write $g(\mathfrak{p})$.

Thus, if $D_{\mathfrak{B}} = 1$, then no $\sigma \in G$ fixes $\mathfrak{B}$ unless it's the identity and $[G : D_{\mathfrak{B}}] = |G| = [L : K]$, and so by the fundamental identity (proposition 1.4.3):

$$D_{\mathfrak{B}} = 1 \iff L_D = L \iff \mathfrak{p} \text{ totally splits}$$

On the other hand, if $D_{\mathfrak{B}} = G$, then every $\mathfrak{B}$ lying over $\mathfrak{p}$ is fixed by σ, and so there must only be one prime ideal above $\mathfrak{p}$ by transitivity giving:

$$D_{\mathfrak{B}} = G \iff L_D = K \iff \mathfrak{p} \text{ totally ramifies}$$

If $1 \subsetneq D_{\mathfrak{B}} \subsetneq G$, there is ambiguity on how $\mathfrak{p}$ splits. For example, if $[L : K] = 50 = 2 \cdot 5^2$ and $|G/D_{\mathfrak{B}}| = 5$, then either

$$\mathfrak{p}\mathcal{O}_L = \mathfrak{B}_1^2\mathfrak{B}_2^2\mathfrak{B}_3^2\mathfrak{B}_4^2\mathfrak{B}_5^2 \qquad \text{or} \qquad \mathfrak{p}\mathcal{O}_L = \mathfrak{B}_1^5\mathfrak{B}_2^5\mathfrak{B}_3^5\mathfrak{B}_4^5\mathfrak{B}_5^5$$

depending on whether the inertia degree is 5 or 2 respectively. The distinction between the two shall be given with the inertia group in proposition 1.4.37. Choosing a different prime does change the resulting index: if we were to choose a different prime $\mathfrak{B}$ lying over $\mathfrak{p}$, then $D_{\mathfrak{B}}$ is conjugate to $D_{\sigma(\mathfrak{B})}$:

$$D_{\sigma(\mathfrak{B})} = \sigma D_{\mathfrak{B}} \sigma^{-1}$$

In particular:

$$\begin{aligned} \tau \in D_{\sigma(\mathfrak{B})} &\iff \tau\sigma\mathfrak{B} = \sigma\mathfrak{B} \\ &\iff \sigma^{-1}\tau\sigma\mathfrak{B} = \mathfrak{B} \\ &\iff \sigma^{-1}\tau\sigma \in D_{\mathfrak{B}} \\ &\iff \tau \in \sigma D_{\mathfrak{B}} \sigma^{-1} \end{aligned}$$

This should make sense, for by the proceeding theorem, the inertia and ramification of each prime lying ove $\mathfrak{p}$ in a galois extension is equal. At the end of this section, we shall show how to partially recover these results when working over non-Galois extensions. A key result when working with Galois extensions is that each prime has the same behavior in it's decomposition:

Proposition 1.4.28: Galois Extension, Inertia Degree And Ramification Index Independent

Let $\mathfrak{p} \subseteq K$ be a prime ideal, L/K a Galois extension, and

$$\mathfrak{p}\mathcal{O}_L = \mathfrak{B}_1^{e_i} \cdots \mathfrak{B}_r^{e_r}$$

Then

$$f_1 = \cdots = f_r = f \qquad e_1 = \cdots = e_r = e$$

***Proof*:**
Let $\mathfrak{B} = \mathfrak{B}_1$ and $\mathfrak{B}_i = \sigma_i(\mathfrak{B})$ (which goes through all primes by transitivity). Then the isomorphism $\sigma_i : \mathcal{O}_L \to \mathcal{O}_L$ induces an isomorphism:

$$\mathcal{O}_L/\mathfrak{B} \xrightarrow{\sim} \mathcal{O}_L/\mathfrak{B}_i \qquad a \bmod \mathfrak{B} \mapsto \sigma_i(a) \bmod \sigma_i(\mathfrak{B})$$

But then:

$$f_i = [\mathcal{O}_L/\sigma_i(\mathfrak{B}) : \mathcal{O}_K/\mathfrak{p}] = [\mathcal{O}_L/\mathfrak{B} : \mathcal{O}_K/\mathfrak{p}]$$

Next, since $\sigma_i(\mathfrak{p}\mathcal{O}_L) = \mathfrak{p}\mathcal{O}_L$ (since $\mathfrak{p} \subseteq K$), we get

$$\mathfrak{B}^n \big| \mathfrak{p}\mathcal{O} \quad \iff \quad \sigma_i(\mathfrak{B}^n) \big| \sigma_i(\mathfrak{p}\mathcal{O}) \quad \iff \quad (\sigma_i(\mathfrak{B}))^n \big| \mathfrak{p}\mathcal{O}$$

showing that $e_1 = \cdots = e_r$.

Hence, the prime decomposition of $\mathfrak{p}$ can be written as :

$$\mathfrak{p} = \left(\prod_{\substack{\sigma \in S \\ S \subseteq G/D_{\mathfrak{B}}}} \sigma(\mathfrak{B}) \right)^e$$

where $S \subseteq G/D_{\mathfrak{B}}$ represents a set of representatives of $G/D_{\mathfrak{B}}$ and $|S| = g_{L/K}(\mathfrak{p})$. This information is already very powerful, the following is a use-case of such information:

Example 1.4.29: Biquadratic Extension That Is Not Monogenic

Let $K = \mathbb{Q}(\sqrt{7}, \sqrt{13})$. Without computing $\mathcal{O}_K$, it can be shown that $\mathcal{O}_K$ is *not* monogenic, that is there does not exists $\alpha \in \mathcal{O}_K$ such that $1, \alpha, \alpha^2, \alpha^3$ generates $\mathcal{O}_K$, in other words $\mathcal{O}_K$ is not isomorphic to a quotient of $\mathbb{Z}[x]$. The key is to look at the behavior of factorizations of small primes (for they can only have so many rood $\bmod p$)

What we shall do is consider the factorization of (3) in $\mathcal{O}_K$, which will be our setup for a contradiction. First, notice that K has 3 intermediary fields[a]:

$$\begin{tikzcd}
& K & \\
\mathbb{Q}(\sqrt{7}) \arrow[ur, "2"] & \mathbb{Q}(\sqrt{91}) \arrow[u, "2"] & \mathbb{Q}(\sqrt{13}) \arrow[ul, "2"'] \\
& \mathbb{Q} \arrow[ul, "2"] \arrow[u, "2"] \arrow[ur, "2"'] &
\end{tikzcd}$$

Since we know the ring of integers of quadratics, by direct computation, we see that (3) splits into two primes in each quadratic extension. For example, in $\mathbb{Q}(\sqrt{7})$, we have:

$$\mathcal{O}_{\mathbb{Q}(\sqrt{7})}/3\mathcal{O}_{\mathbb{Q}(\sqrt{7})} = \frac{\mathbb{F}_3[x]}{(x-1)(x+1)}$$

Hence:

$$(3) = (3, 1+\sqrt{7})(3, 1-\sqrt{7})$$

Doing a similar computation for the other two computation, we get:

$$(3) = (3, 1+\sqrt{13})(3, 1-\sqrt{13})$$
$$(3) = (3, 1+\sqrt{91})(3, 1-\sqrt{91})$$

Thus, over K, there must be at least 2 prime ideals, and since

$$g_{K/\mathbb{Q}}(3) = [G : D_{\mathfrak{B}}] \qquad [G : D_{\mathfrak{B}}] \;\big|\; |\mathrm{Gal}_{\mathbb{Q}}(K)| \qquad |\mathrm{Gal}_{\mathbb{Q}}(K)| = |(\mathbb{Z}/2\mathbb{Z})^2| = 4$$

for any prime ideal $\mathfrak{B}$ over (3), we get that $g_{K/\mathbb{Q}}(3) \big| 4$. I claim that $g_{K/\mathbb{Q}}(3) = 4$. For, if $g_{K/\mathbb{Q}}(3) = 2$, then for any of the two primes lying over (3) in $\mathcal{O}_K$, say $\mathfrak{B}$:

$$\{e\} \neq D_{\mathfrak{B}} = \{\sigma \in \mathrm{Gal}_{\mathbb{Q}}(K) \mid \sigma(\mathfrak{B}) = \mathfrak{B}\} \subsetneq \mathrm{Gal}_{\mathbb{Q}}(K)$$

Since $D_{\mathfrak{B}}$ non-trivial proper subgroup acting transitively on the two primes of K lying over (3), we have $|D_{\mathfrak{B}}| = 2$, implying there is one non-trivial automorphism $\rho \in \mathrm{Gal}_{\mathbb{Q}}(K)$ fixing the prime ideals (since there are two prime ideals, if $\rho(\mathfrak{B}) = \mathfrak{B}$, then $\rho(\mathfrak{B}') = \mathfrak{B}'$, or else ρ wouldn't be bijective), and two non-trivial automorphisms σ, τ not fixing the primes. Consider now $\sigma \in \mathrm{Gal}_{\mathbb{Q}}(K)$ and take the fixed field $K^{\langle\sigma\rangle}$, which by some Galois theory we see must be one of the 3 quadratic extensions found above. Then by the 3rd fundamental theorem of Galois theory:

$$G := \mathrm{Gal}_{\mathbb{Q}}(K^{\langle\sigma\rangle}) \cong \frac{\mathrm{Gal}_{\mathbb{Q}}(K)}{\mathrm{Gal}_{K^{\langle\sigma\rangle}}(K)}$$

Importantly, since $G \cong \mathbb{Z}/2\mathbb{Z}$ and $[\sigma] = [\mathrm{id}]$, the map $\varphi : \mathrm{Gal}_{\mathbb{Q}}(K) \to \mathrm{Gal}_{\mathbb{Q}}(K^{\langle\sigma\rangle})$ maps $\rho, \tau \mapsto \rho|_{K^{\langle\sigma\rangle}}$. Since ρ is an automorphism and fixes both prime ideals in K over (3) (say $\mathfrak{B}, \mathfrak{B}'$), and all prime ideal of $K^{\langle\sigma\rangle}$ over (3) lie below $\mathfrak{B}, \mathfrak{B}'$:

$$\rho|_{K^{\langle\sigma\rangle}}(\mathfrak{B} \cap K^{\langle\sigma\rangle}) = \rho|_{K^{\langle\sigma\rangle}}(\mathfrak{B}) \cap K^{\langle\sigma\rangle} = \mathfrak{B} \cap K^{\langle\sigma\rangle}$$
$$\rho|_{K^{\langle\sigma\rangle}}(\mathfrak{B}' \cap K^{\langle\sigma\rangle}) = \rho|_{K^{\langle\sigma\rangle}}(\mathfrak{B}') \cap K^{\langle\sigma\rangle} = \mathfrak{B}' \cap K^{\langle\sigma\rangle}$$

Hence, G fixes all primes; but we saw that all intermediate fields of K have two primes, thus this contradicts the transitivity of action of the Galois group on the prime ideals. Thus, it must be that $g_{K/\mathbb{Q}}(3) = 4$.

Now, for the sake of contradiction let's say

$$\mathcal{O}_K \cong \frac{\mathbb{Z}[x]}{m_\alpha(x)}$$

for some $\alpha \in \mathcal{O}_K$ and minimal polynomial $m_\alpha(x)$ of degree 4. Then

$$\mathcal{O}_K/3\mathcal{O}_K \cong \frac{\mathbb{F}_3[x]}{(m_\alpha(x) \bmod 3)} \tag{1.4}$$

By our previous calculations, we see that there must be 4 prime ideals in the right hand side. Decomposing $m_\alpha(x) \bmod 3$ into irreducible, by the CRT we get that the right hand side is isomorphic to the direct sum of $\mathbb{F}_3[x]/(m_i(x) \bmod 3)$ for each irreducible $m_i(x) \bmod 3$. Since each $m_i(x) \bmod 3$ is irreducible, each direct summand is a field, and hence by some ring theory we see that the number of primes on the right hand side of equation 1.4 is equal to the number of direct summands, implying $m_\alpha(x)$ splits into linear factors giving us:

$$\frac{\mathbb{F}_3[x]}{(m_\alpha(x) \bmod 3)} \cong \bigoplus_i^4 \mathbb{F}_3$$

However, this is impossible: any degree 4 polynomial over $\mathbb{F}_3$ that breaks down into linear factors has the form:

$$(x-1)^\alpha (x+1)^\beta x^\gamma$$

where one of α, β, γ is 2 and the other two are 1. But then $\mathbb{F}_3[x]/((x-1)^\alpha(x+1)^\beta x^\gamma)$ is not isomorphic to $\bigoplus_i^4 \mathbb{F}_3$ (the quotient contains nilpotent elements), showing that our assumption that $\mathcal{O}_K$ is monogenic is not possible.

[a] this can be seen by taking the fixed fields $\sqrt{7}, \sqrt{13}$, and $\sqrt{91}$, corresponding to the subgroups of $\mathrm{Gal}_{\mathbb{Q}}(K) = \mathbb{Z}_2 \oplus \mathbb{Z}_2$

Our next goal is to break down extension L/K into intermediary extensions which will "capture" the inertial degree and ramification index, similarly to how any field extension F/k may be broken down into intermediary extensions into separable, inseparable, and transcendental extensions. In our case, we shall get the tower:

$$K \xrightarrow[1]{1} L_D \xrightarrow[f]{1} L_I \xrightarrow[1]{e} L$$

where the ramification index of each field extension is on top, and the inertia degree is on the bottom.

Proposition 1.4.30: Prime Ideals in Decomposition Field

Let L/K be a finite field extension, $\mathfrak{p} \subseteq \mathcal{O}_K$ a prime ideal and $\mathfrak{B} \subseteq \mathcal{O}_L$ lying over $\mathfrak{p}$. Let $\mathfrak{B}_Z = \mathfrak{B} \cap L_D$ be the prime ideal of L_D below $\mathfrak{B}$. If $\mathfrak{B}$ has inertia degree f and ramification index e, then:

1. $\mathfrak{B}_Z$ is nonsplit in L, that is $\mathfrak{B}$ is the only prime ideal of L above $\mathfrak{B}_D$
2. $\mathfrak{B}$ over L_D has a ramification index e and inertia index f
3. The ramification index and inertia index of $\mathfrak{B}_D$ over K are both 1

Visually, we have:

$$\begin{array}{c} \mathfrak{B}^e J \\ e \big\uparrow f \\ \mathfrak{B}_D I \\ 1 \big\uparrow 1 \\ \mathfrak{p} \end{array}$$

where I and J are the remainder of the factorization. Note that the factorization of each prime factor of J lying above $\mathfrak{p}$ has the same e and f since L/K is Galois, but L_D/K is not necessarily Galois, and so the factorization of I may be quite different (for ex. another prime $\mathfrak{B}'_D$ in L_D lying ove $\mathfrak{B}$ may have ramification and inertia degree)

***Proof*:**

1. Notice that $\mathrm{Gal}_{L_{\mathfrak{B}}}(L) = D_{\mathfrak{B}}$: the automorphisms of L that fix $L_{\mathfrak{B}}$ are by definition the elements of $D_{\mathfrak{B}}$. Hence, the prime ideals above $\mathfrak{B}_D$ are the $\sigma(\mathfrak{B})$ for $\sigma \in \mathrm{Gal}_{L_{\mathfrak{B}}}(L)$, but $\sigma(\mathfrak{B}) = \mathfrak{B}$ by definition. Hence, $\mathfrak{B}_D$ is nonsplit in L.

2-3. The fundamental identity gives us:

$$n = efr$$

where $n = |\mathrm{Gal}_K(L)| = |G|$ and $r = [G : D_{\mathfrak{B}}]$. hence,

$$|D_{\mathfrak{B}}| = [L : L_{\mathfrak{B}}] = ef \tag{1.5}$$

Now, let e', e'' be the ramification index of $\mathfrak{B}$ over $L_{\mathfrak{B}}$ and $\mathfrak{B}_D$ over K respectively. Then $\mathfrak{p}$ is:

$$\mathfrak{p}\mathcal{O}_{L_{\mathfrak{B}}} = \mathfrak{B}_D^{e''} I$$

for some I and
$$\mathfrak{B}_D\mathcal{O}_L = \mathfrak{B}^{e'}$$
in L. Then
$$\mathfrak{p}\mathcal{O}_L = \mathfrak{B}^{e''e'}J$$
for some J. By the tower property, $e = e'e''$ and $f = f'f''$. Then the fundamental identity for the decomposition of $\mathfrak{B}_D$ in L gives us that the degree of L/L_D is:
$$[L : L_{\mathfrak{B}}] = e'f'$$
in particular, by equation (1.5), we get,
$$e'f' = ef = e'e''f'f''$$
Hence $e' = e$, $f' = f$, and $e'' = f'' = 1$, as we sought to show

This gives us:
$$K \xrightarrow[1]{1} L_{\mathfrak{B}} \xrightarrow[f]{e} L$$
Next, let us see how we can separate out an extension splitting the ramification and the inertia degree into different groups/fields. Since $\sigma(\mathcal{O}_L) = \mathcal{O}_L$ and $\sigma(\mathfrak{B}) = \mathfrak{B}$ for every $\sigma \in D_{\mathfrak{B}}$, there is an induced automorphism on the residue field $\mathcal{O}_L/\mathfrak{B}$:
$$\begin{gathered}\overline{\sigma} : \mathcal{O}_L/\mathfrak{B} \to \mathcal{O}_L/\mathfrak{B}\\ (a \bmod \mathfrak{B}) \mapsto (\sigma(a) \bmod \mathfrak{B})\end{gathered}$$
Letting $\kappa(\mathfrak{B}) = \mathcal{O}_L/\mathfrak{B}$ and $\kappa(\mathfrak{p}) = \mathcal{O}_K/\mathfrak{p}$, we have that
$$f = [\kappa(\mathfrak{B}) : \kappa(\mathfrak{p})]$$
Recall that in Dedekind-Kummer's theorem, when $\mathfrak{p}$ is relatively prime to the conductor, the inertia degree of $\mathfrak{B}$ is given by the degree of $\overline{m_i}(x)$ which within the proof is derived by $\left[R/(\overline{m_i(x)}) : \overline{\mathcal{O}}_K\right]$. This shows that the above map has information about the inertia degree. The following links this quantity to $D_{\mathfrak{B}}$. To understand the result, the reader may see that $\mathrm{Gal}_K(L)$ would naturally surjects onto $\mathrm{Gal}_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{B}))$ via restriction if the restriction was well-defined. The domain of this map can be restricted to the decomposition group to make it well-defined:

Proposition 1.4.31: Decomposition and Inertia Group Relation

The extension $\kappa(\mathfrak{B})/\kappa(\mathfrak{p})$ is normal and admits a surjective homomorphism:
$$D_{\mathfrak{B}} \twoheadrightarrow \mathrm{Gal}_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{B}))$$
where $\mathrm{Gal}_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{B})) = \mathrm{Gal}_{\mathcal{O}_K/\mathfrak{p}}(\mathcal{O}_L/\mathfrak{B}) \cong \mathrm{Gal}_{\mathbb{F}_p}(\mathbb{F}_q)$.

***Proof*:**
As we are working over finite fields, we know that $\kappa(\mathfrak{B}) = \kappa(\mathfrak{p})(\overline{\theta})$ for some θ with minimal polynomial $\overline{g}(x)$ over $\kappa(\mathfrak{p})$ ($\overline{g}(x) \in \kappa(\mathfrak{p})[x]$)). Let $\theta \in \mathcal{O}_L$ be a representative of $\overline{\theta}$, and let $f(x)$ be its minimal polynomial over $\mathcal{O}_K$. Then by definition $\overline{g}(\overline{\theta}) \equiv 0 \bmod \mathfrak{B}$. Since $f(\theta) = 0$,

$\overline{f}(\overline{\theta}) \equiv 0 \bmod \mathfrak{B}$. Hence, taking $\overline{f}(x) \in \kappa(\mathfrak{p})[x]$, we have by minimality that:

$$\overline{g}(x) \big| \overline{f}(x)$$

Next, we know that finite field extensions are normal as we saw in [5, Finite Fields], however here is another way of seeing this worth pointing out: since L/K is normal, $f(x)$ splits over $\mathcal{O}_L$ into linear factors. Thus, $\overline{f}(x)$ splits into linear factors over $\kappa(\mathfrak{B})$, and similarly for $\overline{g}(x)$. But then, $\kappa(\mathfrak{B})/\kappa(\mathfrak{p})$ is a normal extension.

Now, consider:

$$\overline{\sigma} \in \mathrm{Gal}_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{B})) = \mathrm{Gal}_{\kappa(\mathfrak{p})}\left(\kappa(\mathfrak{p})(\overline{\theta})\right)$$

By [5, Order of Fixed Automorphism Group], these homomorphisms are determined by where they send the root θ. Then $\overline{\sigma}(\overline{\theta})$ is a root of $\overline{g}(x)$ and hence of $\overline{f}(x)$, that is there exists a root θ' of $f(x)$ such that $\theta' \equiv \overline{\sigma}(\overline{\theta}) \bmod \mathfrak{B}$. θ' is a conjugate of θ, i.e. $\theta' = \sigma(\theta)$ for some $\sigma \in D_{\mathfrak{B}}$ (as the automorphism in $D_{\mathfrak{B}}$ fix $\mathfrak{B}$ and make the map well-defined). Since $\sigma(\theta) \equiv \overline{\sigma}(\overline{\theta}) \bmod \mathfrak{B}$, the automorphism σ is mapped by the homomorphism in question to $\overline{\sigma}$, proving surjectivity, as we sought to show.

Definition 1.4.32: Inertia Group And Field

The kernel $I_{\mathfrak{B}} \subseteq D_{\mathfrak{B}}$ of the homomorphism

$$D_{\mathfrak{B}} \to \mathrm{Gal}_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{B}))$$

is called the *inertia group* of $\mathfrak{B}$ over K which is a normal subgroup of $D_{\mathfrak{B}}$ (as it is the kernel of a homomorphism). Equivalently:

$$I_{\mathfrak{B}} = \{\sigma \in \mathrm{Gal}_K(L) \mid \sigma(\alpha) \equiv \alpha \bmod \mathfrak{B}\}$$

The fixed field:

$$L_I = L^{I_{\mathfrak{B}}} = \{x \in L \mid \sigma(x) = x,\ \forall \sigma \in I_{\mathfrak{B}}\}$$

is called the *inertia field* of $\mathfrak{B}$ over K.

We thus get the following tower:

$$K \subseteq L_D \subseteq L_I \subseteq L$$

and short exact sequence:

$$1 \to I_{\mathfrak{B}} \to D_{\mathfrak{B}} \to \mathrm{Gal}_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{B})) \to 1$$

Corollary 1.4.33: Isomorphism Of Two Galois Groups

Let $L/K/\mathbb{Q}$ be a finite galois extension, and let $\mathfrak{B}$ be a prime lying over $\mathfrak{p} \subseteq \mathcal{O}_K$. Then:

$$\mathrm{Gal}_{L_D}(L_I) \cong \mathrm{Gal}_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{B}))$$

***Proof*:**
exercise.

Corollary 1.4.34: Quotient Group of Decomposition Group

Let L/K be a finite extension. Then the group $D_{\mathfrak{B}}/I_{\mathfrak{B}}$ is cyclic of order f (hence $[D_{\mathfrak{B}} : I_{\mathfrak{B}}] = f$), and the generator $[\varphi] \in D_{\mathfrak{B}}/I_{\mathfrak{B}}$ that generates the group (and hence $\mathrm{Gal}_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{B})) \cong \mathbb{Z}/m\mathbb{Z}$) is called the *Frobenius Element*. The Frobenius element $\varphi \in D_{\mathfrak{B}}$ can be represented in the simple form

$$\varphi(x) = x^{\mathfrak{N}}(\mathfrak{p})$$

***Proof*:**
This is an immediate consequence of the surjectivity of $D_{\mathfrak{B}}$ onto the Galois group of finite field extensions, which is always cyclic.

The order of $\kappa(\mathfrak{B})/\kappa(\mathfrak{p})$ is f, giving the order of the quotient. As the galois group of a finite field is cyclic, it is generated by a Frobenius element. The Frobenius element can be explicitly described, namely

$$\varphi(x) = x^{\mathfrak{N}(\mathfrak{p})} = x^f$$

giving us the rest of the results.

Notice that this means that Frobenius elements and Cyclotomic extensions naturally appear when studying galois extensions; this motivates the study of Cyclotomic extensions in section 1.5.

Corollary 1.4.35: Equivalent Characterization Of Inertia Group

L/K be a finite Galois extension, $I_{\mathfrak{B}} \leqslant D_{\mathfrak{B}}$. Then:

$$I_{\mathfrak{B}} = \{\sigma \in \mathrm{Gal}_K(L) \mid \sigma(\mathfrak{B}) = \mathfrak{B}, \text{ and } \forall \alpha \in \mathcal{O}_L, \sigma(\alpha) - \alpha \in \mathfrak{B}\}$$

***Proof*:**
Let $\sigma \in I_{\mathfrak{B}}$ so that $\overline{\sigma\overline{\alpha}} = \overline{\alpha}$. Then $\sigma(\alpha) = \alpha + b$ for $b \in \mathfrak{B}$. But then:

$$\sigma(\alpha) - \alpha = b \in \mathfrak{B}$$

as we sought to show.

Corollary 1.4.36: Finding Element Of Galois Group

Let K be a number field, $f(x) \in \mathcal{O}_K[x]$ an irreducible polynomial over $\mathcal{O}_K$, L/K the splitting field extension of $f(x)$. Let $\mathfrak{p} \subseteq \mathcal{O}_K$ such that

$$f(x) \bmod \mathfrak{p} \equiv g_1(x) \cdots g_m(x)$$

for distinct irreducibles with degree $\deg(g_i(x)) = d_i$. Then in $\mathrm{Gal}_K(L) \lesssim S_n$, there exists an element α with cycle type

$$(d_1, d_2, ..., d_m)$$

***Proof*:**
Since L is the splitting field, let $\alpha_1, \alpha_2, ..., \alpha_m$ be the roots of f so that

$$f(\alpha_i) = 0$$

Let $\mathfrak{B}$ in L lie above $\mathfrak{p}$. Then:

$$f(\alpha_i) \bmod \mathfrak{p} \equiv 0$$

Since each each g_i are distinct, each α_i mod $\mathfrak{p}$ are disticnt. Hence, $D_{\mathfrak{B}}$ acts on the roots of f mod $\mathfrak{p}$. Now, taking $g \in D_{\mathfrak{B}}$ that maps to the generator of $\mathrm{Gal}_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{B}))$, we shall get the desired cycle type, completing the proof.

Proposition 1.4.37: Inertia Group Properties

The extension L_I/L_D is normal, and

$$\mathrm{Gal}_{L_D}(L_I) \cong \mathrm{Gal}_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{B}))$$

and $\mathrm{Gal}_{L_I}(L) = I_{\mathfrak{B}}$. If the residue field extension $\kappa(\mathfrak{B})/\kappa(\mathfrak{p})$ is separable, then:

$$|I_{\mathfrak{B}}| = [L : L_I] = e \qquad [D_{\mathfrak{B}} : I_{\mathfrak{B}}] = [L_I : L_D] = f$$

and we can find the prime ideals $\mathfrak{B}_I$ of L_I bellow $\mathfrak{B}$:

1. The ramification index of $\mathfrak{B}$ over $\mathfrak{B}_I$ is e and the inertia degree is 1
2. The ramification index of $\mathfrak{B}_I$ over $\mathfrak{B}_D$ is 1, and the inertia degree if f

Hence:

$$\mathfrak{B}_Z \mathcal{O}_{L_I} = \mathfrak{B}_T I \qquad \mathfrak{B}_T \mathcal{O}_K = \mathfrak{B}^e J$$

for some appropriate ideals I, J.

***Proof*:**
The fact that:

$$|I_{\mathfrak{B}}| = [L : L_I] = e \qquad [D_{\mathfrak{B}} : I_{\mathfrak{B}}] = [L_I : L_D] = f$$

follow from $|D_{\mathfrak{B}}| = [L : L_D] = ef$ and Galois theory (namely $[L : L_I] = |\mathrm{Gal}_{L_I}(L)| = |I_{\mathfrak{B}}|$), hence we need to show $(1)-(2)$. Using the fundamental identity, it all follows from $\kappa(\mathfrak{B}_I) = \kappa(\mathfrak{B})$. Since the inertia group $I_{\mathfrak{B}}$ of $\mathfrak{B}$ over K is also the inertia group of $\mathfrak{B}$ over L_I, it follows from proposition 1.4.31 applied to L/L_I that $\mathrm{Gal}_{\kappa(\mathfrak{B}_I)}(\kappa(\mathfrak{B})) = 1$, thus

$$\kappa(\mathfrak{B}_I) = \kappa(\mathfrak{B})$$

but then the rest follows.

To sum up the results in a diagram

$$K \xrightarrow[1]{1} L_D \xrightarrow[f]{1} L_I \xrightarrow[1]{e} L$$

with a corresponding [contravariant] diagram of groups:

$$G \leftarrow D_{\mathfrak{B}} \leftarrow I_{\mathfrak{B}} \leftarrow \{e\}$$

where the ramification index of each field extension is on top, and the inertia degree is on the bottom. If the residue field extension $\kappa(\mathfrak{B})/\kappa(\mathfrak{p})$ is separable, then:

$$I_{\mathfrak{B}} = 1 \qquad \iff \qquad L_I = L \qquad \iff \qquad \mathfrak{p} \text{ is unramified in } L$$

and in this case the Galois group $\text{Gal}_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{B})) \cong D_{\mathfrak{B}}$, so the galois group of residue field extension can be viewed as a subgroup of $\text{Gal}_K(L)$. This means that the inertia group can be thought of as the "ramification group". In fact, the generalizations of the inertia group (giving more nuanced information on the ramificaion) are called the *higher order ramification group* (which we shall cover soon). The reason the inertia group is not called the 1st ramification group is because the corresponding fixed field only effects the inertia of a prime; ultimately this is due to the contravariant relation the galois correspondence gives, hence we must make a choice on naming convention depending if fields or groups are prioritised. In this case, fields were prioritised.

From the previous proposition, we may think that L_D and L_I are the *maximal* fields with these properties with respect to $\mathfrak{B}$, and this is indeed the case, making them "natural" with respect to these properties:

Proposition 1.4.38: Intermediary Fields Are Maximal

Let L/K be a finite extension of number fields, $\mathfrak{B}$ live over $\mathfrak{p}$. Then:

1. $L_{D_{\mathfrak{B}}}$ is the smallest intermediary field such that $\mathfrak{B}$ is the only prime of $\mathcal{O}_L$ lying over $\mathfrak{B}' = \mathfrak{B} \cap \mathcal{O}_{L_{D_{\mathfrak{B}}}}$
2. $L_{D_{\mathfrak{B}}}$ is the largest intermediate such that $e(\mathfrak{B}_D/\mathfrak{p}) = f(\mathfrak{B}_I/\mathfrak{p}) = 1$
3. $L_{I_{\mathfrak{B}}}$ is the smallest intermediary field such that $\mathfrak{B}$ is totally ramified over $\mathfrak{B}_I$ (that is $e(\mathfrak{B}/\mathfrak{B}_I) = [L : L_I]$)
4. $L_{I_{\mathfrak{B}}}$ is the largest intermediary field such that $e(\mathfrak{B}_I/\mathfrak{p}) = 1$

***Proof*:**
First, L_D and L_I have these properties as we've in proposition 1.4.30 and proposition 1.4.37, hence we prove they are maximal/minimal.

Let $K' = L^H$ for some $H \leqslant \text{Gal}_K(L)$ where only $\mathfrak{B} \subseteq L$ is lying over $\mathfrak{p}' \subseteq K'$. Since H acts transitively on the primes of L over K' but there is only one prime $H \subseteq D_{\mathfrak{B}}$, so $L_D \subseteq K'$, giving (1).

Next, assume $e_{\mathfrak{p}'/\mathfrak{p}} = f_{\mathfrak{p}'/\mathfrak{p}} = 1$. Consider the following diagram:

$$\begin{tikzcd}
L & \\
& L_I K' \arrow[ul, "e'"'] \\
L_I \arrow[uu, "e"] & \\
& L_D K' \arrow[uu, "f'"] \\
L_D \arrow[uu, "f"] \arrow[ur] & \\
& K' \arrow[uu, "g'"] \\
K \arrow[uu, "g"] \arrow[ur] &
\end{tikzcd}$$

Then by multiplicativity of towers, $e = e'$ and $f = f'$. Then by the above diagram, L_D and $L_D K'$ voth ahve the same index in L. Since one contains the other, this means they must be equal, hence

$$K' \subseteq L_D$$

giving (2). A similar arguments works for (3) and (4): if $e_{\mathfrak{p}'/\mathfrak{p}} = 1$, then $e = e'$, so $L_I = L_I K'$, so $K' \subseteq L_I$, and if $\mathfrak{B}$ is totally ramified over $\mathfrak{p}'$, then $[L : K'] = e'$, then by the above diagramwe get $K' = L_I K'$, so $L_I \subseteq K'$, as we sought to show.

Ramification

An importance consequence of this is that we may consider primes that are completely ramified/totally unramified in their galois closure. we first require the following result:

Theorem 1.4.39: Composition Field And Primes

Let L/K and M/K be finite extensions of number fields, and let $\mathfrak{p}$ be a prime of K. Then if $\mathfrak{p}$ is unramified in both L and M, then $\mathfrak{p}$ is unramified in LM. If $\mathfrak{p}$ splits completely in L and M, then $\mathfrak{p}$ splits completely in LM

***Proof*:**
Let $\mathfrak{p}$ be unramified in L and M, and take $\mathfrak{B}$ to be any prime in LM lying over $\mathfrak{p}$. We want to show $e(\mathfrak{B}/\mathfrak{p}) = 1$. Let F be the normal closure of K containing LM, and let $\mathfrak{A}$ be any prime in F lying over $\mathfrak{B}$, namely it lies over $\mathfrak{p}$. Let $I_{\mathfrak{A}/\mathfrak{p}}$ be the inertial group corresponding to the field F_I. Then we see that F_I contains both L and M since the primes $\mathfrak{A} \cap L$ and $\mathfrak{A} \cap M$ are necessarily unramified over $\mathfrak{p}$. Hence, F_I contains LM, implying that $\mathfrak{A} \cap LM$ is unramified over

$\mathfrak{p}$.

The argument for splitting follows exactly the same pattern, replacing $I_{\mathfrak{B}}$ with $D_{\mathfrak{B}}$.

Corollary 1.4.40: Primes In Normal Closure

Let L/K be a finite extension of number fields and $\mathfrak{p}$ a prime in K. Then:

1. If $\mathfrak{p}$ is unramified in L, then it is unramified in the normal closure of M of L over K
2. If $\mathfrak{p}$ splits completely in L, then it splits completely in the normal closure of M of L over K

***Proof*:**
If $\mathfrak{p}$ is unramified in L, then it is unramified in every $\sigma_i(L)$ where i runs over all the automorphism. But since $M = L\sigma_1(L)\cdots\sigma_n(L)$, we use induction on theorem 1.4.39 to get our desired result (similarly for splitting completely).

Example 1.4.41: More Decomposition Of Primes
Let $K = \mathbb{Q}(i, \sqrt{2}, \sqrt{5})$, which is abelian of degree 8. Take the prime $5 \in \mathbb{Z}$. Then it splits into tow primes in $\mathbb{Q}(i)$, stays inert in $\mathbb{Q}(\sqrt{2})$, and becomes a square in $\mathbb{Q}(\sqrt{5})$. Hence in L 5 must have ramification index at least 2, and split into at least 2 primes. Since the inertia field must be a field of degree 4 over $\mathbb{Q}$ in which 5 is unramified, the only choice is $\mathbb{Q}(i, \sqrt{2})$. Now, $(2+i)$ and $(2-i)$ remain prime in $\mathbb{Q}(\sqrt{i}, \sqrt{2})$, and become squares in L, hence 5 must have exactly 2 primes and ramification index 2.

Since normal subgroups over characteristic 0 fields are galois, it is meaningful to ask the behavior of normal subgroups of the galois subgroups:

Proposition 1.4.42: Decomposition Subgroup Is Normal

Suppose $D_{\mathfrak{B}} \trianglelefteq \mathrm{Gal}_K(L)$. Then $\mathfrak{p}\mathcal{O}_L$ splits into r distinct primes in L_D. If $I_{\mathfrak{B}} \trianglelefteq G$, then each prime remains as primes in L_I (i.e. they are inert). Furthermore, each become an eth power in L.

In general, we can only hope for:

$$\mathfrak{B}_D\mathcal{O}_{L_I} = \mathfrak{B}_I I \qquad \mathfrak{B}_I\mathcal{O}_K = \mathfrak{B}^e J$$

for some ideals, while in the normal case of D and I, we get:

$$\begin{aligned}\mathfrak{p}\mathcal{O}_{L_D} &= \mathfrak{B}_{L_1}\cdots\mathfrak{B}_{L_e}\\ \mathfrak{p}\mathcal{O}_{L_I} &= \mathfrak{B}_{L_1}\cdots\mathfrak{B}_{L_e}\\ \mathfrak{p}\mathcal{O}_{L} &= \mathfrak{B}_1^e\cdots\mathfrak{B}_g^e\end{aligned}$$

which also shows better why the groups $D_{\mathfrak{B}}$ and $I_{\mathfrak{B}}$ are called the “decomposition” and “inertia” groups respectively.

***Proof*:**
This is taking advantage of the fact that $L^{D_{\mathfrak{B}}}/K$ would now be a Galois extension since $D_{\mathfrak{B}} \trianglelefteq \mathrm{Gal}_K(L)$. Then we know that

$$f(\mathfrak{B}_D) = e(\mathfrak{B}_D) = 1$$

and that (since $L^{D_{\mathfrak{B}}}$ is Galois), each other prime must also have the same ramification and inertia degree. Thus, $\mathfrak{p}$ splits into exactly g distinct primes in $L^{D_{\mathfrak{B}}}$. Then by a simple counting argument, we see that each $\mathfrak{B}_D$ must lie under a unique $\mathfrak{B}_I$. It may be that some of these ramify. However, if $I_{\mathfrak{B}} \trianglelefteq \mathrm{Gal}_K(L)$, then we have that each prime must have ramification index 1, showing they are all inert. Finally, a counting argument gives that each $\mathfrak{B}_I$ becomes an eth power in L

Corollary 1.4.43: Normal Subgroup And Completely Split Prime

Let $D_{\mathfrak{B}} \trianglelefteq \mathrm{Gal}_K(L)$. Then $\mathfrak{p}$ splits completely in K' if and only if $K' \subseteq L_D$

***Proof*:**
exercise.

In the non-Galois case L/K, take $M/L/K$ where M/K is the Galois closure of L. Then M/L is Galois. Let $G = \mathrm{Gal}_K(M)$ and $H = \mathrm{Gal}_L(M)$ so that $H \leqslant G$. Take the following tower of primes:

$$\begin{tikzcd}
M \arrow[r] & \mathfrak{A} \\
L \arrow[r] \arrow[u] & \mathfrak{B} \arrow[u] \\
K \arrow[r] \arrow[u] & \mathfrak{p} \arrow[u]
\end{tikzcd}$$

Then by definition:

$$D_{\mathfrak{A}/\mathfrak{p}} \cap H = D_{\mathfrak{A}/\mathfrak{B}}$$
$$I_{\mathfrak{A}/\mathfrak{p}} \cap H = I_{\mathfrak{A}/\mathfrak{B}}$$

Proposition 1.4.44: Prime Decomposition Over Finite Extension

Let
$$\mathfrak{p}_M = \prod_{1}^{g_{M/K}} \mathfrak{A}_i^{e_i}$$
Let $\mathfrak{O}_1, \mathfrak{O}_2, ..., \mathfrak{O}_{g_{L/K}}$ be the orbits of H acting on the primes $\{\mathfrak{A}_1, \mathfrak{A}_2, ..., \mathfrak{A}_{g_{M/K}}\}$. Then:
$$\mathfrak{p}_L = \prod_{i}^{g_{L/K}} \mathfrak{B}_i^{e_{\mathfrak{B}_i/\mathfrak{p}}}$$
where each $\mathfrak{B}_i$ comes form an $\mathfrak{O}_i$. Furthermore:
$$e_{\mathfrak{B}_i/\mathfrak{p}} = \frac{|I_{\mathfrak{A}_i/\mathfrak{p}}|}{|\mathrm{stab}_H(\mathfrak{A}_i) \cap I_{\mathfrak{A}_1/\mathfrak{p}}|} \qquad f_{\mathfrak{B}_i/\mathfrak{p}} = \frac{|D_{\mathfrak{A}_i/\mathfrak{B}_i}|}{|e_{\mathfrak{A}_i/\mathfrak{p}}|} = \frac{|\mathrm{stab}_H(\mathfrak{A}_i)|}{|e_{\mathfrak{A}_i/\mathfrak{p}}|}$$

***Proof*:**
exercise

It is also enlightening to have a second perspective here: for subgroup $U, V \leqslant G$, we may consider the equivalence class $\sigma \sim \sigma' \iff \sigma' = u\sigma v$ giving us the double-coset[26]:
$$U\sigma V = \{u\sigma v \mid u \in U, v \in V\}$$
This partition will give us the similar information in the non-Galois case. Let $N/L/K$ be two separable extensions where N/K is Galois with Galois group G. Let $H = \mathrm{Gal}_L(N) \leqslant \mathrm{Gal}_K(N)$. Let $\mathfrak{p}$ be a prime ideal of K and $P_\mathfrak{p}$ the set of all prime ideal of L above $\mathfrak{p}$. Then if $\mathfrak{B}$ is a prime ideal of N above $\mathfrak{p}$, then we get a well-defined bijection:
$$H \setminus G/D_\mathfrak{B} \to P_\mathfrak{p} \qquad H\sigma D_\mathfrak{B} \mapsto \sigma\mathfrak{B} \cap L$$

Exercise 1.4.1

1. Let L/K be a galois extension of Number fields, $G = \mathrm{Gal}_K(L)$, $\mathfrak{B}$ lie over $\mathfrak{p}$.
   a) If $\mathfrak{B}$ is inert, then G Is cyclic
   b) If $\mathfrak{p}$ is totally ramified in each intermediary field but not totally ramified in L, then there does not exist non-trivial proper intermediary field, hence G is cyclic (of prime order)
   c) Simiarly, Suppose every intermediate field contains a unique prime lying over $\mathfrak{p}$, but L does not. Show the same as the previous
   d) Let $\mathfrak{p}$ be unramified in every proper non-trivial intermediate field, but ramified in L. Show that G has unique smallest subgroup H and that $H \trianglelefteq G$. Use this to show that $|G| = p^k$ for some k and prime p, that $|H| = p^l$ for some l, and $H \leqslant Z(G)$.
   e) Similarly, show the same for when $\mathfrak{p}$ splits compeltely in every proper non-trivial intermediate field, but not in L.

[26] warning: double cosets need not all have the same size!

f) Let $\mathfrak{p}$ be inert in every proper non-trivial intermediate field but not in L. Prove that G is a cyclic where $|G| = p^k$.

2. (the non-normal case for Frobenius element by taking normal closure, p.78 Marcus)

1.4.3 Detecting Ramification

(Part of the problem here is that there is some local field theory going on here, so I will need to separate the local field theory for [9])

(also, this strongly motivate having a separate book for number theory. This is imo the best place in my series to have this, but at the same it is too much for this textbook. So once EYNTKA gets its first version so some theorems are set in stone, it will make sense to have ll this detail)

1.4.45 AKLB setup For generality, let A be a Dedekind domain, K the field of fraction of A, L a finite separable extension of K, and B the integral closure of A in L.

Furthermore, let κ be the residue field of A and λ be the residue field for L. Assume λ/κ and B/A are separable extensions.

Definition 1.4.46: Trace Dual of Fractional Ideal

Take AKLB and let $\mathfrak{P} \subseteq B$ be a fractional ideal. Then the *trace dual* of $\mathfrak{P}$ is:

$$^*\mathfrak{P} = \{x \in L \mid \operatorname{tr}(x\mathfrak{P}) \subseteq A\}$$

By proposition 1.1.32, $^*\mathfrak{P}$ is a finitely generated B-module, and hence by definition a fractional ideal. Note that as the trace-pairing is non-degenerate:

$$^*\mathfrak{P} \xrightarrow{\cong} \operatorname{Hom}_A(\mathfrak{P}, A) \qquad x \mapsto (y \mapsto \operatorname{tr}(xy))$$

which works even when $\mathfrak{P} = (1)_B$, that is:

$$^*B \cong \operatorname{Hom}_A(B, A)$$

This particular trace dual shall be very relevant in detecting ramification:

Definition 1.4.47: Different

Assume AKLB. Then the *different* is the ideal

$$\mathfrak{C}_{B/A} = \{x \in L \mid \forall y \in A,\ \operatorname{tr}(xy) \subseteq A\}$$

is called the *Dedekind's complementary module* or *inverse different.* It's inverse:

$$\mathfrak{D}_{B/A} := \mathfrak{C}_{B/A}^{-1}$$

is called the *Different* of B/A

Since $\mathfrak{C}_{B/A} \supseteq \mathcal{O}$, $\mathfrak{D}_{B/A} \subseteq \mathcal{O}$ is an integral ideal of L. Naturally, we will often denote the different as $\mathfrak{D}_{L/K}$ and $\mathfrak{C}_{B/A}$ as $\mathfrak{C}_{L/K}$ given the rings B an A are evident from context.

Proposition 1.4.48: Tower Law For Different

Assume $AKLB$ and let M/L be a finite separable extension. Then

1. If $M/L/K$, then $\mathfrak{D}_{M/k} = \mathfrak{D}_{M/L}\mathfrak{D}_{L/K}$
2. If $S \subseteq \mathcal{o}$ is a multiplicative subset, then $\mathfrak{D}_{S^{-1}\mathcal{O}/S^{-1}\mathcal{o}} = S^{-1}\mathfrak{D}_{\mathcal{O}/\mathcal{o}}$
3. If $\mathfrak{B}/\mathfrak{p}$ is a prime ideal of $\mathcal{O}$ with respect to $\mathcal{o}$, and $\mathcal{O}_{\mathfrak{B}}/\mathcal{o}_{\mathfrak{p}}$ is the associated completion, then $\mathfrak{D}_{\mathcal{O}/\mathcal{o}}\mathcal{O}_{\mathfrak{B}} = \mathfrak{D}_{\mathcal{O}_{\mathfrak{B}}/\mathcal{o}_{\mathfrak{p}}}$

***Proof*:**

1. If we show that
$$\mathfrak{C}_{\mathcal{O}_M/\mathcal{O}_K} = \mathfrak{C}_{\mathcal{O}_M/\mathcal{O}_L}\mathfrak{C}_{\mathcal{O}_L/\mathcal{O}_K}$$
we are done. For $\supseteq$:
$$\begin{aligned}\mathrm{tr}_{M/K}(\mathfrak{C}_{\mathcal{O}_M/\mathcal{O}_L}\mathcal{O}_{L/\mathcal{O}_A}(\mathcal{O}_M)) &= \mathrm{tr}_{L/K}\mathrm{tr}_{M/L}(\mathfrak{C}_{\mathcal{O}_M/\mathcal{O}_L}\mathfrak{C}_{\mathcal{O}_L/\mathcal{O}_K}(\mathcal{O}_M)) \\ &= \mathrm{tr}_{L/K}(\mathfrak{C}_{\mathcal{O}_L/\mathcal{O}_K}\mathrm{tr}_{M/L}(\mathfrak{C}_{\mathcal{O}_M/\mathcal{O}_L)}\mathcal{O}_M) \\ &\subseteq \mathcal{O}_K\end{aligned}$$
For the $\subseteq$, since $\mathcal{O}_L\mathcal{O}_M = \mathcal{O}_M$, we get:
$$\mathrm{tr}_{M/K}(\mathfrak{C}_{\mathcal{O}_M/\mathcal{O}_K}\mathcal{O}_M) = \mathrm{tr}_{L/K}(\mathcal{O}_L\mathrm{tr}_{M/L}(\mathfrak{C}_{\mathcal{O}_M/\mathcal{O}_K}\mathcal{O}_M)) \subseteq \mathcal{O}_K$$
Hence $\mathrm{tr}_{M/L}(\mathfrak{C}_{\mathcal{O}_M/\mathcal{O}_K}\mathcal{O}_M) \subseteq \mathfrak{C}_{\mathcal{O}_L/\mathcal{O}_K}$, thus:
$$\mathrm{tr}_{M/L}(\mathfrak{C}^{-1}_{\mathcal{O}_L/\mathcal{O}_K}\mathcal{O}_{\mathcal{O}_M/C\mathcal{O}_K}\mathcal{O}_M) = \mathfrak{C}^{-1}_{\mathcal{O}_L/\mathcal{O}_K}\mathrm{tr}_{M/L}(\mathfrak{C}_{\mathcal{O}_M/\mathcal{O}_K}C) \subseteq \mathcal{O}_L$$
Thus, we get $\mathfrak{C}^{-1}_{\mathcal{O}_L/\mathcal{O}_K}\mathfrak{C}_{\mathcal{O}_M/\mathcal{O}_K} \subseteq \mathfrak{C}_{\mathcal{O}_M/\mathcal{O}}$, hence
$$\mathfrak{C}_{\mathcal{O}_M/\mathcal{O}_K} \subseteq \mathfrak{C}_{\mathcal{O}_M/\mathcal{O}_K}\mathfrak{C}_{\mathcal{O}_L/\mathcal{O}_K}$$
completing the proof.
2. follows from definition pushing
3. (Neukirch p.195)

Example 1.4.49: Different

1. Let $K = \mathbb{Q}(\sqrt{5})$ so that $\mathcal{O}_K = \mathbb{Z}\left[\frac{1+\sqrt{5}}{2}\right]$. Then:
$$\mathfrak{C}_{K/\mathbb{Q}} = \sqrt{5}^{-1}\mathcal{O}_K \implies \mathfrak{d} = \sqrt{5}\mathcal{O}_K$$

2. Let $K = \mathbb{Q}(\sqrt{3})$. So that $\mathcal{O}_K = \mathbb{Z}[\sqrt{3}]$. Then:

$$\mathfrak{C}_{K/\mathbb{Q}} = \langle \frac{1}{2}, \frac{1}{\sqrt{3}} \rangle_{\mathbb{Z}} = \frac{1}{2\sqrt{3}}\mathcal{O}_K \implies \mathfrak{d}_{K/\mathbb{Q}} = 2\sqrt{3}\mathcal{O}_K$$

Theorem 1.4.50: Norm Of Different

$$\mathfrak{N}(\mathfrak{d}_{K/\mathbb{Q}}) = |\mathrm{disc}(\mathcal{O}_K)|$$

***Proof*:**
good exercise!

(This next proposition gives a special case of the different and the origin of it's name)

Proposition 1.4.51: Different In Monogenic Extension

Let $\mathcal{O} = o[\alpha]$. Then the different is equal to:

$$\mathfrak{D}_{L/K} = (\delta_{L/K}(\alpha))$$

where

$$\delta_{L/K}(\alpha) = \begin{cases} f'(\alpha) & L = K(\alpha) \\ 0 & L \neq K(\alpha) \end{cases}$$

***Proof*:**
Neukirch p.197

Theorem 1.4.52: Generation Of Different

The different $\mathfrak{D}_{L/K}$ is the ideal generated by all the different of the elements $\delta_{L/K}(\alpha)$ for all $\alpha \in \mathcal{O}$

***Proof*:**
Neukirch p.198

Theorem 1.4.53: Detecting Ramification Using Different

A prime ideal $\mathfrak{P}$ of L is ramified over K if and only if

$$\mathfrak{B} \big| \mathfrak{d}_{L/K}$$

***Proof*:**
Neukirch p. 199

(relation between different and discriminant)

Definition 1.4.54: Relative Discriminant

The *discriminant* $\mathfrak{D}_{\mathcal{O}/\mathfrak{o}}$ is the ideal of $\mathfrak{o}$ generated by the discriminants $d(\alpha_1, ..., \alpha_n)$ for all bases $\alpha_1, \alpha_2, ..., \alpha_n$ of L/K contained in $\mathcal{O}$:

$$\mathfrak{D}_{\mathcal{O}/\mathfrak{o}} = \{(d(\alpha_1, \alpha_2, ..., \alpha_n)) \mid \alpha_1, \alpha_2, ..., \alpha_n \text{ basis }, \alpha_i \in \mathcal{O}\}$$

In the case where $K = \mathbb{Q}$ and $L/\mathbb{Q}$ is a number field, then $\mathfrak{d} = (d_L)$ is the principal ideal of the usual discriminant.

Theorem 1.4.55: Relating Different And Relative Discriminant

$$\mathfrak{D}_{L/K} = \mathfrak{N}_{L/K}(\mathfrak{d}_{L/K})$$

***Proof*:**
Neukirch p.201

Corollary 1.4.56: Tower For Relative Discriminant

Let $M/L/K$. Then:

$$\mathfrak{D}_{M/K} = \mathfrak{D}_{L/K}^{[M:L]} \mathfrak{N}_{L/K}(\mathfrak{D}_{M/L})$$

***Proof*:**
Neukirch p.202

Corollary 1.4.57: Discriminant And Ramified Primes

Let $\mathfrak{p} \subseteq K$ be a prime. Then $\mathfrak{p}$ is ramified in L if and only if

$$\mathfrak{p} \big| \mathfrak{D}_{L/K}$$

In particular, the extension L/K is unramified if and only if

$$\mathfrak{D}_{L/K} = (1)$$

***Proof*:**
Immediate consequence

Proposition 1.4.58: Coprime Discriminants And Difference

Let K_1, K_2 be galois over $\mathbb{Q}$ with coprime discriminants and $K = K_1 K_2$. Then:

$$\mathfrak{d}_{K/\mathbb{Q}} = \mathfrak{d}_{K_1/\mathbb{Q}} \mathfrak{d}_{K_2/\mathbb{Q}}$$

***Proof*:**
Using the tower

(Neukirch goes on to prove some important results which are really cool! For example, if K is an algebraic number field and S a finite set of primes of K, then there exists only finitely many extensions L/K of degree n which are unramified outside of S!)

Exercise 1.4.2

1. Show that L_D is the maximal field in which the ramification and inertial degree is 1, and similarly for L_I.
2. Let K be a number fields. Show that if $\text{disc}(K)$ is square free, then K has no intermediary number field (this is a generlization of exercise re:HERE(galPoly))

Wild and Tame Ramification

(see this! here)

(This section is essentially here for the proof of Kronecker-Weber Theorem. May choose to move it down to that section).

We end off by exploring some more about the inertia group, namely we shall like at *wild* and *tame* inertia.

Definition 1.4.59: Wild Inertia Group

Let L/K be an extension of number fields, $I_{\mathfrak{B}} \leqslant D_{\mathfrak{B}}$ be the inertia group. Then $W_{\mathfrak{B}} \trianglelefteq I_{\mathfrak{B}}$ is defined to be:
$$W_{\mathfrak{B}} := \left\{ g \in I_{\mathfrak{B}} \;\middle|\; \forall \overline{\alpha} \in \mathcal{O}_L/\mathfrak{B}^2,\ g(\overline{\alpha}) = \overline{\alpha} \right\}$$

the wild inertia group can be thought of as a parallel to "local coordinates". In particular if we have an extension $K/\mathbb{Q}$ of number fields, then we have an extension $K/\mathbb{Q}_p$ then $W_{\mathfrak{B}}$ is the unique sylow p-subgroup of the inertia group. If you know some algebraic geometry, an intuitive reason to see why the quotient $\mathfrak{B}^2$ shows up is since the tangent space of a variety/scheme is given by $\mathfrak{m}/\mathfrak{m}^2$.

Lemma 1.4.60: Wild Inertia And Uniformizer

Let $\pi \in \mathcal{O}_L$ be the uniformizer for Q (i.e. $Q \| \pi$). Let $\sigma \in I_Q$. Then:
$$\sigma \in W_Q \qquad \Leftrightarrow \qquad \sigma(\pi) - \pi \in Q^2$$
that is, $\sigma(\pi) = \pi \bmod Q^2$

***Proof*:**
The $\Rightarrow$ direction is by definition. For $\Leftarrow$, note that we have calculated before that:
$$\left|(\mathcal{O}_L/Q^2)^\times\right| = \left|(\mathcal{O}_L/Q)^\times\right| \left|Q\mathcal{O}_L/Q^2\right| = (q-1)q$$
where $q = |\mathbb{F}_{p^r}|$ for appropriate p^r. Hence, there exists $H \leqslant (\mathcal{O}_L/Q^2)^\times$ such that $|H| = q - 1$

and $H \xrightarrow{\sim} \mathbb{F}_Q^\times$. Hence

$$(\mathcal{O}_L/Q^2)^\times \cong H \oplus (1+\pi\mathcal{O}_L)_{Q^2} \bmod Q^2$$

Hence $\sigma \in I_G$, then σ acts trivially on $\mathbb{F}_Q^\times$, and hence acts trivially on $H>$ Thus $\sigma \in W_Q$ if and only if σ fixes $(1+\pi\mathcal{O}_L)_{Q^2}$ if and only if $(\pi\mathcal{O}_L)_{Q^2}$ if and only if σ fixes π mod Q^2

Example 1.4.61: Example from Class for wild inertia

$K = \mathbb{Q}(\sqrt[8]{1})$, $i^4 = 1$, $\left(\frac{1+i}{\sqrt{2}}\right)^2 = i$. $[K : \mathbb{Q}] = |\mathbb{Z}/8\mathbb{Z}^\times| = 4$. and $(\mathbb{Z}/8\mathbb{Z})^\times \cong (\mathbb{Z}/2\mathbb{Z}) \oplus \mathbb{Z}/2\mathbb{Z}$. So K is biquadratic. Note that

$$K = \mathbb{Q}\left(\frac{1+i}{\sqrt{2}}\right) = \mathbb{Q}(\alpha) = \mathbb{Q}(i, \sqrt{2})$$

(diagram of the 3 quadratic extensions). Notice that 2 is the only prime that ramifies. In $\mathbb{Q}(\sqrt{d})$ where $d = -1, 2, 2$, so 2 ramifies

$$(e, f, g) = (2, 1, 1)$$

It may not be immediately clear, but in K, $2\mathcal{O}_K = \mathfrak{B}^4$, i.e. 2 totally ramifies (we have not seen cases where e increases when it's coprime). This cannot happen with unramified fields. What is $\mathfrak{B}$? Explicitly, we know that

$$\mathfrak{B} = (1-\alpha)$$

which we can directly see because $\mathfrak{B}^2 = (1 - 2\alpha + \alpha^2) = (1 + i - 2\alpha) = (1 + i)(1 - (1 - i)\alpha)$, and we see that $(1 - (1 - i)\alpha)$ is a unit.

Let $\pi = 1 - \alpha$, which is a uniformizer of $\mathfrak{B}$. Recall that we looked at $\sigma(\pi) - \pi$ for $\sigma]in\mathrm{Gal}_{\mathbb{Q}}(K)$. Computing it's valuation at $\mathfrak{B}$

$$v_{\mathfrak{B}}(\sigma(\pi) - \pi)$$

we can do this by independence of valuation on the prime. First

$$\sigma(\alpha) = \alpha^3 \implies (1 - \alpha^3) - (1 - \alpha) = v_{\mathfrak{B}}(\alpha^2 - 1) = 1 + v_{\mathfrak{B}}((\alpha - 1) + 2) = 2$$

Next, we do

$$\sigma(\alpha) = \alpha^7 = \alpha^{-1} \implies v_{\mathfrak{B}}(\alpha^7 - \alpha) = v_{\mathfrak{B}}(\alpha^{-1} - \alpha) = v_{\mathfrak{B}}(\alpha^2 - 1) = 2$$

Finally, we have

$$\sigma(\alpha) = \alpha^5 \implies v_{\mathfrak{B}}(\alpha^5 - \alpha) = v_{\mathfrak{B}}(\alpha^4 - 1) = v_{\mathfrak{B}}(\alpha - 1) + v_{\mathfrak{B}}(\alpha + 1) + v_{\mathfrak{B}}(\alpha^2 + 1) = 1 + 1 + 2 = 4$$

Hence, we have a chain:

$$\langle 1\rangle \subseteq \langle 1, 5\rangle \subseteq (\mathbb{Z}/\gamma\mathbb{Z})^\times = \mathrm{Gal}_{\mathbb{Q}}(K)$$

It may not jump out, but notice that 2 divides the difference (linked to a theorem called *Hasse-Arf* Theorem).

Theorem 1.4.62: Abelian Extension Wild Map

Let L/K Galois and abelian, $G = \mathrm{Gal}_K(L)$, $P \subseteq \mathcal{O}_K$ a prime ideal of $\mathcal{O}_K$, Q a prime lying over P W_Q the wild inertia group, and π a uniformizer so that $Q \| \pi$. Then:

$$\varphi : I_Q/W_Q \hookrightarrow \mathbb{F}_Q^\times \qquad [\sigma] \mapsto \frac{\sigma(\pi)}{\pi} \bmod Q$$

Assume G is abelian. Then $\mathrm{im}\varphi \subseteq \mathbb{F}_P^\times$.

***Proof*:**
Look at $\varphi(\sigma) = \varphi(\tau^{-1}\sigma\tau)$ (because abelian), so:

$$\frac{\tau^{-1}\sigma\tau(\pi)}{\pi} \bmod Q$$

and write it as:

$$\frac{\tau{-}1(\sigma(\tau(\pi)))}{\tau(\pi)} \bmod Q$$

Now, it doesn't matter what uniformizer we pick. If $\tau \in D_Q$, then we can mod by Q before or after, so

$$\tau^{-1}\varphi(\sigma)$$

What kind of element can be written as τ^{-1}? We know the decomposition group surjects onto $\mathrm{Gal}_{\mathbb{F}_p}(\mathbb{F}_Q)$, we can pick τ^{-1} to be the Frobenius operator, that is

$$\varphi(\sigma) = \varphi(\sigma)^{\mathrm{Nm}(P)} \implies \varphi(\sigma) \in \mathbb{F}_p$$

Theorem 1.4.63: Wild Inertia Group Is Sylow Subgroup

Take $W_Q \leqslant \mathrm{Gal}_{\mathbb{Q}}(L)$ where Q is a prime lying over p. Then W_Q the unique p-sylow subgroup, hence $W_Q \trianglelefteq I_Q$.

***Proof*:**
check notes for now

Exercise 1.4.3

1. (Define higher-order ramification. Show that the inertia group is solvable, and when the residue fields are finite then the decomposition group is solvable).

1.4.4 Frobenius Automorphism

This is a short section to single out the Frobenius automorphisms from corollary 1.4.34 due to its importance in Class Field Theory (CFT); namely the Frobenius automorphism is central to relating the structure of abelian galois groups to the structure of the class group. The full exposition is beyond the scope of this textbook, but the reader can still get an appreciation behind why the Frobenius automorphism is so important. For a full exposition, see [9].

Assume $\mathfrak{B}$ over $\mathfrak{p}$ is unramified in $L/K/\mathbb{Q}$, so that $I_{\mathfrak{B}}$ is trivial (which by proposition 1.4.12 and proposition 1.4.16 can be shown to be the case for all put finitely many primes). Then $D_{\mathfrak{B}} \cong \mathrm{Gal}_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{B})) \cong (\mathbb{Z}/n\mathbb{Z})$. By finite field theory, we know that there exists the *Frobenius generator* sending $x \mapsto x^{\mathfrak{N}(\mathfrak{B})}$ since $\mathfrak{N}(\mathfrak{B}) = f$ and the order of this group is f. Thus, for each prime $\mathfrak{B}/\mathfrak{p}$, have a corresponding automorphism in the decomposition group, and hence the Galois group:

Definition 1.4.64: Frobenius Automorphism of a Prime

Let $\sigma \in D_{\mathfrak{B}}$. Then σ is called the *Frobenius* if $\sigma(\alpha) \equiv \alpha^f \mod \mathfrak{B}$ where f is the inertia degree of $\mathfrak{B}$ over $\mathfrak{p}$. The collection of element $\sigma \in D_{\mathfrak{B}}$ with this property is denoted $\mathrm{Frob}_{\mathfrak{B}/\mathfrak{p}}$ so that is $\sigma \in \mathrm{Frob}_{\mathfrak{B}/\mathfrak{p}}$:

$$\sigma(\alpha) \equiv \alpha^f \mod \mathfrak{B}$$

Note that if $\mathfrak{B}$ were ramified, then the Frobenius automorphism is defined up to the coset $I_{\mathfrak{B}}$: $\sigma + I_{\mathfrak{B}}$. If another $\mathfrak{B}' = \sigma(\mathfrak{B})$[27] is considered, the Frobenius element differ conjugation:

$$\mathrm{Frob}_{\sigma(\mathfrak{B}/\mathfrak{p})} = \sigma \mathrm{Frob}_{\mathfrak{B}/\mathfrak{p}} \sigma^{-1}$$

If $D_{\mathfrak{B}}$ is abelian, then the Frobenius element is *only* determined by the prime $\mathfrak{p}$ as

$$\mathrm{Frob}_{\sigma(\mathfrak{B}/\mathfrak{p})} = \sigma \mathrm{Frob}_{\mathfrak{B}/\mathfrak{p}} \sigma^{-1} = \mathrm{Frob}_{\mathfrak{B}/\mathfrak{p}}$$

Thus, we may denote it as $\mathrm{Frob}_{\mathfrak{p}}$. In this case, we have:

$$\sigma(\alpha) = \alpha^{\mathfrak{N}(\mathfrak{p})} \mod \mathfrak{p}\mathcal{O}_L$$

because $\mathfrak{p}\mathcal{O}_L$ is the product of all primes $\mathfrak{B}$ over $\mathfrak{p}$. Another immediate observation is that if $\mathfrak{p}$ spits completely, then $f = 1$, and the Frobenius element for $\mathfrak{p}$ shall be the identity, and hence the Frobenius element captures behaviour in the case where a prime doesn't completely split.

The important realization is that if $\mathrm{Gal}_k(F)$ is abelian, there exist $\sigma \in \mathrm{Gal}_k(F)$ that are associated to primes $\mathfrak{p}$ in the sense that the structure of $\sigma \mod \mathfrak{B}/\mathfrak{p}$ is simply the power map (i.e. cyclic)! If each element $\sigma \in \mathrm{Gal}_k(F)$ can have such an association, then that means that primes have a strong link to the structure of an abelian galois group! Even better, the decomposition of primes is regulated by the class group. The culminating result of 20th century number theory is that any abelian galois group is isomorphic to a class group! Thus, abelian galois theory and number theory have a strong correspondence to each other.

When $\mathrm{Gal}_k(F)$ is non-abelian, the situation become more tenuous. Robert Langland is credited for finding the right framework to extend this correspondence through the use of *representation theory*. This is the origins of the famous *Langland's Conjecture*.

1.4.5 Example: Cyclotomic Fields

As we saw in example 1.1.31, the discriminant of $\mathbb{Z}[\zeta_{p^n}]$ is a prime power of p. We now generalize this result for arbitrary n, and find how p, and will find the discriminant is a power of p. Thus, p is the only prime that ramifies in K, and we shall find how p ramifies:

[27] by transitivity and definition of the decomposition group, any $\mathfrak{B}'$ is of this form

Lemma 1.4.65: Cyclotomic Fields Lemma

Let $n = p^k$ be a prime power, ζ_{p^k} a root of unity, and $\lambda = 1 - \zeta_{p^k}$. Then the principal ideal $(\lambda) \subseteq \mathcal{O}$ in the ring of integers of $\mathbb{Q}(\zeta_{p^k})$ is a prime ideal of degree 1, and:

$$p\mathcal{O} = (\lambda)^d$$

where $d = \varphi(n) = [\mathbb{Q}(\zeta_{p^k}) : \mathbb{Q}]$. Furthermore, the basis $1, \zeta_{p^k}, ..., \zeta_{p^k}^{d-1}$ of $\mathbb{Q}(\zeta_{p^k})/\mathbb{Q}$ has discriminant:

$$d_{\mathbb{Q}(\zeta_{p^k})/\mathbb{Q}}(1, \zeta_{p^k}, ..., \zeta_{p^k}^{d-1}) = \pm p^s$$

where $s = p^{k-1}(kp - k - 1)$

Note that in order for the above discriminant to be the discriminant of the ring of integers, we would have to insure the ring of integers has the above basis up to change of coordinates, which shall be the result of the next theorem.

***Proof*:**
Recall that the minimal polynomial of ζ_{p^k} over $\mathbb{Q}$ is:

$$m_{\zeta_{p^k},\mathbb{Q}}(x) = \frac{x^{l^n} - 1}{x^{l^{n-1}} - 1} = x^{l^{n-1}(l-1)} + \cdots + x^{l^{n-1}} + 1$$

Letting $x = 1$, we get

$$n = p^k = \prod_{r \in (\mathbb{Z}/n\mathbb{Z})^\times} (1 - \zeta_{p^k}^r)$$

Next, we know that

$$(1 - \zeta_{p^k}^r) = (1 - \zeta_{p^k})(1 + \zeta_{p^k} + \cdots + \zeta_{p^k}^{r-1}) = (1 - \zeta_{p^k})\epsilon_r$$

If r' is an integer such that $rr' = 1 \bmod n$, then

$$\frac{1 - \zeta_{p^k}}{1 - \zeta_{p^k}^r} = \frac{1 - (\zeta_{p^k}^r)^{r'}}{1 - \zeta_{p^k}^r} = 1 + \zeta_{p^k}^r + \cdots + (\zeta_{p^k}^r)^{r'-1}$$

is integral as well, namely ϵ_r is a unit. Thus,

$$p = \epsilon(1 - \zeta_{p^k})^{\varphi(n)}$$

with unit $\epsilon = \prod_r \epsilon_r$. Thus:

$$p\mathcal{O} = (\lambda)^{\varphi(n)}$$

Since $[\mathbb{Q}(\zeta_{p^k}) : \mathbb{Q}] = \varphi(n)$, the fundamental identity ($\sum e_i f_i$) shows that (λ) is a prime ideal of degree 1.

Now, take $\zeta_{n,1}, ..., \zeta_{n,d}$ be the conjugates of ζ_{p^k} (i.e. powers of ζ_{p^k} that are coprime with the order of ζ_{p^k}). Then the cyclotomic polynomial:

$$\Phi_{p^k}(x) = \prod_{i=1}^{d} (x - \zeta_{n,i})$$

and

$$\pm\text{disc}(1,\zeta_{p^k},...,\zeta_{p^k}^{d-1}) = \prod_{i\neq j}(\zeta_{n,i}-\zeta_{n,j}) = \prod_{i=1}^{d}\Phi'_{p^k}(\zeta_{n,i}) = \text{Nm}_{\mathbb{Q}(\zeta_{p^k})/\mathbb{Q}}(\Phi'_{p^k}(\zeta_{p^k}))$$

Differentiating the equation:

$$(x^{p^{k-1}}-1)\Phi_{p^k}(x) = x^{p^k}-1$$

and substituting ζ_{p^k} for x gives:

$$(\zeta_{p^k}^{p^{k-1}}-1)\Phi'_{p^k}(\zeta_{p^k}) = (\eta-1)\Phi'_{p^k}(\zeta_{p^k}) = n\zeta_{p^k}^{-1}$$

But we know $\text{Nm}_{\mathbb{Q}(\eta)/\mathbb{Q})}(\epsilon-1) = \pm p$, thus

$$\text{Nm}_{\mathbb{Q}(\zeta_{p^k})/\mathbb{Q}}(\eta-1) = \text{Nm}_{Q(\eta)/\mathbb{Q}}(\eta-1)^{p^{k-1}} = \pm p^{p^{k-1}}$$

since $\zeta_{p^k}^{-1}$ has norm ± 1, we get:

$$d(1,\zeta_{p^k},..,\zeta_{p^k}^{d-1}) = \pm\text{Nm}_{\mathbb{Q}(\zeta_{p^k})/\mathbb{Q}}\left(\Phi'_{p^k}(\zeta_{p^k})\right) = \pm p^{k^{p^{k-1}(k-1)-p^{k-1}}} = \pm p^s$$

where s is the exponent

For completeness of this section, here is a quick proof of the ring of integers of cyclotomic fields:

Proposition 1.4.66: Ring Of Integers Of Cyclotomic Fields

Let $K = \mathbb{Q}(\zeta_n)$ be a cyclotomic field. Then

$$\mathcal{O}_{K_n} \cong \mathbb{Z}[\zeta_n]$$

With discriminant:

$$\text{disc}(\mathcal{O}_{K_n}) = (-1)^{\varphi(n)/2}\frac{n^{\varphi(n)}}{\prod_{p|n} p^{\frac{\varphi(n)}{p-1}}}$$

***Proof*:**
Let's start with n being a prime power p^k. Since $\text{disc}(1,\zeta,...,\zeta^{d-1}) = \pm p^s$, we have:

$$p^s\mathcal{O}_K \subseteq \mathbb{Z}[\zeta_n] \subseteq \mathcal{O}$$

Putting $\lambda = 1-\zeta_n$, by lemma 1.4.65 we have $\mathcal{O}_K/\lambda\mathcal{O}_K \cong \mathbb{Z}/p\mathbb{Z}$, thus $\mathcal{O}_K = \mathbb{Z}+\lambda\mathcal{O}_K$, hence we *a fortiori* get

$$\lambda\mathcal{O}_K + \mathbb{Z}[\zeta] = \mathcal{O}$$

Multiplying by λ, and substituting $\lambda\mathcal{O}_K = \lambda^2\mathcal{O}_K + \lambda\mathbb{Z}[\zeta]$, we get

$$\lambda^2\mathcal{O}_K + \mathbb{Z}[\zeta] = \mathcal{O}$$

Iterating this procedure, we get:

$$\lambda^t\mathcal{O}_K + \mathbb{Z}[\zeta] = \mathcal{O} \qquad \forall t \geq 1$$

When $t = s\varphi(p^k)$, then given $p\mathcal{O}_K = \lambda^{\varphi(p^k)}\mathcal{O}_K$, we get

$$\mathcal{O} = \lambda^t\mathcal{O}_K + \mathbb{Z}[\zeta] = p^s\mathcal{O}_K + \mathbb{Z}[\zeta] = \mathbb{Z}[\zeta]$$

For the general case, let $n = p_1^{k_1}\cdots p_r^{k_r}$. Then $\zeta_i = \zeta^{n/p_i^{k_i}}$ is a primitive $p_i^{k_k}$-th root of unity, and hence:

$$Q(\zeta) = \mathbb{Q}(\zeta_1)\cdots\mathbb{Q}(\zeta_r)$$

and $\mathbb{Q}(\zeta_1)\cdots\mathbb{Q}(\zeta_{i-1})\cap\mathbb{Q}(\zeta_i) = \mathbb{Q}$. Then for each $i \in \{1, ..., r\}$, the elements $1, \zeta_i, ..., \zeta_i^{d_i-1}$, where $d_i = \varphi(p_i^{k_i})$ form an integral basis of $\mathbb{Q}(\zeta_i)/\mathbb{Q}$. Since the discriminants $d(1, \zeta_i, ..., \zeta_i^{d_i-1}) = \pm p_i^{s_i}$, are relatively prime, we get from successive uses of (Proposition 2.11 in Neukirch, I should add it to my notes!!) we get that

$$\zeta_1^{j_1}\cdots\zeta_r^{j_r}$$

where $j_i \in \{0, ..., (d_i - 1)\}$ form an integral basis of $\mathbb{Q}(\zeta)/\mathbb{Q}$. But then each of these elements is a power of ζ. Thus, every $\alpha \in \mathcal{O}$ may be written as a polynomial $\alpha = f(\zeta)$ with coefficients in $\mathbb{Z}$. Since ζ has degree $\varphi(n)$ over $\mathbb{Q}$, the degree of $f(\zeta)$ may be reduced to $\varphi(n) - 1$. But then, we may obtains:

$$\alpha = a_1 + a_1\zeta + \cdots + a_{\varphi(n)-1}\zeta^{\varphi(n)-1}$$

Thus, $1, \zeta, ..., \zeta^{\varphi(n)-1}$ is an integral basis, as we sought to show.

by theorem 1.4.53 commented this theorem out! , we know that the primes that ramify are those that divide $\mathrm{disc}(\mathcal{O}_{K_n})$, hence by the above the primes that ramify are the factors of n. The following gives exactly how they factor:

Proposition 1.4.67: Prime Decomposition In Cyclotomic Field

Enumerates all the primes $p_1, p_2, ...$, then let $n = \prod_n p_n^{n_k}$ be a prime factorization of $n \in \mathbb{Z}$. Let f_k be the smallest positive integer such that

$$p_k^{f_k} \equiv 1 \bmod (n/p_k^{n_k})$$

Then in $\mathbb{Q}(\zeta_n)$, we have:

$$p_k = (\mathfrak{p}_1\cdots\mathfrak{p}_r)^{\varphi(p_k^{n_k})}$$

where $\mathfrak{p}_1, ..., \mathfrak{p}_r$ are distinct prime ideals all of degree f_k.

Note that if $p_k \nmid n$, then $n_k = 0$, so the condition becomes

$$p_k^{f_k} \equiv 1 \bmod n$$

and $\varphi(p_k^{n_k}) = 1$, giving us:

$$p_k = \mathfrak{p}_1\cdots\mathfrak{p}_r$$

***Proof*:**
Since $\mathcal{O}_K = \mathbb{Z}[\zeta]$, we have that the conductor of $\mathbb{Z}[\zeta]$ is 1, and hence we may apply theorem 1.4.7 to any prime number p, in particular every p decompose into prime ideals in the exact same way that $\Phi_n(x)$ (the minimal polynomial of ζ_n) factors into irreducible mod p. Thus, we are reduced to showing that:

$$\Phi_n(x) \equiv (p_1(x)\cdots p_r(x))^{\varphi(p^{\nu_p})} \bmod p$$

where $p_1(x), p_2(x), ..., p_r(x)$ are distinct irreducible polynomials over $\mathbb{Z}/p\mathbb{Z}$ of degree f_p. Write n as $p^{\nu_p}m$ ($p \nmid m$). Let ξ_i and η_j vary over the primitive roots of unity over order m and p^{ν_p} respectively. Then the product $\xi_i\eta_j$ vary over the nth root of unity, namely we have the decomposition:

$$\Phi_n(x) = \prod_{i,j}(x - \xi_i\eta_j)$$

Since $x^{p^{\nu_p}} - 1 \equiv (x-1)^{p^{\nu_p}} \bmod p$, we have that $\eta_j \equiv 1 \bmod \mathfrak{p}$ for any prime ideal where $\mathfrak{p}\big|p$. This, we get:

$$\Phi_n(x) \equiv \prod_i (x - \xi_i)^{\varphi(p^{\nu_p})} = \Phi_n(x)^{\varphi(p^{\nu_p})} \bmod \mathfrak{p}$$

Hence, we have when modding by p:

$$\Phi_n(x) \equiv \Phi_n(x)^{\varphi(p^{\nu_p})} \bmod p$$

Now, notice that f_p is the smallest positive integer such that $p^{f_p} \equiv 1 \bmod m$, hence the congruence reduces to th case where $p \nmid n$, and then $\varphi(p^{\nu_p}) = \varphi(1) = 1$.

Now, since the characteristic of $\mathcal{O}_K/\mathfrak{p}$ (i.e. p) does not divide n, the polynomial $x^n - 1$ and nx^{n-1} have no common roots in $\mathcal{O}_K/\mathfrak{p}$ by the derivative test. Thus, $x^n - 1 \bmod p$ has *no multiple roots*. Thus, we get that $\mathcal{O}_K \to \mathcal{O}_K/\mathfrak{p}$ maps the group of nth roots of unity μ_n bijectivity onto the nth roots of unity of $\mathcal{O}_K/\mathfrak{p}$. In particular ,the primitive nth roots of unity modulo $\mathfrak{p}$ ($\zeta \bmod \mathfrak{p}$) remain primitive nth roots of unity. Since the extension field of $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ containing it is the field $\mathbb{F}_{p^{f_p}}$, since the multiplicative group $\mathbb{F}^*_{p^{f_p}}$ is cyclic of order $p^{f_p} - 1$. Thus, $\mathbb{F}_{p^{f_p}}$ is the field decomposition of the reduced cyclotomic polynomial:

$$\overline{\Phi}_n(x) = \Phi_n(x) \bmod p$$

Since it is a divisor of $X^n - 1 \bmod p$, this polynomial has no multiple roots, and if

$$\overline{\Phi}_n(x) = \overline{p}_1(x) \cdots \overline{p}_r(x)$$

i it's factorization into irreducible over $\mathbb{F}_p$, then every $\overline{p}_i(x)$ is the minimal polynomial of a primitive nth root of unity $\overline{\xi}\ in \mathbb{F}^*_{p^{f_p}}$. It's degree is thus f_p, as we sought to show.

Corollary 1.4.68: Special Case Of Factorization

A prime number p is ramified in $\mathbb{Q}(\zeta_n)$ iff

$$n \equiv 0 \bmod p$$

except in the case where $p = 2 = (4, n)$. A prime number $p \neq 2$ is totally split in $\mathbb{Q}(\zeta)$ iff

$$p \equiv 1 \bmod n$$

***Proof*:**
direct consequence of last proposition

Corollary 1.4.69: Subfield Of Cyclotomic Field

Let $K \subseteq \mathbb{Q}(\zeta_n) = L$ with corresponding subgroup $H \leqslant \mathrm{Gal}_K(L) \cong (\mathbb{Z}/n\mathbb{Z})^\times$ considered as a subgroup of $(\mathbb{Z}/n\mathbb{Z})^\times$. For prime p where $p \nmid n$, let f be the least positive integer such that $\overline{p^f} \in H$. Then for prime $\mathfrak{p}$ lying above p in K:

$$f = f_{\mathfrak{p}/p}$$

***Proof*:**
Notice hat $f_{\mathfrak{p}/p}$ is the order of the Frobenius automorphism.

Overall, the ring of integers of a cyclotomic field is monogenic, the only primes that ramify are those that divide n (where n is the order of the root of unity) since the divisors of the discriminant are completely dependent on n, the discriminant is a product of the powers of primes of n, and we know exactly how the each element $n \in \mathbb{Z}$ factors in cyclotomic fields!

1.4.6 Application: Quadratic Extensions and Quadratic Reciprocity

We now take a moment to solve a classical number theory problems with the developed tools. Let $p \in \mathbb{Z}$ be a prime number. In [5, Application: Sums of Primes and Gaussian Integers], it was shown that p can be written as the sum of two squares if and only if $p \equiv 1 \bmod 4$. Let us now ask another question that may shed further light on the nature of primes: the square-root of primes is always an irrational number. However, when modding by $n \in \mathbb{N}$, it is possible that there is an answer. By the Chinese Remainder Theorem, it suffices to check when $n = q$ is a prime. Thus:

what primes p have the property that there exists an n such that $n^2 \equiv p \bmod q$

Example 1.4.70: Concrete Example
Let $p = 3$. Then:

mod	1^2	2^2	3^2	4^2	5^2	6^2
2	1					
3	1	1				
5	1	4	4	1		
7	1	4	2	2	4	1
11	1	4	9	5	***3***	***3***

Note the table is mirrored as $(-n)^2 = n^2$. Hence, for a reason that is yet to be explained in this book, the prime 3 is somehow tied to 11 when looking for square-roots of 3. The reader can verify that 13, 23, 37, and so forth also have a square-root of 3.

The reader should play around with different pairs (p, q) and flip them around to consider (q, p), there shall appear some symmetry that at first is not obvious.

This relation between primes and square-root in modular arithmetics was first documented to be discovered by Euler, however it took a generation before a complete proof by Gauss appeared. As it turns out, the condition translates into a condition on rings of integers of quadratic extensions!

Definition 1.4.71: Quadratic Residue

Let $n \in \mathbb{Z}$. Then if there exists an $m \in \mathbb{Z}$ such that $m^2 \equiv n \mod p$, then n is called a *quadratic residue* mod p.

First, to as whether p has a square-root $\mod q$ is is equivalent to solving $x^2 - p \equiv 0 \mod q$. Let $K = \mathbb{Q}(\sqrt{p})$. Then we we may consider $\mathcal{O}_K/q\mathcal{O}_K$; this ring is isomorphic to $(\mathbb{Z}/q\mathbb{Z})^n$ for some $n \in \mathbb{N}_{>0}$ Note that having a root in $\mathcal{O}_K$ implies having a root in $\mathcal{O}_K/p\mathcal{O}_K$ for any p (as shown in [5, Limitation of reduction method], the converse does not hold even if an equation has a solution mod every prime). This shall in fact be the key property to detecting whether p has a square-root mod q. This question in a slightly more general framework where $K = \mathbb{Q}(\sqrt{m})$ where m is square-free. To start off, we have found the integral basis of the ring of integers of quadratic extension in 1.1.6[4], namely we have integral basis

$$\begin{array}{cc} \{1, \sqrt{m}\} & m \not\equiv 1 \mod 4 \\ \left\{1, \dfrac{1+\sqrt{m}}{2}\right\} & m \equiv 1 \mod 4 \end{array}$$

and $\operatorname{disc}(\mathcal{O}_K) = 4m$ in the first case and $\operatorname{disc}(\mathcal{O}_K) = m$ in the second. Now, take $(p) \subseteq \mathbb{Z}$ a prime ideal. Then we know that either p can split as:

$$p\mathcal{O}_K = \mathfrak{B} \qquad p\mathcal{O}_K = \mathfrak{B}^2 \qquad p\mathcal{O}_K = \mathfrak{B}_1\mathfrak{B}_2$$

and that only m and possibly 2 ramifies.

Theorem 1.4.72: Primes In Quadratic Extensions

Let $p \subseteq \mathbb{Z}$ be a prime ideal and $K = \mathbb{Q}(\sqrt{m})$ a quadratic extension where m is square-free. Then:

1. If $p\big|m$:
$$p\mathcal{O}_K = (p, \sqrt{m})^2$$
2. if m is odd, then 2 splits as:
$$2\mathcal{O}_K = \begin{cases} (2, 1+\sqrt{m})^2 & m \equiv 3 \mod 4 \\ \left(2, \frac{1+\sqrt{m}}{2}\right)\left(2, \frac{1-\sqrt{m}}{2}\right) & m \equiv 1 \mod 8 \\ (2) & m \equiv 5 \mod 8 \end{cases}$$
and if p is odd and $p \nmid m$:
$$p\mathcal{O}_K = \begin{cases} (p, n+\sqrt{m})(p, n-\sqrt{m}) & m = n^2 \mod p \\ (p) & m \neq n^2 \mod p \end{cases}$$

***Proof*:**

1. Assume $p|m$. Then as it divides the discriminant, it certainly ramifies. More concretely, note this implies $m \equiv 0 \bmod p$ so $\sqrt{m} \equiv 0 \bmod p$, and $(p, \sqrt{m})$ is certainly prime by corollary 1.2.19. With this, it can be verified by hand that $(p, \sqrt{m})^2 = (p^2, p\sqrt{m}, m)$ and that (since $p|m$):
$$(p, \sqrt{m})^2 \subseteq (p)_K$$

2. let us first deal with the single prime $p = 2$ where $p \nmid m$ so that m is odd.

   (a) If $m \equiv 4 \bmod 4$, the discriminant is $4m$, hence 2 ramifies. Then by direct computation it can be verified that:
   $$(2, 1 + \sqrt{m})^2 = (2)_L$$
   giving us the splitting prime.

   (b) Assume $m \equiv 1 \pmod 8$. Then $m \equiv 1 \bmod 4$, so $\mathcal{O}_K = \mathbb{Z}\left[\frac{1+\sqrt{m}}{2}\right] = \mathbb{Z}[\omega]$. Then ω satisfies $\omega^2 - \omega + \frac{1-m}{4} = 0$. Then as $m \equiv 1 \pmod 8$, we have $\frac{1-m}{4} \equiv 0 \pmod 2$, so
   $$\omega^2 \equiv \omega \pmod 2$$
   The two prime ideals above 2 are generated by the two roots of $x^2 - x \equiv 0 \pmod 2$, which are $x \equiv 0$ and $x \equiv 1 \pmod 2$. Thus, the two prime ideals are $(2, \omega)$ and $(2, \omega - 1)$ (the reader can proof these are distinct primes)

   (c) Assuming $m \equiv 5 \bmod 8$, a similar process shows that 2 is inert.

   Now let p be any odd prime and $p \nmid m$. Note that as $p \nmid m$, $p \nmid \mathrm{disc}(\mathcal{O}_K) \in \{4m, m\}$, hence p will not ramify. Thus by corollary 1.4.8 The behavior of p in $\mathcal{O}_K$ depends on whether $f(x)$ factors over $\mathbb{F}_p$.

   (a) Assume $m \equiv n^2 \bmod p$ so that x^2-m has two roots in $\mathbb{F}_p$. Then $x^2-m \equiv (x-n)(x+n) \pmod p$ In this case, we can show that p splits as:
   $$p\mathcal{O}_K = \mathfrak{p}_1\mathfrak{p}_2$$
   where $\mathfrak{p}_1 = (p, \sqrt{m} - n)$ and $\mathfrak{p}_2 = (p, \sqrt{m} + n)$. Indeed, $\mathcal{O}_K/\mathfrak{p}_1 \cong \mathbb{F}_p$ since $\sqrt{m} \equiv n \pmod{\mathfrak{p}_1}$ (Similarly for $\mathfrak{p}_2$). Then it is not too hard to notice they are distinct; notice that $\mathfrak{p}_1 + \mathfrak{p}_2 = (p, 2n) = \mathcal{O}_K$ (when $p \neq 2$)

   (b) Assume $m \not\equiv n^2 \bmod p$. Then $x^2 - m$ is irreducible over $\mathbb{F}_p$, so $\mathcal{O}_K/p\mathcal{O}_K \cong \mathbb{F}_{p^2}$ is a field, and so by the fundamental identity (p) must remain innert.
   This means $p\mathcal{O}_K$ is a prime ideal, so p remains inert.

Coming back to the question of the existence of quadratic reciprocity, the following notation shall be useful:

Definition 1.4.73: Legendre Symbol

Let p be an odd prime in $\mathbb{Z}$. Then for $n \in \mathbb{Z}$, $p \nmid n$, the *Legendre symbol* is defined to be:

$$\left(\frac{n}{p}\right) = \begin{cases} 1 & \exists m,\ m^2 \equiv n \bmod p \\ -1 & \text{otherwise} \end{cases}$$

The intuition behind the notation is that if n is the square mod p, it is "divisible" in mod p, motivating the $\left(\frac{n}{p}\right)$ notation.

Theorem 1.4.74: Quadratic Reciprocity

Let p, q be two distinct odd prime numbers. Then:

$$\left(\frac{q}{p}\right)\left(\frac{p}{q}\right) = (-1)^{\frac{q-1}{2}\frac{p-1}{2}}$$

In other words, if:

$$p^* = \begin{cases} 1 & p \equiv 1 \bmod 4 \\ -p & p \equiv -1 \bmod 4 \end{cases}$$

Then q is a quadratic residue modulo p if and only if p' is a quadratic residue modulo p

Quadratic reciprocity can also be interpreted as:

$$\left(\frac{2}{p}\right) = \begin{cases} 1 & p \equiv \pm 1 \bmod 8 \\ -1 & p \pm 3 \bmod 8 \end{cases}$$

and for odd $p \neq q$:

$$\left(\frac{p}{q}\right) = \begin{cases} \left(\frac{p}{q}\right) & p \text{ or} q \equiv 1 \bmod 4 \\ -\left(\frac{p}{q}\right) & p \equiv q \equiv 3 \bmod 4 \end{cases}$$

***Proof*:**
Let $q \neq p$ be odd primes. We want to show that

$$n^2 \equiv q \bmod p \qquad \text{if and only if} \qquad m^2 \equiv p' \bmod q$$

for some numbers $n, m \in \mathbb{Z}$. By theorem 1.4.72 it suffices to show hat q splits if and only if $m^2 \equiv p' \bmod q$. Let's first say that q splits, so we have $\mathfrak{B}_1, \mathfrak{B}_2$ lying above q, and $(q)_L \subsetneq \mathfrak{B}_1, \mathfrak{B}_2$. Since $p' \equiv 1 \bmod 4$, we know the number ring of L is:

$$\mathbb{Z}\left[\frac{1+\sqrt{p'}}{2}\right]$$

Hence, we have that there is an element

$$a + b\sqrt{p'} \in \mathfrak{B} - (q) \qquad a, b \in \frac{1}{2}\mathbb{Z} \tag{1.6}$$

and since q is odd, we may without loss of generality choose a, b to be in $\mathbb{Z}$ by multiplying by 2. Now, Consider the norm of the element:

$$(a + b\sqrt{p'})(a - b\sqrt{p'}) = a^2 - b^2 p' \in \mathfrak{B} \cap \mathbb{Z} = (q)$$

Hence, $q \big| a^2 - b^2 p'$. Notice that q cannot divide a or b by our choice in equation (1.6). Hence, get that:

$$a^2 - b^2 p' \equiv 0 \bmod q$$

re-arranging (and dividing since b is invertible), we get:

$$(ab^{-1})^2 \equiv p' \bmod q$$

showing one direction of the implication.

Now, for the other direction, it suffices that q splits in L. By assumption:

$$m^2 \equiv p' \bmod q \qquad \Leftrightarrow \qquad m^2 - p' = (m - \sqrt{p'})(m + \sqrt{p'}) = cq$$

for some constant $c \in \mathbb{Z}$. I claim that $(m + \sqrt{p'}, q)$ divides q in L. Certainly

$$(q) \subsetneq (m + \sqrt{p'}, q)$$

since $m + \sqrt{p'} \notin (q)$ (since $m + \sqrt{p'}$ is not divisible by q). Furthermore, $(m + \sqrt{p'})$ is not a unit, for if it were, then if $\alpha = (m - \sqrt{p'})$ and $u = (m + \sqrt{p'})$:

$$u\alpha = (m + \sqrt{p'}, q)(m - \sqrt{p'}) = (x^2 - p', q) = (cq, q)$$

hence $q \big| u\alpha$, implying $q \big| \alpha$, but $q \not| m - \sqrt{p'}$. Thus, it must be that q splits.

But then,this implies that q is a quadratic residue modulo p if and only if p' is a quadratic residue modulo q, as we sought to show.

1.5 Application: Kronecker Weber Theorem

We now apply our finding to Field Theory and find some important consequences' on *Abelian Extension.* This shall also lead to proving one of the central theorems in field theory, The *Kronecker-Weber Theorem*, which states that all finite abelian extensions show up as sub-extensions of Cyclotomic extensions over $\mathbb{Q}$! Since Cyclotomic extensions are generated by roots of unity, every algebraic integers whose Galois group is abelian can be written as a $\mathbb{Q}$-linear combination of roots of unity! For example:

$$\sqrt{5} = e^{2\pi i/5} - e^{4\pi i/5} - e^{6\pi i/5} + e^{8\pi i/5}$$

namely, all square roots of integers may be written as $\mathbb{Q}$-linear combinations of roots of unity! We first give the setting in which we work in:

Definition 1.5.1: Universal Cyclotomic Field and Abelian Closure

1. Let $K^{\mathbf{cycl}} = \bigcup_{n\in\mathbb{N}} K(\zeta_n)$. If $K = \mathbb{Q}$, then:
$$\mathrm{Gal}(\mathbb{Q}^{\mathbf{cycl}}) = \varprojlim \mathrm{Gal}(\mathbb{Q}(\zeta_n)) = \varprojlim (\mathbb{Z}/n\mathbb{Z})^\times = \hat{\mathbb{Z}}^\times$$
is the pro-finite group given by the limit of units of cyclic groups.
2. Let K be a field. Then
$$K^{\mathbf{ab}} \subseteq \overline{K}$$
is the maximal extension for which $\mathrm{Gal}_K(K^{\mathbf{ab}})$ is abelian.

Proposition 1.5.2: Properties of Abelian Extensions

1. If $L_1, L_2 \subseteq K^{\mathrm{ab}}$ have finite abelian groups, then $\mathrm{Gal}_K(L_1L_2)$ is a finite abelian group.
2. If $G_K = \mathrm{Gal}_K(\overline{K})$ and $H = [G, G]$ (the commutator [closed] subgroup), then
$$K^{\mathrm{ab}} = \overline{K}^H$$
with Galois group G_K/H.

***Proof*:**

1. See [5, Galois Composite Fields II]
2. This comes from the universal property of abelian quotients, see [5, Commutator Subgroup Properties]

Note that in general, $K^{\mathbf{cycl}} \subsetneq K^{\mathrm{ab}}$:

Example 1.5.3: Abelian, But Not Cyclotomic

Let $k = \mathbb{Q}(i)$ and $F = \mathbb{Q}(\sqrt{1+2i})/\mathbb{Q}(i)$. Since every quadratic extension is abelian, F is an abelian extension. However, it is not a cyclotomic extension: for if it was a sub-cyclotomic extension, then since $\mathbb{Q}(i)$ is a cyclotomic extension over $\mathbb{Q}$, then $\mathbb{Q}(\sqrt{1+2i})$ would be a cyclotomic extension, but that is not possible since $\mathbb{Q}(\sqrt{1+2i})/\mathbb{Q}$ is not Galois (but all sub-extensions of a cyclotomic extension are necessarily Galois).

It is thus quite interesting that if $K = \mathbb{Q}$, we have the following:

Theorem 1.5.4: Kronecker Weber's Theorem

Let $K = \mathbb{Q}$. Then:
$$\mathbb{Q}^{\mathbf{cycl}} = \mathbb{Q}^{\mathrm{ab}}$$
In other words, every abelian extension of $\mathbb{Q}$ is contained in a cyclotomic extension.

We'll first show we can assume $\mathrm{Gal}_{\mathbb{Q}}(K)$ is a p-group for some prime p. For if $\mathrm{Gal}_{\mathbb{Q}}(K)$ is abelian, we have

$$\mathrm{Gal}_{\mathbb{Q}}(K) \cong \bigoplus_p G_p$$

where G_p is a p-group.

Proof. **Reduction #1**:

Define $K_{p'} = K^{(\bigoplus_{p \neq p'} G_p)}$. Then $K = K_2 K_3 \cdots$ is the join of all these fields. Hence, if all such fields is inside a Cyclotomic extension, so is K. This can naturally be extended to reduce to cyclic groups in the same manner. $\square$

Next, we show that p is the *only* prime which ramifies in K

Proof. **Reduction #2**:

Assume not. Assume ℓ ramifies in K as well. Let I be the inertia group (doesn't matter what prime we pick). Then the wield inertia group W is trivial (by group theory). Then by theorem 1.4.62:

$$I \hookrightarrow \mathbb{F}_\ell^\times \qquad I \cong \mathbb{Z}/\ell\mathbb{Z}, \quad e \big| \ell - 1$$

Let $L \in \mathbb{Q}(\mu_\ell)$ be the (unique) extension such that $[L : \mathbb{Q}] = e$. So ℓ totally ramifies in L. Then take:

$$\begin{tikzcd}
 & M = KL & \\
L \arrow[ur] & & K \arrow[ul] \\
 & \mathbb{Q} \arrow[ul] \arrow[ur] &
\end{tikzcd}$$

K and L are abelian, thus so is M. Let I_M be the inertia group of a prime over ℓ. Then

$$I_M \hookrightarrow \mathbb{Z}/e\mathbb{Z} \oplus \mathbb{Z}/\ell\mathbb{Z}$$

Hence

$$I_M \cong \mathbb{Z}/e\mathbb{Z}$$

Let $K' = M^{I_M}$. Then $K' \cap L = \mathbb{Q}$ since nothing ramifies in $K' \cap L$.

Now, $K'L/K'$ has degree at least e, in particular$\geq e$, since ℓ has to ramify (ℓ doesn't ramify in K', and ramifies in L by e). But then,

$$K'L = M = KL$$

Note that we have not introduced any new ramifications, since no other primes ramify that ramify in L, K. We can afterward inductively remove (if it ramifies, it has to have some inertia group, but it will be trivial in the intermediate extensions).

With this, let's now prove the $p = 2$ case.

Assume $K/\mathbb{Q}$ is abelian $[K : \mathbb{Q}] = 2^n$ and 2 is the only prime which ramifies. I claim that K is cyclotomic.

Without loss of generality, let's assume $\mathbb{Q}(i) \subseteq K$ (just adjoin it if need be). Then $K_0 = K \cap \mathbb{R}$ (fixed field under conjugation). This is still abelian since K is abelian, and $[K : K_0] = 2$, so $K = K_0(i)$. Now, let $K_n/\mathbb{Q}$ be the real cyclotomic field where $\mathrm{Gal}(K_n/\mathbb{Q}) \cong \mathbb{Z}/2^n\mathbb{Z}$ (done it in comments). Consider now

$$M = KK_n$$

Then $\mathrm{Gal}_{\mathbb{Q}}(M) \subseteq \mathbb{Z}/2^n\mathbb{Z} \oplus \mathbb{Z}/2^n\mathbb{Z}$. If $K_n \neq K$, then this Galois group is *not* cyclic (too large order). Therefore, M has ≥ 3 real quadratic subfields (real, since everything is real). But there is only 2 with ramified primes – a contradiction ($\mathbb{Q}(\sqrt{2})$ is the only real quadratic extension in which 2 ramifies). ☐

Now, for the other primes, we have the following key proposition:

Proposition 1.5.5: Ramification Of Primes in $\mathbb{Z}/n\mathbb{Z}$-Extensions

Let p be an odd prime. Then there exists a unique field $K/\mathbb{Q}$ which is Galois with Galois group $\mathbb{Z}/p\mathbb{Z}$ in which p is the only ramified prime.

***Proof*:**
Let ζ be a pth primitive root of 1, let $L = \mathbb{Q}(\zeta)$. $\pi = 1 - \zeta$. So that $(\pi)^{p-1} = (p)$. Let $\gamma \in \mathcal{O}_L$ be a non pth power. Assume $\gamma \equiv 1 \bmod \pi^p$. Take the field $M = L(\sqrt[p]{\gamma})/L$. Then π is unramified. Suppose not. Then $\pi\mathcal{O}_M = Q^p$ for some prime Q where $\mathbb{F}_Q \cong \mathbb{F}_{(\pi)} \cong \mathbb{F}_p$. Let δ be such that $\delta^p = \gamma$. Let $r = \frac{\delta - 1}{\pi}$. Claim that $r \in \mathcal{O}_M$. Take the conjugates of r over L, i.e. they are

$$\frac{\zeta^i\delta - 1}{\pi} \qquad i \in \mathbb{Z}/p\mathbb{Z}$$

So the minimal polynomial is the product:

$$\prod\left(x - \frac{\zeta^i\delta - 1}{\pi}\right) = \frac{1}{\pi^p}\prod_{i=1}^{p-1}((\pi x - 1) - \zeta^i\delta) = \frac{(\pi x - 1)^p - \gamma}{\pi^p}$$

So this is inside $\mathcal{O}_L[x]$, so r is inside of integers. Thus, there exists some $m \in \mathbb{Z}$ such that $r - m \in \mathbb{Q}$ (because $\mathbb{F}_Q$ is just $\mathbb{F}_p$). But then for all $\sigma \in \mathrm{Gal}_K(M)$, $\sigma(r - m) \in \mathbb{Q}$ since it's the inerta group. But

$$\frac{\zeta\gamma - 1}{\pi} - m \in \mathbb{Q}$$

and so is $\frac{\gamma - 1}{\pi} \in \mathbb{Q}$. Subtracting we get

$$\frac{1 - \gamma}{\pi}\gamma = \frac{\pi}{\pi}\gamma = \gamma \in \mathbb{Q}$$

But this is a contradiction of our $\gamma \equiv 1 \bmod Q$, as we sought to show.

With this, we can continue the second reduction

Proof. **Reduction #2 cont.**:

By reduction # 2, we only need to work with $K/\mathbb{Q}$ abelian of prime power order p^n, and without loss of generality $K/\mathbb{Q}$ may be assumed to be cyclic since every field is a compositum which have cyclic subgroups.

Now consider $K_n \subseteq \mathbb{Q}(\zeta_{n+1})$ be the field such that

$$[K_n : \mathbb{Q}] = p^n \qquad \mathrm{Gal}(K_n/\mathbb{Q}) \cong \mathbb{Z}/p^n\mathbb{Z}$$

Let $M + KK_n$. In general $\mathrm{Gal}(M/\mathbb{Q}) \hookrightarrow \mathbb{Z}/p^n\mathbb{Z} \oplus \mathbb{Z}/p^n\mathbb{Z}$. If $K \neq K_n$, then like for the $p = 2$ case, this Galois group is not cyclic. Thus, M contains at least $p+1$ subfields which have Galois group $\mathbb{Z}/p\mathbb{Z}$ over $\mathbb{Q}$ in which only p ramifies, a contradiction $\square$

***Proof*:**
Of Kronecker-Weber's Theorem

We shall show that if p is an odd prime, $K/\mathbb{Q}$ is a $\mathbb{Z}/p\mathbb{Z}$ extension, and p is the only prime that ramified in K, then $K \subseteq \mathbb{Q}(\zeta_{p^2})$, which by our earlier reduction shall complete the proof.

Let $\mathrm{Gal}_{\mathbb{Q}}(M) = \langle \sigma \mid \sigma^{p(p-1)} = \mathrm{id}\rangle$, $K = M^{\langle \sigma^p \rangle}$ and $L = M^{\langle \sigma^{p-1}\rangle}$. Consider:

$$\begin{tikzcd}
& M = KL & \\
L = \mathbb{Q}(\zeta_p) \arrow[ur] & & K \arrow[ul] \\
& \mathbb{Q} \arrow[ul, "\mathbb{Z}/(p-1)\mathbb{Z}"] \arrow[uu, "\mathbb{Z}/p(p-1)\mathbb{Z}"] \arrow[ur, "\mathbb{Z}/p\mathbb{Z}"'] &
\end{tikzcd}$$

We hat that p is the only prime that ramifies in M. Thus, by Kummer theory, we know that $M = L(\alpha)$ where $\alpha^p = \beta \in L^\times$. Next, we know that

$$\frac{\sigma(\alpha)}{\alpha}$$

is a non-trivial rot of unity. Without loss of generality say that $\frac{\sigma^{p-1}(\alpha)}{\alpha} = \zeta_p$ (so $\langle \sigma^{p-1}\rangle = \mathrm{Gal}_L(M)$). Note that $\sigma(\zeta_p) = \zeta_p^n$, where $\overline{n} \in (\mathbb{Z}/p\mathbb{Z})^\times$ (i.e. is a primitive root).

I claim that

$$M = L\left(\frac{\sigma(\alpha)}{\alpha}\right)$$

For suppose not. Then $\frac{\sigma(a)}{\alpha} \in L$, and

$$\frac{\sigma(\alpha)}{\alpha} = \sigma^{p-1}\left(\frac{\sigma(\alpha)}{\alpha}\right) = \frac{\sigma^p(\alpha)}{\sigma^{p-1}(\alpha)} = \frac{\sigma(\zeta_p\alpha)}{\zeta_p\alpha} \implies 1 = \frac{\sigma(\zeta_p)}{\zeta_p}$$

but that is a contradiction, hence $M = L\left(\frac{\sigma(\alpha)}{\alpha}\right)$.

For the rest of this proof, we write

$$\alpha \mapsto \frac{\sigma(\alpha)}{\alpha} = \alpha' \qquad \beta \mapsto \frac{\sigma(\beta)}{\beta} = \beta'$$

Let P_k, P_L, P_M be the primes ideals above $p\mathbb{Z}$. Take the prime decomposition:

$$\beta\mathcal{O}_L = \prod Q^{n_Q} = P_L^r \cdot \prod_{Q\neq P} Q^{n_Q} \qquad n_Q \in \mathbb{Z}$$

Then:

$$\sigma(\beta)\mathcal{O}_L = P_L^r \prod_{Q\neq P_L} \sigma(Q)^{n_Q}$$
$$\frac{\sigma(\beta)\mathcal{O}_L}{\beta} = \prod_{Q\notin P_L} Q^{m_Q}$$

Hence, P_L does not occur in $\beta\mathcal{O}_L$. Take $\beta'' = \beta' N^P$ and $\alpha'' = \alpha' N$ for $N \in \mathbb{Z}$ such that $\beta'' \in \mathcal{O}_L$ Then:

$$M = L(\alpha'') \qquad (\alpha'')^P = \beta''$$

Now let $\pi = 1 - \zeta_p \in \mathcal{O}_L$ so that $P_L = (\pi)$. Then $\mathbb{F}_{P_L} \cong \mathbb{F}_P$ implies β'' mod $P_L = m$ mod P_L for $m \leq N$. Let $m_0 = m^{-1}$ mod p, so $\beta'' \to m_0\beta''$ and $\alpha'' \to m_0\alpha''$. Without loss of generality say $\beta'' = 1$ mod P_L. Then for some $c \in \mathbb{Z}/p\mathbb{Z}$

$$\beta'' = 1 + c\pi \bmod P_L^2 \qquad \zeta_p = 1 - \pi \bmod P_L^2 \qquad \zeta_p^r = 1 - r\pi \bmod P_L^2 \implies \beta'' = \zeta_p^{-c}\nu$$

where $\nu \equiv 1$ mod P_L^2. I claim that $\nu \in (L^\times)^P$, which will imply $M = \mathbb{Q}(\zeta_{p^2})$ and complete the proof. First:

$$\begin{aligned}
(\alpha'')^p &= \beta'' = \zeta_p^{-c}\nu \\
\sigma^{p-1}(\alpha'') &= \alpha''\zeta_p \\
\implies \sigma(\zeta_p) &= \zeta_p^n && n \in (\mathbb{Z}/p\mathbb{Z})^\times \\
\implies \sigma^p(\alpha'') &= \sigma(\sigma^{p-1}(\alpha'')) = \sigma(\alpha''\zeta_p) = \sigma(\alpha'')\cdot\zeta_p^n \\
\implies \sigma^{p-1}((\alpha'')^n) &= (\alpha'')^n\zeta_p^n \\
\implies \sigma^{p-1}\left(\frac{\sigma(\alpha'')}{(\alpha'')^n}\right) &= \frac{\sigma(\alpha'')}{(\alpha'')^n} \\
\implies \frac{\sigma(\alpha'')}{(\alpha'')^n} &\in M^{\langle\sigma^{p-1}\rangle} = L \\
\implies \frac{\sigma(\beta'')}{(\beta'')^n} &\in (L^\times)^P \\
\implies \frac{\sigma(\nu\zeta_p^{-c})}{\nu^n\zeta_p^{-nc}} &\in (L^\times)^p
\end{aligned}$$

Thus $\frac{\sigma(\nu)}{\nu^n} \in (L^\times)^p$.

Next, we claim that $\nu = 1$ mod P_L^p. Suppose not. Then let $p^M \| \nu - 1$ where $1 \leq m \leq p-1$, and $\nu = 1 + c\pi^m$ mod P_L^{m+1} (where $c \in (\mathbb{Z}/p\mathbb{Z})^\times$). Then by this:

$$\nu^n = 1 + nc\pi^m \bmod P_L^{m+1}$$

hence,

$$\sigma(\pi) = \sigma(1-\zeta_p) = 1 - \zeta_p^n = (1-\zeta_p)(1 + \zeta_p + \cdots + \zeta_p^{n-1}) = \pi n \bmod P_L^2$$

implying
$$\sigma(\pi^m) = \pi^m n^m \bmod P_L^{m+1} \implies \sigma(\nu) = 1 + cn^m\pi^m \bmod P_L^{m+1}$$
Note that $n^m \not\equiv n \bmod p$ for $n \in (\mathbb{Z}/p\mathbb{Z})^\times$. Now as a corollary:
$$\frac{\sigma(\nu)}{\nu^m} \equiv 1 + d\pi^m \bmod P_L^{m+1} \qquad d \in \mathbb{Z}/p\mathbb{Z}^\times$$
However,
$$\frac{\sigma(\nu)}{n\nu} = S^p \qquad S \in L$$
where P_L does not occur in S, and:
$$S^p \equiv 1 \bmod P_L \implies S = 1 \bmod P_L$$
Giving us $S = 1 + t$, $P_L \big| t$, so
$$S^P = 1 + pt + \cdots + t^p \equiv 1 \bmod P_L^P$$
a contradiction!

Finally, $L(\sqrt[p]{\nu})/L$ is unramified above P_L. Let
$$F = L(\sqrt[p]{\nu}),\ F \subseteq M(\zeta_{p^2}) \qquad F/Q \text{ abelian}$$
and $I \leqslant \mathrm{Gal}_{\mathbb{Q}}(\mathbb{F})$ where I is the inertia group above P, and $|I| = p-1$, so $F^I/\mathbb{Q}$ is unramified everywhere, namely $F^I = \mathbb{Q}$, so $F = L$. This implies
$$\nu \in (L^\times)^P$$
But then
$$M = \mathbb{Q}(\zeta_{p^2})$$
as we sought to show.

Let us use it in an actual example:

Example 1.5.6: Using Kronecker-Weber

Let $K/\mathbb{Q}$ be a galois extension with Galois group $\mathbb{Z}/5\mathbb{Z}$ whose discriminant is only divisible by $5, 11$, and 31. How many such number fields are there?

Proof. By Kronecker Weber, we know that K is an intermediary field of a Cyclotomic extension:
$$K \subseteq \mathbb{Q}(\zeta_n)$$
We may without loss of generailty assume n is minnimal for which this containment makes sense. Since only $5, 11, 31$, divide the discriminant, we know the discriminant of K is of the form:
$$\mathrm{disc}(\mathcal{O}_K) = 5^a 11^b 31^c$$
which also mean only $5, 11, 31$ ramify in K. Now, I claim that only $5, 11, 31$ divide n and no other number those. For the sake of contradiction, let's say $p \big| n$ but $p \notin \{5, 11, 31\}$. By minimality, we

have:

$$\begin{tikzcd} & \mathbb{Q}(\zeta_n) \\ \mathbb{Q}(\zeta_{n/p}) \arrow[ur] & K \arrow[u] \\ & \mathbb{Q} \arrow[u] \arrow[ul] \end{tikzcd}$$

Let $I_{\mathfrak{B}}$ be the inertia group (for any prime $\mathfrak{B}$ lying above p). Since $p \nmid \operatorname{disc}(\mathcal{O}_K)$, p is unramified. If p is unramified in $\mathbb{Q}(\zeta_{n/p})$, then since $\mathbb{Q}(\zeta_{n/p}) \subseteq \mathbb{Q}(\zeta_n)$, we get that $\mathbb{Q}(\zeta_{n/p}) \subseteq \mathbb{Q}(\zeta_n)^{I_p} \subseteq \mathbb{Q}(\zeta_n)$

$$K \subseteq \mathbb{Q}(\zeta_n)^{I_{\mathfrak{B}}} \subseteq \mathbb{Q}(\zeta_{n/p})$$

contradicting minimality of n. Hence, we get $n = 5^a 11^b 31^c$. Thus:

$$\operatorname{Gal}_{\mathbb{Q}}(\mathbb{Q}(\zeta_n)) \cong (\mathbb{Z}/n\mathbb{Z})^\times \cong \mathbb{Z}/4\mathbb{Z} \times (\mathbb{Z}/5^{a-1}\mathbb{Z}) \times \mathbb{Z}/10\mathbb{Z} \times (\mathbb{Z}/11^{b-1}\mathbb{Z}) \times \mathbb{Z}/30\mathbb{Z} \times (\mathbb{Z}/31^{c-1}\mathbb{Z})$$

which now become a counting argument of how many subgroups of index 5 there are, counting the number of intermediary subfields, thus completing the proof. □

1.6 *Modules Over Dedekind Domains

(Useful for alg-geo, theory itself is also really nice. Also $\mathcal{O}_K$-modules appear in the next section)

Lemma 1.6.1: all Ideal co-maximal up to Module Isomorphism

Let $I, J \subseteq R$ be fractional ideals of a Dedekind domain R. Then $I \oplus J \cong_R R$

***Proof*:**
By corollary 1.2.12, we may choose in fact choose I, J to be integral ideals. We shall show that $I + j^{-1}J = R$ for some appropriate j^{-1}, which since $j^{-1}J \cong_R J$, we have $I \oplus J \cong_R R$

In particular, we shall show that for any ideal J, there exists an element $j \in J$ such that $I + jJ^{-1} = R$ (that is, every ideal is "almost" comaximal with any fractional ideal). It is equivalent to show that $IJ = (j) = J$. By corollary 1.2.19, IJ is generated by at most 2 elements, say $IJ = (a, b)$. Then since $a \in IJ \subseteq J$, we may choose a c such that $(a, c) = J$ (see proof of corollary 1.2.19 if you're not convinced we can choose such a c). Then since $b \in J$ (since $b \in IJ$)

$$IJ + (c) = (a, b) + (c) = (a, b, c) = (a, c) = J$$

hence, $I + jJ^{-1} = R$, which by corollary 1.2.12 we get $I + J \cong_R R$, completing the proof.

Proposition 1.6.2: Ideals Are Projective

Let R be a Dedekind domain and $I \subseteq R$ an ideal. Then I is a projective R-module

***Proof*:**
By lemma 1.6.1, we may choose any J such that $I + jJ^{-1} = R$, or equivalently $IJ + (j) = J$. We shall define an isomorphism $I \oplus J \to IJ \oplus R$ via

$$(x, y) \mapsto (iy - jx, x + ky)$$

This map is certainly well-defined since $iy, jx \in IJ$, so $iy - jx \in IJ$, and it is certainly R-linear. It is bijective, since it's inverse is $IJ \oplus R \to I \oplus J$

$$(x, y) \mapsto (jx + y, ix - ky)$$

Hence, we have $I \oplus J \cong IJ \oplus R$. Finally, by corollary 1.2.18, choose J so that $IJ = (a)$ is principal. Then it is a free R-module of rank 1, and so

$$I \oplus J \cong IJ \oplus R \oplus R^2$$

as we sought to show.

Proposition 1.6.3: IBN equivalent for Dedekind Domains

Let $I_1, ..., I_n$ and $J_1, ..., J_m$ be fractional ideals of a Dedekind domain R. Then

$$I_1 \oplus I_2 \oplus \cdots \oplus I_n \cong_r J_1 \oplus J_2 \oplus \cdots \oplus J_m$$

if and only if $n = m$ and

$$I_1 I_2 \cdots I_n = (a) J_1 J_2 \cdots J_n$$

that is, the product differs by a principal ideal (they belong to the same equivalence class in the class group). In particular,

$$I_1 \oplus I_2 \oplus \cdots \oplus I_n \cong_R \underbrace{R \oplus R \oplus \cdots \oplus R}_{n-1 \text{ times}} \oplus (I_1 I_2 \cdots I_n)$$

and $R^n \oplus I \cong R^n \oplus J$ if and only if $I = (a)J$ for some $a \in K(R)$.

***Proof*:**
First, let $I, J \subseteq R$ be two fractional ideals. By lemma 1.6.1, we may choose I, J to be integral ideals that are up to R-isomorphism relatively prime: $I + J = R$. Then $I \cap J = IJ$, and $\varphi : I \oplus J \to R$ given by $(x, y) \mapsto x + y$ has kernel $I \cap J = IJ$. Then:

$$0 \to IJ \to I \oplus J \to R \to 0$$

is a short exact sequence. Since R is a free R-module, it is projective, and hence the sequence splits, giving us

$$I \oplus J = IJ \oplus R$$

Theorem 1.6.4: Classifying Torsion-Free Modules Over Dedekind Domain

Let M be a torsion-free R-module over a Dedekind domain R of rank n. Then

$$M \cong R^{n-1} \oplus I$$

where we I may be replaced with any ideal J such that $I = (a)J$ (they belong in the same element in the class group). As a consequence, any torsion-free module over a Dedekind domain is projective.

***Proof*:**
If there exists a non-trivial map $\varphi : M \to R$, then since R-submodules of R are projective, we may iteratively split M as $M \cong N \oplus I$ to get a direct sum of ideals of R, which we may then use to conclude by using proposition 1.6.3.

Since M is torsion-free of rank n, we may extend scalars we get a n-dimensional $K = K(R)$ vector space $K \otimes_R M$ where $M \hookrightarrow K \otimes_R M$ naturally embeds injectively. Let $\beta_1, \beta_2, ..., \beta_n$ be a basis for $K \otimes_R M$ and $m_1, m_2, ..., m_k$ be a set of generators for M. Then

$$m_i = \sum_j (a_{ij}/b_{ij})\beta_j$$

Let $d = \prod_{ij} b_{ij}$ and $y_j = \beta_j/d$. Then

$$M = Ry_1 + \cdots + Ry_n \subseteq K\beta_1 + \cdots + K\beta_n$$

Hence, M is a R-submodule of a free R-module! Importantly, for each $m \in M$, it can be written as

$$m = \sum a_i y_i \qquad a_i \in R$$

hence, we may define $\varphi : M \to R$ by $\varphi(m) = a_n$. Then $\varphi(M) \subseteq R$ is an ideal of R, and so

$$M = \ker(M) \oplus I$$

Notice that $\mathrm{Tor}(\ker(M)) = 0$, and so we may inductively repeat this procedure until we have products of prime ideals!

since the product of projective module is projective, any torsion-free module over a Dedekind domain is projective, as we sought to show.

1.7 *Geometric Idea: 1-Dimensional Schemes

See Neukirch, and put orders here too for their geometric interpretation. I think it is important to include (especially once this chapter becomes its own book) since (to me) this explains why ramification happens. We naturally can't prove the Riemann-Hurwitz Formula here, but this still is very interesting.

1.7.1 Localization of Dedekind Domains

For the motivation to CFT

1.7.2 Orders: Curves with Singularities

This entire section is completely inspired by Neukirch p.72

should also include how if finite L/K then $\operatorname{Spec}(\mathcal{O}_L)$ is the normalization of $\operatorname{Spec}(\mathcal{O}_K[\alpha])$ where $L = K(\alpha)$. And I would really love a discussion linking $\operatorname{Spec}(\mathcal{O}_L)$ to the "generic point" $\operatorname{Spec}(L)$ and to $y - m_{\alpha,K}(x)$ where $m_{\alpha,K}(x)$ is the minimal polynomial of L/K

All smooth 1-dimensional curves correspond to Dedekind domains. However, there are some curves with singularities. These curves are associated to sub-rings of Dedekind domains called *orders*

Definition 1.7.1: Order

Let $K/\mathbb{Q}$ be a number field of degree n. Then an order of K is a subring of $\mathcal{O}_K$ which contains an integral basis of length n. The ring $\mathcal{O}_K$ is called a *maximal order*.

By the above definition, an order is of the form:

$$\mathbb{Z}[\alpha_1, ..., \alpha_n]$$

where $K = \mathbb{Q}(\alpha_1, ..., \alpha_n)$, which motivates why $\mathcal{O}_K$ is the maximal order; it is the maximal ring with this property. This also implies that orders need not be normal, hence an order is a one-dimensional Noetherian integral domain (prime remain maximal as if $a \in \mathfrak{p} \cap \mathbb{Z}$ then $a\mathcal{O} \subseteq \mathfrak{p} \subseteq \mathcal{O}$ that is $\mathfrak{p}$ and $\mathcal{O}$ have the same rank n, and thus $\mathcal{O}/\mathfrak{p}$ is a finite integral domain, showing $\mathfrak{p}$ is maximal) . An example of an order is $\mathbb{Z} = \mathbb{Z}(\sqrt{5})$. Many order's naturally are as "rings of multipliers. For example if $\alpha_1, .., \alpha_n$ is a basis for $K/\mathbb{Q}$ and $M = \sum_i \mathbb{Z}\alpha_i$, then

$$\mathcal{O}_M = \{\alpha \in K \mid \alpha M \subseteq M\}$$

is an order.

It is important that we re-prove the Chinese remainder

Here

As orders are not Dedekind domain in general, *not* all ideals are invertible. This also means that the collection of fractional ideals does not form a group.

Proposition 1.7.2: Invertible Fracitonal Ideals

Let I be a fractional ideal of a Noetherian domain R. Then I is invertible iff its localization at every maximal ideal of R is invertible, equivalently, iff its localization at every prime ideal of R is invertible.

Proof:
MIT notes p.17

Example 1.7.3: non-invertible ideal of an order
put this example here

(I want this build up here)

Definition 1.7.4: Picard Group Of An Order

Let $J(\mathcal{O})$ be the collection of invertible ideals and $P(\mathcal{O})$ the fractional principal ideals. Then

$$\operatorname{Pic}(\mathcal{O}) = \frac{J(\mathcal{O})}{P(\mathcal{O})}$$

A lot more to be said.

The Picard group ignore s ideals that are not invertible. Another approach that gives some useful results is to define the following for A Dedekind domain:

$$\operatorname{Div}(\mathcal{O}) = \mathcal{O}_{\mathfrak{p}}\mathbb{Z}\mathfrak{p}$$

Then $\operatorname{Div}(\mathcal{O})$ is isomorphic to the ideal group. For an order, the divisor group shall be a quotient of this group. We would then need to define the divisor elements, which gives the divisor class group/chow group.

1.8 *Profinite Groups

(original title: infinite galois extensions)

I moved this from the galois theory chapter. I actually want to make this its own chapter when I extract these notes into their own book so that the many books that use profinite groups can reference this namely: the class field theory book, the p-adic rep book, and the galois cohomology book.

In this section, we shall explore the case of infinite separable normal extensions L/K. This section shall require the notion of topological groups and topological completion, which is covered in [4, Basic Topology and Topological Groups]. Most of this material will be important for a course in *Class Field Theory* (see cite:HERE for a reference where this material is used).

In this section, a Galois extension is a normal separable extension, where the finiteness condition is dropped. We may want to keep the galois correspondence between the fixed fields and the subgroups, however there are simply way too many subgroups. Consider

$$\overline{\mathbb{Q}}/\mathbb{Q} \qquad \text{and} \qquad \operatorname{Gal}_{\mathbb{Q}}(\overline{\mathbb{Q}})$$

Then the reader should check that there are only countably many intermediary extension of degree 2 corresponding to the extensions of the form $\mathbb{Q}(\sqrt{\alpha})/\mathbb{Q}$ where α is either a prime p, ip, or a finite compositum of such elements. However, there are uncountably many index two subgroups. To see this, consider $\varphi_S : \operatorname{Gal}_{\mathbb{Q}}(\overline{\mathbb{Q}}) \to \mathbb{Z}/2\mathbb{Z}$ where S is a finite collection of primes where if σ fixes each $\sqrt{p}$. Then $\varphi_S(s) = 0$, and it is 1 otherwise. This is a group-homomorphism, as the composition of two maps that fix each $\sqrt{p}$ for $p \in S$ shall still fix those primes. Furthermore, There are certainly always morphisms τ such that $\varphi_S(\tau) = 1$ for each finite subset S. Thus, each φ_S is a surjective homomorphism, whose kernel is index 2. But then, there are uncountably distinct $\ker(\varphi_S)$. In fact, we don't even need S to be finite (exercise). In this case, we shall even have index two subgroups whose fixed field corresponds to $\mathbb{Q}$!

The reader may feel that there is some simple solutions that can be found, as it seems there are many galois subgroups corresponding to an intermediary. Indeed, we may still recover the galois correspondence if we appropriately limit which subgroups of $\mathrm{Gal}_{\mathbb{Q}}(\overline{\mathbb{Q}})$ we consider (and more generally which subgroups of $\mathrm{Gal}_K(L)$ we consider). This is done by taking the *Krull topology* on the galois group, which we shall naturally derive by first taking the *profinite topology*. The profinite topology shall make us limit our attention to normal subgroups of finite index (namely, such subgroups shall be open in the profinite topology), while in the Krull topology we shall limit our attention to normal subgroups of finite index which also have a [finite] galois group (namely, such subgroups shall be open in the Krull topology). Both of these topologies are achieved by showing that the infinite galois group, as a topological group, is a profinite group. The main idea is that all of the information of infinite galois extension is in fact fully captured by the finite normal extension, due to the fact galois groups represent the relationship of polynomials (where an infinite galois groups stores the information of infinitely many polynomials).

1.8.1 Profinite Groups

In this section, a brief introduction to profinite groups is given. This is not an in-depth coverage, but rather it gives enough information to describe infinite galois theory.

Definition 1.8.1: Profinite Group

Take an inverse system of finite groups with the discrete topology. Then a *profinite group* is a topological group that is the limit of the system. Given any topological group G, we can take its *profinite completion* by taking:

$$\widehat{G} = \varprojlim_N G/N$$

where N ranges over all finite index open normal subgroups ordered by containment.

Any group G can be given a topology by given it the profinite topology given by taking the cosets of finite-index normal subgroups as a basis. The reader should check that if G is a topological group and $\widehat{G}$ is its profinite completion, then the continuous homomorphism $\varphi : G \to \widehat{G}$ can be shown that G is dense in $\widehat{G}$.

Example 1.8.2: Profinite Groups

1. Let G be finite. Then all normal subgroups have finite index and there are finitely many. It follows that $G \cong \widehat{G}$.

2. Let $G = \mathbb{Z}$. Then
$$\widehat{Z} = \varprojlim_n \mathbb{Z}/n\mathbb{Z} \cong \prod_p \mathbb{Z}_p$$
where $\mathbb{Z}_p = \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$ is the p-adic completion of $\mathbb{Z}$.

3. Take $G = \mathbb{Q}$. Then $\mathbb{Q}$ has only one finite-index normal subgroup, namely itself. Hence $\widehat{\mathbb{Q}} = \{1\}$.

The above examples show that the map $\varphi : G \to \widehat{G}$ could be injective, surjective, both. The map will be injective if the intersections of all finite-index open normal subgroups of G is trivial (where we would then say G is residually finite). The completion comes equipped with a natural universal property where a continuous homomorphism $\varphi : G \to H$ with H a profinite group has a unique continuous homomorphism such that the following diagram commutes:

$$\begin{tikzcd} G \arrow[r, "\varphi"] \arrow[dr, "\varphi"'] & \widehat{G} \arrow[d, "\exists!"] \\ & H \end{tikzcd}$$

A word on strongly complete? MIT notes p.275 Remark 26.15

Theorem 1.8.3: Characterizing Profinite Groups

A topological group is profinite if and only if it is a totally disconnected compact group

***Proof*:**
exercise: Compactness comes from Tychnoff's Theorem, while totally disconnected shall come from the discreteness in each finite quotient

Corollary 1.8.4: Distinguishing Profinite Completion

Let G be a profinite group. Then G is naturally isomorphic to its profinite completion, or even stronger:

$$G \cong \varprojlim G/U$$

where U ranges over all open normal subgroups ordered by contain need.
If G is isomorphic to its profinite completion *as a group* (i.e. is strongly complete) if and only if every finite index subgroup is open

***Proof*:**
MIT notes p.276

1.8.2 Infinite Galois Theory

Let us now find which topology to put on galois groups. Let us first make sure that the notion of there being a correspondence between subfields and fixed fixed fields of subgroups of galois group is still well-defined

Lemma 1.8.5: Fixed Field Infinite Extension

Let L/K be a galois extension with galois group $G = \text{Gal}_K(L)$. Then if F/K is a normal subextension of L/K, then $\text{Gal}_F(L) = H \trianglelefteq G$ is a normal subgroup of G wit fixed field F. As the subgroup is normal,we have the short exact sequence:

$$1 \to \text{Gal}_F(L) \to \text{Gal}_K(L) \to \text{Gal}_K(F) \to 1$$

which implies:

$$G/H \cong \text{Gal}_K(F)$$

***Proof*:**
As F/K is normal, the map $\sigma \mapsto \sigma|_F$ is a well-defined surjective map (as every $\tau \in \text{Gal}_K(F)$ can be extended to an element $\sigma \in \text{Gal}_K(L)$, see exercise ref:HERE) and has kernel $\text{Gal}_F(L)$ by definition. Then by definition $F \subseteq L^H$, and $L^H \subseteq F$ since if $\alpha \in L^H \setminus F$, then we can construct an element of H that sends α to a distinct root $\alpha' \neq \alpha$ of its minimal polynomials f over F, completing the proof.

Hence, every normal subextension has (at least one) corresponding normal subgroup. However, there are many normal subgroups whose fixed field F is *not* equal to $\text{Gal}_F(L)$, but is contained in $\text{Gal}_F(L)$ The index two example of $\text{Gal}_{\mathbb{Q}}(\overline{\mathbb{Q}})$ is the prototypical example, where there are uncountably many index 2 subgroups $H \leqslant \text{Gal}_{\mathbb{Q}}(\overline{\mathbb{Q}})$ whose fixed field is $\mathbb{Q}$ but naturally H is not the galois subgroup of $\overline{\mathbb{Q}}/\mathbb{Q}$, nor is it the galois group of $\overline{\mathbb{Q}}/K$ for any subextension $K/\mathbb{Q}$. Hence, the following topology limits our attention to the right groups:

Definition 1.8.6: Krull Topology On Galois Group

Let L/K be a Galois (i.e. normal separable) extension with $G = \text{Gal}_K(L)$. Then the *Krull Topology* on G is the basis given by the basis consisting of cosets of the subgroups

$$H_F = \text{Gal}_F(L)$$

where F ranges over finite normal extensions of K in L.

In this topology, every open normal subgroups has finite index, but not every normal subgroup of finite index is open (think $\text{Gal}_{\mathbb{Q}}(\overline{\mathbb{Q}})$ again). Hence, this topology is (strictly) coarser than the porfinite topology, however it is still a profinite group:

Proposition 1.8.7: Galois Group is Profinite Group

Let L/K be a galois extension. Then under the Krull topology, the restriction maps induce a natural isomorphism of topological groups:

$$\mathrm{Gal}_K(L) \cong \varprojlim_F \mathrm{Gal}_K(F)$$

where F ranges over finite galois extension of K in L. Hence, $\mathrm{Gal}_K(L)$ is a profinite group whose open normal subgroups are those of the form $\mathrm{Gal}_F(L)$ for some finite normal extension F/K.

***Proof*:**
Let's first show the map is bijective. For injectivity, for every $\alpha \in L$ it is algebraic over K, and hence liens ins some finite normal subextensions F/K, and so every automorphism in $\mathrm{Gal}_K(L)($ is uniquely determined by all its restrictions to finite normal F/K, showing φ is injective. For surjectivity, given any $(\sigma_F) \in \varprojlim_F \mathrm{Gal}_K(F)$, we may define a corresponding $\sigma \in \mathrm{Gal}_K(L)$ that maps to it by point-wise defining:

$$\sigma(\alpha) := \sigma_F(\vec{\alpha})$$

where F is the normal closure of $K(\alpha)$. This map is well-defined by construction of the inverse limit, giving surjectivity.

For continuity, Letting $G = \mathrm{Gal}_K(L)$ and $H_F = \mathrm{Gal}_F(L)$, we can view φ as the natural continuous map:

$$\varphi : G \to \varprojlim G/H_F$$

Finally, to show the inverse is continuous, we'll show the equivalent of φ being an open map, which translates to showing that φ maps open subgroups $H \subseteq G$ (i.e. the basis) to open subsets in $\varprojlim G/H_F$. Taking $H = \mathrm{Gal}_F(L)$, we get:

$$\varphi(H) = \{(\sigma_E) \mid \sigma_{E_{E\cap F}} = \mathrm{id}_{E\cap F}\} = \pi_F^{-1}(\mathrm{id}|_F)$$

as E/K ranges over all finite normal subextension of L/K and π_F is the projection from the limit to $\mathrm{Gal}_K(F)$. Then the singleton set $\{\mathrm{id}|_F\}$ is open in the discrete group $\mathrm{Gal}_K(F)$, and so its inverse is open, completing the proof.

Theorem 1.8.8: Fundamental Theorem of Galois Theory, Infinite Extensions

Let L/K be a galois extension with galois group $G = \mathrm{Gal}_K(L)$ endowed with the Krull topology. Then the maps

$$F \mapsto \mathrm{Gal}_F(L) \qquad H \mapsto L^H$$

define inclusion reversing bijections between subextensions F/K of L/K and closed subgroups H of G. Thus, subextensions of finite degree n correspond to subgroups of finite index n, and normal subextensions F/K correspond to normal subgroups $H \leqslant G$ such that as topological groups:

$$\mathrm{Gal}_K(F) \cong G/H$$

***Proof*:**
First off, recall that every open subgroup is clopen, as the compliment is the union of non-trivial cosets, and so closed subgroups of finite index are open. Next, the correspondence between finite galois subextensions F/K and finite index closed normal subgroups H is given by proposition 1.8.7, and furthermore we have $[F : K] = [G : H]$ as $G/H \cong \mathrm{Gal}_K(F)$ by lemma 1.8.5.

For the correspondence maps, if F/K is any finite subextension with normal closure E, then $H = \mathrm{Gal}(L/F)$ contains the normal subgroup $N = \mathrm{Gal}(L/E)$ with finite index. The subgroup N is open and therefore closed, thus H is closed since it is a finite union of cosets of N. The fixed field of H is F (by the same argument as in the proof of lemma 1.8.5), thus finite subextensions correspond to closed subgroups of finite index. Conversely, every closed subgroup H of finite index has a fixed field F of finite degree, since the intersection of its conjugates is a normal closed subgroup $N = \mathrm{Gal}(L/E)$ of finite index whose fixed field E contains F and has finite degree. The degrees and indices match because $[G : N] = [G : H][H : N]$ and $[E : K] = [F : K][E : F]$; by the previous argument for finite normal subextensions, $[E : K] = [G : N]$ and $[E : F] = [H : N]$ (for the second equality, replace L/K with L/F and G with H).]

Next, for the closed part, any subextension F/K is the union of its finite subextensions E/K. The intersection of the corresponding closed finite index subgroups $\mathrm{Gal}(L/E)$ is equal to $\mathrm{Gal}(L/F)$, which is therefore closed. Conversely, every closed subgroup H of G is an intersection of basic closed subgroups, all of which have the form $\mathrm{Gal}(L/E)$ for some finite subextension E/K, thus $H = \mathrm{Gal}(L/F)$, where F is the union of the E, completing the proof.

Finally, we see that the isomorphism $\mathrm{Gal}_K(F) \cong G/H$ of topological groups follows from lemma 1.8.5, completing the proof.

Corollary 1.8.9: Closure in Krull Topology

Let L/K be a galois extension and $H \leqslant \mathrm{Gal}_K(L)$ with fixed field F. Then the closure $\overline{H}$ in the Krull Topology is $\mathrm{Gal}_F(L)$:

$$\overline{H} = \mathrm{Gal}_F(H)$$

***Proof*:**
First, $H \subseteq \mathrm{Gal}_F(L)$ since by definition $\mathrm{Gal}_F(L)$ contains every $\sigma \in \mathrm{Gal}_K(L)$ that fixes F . Next, $\mathrm{Gal}_F(L)$ is closed and is the group characterized by $L^{\mathrm{Gal}_F(L)} = F$ by theorem 1.8.8. Thus

$$H \subseteq \overline{H} \subseteq \mathrm{Gal}_F(L)$$

Then in the bijection presented in theorem theorem 1.8.8, we have that they are equal, completing the proof.

Let us conclude with a result proven by Waterhouse in 1973 that generalizes [5, Finite Group Represented by Galois Group]:

Theorem 1.8.10: Profinite Group Representable By Galois Group

Let G be a profinite group. Then G is isomorphic to the galois group of some galois extension L/K

> ***Proof*:**
> MIT notes p.279

Just like for finite groups where $K = \mathbb{Q}$ is a difficult (as of writing this book unsolved) problem called the *inverse Galois Problem*, there is little control on what base field K we can choose.

Exercise 1.8.1

1. Find $G = \mathrm{Gal}_{\mathbb{F}_p}(\overline{\mathbb{F}}_p)$ recalling that for each degree n, there is only one extension of $\mathbb{F}_p$. Show that there are too many subgroups to correspond to the fixed fields. Define the Krull topology on G.